Fruit Market · Decision register

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Approval status and delivery status are separate. Only the owner's explicit words set "approved".

Pending approvals

ID	What	Status
PLAN-003	Package C, paid part: capacity and cost evidence	Proposed
PLAN-018	Data insights: balance, retention and online health from anonymous counters	Proposed

Current decisions

ID	Decision	Approval	Delivery	Reports
DEC-001	Adopt the personal workflow guide 1.2 for Fruit Market	Approved (owner, 2026-10-27)	In progress	REP-001, REP-038
DEC-002	Package A scope = PLAN-001	Approved (owner, 2026-10-26)	Verified	REP-001, REP-002
DEC-003	Live deploys only through the manual Deploy workflow after verify	Approved (owner, 2026-10-26)	Verified	REP-001, REP-002
DEC-004	PDF exports with pandoc and the existing Playwright browser	Approved (owner, 2026-10-27)	Implemented	REP-001, REP-038
DEC-005	Budget interpretation: RM500 per month across all games	Approved (owner, 2026-10-27)	Not started	REP-001, REP-038
DEC-006	Preview = separate staging Worker	Approved (owner, 2026-10-26)	Verified	REP-002, REP-003
DEC-007	Package B scope = PLAN-002	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-008	Canonical record format: JSON records plus Markdown narrative reports	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-009	Records check runs in verify and CI	Approved (owner, 2026-10-26)	Verified	REP-003, REP-004
DEC-010	Reconnect fix scope = PLAN-004 (evidence first, then fix)	Approved (owner, 2026-10-26)	Verified	REP-005, REP-006
DEC-011	Stress test runs on demand, not on every push	Approved (owner, 2026-10-26)	Verified	REP-005, REP-006
DEC-012	4-player clarity scope = PLAN-005 (screen text only)	Approved (owner, 2026-10-26)	Verified	REP-007, REP-008
DEC-013	Publish version 1.1.1 to players	Approved (owner, 2026-10-26)	Released	REP-009, REP-013

ID	Decision	Approval	Delivery	Reports
DEC-014	Use the Concept and Theme Companion for creative work	Approved (owner, 2026-10-27)	In progress	REP-010, REP-038
DEC-015	Adopt three design pillars	Approved (owner, 2026-10-26)	Implemented	REP-010, REP-011, REP-013
DEC-016	First creative test: reveal-comprehension playtest	Approved (owner, 2026-10-26)	Not started	REP-010, REP-011, REP-013
DEC-017	Setting: a market where the whole world shops	Approved (owner, 2026-10-26)	Implemented	REP-011
DEC-018	Rename Panda Lim while keeping the cast coherent	Superseded	Verified	REP-011, REP-012
DEC-019	Rename scope = PLAN-006, shipped in 1.1.2	Approved (owner, 2026-10-26)	Verified	REP-011, REP-012
DEC-020	UI and UX polish in three phases	Superseded	Not started	REP-014, REP-015
DEC-021	Art direction: B, toy-box tabletop	Approved (owner, 2026-10-26)	In progress	REP-015, REP-016
DEC-022	Look and tone: cunning edition of direction B	Approved (owner, 2026-10-27)	Implemented	REP-017
DEC-023	Look: cosy pixel edition of direction B	Approved (owner, 2026-10-27)	Implemented	REP-018, REP-020
DEC-024	Layout: one shared market table (board game, not menus)	Approved (owner, 2026-10-27)	Implemented	REP-019, REP-020
DEC-025	Market board: vertical decay tracker and a delivery lorry	Approved (owner, 2026-10-27)	Implemented	REP-021
DEC-026	Freeze rules version 4 until the first human playtest round is done	Superseded	Not started	REP-038
DEC-027	One game, two editions: tabletop and digital with identical mechanics	Approved (owner, 2026-10-27)	Not started	REP-038
DEC-028	Lift the rules freeze; plan fixes for supply, session length and UI requests	Approved (owner, 2026-10-27)	Implemented	REP-047
DEC-029	Supply and demand: option A + C (rules version 5)	Approved (owner, 2026-10-27)	Implemented	REP-050
DEC-030	Tabletop fixes: tie-break by buying order, and table procedures (no price ladder)	Approved (owner, 2026-10-28)	Implemented	REP-051
DEC-031	Character cards: pick one of six characters on entering a room or the set-up screen	Approved (owner, 2026-10-28)	Implemented	REP-052
DEC-032	Customer deck grows from 24 to 36 cards	Approved (owner, 2026-10-28)	Implemented	REP-053
DEC-033	Art direction: Look C, calm flat world with pixel objects	Superseded	Not started	REP-054
DEC-034	Board look: which of four looks the rest of the game follows	Approved (owner, 2026-10-28)	In progress	REP-056

ID	Decision	Approval	Delivery	Reports
DEC-035	Naming: adopt the names in Naming_Audit	Approved (owner, 2026-10-28)	Not started	REP-057
DEC-036	Replace Uncle Toad with Mak Jenab	Superseded	Not started	REP-061
DEC-037	Jenab is a customer: Auntie Jenab replaces Chris the Clown	Superseded	Implemented	REP-062
DEC-038	Rara Jenab the Cheerleader replaces Chris the Clown	Approved (owner, 2026-10-28)	Implemented	REP-063
DEC-039	Cast theme: one coherent theme for every character	Approved (owner, 2026-10-28)	Implemented	REP-063
DEC-040	Shopkeepers become humans (replaces the animal rivals rule)	Approved (owner, 2026-10-28)	Implemented	REP-063
DEC-041	Cast coherence rules (creative bible) approved	Approved (owner, 2026-10-28)	Implemented	REP-066
DEC-042	Order of work, the lorry direction, no portrait frames, and how design work is approved	Approved (owner, 2026-10-29)	In progress	REP-069
DEC-043	Table screen design approved (look D)	Approved (owner, 2026-10-29)	Implemented	REP-071
DEC-044	PLAN-021 (session length) cancelled	Approved (owner, 2026-10-29)	Implemented	REP-073
DEC-045	Other screens and printed mat in look D approved, results show stats and best moments	Approved (owner, 2026-10-29)	Released	REP-074
DEC-046	Overnight work: scope and preview-push authority	Approved (owner, 2026-10-29)	In progress	REP-075, REP-076, REP-077
DEC-047	Opening animation: Market morning (option 1)	Approved (owner, 2026-10-29)	Released	REP-078
DEC-048	Opening animation: plays on launch, 2 to 3 seconds	Approved (owner, 2026-10-29)	Released	
DEC-049	Bulk working session: QA, accessibility, print backs, rulebook and concept tutorial	Approved (owner, 2026-10-29)	Released	REP-079
DEC-050	Scout replaces Good Riddance (rules version 6)	Approved (owner, 2026-10-29)	Released	REP-080
DEC-051	Shopkeeper perks unchanged until the first playtest; release 1.2.0 now	Approved (owner, 2026-10-29)	Released	REP-080
DEC-052	New lorry (Sunny Pickup, bunting roof) and market stalls (tiled roof and sign) in pixel art	Approved (owner, 2026-10-29)	Released	REP-081
DEC-053	Full-screen play option	Approved (owner, 2026-10-29)	Released	REP-082
DEC-054	How to play rework and skippable tutorial	Approved (owner, 2026-10-30)	Released	REP-083

ID	Decision	Approval	Delivery	Reports
DEC-055	Tabletop review fixes	Approved (owner, 2026-10-30)	Released	REP-084
DEC-056	Optional accounts with ID and password	Approved (owner, 2026-10-31)	Implemented	REP-085
DEC-057	Achievements, free colours and collectible looks	Approved (owner, 2026-10-31)	Implemented	REP-085
DEC-058	Balance round 1, readable numbers, privacy page and payout display	Approved (owner, 2026-11-05)	Released	REP-086
DEC-059	Balance round 2, screen polish, change password, table kit	Approved (owner, 2026-11-05)	Released	REP-086
DEC-060	Printed card backs approved	Approved (owner, 2026-11-05)	Implemented	REP-079
DEC-061	Shopfront home screen, popups and rounder numbers	Approved (owner, 2026-11-05)	Released	
DEC-062	Rules 9: final customer deck, lorry deck, events and shopkeepers	Approved (owner, 2026-11-07)	Released	
DEC-063	Art direction for the Kickstarter edition: Look E, crate-label world with townsfolk characters	Approved (owner, 2026-11-07)	In progress	REP-087
DEC-064	Shopkeeper and guest names: Uncle Benny, Ah Chen and Kiki; guests unchanged	Approved (owner, 2026-11-07)	Not started	REP-088
DEC-065	Customer names: the owner's own playful list (36 cards)	Approved (owner, 2026-11-07)	Not started	REP-088
DEC-066	Implement the owner's Great Fruit Exchange deck art throughout the game	Approved (owner, 2026-11-07)	Released	

Work packages

ID	Package	Approval	Delivery	Release
PLAN-001	Package A: baseline, one verification path, preview and owner brief	Approved (owner, 2026-10-26)	Released	1.1.1
PLAN-002	Package B: canonical records, generated views and checks	Approved (owner, 2026-10-26)	Released	1.1.1
PLAN-003	Package C, paid part: capacity and cost evidence	Proposed	Not started	Unreleased
PLAN-004	Reconnect fix: the game screen sometimes doesn't return after rejoining an online room	Approved (owner, 2026-10-26)	Released	1.1.1
PLAN-005	4-player clarity: make it obvious that up to 4 people can play	Approved (owner, 2026-10-26)	Released	1.1.1
PLAN-006	Rename the panda rival (cosmetic, within the animal-trader cast)	Approved (owner, 2026-10-26)	Released	1.2.0
PLAN-007	UI polish phase 1: fix what looks broken	Superseded	Not started	Unreleased

ID	Package	Approval	Delivery	Release
PLAN-008	UI polish: art-direction style test and layout mockups (design only)	Approved (owner, 2026-10-26)	Released	1.2.0
PLAN-009	Cosy table build 1: seated rivals and the new market board	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-010	Cosy table build 2: the delivery lorry (outline)	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-011	Cosy table build 3: pricing, reveal and evening on the table (outline)	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-012	Market rebalance and cosy home screen: tighter town, fair Handcart, better apples, optional tweaks	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-013	Full-screen pricing and event sheets; clearer rival status while pricing	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-014	Tabletop edition rules: printed supply and demand, posted prices, market pulse, fruit payouts	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-015	Readable price duel: five-beat day, prototypes A/B/C and the Market Gossip card	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-016	Speciality price cut: Cold Room $4, Juice Bar $3	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-017	More fruit on the farmer cards; computers that use Cold Room, Juice Bar and Market Days	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-018	Data insights: balance, retention and online health from anonymous counters	Proposed	Not started	Unreleased
PLAN-019	First human playtest: digital and paper rounds (no game code)	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-020	Supply and customer demand in 4-player games: measure first, then propose one fix	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-021	Session length: shorter default and faster turns before touching the rules	Superseded	Not started	Unreleased
PLAN-022	Blank name field that opens the keyboard	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-023	Undo a buy in online rooms, and rename inside the room	Approved (owner, 2026-10-27)	Released	1.2.0
PLAN-024	Tabletop fixes: tie-break by buying order and table procedures	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-025	Character cards: six characters with fixed perks	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-026	Customer deck: 36 cards	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-027	Look C style guide and one palette	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-028	One Farmers Market board for the screen and the table (superseded by PLAN-030)	Superseded	Not started	Unreleased

ID	Package	Approval	Delivery	Release
PLAN-029	Animated market opening on the home screen	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-030	One Farmers Market board for the screen and the table (after the board look is chosen)	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-031	Rename pass: adopt the Naming_Audit names	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-032	Free evidence drills: backup restore, crash recovery, device checklist	Approved (owner, 2026-10-29)	Released	1.2.0
PLAN-033	Restyle the screens to the board look (the kit)	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-034	Replace Uncle Toad with Mak Jenab	Superseded	Not started	Unreleased
PLAN-035	Rara Jenab replaces Chris the Clown	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-036	Character coherence: one human cast, one theme, tied to the design style	Approved (owner, 2026-10-28)	Released	1.2.0
PLAN-037	Scout favour card replaces Good Riddance (rules version 6)	Approved (owner, 2026-10-29)	Released	1.2.0
PLAN-038	New lorry and stall roofs in pixel art	Approved (owner, 2026-10-29)	Released	1.2.0
PLAN-039	Full-screen play option	Approved (owner, 2026-10-29)	Released	1.2.0
PLAN-040	How to play rework and skippable tutorial	Approved (owner, 2026-10-30)	Released	1.2.1
PLAN-041	Tabletop review fixes	Approved (owner, 2026-10-30)	Released	1.2.1
PLAN-042	Optional accounts	Approved (owner, 2026-11-04)	Implemented	1.3.0
PLAN-043	Achievements, free colours and looks	Approved (owner, 2026-11-04)	Implemented	1.3.0
PLAN-044	Balance round 1 and polish	Approved (owner, 2026-11-05)	Released	1.4.0
PLAN-045	Balance round 2 and polish	Approved (owner, 2026-11-05)	Released	1.5.0
PLAN-046	Shopfront home screen, popups and rounder numbers	Approved (owner, 2026-11-05)	Released	1.6.0
PLAN-047	Rules 9: final customer deck, lorry deck, events and shopkeepers	Approved (owner, 2026-11-07)	Released	1.7.0
PLAN-048	Look E design package: naming review, pilot Character Bible, reference sheets, card mock-ups and mobile specifications (design only)	Approved (owner, 2026-11-07)	In progress	Unreleased

ID	Package	Approval	Delivery	Release
PLAN-049	Implement the Great Fruit Exchange art across the app and the printed kit	Approved (owner, 2026-11-07)	Released	1.8.0

Decision details

DEC-001 · Adopt the personal workflow guide 1.2 for Fruit Market

- **Question:** Should Fruit Market be run with the workflow guide (one approved package at a time, evidence-based checks, decision records)?
- **Choice:** Yes; one package at a time, starting with Package A.
- **Rationale:** The game is complete but releases were unreviewed and records lived in chat history.
- **Player impact:** None directly; fewer broken releases.
- **Technical impact:** Adds checks, records and a release gate.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “2a (2A = approve DEC-001, DEC-004, DEC-005 and DEC-014, answering the lifecycle trial decision card)” (2026-10-27)
- **Delivery:** In progress
- **Review trigger:** If the process costs more time than it saves after three packages.
- **Reports:** REP-001, REP-038

DEC-002 · Package A scope = PLAN-001

- **Question:** What should the first setup package contain?
- **Choice:** One verify command, preview before publishing, short records and an owner brief (PLAN-001).
- **Rationale:** Smallest package that stops unreviewed releases and gives the owner a clear way to try builds.
- **Player impact:** None until a release; protects players from silent breakage.
- **Technical impact:** package.json scripts, CI and deploy workflows, new docs.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** Acceptance A4 (owner preview playtest) still pending.
- **Reports:** REP-001, REP-002

DEC-003 · Live deploys only through the manual Deploy workflow after verify

- **Question:** When should the live game change?
- **Choice:** Only when the owner runs Deploy LIVE game and types PUBLISH; it runs verify first and a smoke test after. Never on push.

- **Rationale:** Makes 'a preview is not a release' true for this repository.
- **Player impact:** Updates arrive only after the owner approves them.
- **Technical impact:** deploy-web.yml is manual only.
- **Data and security:** None.
- **Cost:** RM0; one extra tap per release.
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-001 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If urgent fixes are delayed because nobody pressed Deploy.
- **Reports:** REP-001, REP-002

DEC-004 · PDF exports with pandoc and the existing Playwright browser

- **Question:** Which tool builds the decision PDFs?
- **Choice:** pandoc (Markdown to HTML) plus the project's existing Playwright Chromium (HTML to PDF). No installs.
- **Rationale:** Both are already available; no new dependency.
- **Player impact:** None.
- **Technical impact:** PDF export runs locally, not in CI.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “2a (2A = approve DEC-001, DEC-004, DEC-005 and DEC-014, answering the lifecycle trial decision card)” (2026-10-27)
- **Delivery:** Implemented
- **Review trigger:** If pandoc is unavailable on the machine that exports.
- **Reports:** REP-001, REP-038

DEC-005 · Budget interpretation: RM500 per month across all games

- **Question:** How should the RM500 monthly ceiling be applied?
- **Choice:** Treat it as portfolio-wide. Fruit Market's current spend is unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5).
- **Rationale:** The workflow guide's conservative default.
- **Player impact:** None.
- **Technical impact:** Paid services need a quote and approval.
- **Data and security:** None.
- **Cost:** Unknown; to be recorded.
- **Owner:** Murshidi
- **Approval:** Approved: “2a (2A = approve DEC-001, DEC-004, DEC-005 and DEC-014, answering the lifecycle trial decision card)” (2026-10-27)
- **Delivery:** Not started
- **Review trigger:** When the owner states the Cloudflare plan and other paid accounts.
- **Reports:** REP-001, REP-038

DEC-006 · Preview = separate staging Worker

- **Question:** How should changes be previewed, given that Cloudflare preview and version URLs are not generated for Workers with Durable Objects?
- **Choice:** A separate Worker, fruit-market-staging, with its own rooms, leaderboard and stats, updated by CI after checks pass on main.
- **Rationale:** Closest free option to a per-change preview URL; keeps the live data untouched.
- **Player impact:** None; players never see the preview.
- **Technical impact:** A second Worker and a preview job in CI.
- **Data and security:** Separate storage; preview link is unlisted, not private.
- **Cost:** No new paid service; tiny usage within the existing plan.
- **Owner:** Murshidi
- **Approval:** Approved: “Confirm DEC-006, and plan PLAN-002” (2026-10-26)
- **Delivery:** Verified (PLAN-001)
- **Review trigger:** If Cloudflare adds preview URLs for Durable Object Workers.
- **Reports:** REP-002, REP-003

DEC-007 · Package B scope = PLAN-002

- **Question:** What should the records package contain?
- **Choice:** Canonical JSON records, generated views (register, changelog, state block, PDFs), a records check in verify, release tagging on Deploy, and migration of existing records.
- **Rationale:** Records were kept by hand in several places with nothing checking that they agreed.
- **Player impact:** None directly; releases become traceable.
- **Technical impact:** New records, a generator and checker script, and a deploy step.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If keeping records costs more than it saves.
- **Reports:** REP-003, REP-004

DEC-008 · Canonical record format: JSON records plus Markdown narrative reports

- **Question:** Where is the single source of truth for decisions and packages?
- **Choice:** JSON files in docs/workflow/records; hand-written REP reports stay as narrative snapshots; Markdown, changelog and PDF views are generated.
- **Rationale:** Node reads JSON without dependencies and the check can be strict. YAML needs a parser; plain Markdown can't be validated; a database is ruled out by the guide.
- **Player impact:** None.
- **Technical impact:** Claude edits the JSON; the owner reads the generated views.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)

- **Review trigger:** If hand edits to JSON become a frequent source of errors.
- **Reports:** REP-003, REP-004

DEC-009 · Records check runs in verify and CI

- **Question:** Should record disagreements fail the build?
- **Choice:** Yes: npm run verify runs records:check, so CI fails on broken records.
- **Rationale:** Once branch protection is on, broken records can't be merged.
- **Player impact:** None.
- **Technical impact:** verify is about 1 second slower.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-002 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-002)
- **Review trigger:** If the check blocks urgent fixes for record reasons.
- **Reports:** REP-003, REP-004

DEC-010 · Reconnect fix scope = PLAN-004 (evidence first, then fix)

- **Question:** How should the intermittent rejoin failure be fixed?
- **Choice:** Collect evidence and reproduce it with a stress test before changing game code; fix only the measured root cause, with a regression test.
- **Rationale:** Reading the code didn't identify a certain cause, and two failures in five runs are too rare to debug by guessing. The workflow guide says to reproduce and isolate before fixing.
- **Player impact:** Fewer stuck screens after a reload or dropped connection, once the fix is published.
- **Technical impact:** New stress test and diagnostics; a small fix in the online client or room server.
- **Data and security:** Diagnostics stay in CI artifacts for 7 days; they contain test players only.
- **Cost:** RM0 (GitHub Actions minutes are free for public repositories)
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If not reproduced after 300 cycles: stop and report.
- **Reports:** REP-005, REP-006

DEC-011 · Stress test runs on demand, not on every push

- **Question:** Should the long reconnect stress test run in every CI run?
- **Choice:** No: a manual CI workflow plus local runs. Every-push CI keeps the existing tests (about 15 minutes).
- **Rationale:** Hundreds of cycles would make every CI run far slower; the stress test is for investigation and before releases.
- **Player impact:** None.

- **Technical impact:** One extra manual workflow.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-004 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-004)
- **Review trigger:** If reconnect bugs slip through between releases.
- **Reports:** REP-005, REP-006

DEC-012 · 4-player clarity scope = PLAN-005 (screen text only)

- **Question:** How should the game make 4-player play discoverable?
- **Choice:** Clearer screen text and open-seat rows in the lobby, short hints on the home, quick-match and new-game screens, plus an automated 4-person test. No rule or server changes.
- **Rationale:** 4-player play already works; only discoverability is missing. Text changes are low risk and fast to verify.
- **Player impact:** Groups of 3 or 4 can find how to play together.
- **Technical impact:** src/ui/Online.tsx, src/ui/App.tsx, styles; one new browser test.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-005 only, including its listed tests and documentation” (2026-10-26)
- **Delivery:** Verified (PLAN-005)
- **Review trigger:** If playtesters still miss how to invite more than one friend.
- **Reports:** REP-007, REP-008

DEC-013 · Publish version 1.1.1 to players

- **Question:** Should the 1.1.1 candidate (reconnect fix, 4-player clarity, release process) replace 1.1.0 on the live game?
- **Choice:** Recommended after the owner's 15-minute preview playtest passes. Publication happens only when the owner runs Deploy LIVE game and types PUBLISH.
- **Rationale:** Fixes a bug players can hit (stuck on Connecting after a reload or rejoin) and makes group play discoverable. The game-code change is small (4 files, about 40 lines), the server code is unchanged, and all automated checks pass.
- **Player impact:** Fewer stuck online screens; clearer 2 to 4 player options.
- **Technical impact:** Live Worker updated to the candidate commit and tagged v1.1.1.
- **Data and security:** No data or storage format change; rolling back is safe.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes to 1 and 2 (1 = release 1.1.1: approve DEC-013, then the owner presses Deploy LIVE game)” (2026-10-26)
- **Delivery:** Released
- **Review trigger:** If the preview playtest finds a problem, or the post-deploy smoke test fails.
- **Reports:** REP-009, REP-013

DEC-014 · Use the Concept and Theme Companion for creative work

- **Question:** Should Fruit Market's creative decisions follow the Game Concept and Theme Companion (edition 1.0) alongside the development workflow?
- **Choice:** Yes, with one shared record system (same reports, register and PDFs); creative approval never implies implementation approval.
- **Rationale:** Keeps creative and development decisions in one history and avoids a parallel register.
- **Player impact:** None directly.
- **Technical impact:** None; documentation process only.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “2a (2A = approve DEC-001, DEC-004, DEC-005 and DEC-014, answering the lifecycle trial decision card)” (2026-10-27)
- **Delivery:** In progress
- **Review trigger:** If creative work starts needing its own package type.
- **Reports:** REP-010, REP-038

DEC-015 · Adopt three design pillars

- **Question:** Which design pillars guide Fruit Market's creative choices?
- **Choice:** 1) Fresh fruit, fast money. 2) Outprice rivals in secret. 3) A cheerful town market. Each with inclusion and exclusion rules (REP-010 section 3).
- **Rationale:** Derived from the actual rules and the strongest theme-to-mechanic matches found in the review.
- **Player impact:** Future features and content are judged against these.
- **Technical impact:** None now.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes to 1 and 2 (2 = approve DEC-015 and DEC-016)” (2026-10-26)
- **Delivery:** Implemented
- **Review trigger:** If playtests show players value something the pillars exclude.
- **Reports:** REP-010, REP-011, REP-013

DEC-016 · First creative test: reveal-comprehension playtest

- **Question:** What is the smallest useful test of the riskiest creative hypothesis?
- **Choice:** 3 to 5 first-time players, Quick game vs 1 computer shop, no coaching; continue if at least 3 of 4 can explain a lost sale by day 3 (provisional, owner-set). No code.
- **Rationale:** Pillar 2 depends on players understanding why they won or lost a sale; no player evidence exists yet.
- **Player impact:** Testers play the existing game.
- **Technical impact:** None.
- **Data and security:** Notes only; no personal data kept.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “yes to 1 and 2 (2 = approve DEC-015 and DEC-016)” (2026-10-26)
- **Delivery:** Not started
- **Review trigger:** After the first playtest round.
- **Reports:** REP-010, REP-011, REP-013

DEC-017 · Setting: a market where the whole world shops

- **Question:** Where is Fruit Market's market?
- **Choice:** Option C: a deliberate 'the whole world shops here' mix. A sunny harbour market; customers from every culture and fantasy role; seaside palette; plain market dollars.
- **Rationale:** Keeps the existing art and cast, turns the mix into an intentional identity, and costs nothing now.
- **Player impact:** No visible change now; guides future content.
- **Technical impact:** None now; recorded in the creative bible.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “1. Where is this market? C: a deliberate “the whole world shops here” mix.” (2026-10-26)
- **Delivery:** Implemented
- **Review trigger:** If playtesters find the setting confusing.
- **Reports:** REP-011

DEC-018 · Rename Panda Lim while keeping the cast coherent

- **Question:** Should the 'Panda Lim' rival be renamed?
- **Choice:** Yes: an animal nickname that fits beside Foxy Rosa, Big Benny, Uncle Toad, Rocky and Pinky; picture and lines unchanged. Cast rules recorded in the creative bible.
- **Rationale:** Avoids pairing a surname with a national symbol while preserving the animal-trader theme.
- **Player impact:** New games show the new name.
- **Technical impact:** One data line plus tests (PLAN-006).
- **Data and security:** Saved games keep the old name until they end.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded: “Rename Panda Lim? - sure, but dont run away from the general theme coherence” (2026-10-26)
- **Delivery:** Verified (PLAN-006)
- **Review trigger:** If the new name tests poorly with players.
- **Superseded by:** DEC-040
- **Reports:** REP-011, REP-012

DEC-019 · Rename scope = PLAN-006, shipped in 1.1.2

- **Question:** How and when should the rename be built?
- **Choice:** As PLAN-006 (display name, reserved names, compatibility tests), released as 1.1.2 after 1.1.1 is live.

- **Rationale:** Keeps the 1.1.1 candidate unchanged; the rename is small and safe but still needs its own checks.
- **Player impact:** New name from 1.1.2.
- **Technical impact:** Small data change plus tests.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve PLAN-006 only, including its listed tests and documentation, name: Patches” (2026-10-26)
- **Delivery:** Verified (PLAN-006)
- **Review trigger:** If 1.1.1 is delayed for long.
- **Reports:** REP-011, REP-012

DEC-020 · UI and UX polish in three phases

- **Question:** How should the game's look and feel be made to feel finished?
- **Choice:** Phase 1 fixes 9 visible defects (PLAN-007, ships in 1.1.2); phase 2 calms the table layout and phase 3 adds a design system, motion and sound, both designed as phone-viewable mockups first (PLAN-008) and built only after the owner picks; playtest before and after.
- **Rationale:** The review found concrete causes (defects, clutter, no focus, inconsistent system). Fixing defects is cheap and safe; layout changes need the owner's eye and player evidence.
- **Player impact:** A cleaner, clearer game in steps.
- **Technical impact:** Presentation code only; no rules, saves or server.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded
- **Delivery:** Not started
- **Review trigger:** If the baseline playtest shows a different priority.
- **Superseded by:** DEC-033
- **Reports:** REP-014, REP-015

DEC-021 · Art direction: B, toy-box tabletop

- **Question:** Which art direction should the polish follow, given the owner dislikes the current art style and points to It Takes Two?
- **Choice:** Direction B: a smooth 2D toy-like tabletop style (rounded shapes, wood, felt and clay materials, warm light, soft shadows, motion on every action), drawn by Claude as vector art; a paid illustrator stays a later, separately priced option. The A/B/C style test is dropped.
- **Rationale:** B addresses clutter, feel and art style, keeps phones fast and costs nothing to test; C is too large and risky now; A does not answer the art-style concern.
- **Player impact:** A warmer, toy-like look with more motion, if chosen and built.
- **Technical impact:** Redraw of fruits, about 30 characters, rivals, about 40 icons and the UI (only after approval of a build plan).
- **Data and security:** None.
- **Cost:** RM0 for the test; any illustrator needs a quote and approval

- **Owner:** Murshidi
- **Approval:** Approved: “go with plan b” (2026-10-26)
- **Delivery:** In progress (PLAN-008)
- **Review trigger:** After the style test, or if the owner wants a paid illustrator.
- **Reports:** REP-015, REP-016

DEC-022 · Look and tone: cunning edition of direction B

- **Question:** How should direction B reflect the game's undercutting mechanic?
- **Choice:** Keep the toy-box tabletop style and add a cheeky, scheming layer: plum and gold for secret moments (pricing, reveal, evening), wax-sealed face-down price boards, 'Psst' whisper bubbles that restate public facts, sly rival faces (Foxy Rosa winks), a spotlight reveal with an 'Outfoxed by $1!' headline, and the tagline 'Undercut. Outfox. Sell it fresh.'
- **Rationale:** The owner asked for the look to match the undercut mechanic. Cheeky rather than mean keeps pillar 3 (a cheerful town market) and pillar 2 (outprice rivals in secret) aligned; the market stays sunny by day.
- **Player impact:** Secret phases feel secret and the reveal feels like a showdown, if built.
- **Technical impact:** Same redraw as DEC-021 plus a plum secret-phase theme; only after a build plan is approved.
- **Data and security:** None. Whispers only restate information the game already shows; no hidden prices or choices.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes but align more with the cunning aesthetic given the undercut mechanism of the game” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** If playtesters read the rivals as mean, or the plum screens feel too dark.
- **Reports:** REP-017

DEC-023 · Look: cosy pixel edition of direction B

- **Question:** The cunning edition lost the cosy feel and the toy-box art moved too far from pixel art. What should the look be?
- **Choice:** Go back to the first toy-box mockups (warm cream, teal and wood; the owner's own Evening layout edit) and use the game's own pixel characters, fruit and icons with its pixel heading font, inside the smooth, polished frame. Keep a light sly touch in warm colours: dotted 'Psst' whispers from Foxy Rosa, a sly-eyed Foxy Rosa and Patches, sealed prices, 'Outfoxed by $1!'. Drop the dark plum screens, curtain and spotlight.
- **Rationale:** Owner feedback: the first toy-box version felt cosy; the cunning edition did not; balance the art back toward pixel art. Reusing the game's sprites also means far less redraw work than a full vector cast.
- **Player impact:** Familiar pixel cast and fruit, cleaner screens, cosy warm colours, if built.
- **Technical impact:** Much smaller art job than DEC-021: reuse and tidy the existing sprites; the UI frame is the main build.
- **Data and security:** None. Whispers repeat only public information.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** Owner review of the cosy pixel mockups; if confirmed, DEC-022's plum layer is superseded.
- **Reports:** REP-018, REP-020

DEC-024 · Layout: one shared market table (board game, not menus)

- **Question:** The cosy pixel mockups still felt like menus. How should the game screens be laid out?
- **Choice:** Every game screen is the same market table, like the current game: customers queue along the street at the top, Foxy Rosa's and Patches' stalls flank the market in the middle, your stall sits at the bottom. Pricing seals boards on each stall; the reveal flips the boards on the stalls themselves, the winner's stall glows, coins fly to it and the customer walks over; at night the stalls light lanterns and the market is covered. Keeps DEC-023's pixel art, warm colours and smooth buttons.
- **Rationale:** Owner: "revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" and "the one currently feels very menu based, instead of board game". Seeing everyone at one table is what made the current game feel cosy.
- **Player impact:** The game feels like a board game you sit at, with less clutter than today.
- **Technical impact:** Reuses the current game's table structure; the build restyles it rather than replacing it.
- **Data and security:** None. Rival boards stay sealed until the bell; whispers repeat only public information.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** Owner review of the table mockups.
- **Reports:** REP-019, REP-020

DEC-025 · Market board: vertical decay tracker and a delivery lorry

- **Question:** How should the market look and how should fruit arrive each day?
- **Choice:** The market is a vertical decay tracker: one column per fruit, left to right (blueberry, watermelon, apple, mango); new and ripening fruit at the top, spoiling fruit at the bottom; each slot shows its price, and empty slots stay visible. Your stall uses the same tracker in small strips. At the start of buying, a lorry drives in along the street, customers shuffle aside, and crates drop onto the board; it drives off when buying ends, before selling.
- **Rationale:** Owner's explicit instruction. The tracker is the game's existing decay rule made visible, and the lorry turns the daily delivery into a moment.
- **Player impact:** Easier to see what is fresh and what is about to spoil; a lively start to each day.
- **Technical impact:** Visual only: the tracks follow src/engine/data.ts exactly; no rule or balance change.
- **Data and security:** None.
- **Cost:** RM0

- **Owner:** Murshidi
- **Approval:** Approved: “change the board layout to a vertical layout arranging the fruits from left to right instead, revert back to the decay style tracker for the game board, at the start of the buying turn, animate a lorry offloading the fruit on to the board, the customers are to make space accordingly, the lorry would leave before the selling turn” (2026-10-27)
- **Delivery:** Implemented (PLAN-008)
- **Review trigger:** If the vertical board is too tall on small phones, or the lorry slows down play (keep it skippable).
- **Reports:** REP-021

DEC-026 · Freeze rules version 4 until the first human playtest round is done

- **Question:** Should rules stop changing until real players have tried rules version 4?
- **Choice:** Yes. No rule, balance or content change until the first human playtest round (DEC-016) is complete and its result is recorded. Allowed during the freeze: fixes for bugs that stop play, lose data or break security (S1), documentation, records and playtest materials.
- **Rationale:** Rules changed from version 2 to 4 in two days, tuned only against computer players. No unfamiliar player has played version 4, so further tuning has no human evidence behind it (lifecycle trial, G1 CONCERNS).
- **Player impact:** No gameplay change while testing; testers play the preview as it is.
- **Technical impact:** Pauses PLAN-016/017 follow-ups and any new rule work; PLAN-018 (data handling) is not part of the freeze unless it changes rules.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded: “1a (1A = freeze rules version 4 and run the first human playtest: 3 to 5 strangers plus a paper round)” (2026-10-27)
- **Delivery:** Not started (PLAN-019)
- **Review trigger:** When the first playtest round is recorded, or if an S1 bug forces a rules fix.
- **Superseded by:** DEC-028
- **Reports:** REP-038

DEC-027 · One game, two editions: tabletop and digital with identical mechanics

- **Question:** What is the commercial and design direction for Fruit Market?
- **Choice:** Fruit Market is developed as a tabletop game and a digital game at the same time, with the same mechanics. The tabletop rules (RULES.md, rules version 4 and later) are the master; the digital game implements exactly those rules. The digital edition adds single-player capability (computer shops, with modes such as the League and daily challenge) but no digital-only mechanics. A rule change is made once and applies to both editions.
- **Rationale:** Owner direction, in their words: "tabletop which can be translated into the game at the same time, both with the same mechanics but the digital version to have single player capability". Interpretation to confirm: single-player already exists in the digital game; the requirement is that it follows the tabletop rules rather than adding its own.
- **Player impact:** Both editions play the same; the digital game keeps solo play against computer shops.
- **Technical impact:** Every future rule must be playable by hand. Open: nightly ageing is too fiddly by hand (REP-036; date-stamped crates proposed, undecided). Digital-only features

are limited to presentation, modes and computer players.
- **Data and security:** None.
- **Cost:** RM0 now. Physical-edition costs (printing, components, manufacturing, shipping, storage) are unknown and need a dated quote and a separate approval.
- **Owner:** Murshidi
- **Approval:** Approved: “1a 2a 3 tabletop which can be translated into the game at the same time, both with the same mechanics but the digital version to have single player capability” (2026-10-27)
- **Delivery:** Not started
- **Review trigger:** If a rule that makes the game fun cannot be done by hand, or if physical-edition costs are not viable.
- **Reports:** REP-038

DEC-028 · Lift the rules freeze; plan fixes for supply, session length and UI requests

- **Question:** Should rules version 4 stay frozen after PLAN-019 ended without a playtest result?
- **Choice:** No. The freeze (DEC-026) is lifted. Fixes for supply, session length and the UI requests are to be planned. Nothing is built until the owner approves a named plan; every rule or balance change still needs its own plan, rules-version bump and approval.
- **Rationale:** PLAN-019 ended early (REP-046), so the condition for the freeze (a recorded playtest round) cannot be met soon. The owner wants the issues from the informal feedback (REP-045) addressed. Risk accepted: further tuning rests on computer-player simulations plus one informal game, not on a structured human test.
- **Player impact:** None until a plan is approved and built.
- **Technical impact:** Allows plans PLAN-020 to PLAN-023 to be proposed and, once approved, built. DEC-027 (tabletop rules are the master; same mechanics in both editions) still applies to any rule change.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Lift the rules freeze (DEC-026). Plan fixes for supply, session length and the UI requests.” (2026-10-27)
- **Delivery:** Implemented
- **Review trigger:** If a rule change is made without a human playtest, or after the first structured playtest.
- **Reports:** REP-047

DEC-029 · Supply and demand: option A + C (rules version 5)

- **Question:** Which fix for the empty market on Market Days and the late-game customer surge (PLAN-020 step 2)?
- **Choice:** Option A + C. A: the lorry comes every day, Market Days included. C: no extra walk-in in the second half; walk-ins are players minus 2 every day, plus 1 on a Market Day. Rules version 5. Both are printed numbers, playable by hand (DEC-027). Target (owner): the market should not be empty, so that people can always buy something.
- **Rationale:** Simulation with computer players (REP-049): in rules 4 with 4 shops, 22% of shop turns in the last 3 days faced an empty market, rising to 49% on day 9, and late-game

customers nearly doubled. A + C removed the empty market and kept demand level. The owner asked that any rule be implementable on a real tabletop; both parts are.

- **Player impact:** Fruit arrives every day, so shops can always buy something. The last days are no longer flooded with customers. Market Days keep only the extra walk-in and Last Call.
- **Technical impact:** Rules version 5: saves from version 4 cannot resume, daily boards start fresh (keyed by rules version), and online players need the same version. RULES.md updated as the master for the tabletop.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “why is the number in decimal? remember that rules should be able to be implemented on a tabletop elsewhere, else do plan a + c, target is so that the board would not be empty / ppl cant buy anything” (2026-10-27)
- **Delivery:** Implemented (PLAN-020)
- **Review trigger:** After a structured human playtest, or if 2-shop games feel empty of customers (about 2 customers a day in the simulation).
- **Reports:** REP-050

DEC-030 · Tabletop fixes: tie-break by buying order, and table procedures (no price ladder)

- **Question:** Which of the tabletop fixes (Tabletop_Fixes) should be built into the rules and the app?
- **Choice:** All except the price ladder board and the colour zones that depend on it. One rule line changes: a price tie, after Favoured Trade, Haggler and fresher fruit, goes to whoever bought earlier today (the poorer-shop-by-cash step is removed). The rest are table procedures written into RULES.md: order of customers, the reader, order circles, the sole-supplier cube, the dawn line-up, evening for card holders only, day chips with age cards, and a waiting pile. Kept as rules version 5 because version 5 is not released.
- **Rationale:** Owner instruction, in their words. Cash changes while customers are paid, so the old tie-break was awkward on a table and decided about half of all ties. In simulation (same 300 games) the average final cash per shop did not move (4 shops $146.9 to $146.3; 2 shops $145.9 to $145.9). The price ladder was left out because the owner did not see the problem it solves.
- **Player impact:** A tie after fresher fruit now goes to the shop that bought earlier today. Nothing else in the app changes; the table procedures are for the tabletop edition.
- **Technical impact:** game.ts tie-break, story text and How to play text; RULES.md gains a "Playing on a table" section. The day-chip method is untested by hand.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “implement all the fix except for the price ladder, i dont get the problem and your solution” (2026-10-28)
- **Delivery:** Implemented (PLAN-024)
- **Review trigger:** After a hand-played tabletop test, or if tie-breaks feel unfair in human games.
- **Reports:** REP-051

DEC-031 · Character cards: pick one of six characters on entering a room or the set-up screen

- **Question:** Should each shop pick a character with a fixed perk instead of being dealt three traits?
- **Choice:** Yes. Six characters (the six animals), each with one fixed perk (the six existing traits): Foxy Rosa Haggler, Big Benny Big Basket, Rocky Bargain Hunter, Patches Charmer, Pinky Local Favourite, Uncle Toad Nest Egg. Each shop picks a different character when it enters an online room (changeable in the lobby, taken ones greyed out) or on the set-up screen; computer shops take the rest. This replaces the first-morning "deal 3, keep 1" trait pick. The daily challenge and league keep the old trait pick for the person. Kept as rules version 5.
- **Rationale:** Owner instruction, in their words, after the character-card proposal (Characters_and_Art). It also gives every seat a different icon. The perks were balanced in simulation (900 games, 4 shops, each trait first 22% to 28% against a fair 25%).
- **Player impact:** Pick a character before the game; each has one perk for the whole game, visible to everyone. No two shops share a character. The first-morning trait pick no longer appears for characters.
- **Technical impact:** New character data, fixed-trait start in the engine, a character message in the room protocol and server, a character picker in the room entry, lobby and set-up screens. Rules version stays 5 (unreleased).
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “make it so that character is selected when you enter the room instead apply all the changes and push it the game” (2026-10-28)
- **Delivery:** Implemented (PLAN-025)
- **Review trigger:** After a human playtest, or if one character proves too strong or too weak.
- **Reports:** REP-052

DEC-032 · Customer deck grows from 24 to 36 cards

- **Question:** Should the customer deck have 36 cards instead of 24, with the same demand profile?
- **Choice:** Yes. Twelve new customers (Jia Wen, Rizal, Devi, Marcus, Fatimah, Ben, Yuki, Hadi, Lakshmi, Kenji, Mira, Hock Seng) join the original 24, written to match them: average card 4.4 pieces and $31.8 fresh value (original 4.4 and $31.9), the same share of each fruit and the same mix of card sizes and fruits per card. The original 24 are unchanged.
- **Rationale:** Owner instruction after the demand review ("expand customer card to 36, keep the demand profile and design language in line with the rest of the game"). A bigger deck means fewer repeats in 5 to 6 shop games and on the table.
- **Player impact:** More variety of customers; each new customer has a pixel portrait, a title and a one-line quirk.
- **Technical impact:** Twelve entries in CUSTOMER_CARDS, looks and portraits; a test that depended on the dealt traits for one seed was made independent of them.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “expand customer card to 36, keep the demand profile and design language in line with the rest of the game” (2026-10-28)
- **Delivery:** Implemented (PLAN-026)
- **Review trigger:** After a human playtest, or if a new customer proves too strong or weak a draw.
- **Reports:** REP-053

DEC-033 · Art direction: Look C, calm flat world with pixel objects

- **Question:** Which single coherent look should the game follow: pixel everywhere (A), flat vector everywhere (B), or calm flat scenery with pixel objects (C)?
- **Choice:** Look C. Scenery and structure (home scene, sky, ground, signs, awnings, the lorry, board frames) are flat, smooth and calm in the harbour picture's palette and softness; everything the player counts, taps or prints as a token (fruit, people, icons) stays pixel art. The Farmers Market board is drawn once and shared by the screen and the printed table mat. The lorry style and the board size are still to be decided.
- **Rationale:** Owner choice ("Look C") after the three-way comparison (Look_Comparison.html). It keeps the pixel identity and 49 portraits, takes the calm of the flat harbour picture, and is the cheapest route to a coherent look.
- **Player impact:** No change yet. Later: a calmer look, a market board that looks like the printed one, an animated market on the home screen.
- **Technical impact:** Nothing built. Three plans proposed: PLAN-027 style guide and one palette, PLAN-028 shared board, PLAN-029 animated opening.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded: “Look C” (2026-10-28)
- **Delivery:** Not started (PLAN-027)
- **Review trigger:** After the board prototype is printed and seen on a phone.
- **Superseded by:** DEC-066
- **Reports:** REP-054

DEC-034 · Board look: which of four looks the rest of the game follows

- **Question:** Which look should the Farmers Market board have, so the rest of the game is designed back from it?
- **Choice:** D, the soft tabletop mat (Board_Look_Options.html): pastel lanes by fruit, paper slots with thin ink lines, a coin-shaped price and a freshness flag at each slot, cream mat surfaces, pill buttons with an ink outline. The market character is carried by the scene around the board (home scene, street, lorry), drawn flat and smooth; fruit, people and icons stay pixel.
- **Rationale:** Owner: "i want to decide on the board look first, then design the game back because that the proof of concept". Recommendation: D (calmest, cheapest, prints lightest), with the market character carried by the scene around it. Owner approved the recommendation ("approve everything above", after the recap that named D as the pick).
- **Player impact:** None yet. The chosen look becomes the palette, corner shape, line weight and button style for the whole game.
- **Technical impact:** Nothing built.
- **Data and security:** None.

- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “approve everything above” (2026-10-28)
- **Delivery:** In progress (PLAN-030)
- **Review trigger:** When the owner chooses.
- **Reports:** REP-056

DEC-035 · Naming: adopt the names in Naming_Audit

- **Question:** Should the game adopt the names proposed in the naming audit, including the optional ones?
- **Choice:** Yes, all of them (Naming_Audit.docx): Wholesale Market (sign may read FARM GATE); stall everywhere; shopkeepers for the six picks; Tired instead of "Not so" (Green, Fresh, Okay, Tired); piece for one fruit; cost for buying and price for selling; track, slot, shelf, crate, pile; phases Morning, Buying, Stocking, The bell, The rush, Evening; buttons Start buying and Start game; Delivery card; Gossip; Clearance; Tip Jar; perk for trait; stall upgrade and Handcart; the queue, regular and special guest; the crier; Closing Day; Daily market; Demand, Price drift, Fixed prices and Basic game for the options.
- **Rationale:** Owner instruction ("approve everything") straight after the naming audit. The audit found "market" meant five things, stall and shop were the same thing, and the rules and app used different names for the same beats.
- **Player impact:** Words only: the rules, how-to text, screens, cards and board use the new names. No rule changes.
- **Technical impact:** A rename across RULES.md, the app text, HowTo, data titles, the printed cards and the board. Code identifiers may stay as they are.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “approve everything” (2026-10-28)
- **Delivery:** Not started (PLAN-031)
- **Review trigger:** After a human playtest: do new players understand the names without help?
- **Reports:** REP-057

DEC-036 · Replace Uncle Toad with Mak Jenab

- **Question:** Which character becomes Jenab, and what is she like?
- **Choice:** Uncle Toad is replaced by Mak Jenab, a lady stallholder. She keeps the Nest Egg perk (start with $15 instead of $10): a thrifty, experienced trader who saves every sen. Look: a human portrait with a purple headscarf (tudung) and a teal baju; her computer-shop lines are warm and thrifty ("Every sen counts!", "Jimat, jimat!", "Terima kasih!"). The other five characters and all perks are unchanged.
- **Rationale:** Owner instruction ("replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady"). The owner left the choice of character and her attributes to the assistant. Uncle Toad (Nest Egg) was replaced because the saver perk fits her best and keeping a perk already balanced in simulation (22% to 28% first place) avoids a rebalance. A male animal was the natural one to replace for a lady.
- **Player impact:** The sixth character on the character pick is Mak Jenab (human, headscarf). The Nest Egg perk is the same.

- **Technical impact:** RIVALS and CHARACTER_TRAIT entries, a new headscarf part and portrait, RULES.md table, two tests updated. The frog sprite is removed. A saved game or room that used the frog avatar falls back to a plain shopper picture.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded: “replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady” (2026-10-28)
- **Delivery:** Not started (PLAN-034)
- **Review trigger:** If the owner wants a different perk or a different look for her, or after a playtest.
- **Superseded by:** DEC-037
- **Reports:** REP-061

DEC-037 · Jenab is a customer: Auntie Jenab replaces Chris the Clown

- **Question:** Which customer becomes Jenab, and what is she like?
- **Choice:** Chris the Clown becomes Auntie Jenab, a lady customer. She keeps Chris's order (4 mangoes at $9 and 1 blueberry at $6 for a fresh piece), so the customer demand profile does not change. Her quirk is about what she does: "Sniffs every mango twice, then asks for a discount anyway." Her portrait is a human with a purple headscarf (tudung) and a teal baju. The six shopkeepers (the animals) are unchanged, Uncle Toad included.
- **Rationale:** Owner instruction ("replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady; sorry jenab is a customer"). The owner left the choice of customer and her attributes open. Chris was chosen because his order is mango-heavy, which her mango quirk matches, and because the creative bible names Grandpa Tan as its sample customer and vampires, cowboys, astronauts and mermaids as examples, so Chris is the least disruptive to replace. Cast rules (DEC-018): customer jokes are about what someone does, never who they are; rivals stay animals.
- **Player impact:** A new customer, Auntie Jenab, in the customer deck in place of Chris the Clown. Her order is the same as his.
- **Technical impact:** One customer card, one look entry and one portrait (a new headscarf part); the shopkeeper change in DEC-036 was reverted.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Superseded: “replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady; sorry jenab is a customer” (2026-10-28)
- **Delivery:** Implemented (PLAN-035)
- **Review trigger:** If the owner wants a different customer replaced, or a different look for her.
- **Superseded by:** DEC-038
- **Reports:** REP-062

DEC-038 · Rara Jenab the Cheerleader replaces Chris the Clown

- **Question:** What should the Jenab customer be called and how should she be designed around her name?

- **Choice:** Rara Jenab (the title is just her name; the word "cheerleader" was dropped on the owner's review). "Rara" gives the idea: a cheerful "ra-ra" cheerleader (and in Javanese "rara" means a young lady). Her quirk is about what she does: "Cheers every mango she buys. 'Ra-ra, Jenab!'" Portrait: a purple headscarf (tudung), a teal baju and gold-and-red pom-poms at her sides. She keeps Chris the Clown's order (4 mangoes and 1 blueberry), so demand is unchanged.
- **Rationale:** Owner instruction ("change it to Rara Jenab, design it around the name"). Replaces the first design (Auntie Jenab, DEC-037).
- **Player impact:** A customer called Rara Jenab, a cheerleader, in place of Chris the Clown.
- **Technical impact:** Her name, look and portrait entries; a new pom-poms part.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “change it to Rara Jenab, design it around the name” (2026-10-28)
- **Delivery:** Implemented (PLAN-035)
- **Review trigger:** If the owner wants a different role or look for her, or when PLAN-036 sets the cast theme.
- **Reports:** REP-063

DEC-039 · Cast theme: one coherent theme for every character

- **Question:** Which single theme should every shopkeeper and customer follow?
- **Choice:** T1, Townsfolk of the harbour market: every character is an ordinary person with a job or hobby and one playful detail. Nobody is an animal, creature, robot or monster. The 10 non-human and 8 costume customers became everyday roles (for example Night Baker Robert, Diver Mei Ling, Magician Kumar, Florist Nurul, Chemist Daniel, Delivery Rider Ravi). On the owner's review four roles were replaced: Tailor Siti, Fisherman Marcus, Librarian Robert and Photographer Sarah (names and orders unchanged).
- **Rationale:** Owner: "customers should all be humans as well, design around the same coherent theme". Today the cast is 6 animal shopkeepers, 10 non-human customers, 8 costume customers and 21 everyday ones (Cast_Coherence_Audit.html). T1 fits the calm flat look and the cast rule that a joke is about what someone does.
- **Player impact:** Every customer a human; titles and quirks follow one pattern.
- **Technical impact:** Nothing built; portraits and titles change when approved.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “t1 and proceed with dec040” (2026-10-28)
- **Delivery:** Implemented (PLAN-036)
- **Review trigger:** When PLAN-036 refines the portraits, or after a playtest.
- **Reports:** REP-063

DEC-040 · Shopkeepers become humans (replaces the animal rivals rule)

- **Question:** Should the six shopkeepers be humans instead of animals, keeping their perks?
- **Choice:** Yes. The six shopkeepers are humans with plain first names and the perk unchanged: Rosa (Haggler), Benny (Big Basket), Chen (Charmer), Saleh (Nest Egg), Raj (Bargain Hunter), Kiran (Local Favourite). The manner nicknames were dropped on the owner's review. DEC-

018 (rivals are animal traders) is superseded. The internal avatar keys are unchanged so saves and rooms keep working; they are never shown. The home screen fox mascot stays until the animated opening replaces it (PLAN-029).
- **Rationale:** Owner: "i dont like the fact that the characters are animal". The perks were balanced in simulation and do not depend on the animal.
- **Player impact:** The character pick shows six humans; the computer shops are human too.
- **Technical impact:** Six new portraits, names and lines; RIVALS and the avatar keys change; tests and the rules table update.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “t1 and proceed with dec040” (2026-10-28)
- **Delivery:** Implemented (PLAN-036)
- **Review trigger:** When PLAN-036 refines the portraits, or after a playtest.
- **Reports:** REP-063

DEC-041 · Cast coherence rules (creative bible) approved

- **Question:** Do the Cast coherence rules in the creative bible stand as the rules for every character?
- **Choice:** Yes. Theme: townsfolk of the harbour market (an ordinary person with a job or hobby and one playful detail; no animals, creatures, robots or monsters). Shopkeepers: six humans with plain first names. Customers: a title "Role Name" (or just the name when the name carries the joke) and a quirk about what they do, never who they are. Names and roles mix across cultures with no culture tied to a role. Portraits: 16×16 head and shoulders, 1 pixel ink outline, one face template, told apart by hair, one accessory, skin tone and shirt colour, every colour from src/ui/palette.ts, light from the top left, no text or creature parts. One source of portraits for the screen, the printed cards, the character pick and the gallery.
- **Rationale:** Owner approval ("Approve rules") after reviewing the cast sheet and the rules.
- **Player impact:** Every character follows one theme and one portrait style.
- **Technical impact:** The rules are in docs/workflow/CREATIVE-BIBLE.md and are enforced in part by tests/cast.test.ts.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Approve rules” (2026-10-28)
- **Delivery:** Implemented (PLAN-036)
- **Review trigger:** When a new character is added, or after a playtest.
- **Reports:** REP-066

DEC-042 · Order of work, the lorry direction, no portrait frames, and how design work is approved

- **Question:** In what order are the approved plans built, and what needs the owner's approval before it is built?
- **Choice:** Order: PLAN-031 (rename pass), then PLAN-030 (the screen board in look D, ahead of the printed-mat test), then PLAN-033 (restyle the screens), then PLAN-029 (animated

opening). The lorry is redrawn flat and smooth as part of the scenery. Happy and sad portrait frames: no for now. PLAN-032 (free evidence drills) and the PLAN-036 quirk check are approved. PLAN-021 (session length) is NOT approved by this decision: its time target is still open. Rule for design work: anything about how the game looks (the board, the lorry, the restyle, the opening scene) is shown to the owner as a design first and built only after the owner approves that design; everything else is simply built.

- **Rationale:** Owner instruction after the review list ("approve everything except plan 21, for the design related, require my approval on the design, else just run").
- **Player impact:** Words change first (rename pass); then the new board, the restyled screens and the opening scene, each after the owner approves its design.
- **Technical impact:** Sets the build order and the approval rule; nothing visual is built before a design is approved.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “approve everything except plan 21, for the design related, require my approval on the design, else just run” (2026-10-29)
- **Delivery:** In progress (PLAN-031)
- **Review trigger:** When the owner changes the order or the rule.
- **Reports:** REP-069

DEC-043 · Table screen design approved (look D)

- **Question:** Is the proposed table screen in look D approved for building?
- **Choice:** Yes, as shown in Design_Proposal_Table_Screen.html: the Wholesale Market as pastel lanes with paper slots, coin-shaped prices, freshness corner flags and a gold outline on the slots the lorry delivers into; a flat smooth lorry on a road strip under the queue; the new human cast in the queue; paper rival stalls and your stall with striped awnings; ink-outlined pill buttons and top bar; a warm cream mat for everyday play. The trays and pop-ups keep their dark look until PLAN-033.
- **Rationale:** Owner approval ("okay approve this design") of the design mock-up.
- **Player impact:** The main game screen looks like the approved design: cream mat, paper slots, coin prices, flat lorry.
- **Technical impact:** A look-D style block, the market markup (lane class, ghost fruit, delivery slots) and an inline flat lorry replace the wood board and the lorry picture; two scenery colours added to the palette.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “okay approve this design” (2026-10-29)
- **Delivery:** Implemented (PLAN-030)
- **Review trigger:** After the owner looks at the preview on a phone.
- **Reports:** REP-071

DEC-044 · PLAN-021 (session length) cancelled

- **Question:** Should the shorter-session plan (PLAN-021) go ahead?

- **Choice:** No. PLAN-021 (a shorter default session and faster turns) is cancelled. No time target was ever set and nothing was built for it. Session length can come back as a new plan later, for example as part of the animation polish.
- **Rationale:** Owner instruction ("cancel 21 and lets work on 42").
- **Player impact:** No change.
- **Technical impact:** PLAN-021 closed as superseded; nothing to undo.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “cancel 21 and lets work on 42” (2026-10-29)
- **Delivery:** Implemented (PLAN-021)
- **Review trigger:** If sessions feel too long in a playtest.
- **Reports:** REP-073

DEC-045 · Other screens and printed mat in look D approved, results show stats and best moments

- **Question:** Approve the two design proposals (other screens in look D; printed mat and tokens)?
- **Choice:** Yes. The other screens follow look D, and the results screen shows stats and best moments. The printed mat and tokens follow the approved proposal.
- **Rationale:** Owner instruction ("okay with the design, but include stats like the best moments fir the results").
- **Player impact:** Home, set-up, results and trays match the table screen; results show per-player stats and "Your best moments".
- **Technical impact:** CSS block and flat home scene; no rules change.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “okay with the design, but include stats like the best moments fir the results” (2026-10-29)
- **Delivery:** Released (PLAN-033)
- **Review trigger:** After a phone playtest.
- **Reports:** REP-074

DEC-046 · Overnight work: scope and preview-push authority

- **Question:** What may be done unattended, and may finished work be pushed to the preview?
- **Choice:** Yes to: restyle of the remaining screens (covered by DEC-045), print kit, full QA sweep, balance simulations, tabletop playtest pack, and two opening-animation mock-ups for approval. Finished chunks may be pushed to the PREVIEW after green CI. Never live. No rule changes: simulations only. Good Riddance stays as it is until the owner decides.
- **Rationale:** Owner instruction ("give me 5 decisions which you can work for hours cause i wanna sleep do heavy bulk work as much as possible (answers: push to preview only; print kit, full QA sweep, balance simulations, tabletop playtest pack; two opening designs to approve; Good Riddance no preference)").
- **Player impact:** Preview may change (restyled screens, QA fixes). Rules unchanged.

- **Technical impact:** Branches workflow/overnight-*, reports REP-075 onward.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “give me 5 decisions which you can work for hours cause i wanna sleep do heavy bulk work as much as possible (answers: push to preview only; print kit, full QA sweep, balance simulations, tabletop playtest pack; two opening designs to approve; Good Riddance no preference)” (2026-10-29)
- **Delivery:** In progress
- **Review trigger:** Morning review.
- **Reports:** REP-075, REP-076, REP-077

DEC-047 · Opening animation: Market morning (option 1)

- **Question:** Which opening design: 1 Market morning or 2 The mat is laid?
- **Choice:** Option 1, Market morning: sun, sign, three stalls, the lorry brings fruit, the bell rings. Played when a new game starts; tap to skip.
- **Rationale:** Owner answer "1" to the question which opening design to build.
- **Player impact:** A short scene (about 8 seconds, skippable) plays when a new game starts. It does not play on Continue or Rematch.
- **Technical impact:** New Opening component; automated browser tests skip it.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “1” (2026-10-29)
- **Delivery:** Released (PLAN-029)
- **Review trigger:** After a phone playtest: if it feels long, shorten it.
- **Reports:** REP-078

DEC-048 · Opening animation: plays on launch, 2 to 3 seconds

- **Question:** When should the opening play and how long should it be?
- **Choice:** The market opening plays when the game is opened from its link, not when a new game starts. It runs about 2.5 seconds. Tap to skip stays.
- **Rationale:** Owner: "the animation should play when you click on the link to access the game not when u open the wholesale market, it should be between 2-3 secs only".
- **Player impact:** A 2.5 second scene plays each time the game is opened. It no longer plays when starting a game.
- **Technical impact:** Opening moved to the App root; timings shortened.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “the animation should play when you click on the link to access the game not when u open the wholesale market, it should be between 2-3 secs only” (2026-10-29)
- **Delivery:** Released
- **Review trigger:** After a phone check: confirm the length feels right.
- **Reports:**

DEC-049 · Bulk working session: QA, accessibility, print backs, rulebook and concept tutorial

- **Question:** Which unattended work may be done while the owner is at the office?
- **Choice:** All seven items (width pass, online lobby check, accessibility audit, card backs, soak test, 1.2.0 draft, cleanup) plus an updated rulebook and a concept tutorial, with preview pushes only. No rule or balance change.
- **Rationale:** Owner answer: "yes do all 7 with preview pushes, together with updating the rulebook and including a tutorial explaining the concept of the game".
- **Player impact:** Trophies, chips, tips and several buttons are readable; terracotta is a little darker; How to play opens with two concept slides.
- **Technical impact:** New qa-sweep script, soak script and a QA workflow that runs only on qa/ branches or on request; unused styles removed.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “yes do all 7 with preview pushes, together with updating the rulebook and including a tutorial explaining the concept of the game” (2026-10-29)
- **Delivery:** Released
- **Review trigger:** Owner reads REP-079 and the new How to play wording.
- **Reports:** REP-079

DEC-050 · Scout replaces Good Riddance (rules version 6)

- **Question:** What should replace the dead Good Riddance favour card? A) keep it, B) remove it, C) replace it with Scout
- **Choice:** C: Scout. On your buying turn, look at the front of one walk-in who arrives today. Everyone sees it was played, not what was seen. Rules version 6.
- **Rationale:** Owner answer "1c" to the decision card (recommended option).
- **Player impact:** A favour card that was never worth playing becomes a peek at one of today's customers.
- **Technical impact:** Rules version 6; Scout in the engine, a tap-to-choose step in the UI, tests; old phones cannot join new online rooms.
- **Data and security:** Saved games and daily boards from rules version 5 are not carried over.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “1c” (2026-10-29)
- **Delivery:** Released (PLAN-037)
- **Review trigger:** After the first outside playtest: does anyone play Scout?
- **Reports:** REP-080

DEC-051 · Shopkeeper perks unchanged until the first playtest; release 1.2.0 now

- **Question:** 2) Shopkeeper perks: A) leave for the first playtest, B) tweak now. 3) Release 1.2.0: A) now, B) after a phone check and a friend's playtest.
- **Choice:** 2A: no perk change. 3A: release 1.2.0 now. Claude prepares the release; the owner runs Deploy LIVE game and types PUBLISH.
- **Rationale:** Owner answer "2a 3a" on the decision card.

- **Player impact:** Everything on the preview since 1.1.1 goes live once the owner runs the deploy.
- **Technical impact:** Version 1.2.0; release candidate entry; no perk or balance change.
- **Data and security:** Rules version 6: saved games in progress and old daily boards do not carry over.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “2a 3a” (2026-10-29)
- **Delivery:** Released
- **Review trigger:** After publishing: smoke test passes and the owner has played the live game.
- **Reports:** REP-080

DEC-052 · New lorry (Sunny Pickup, bunting roof) and market stalls (tiled roof and sign) in pixel art

- **Question:** Which roof and stalls? Five mockups were shown (REP-081).
- **Choice:** Stalls: option 4, Tiles and Sign. Lorry: the Sunny Pickup with option 2, the bunting and boards roof. Both in pixel art like the characters.
- **Rationale:** Owner answer: "lets go for option tiles and sign for stall, lorry one use bunting".
- **Player impact:** The farmer's lorry is a pixel-art pickup with a wooden roof and a string of flags; stall panels get a tiled roof band in each player's colour; the market opening shows the new stalls and lorry.
- **Technical impact:** New src/ui/propsArt.tsx; table, opening and styles updated; ART-STYLE.md updated so props are pixel art.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “lets go for option tiles and sign for stall, lorry one use bunting” (2026-10-29)
- **Delivery:** Released (PLAN-038)
- **Review trigger:** Owner looks at it on a phone: size of the lorry in the street, readability of the flags.
- **Reports:** REP-081

DEC-053 · Full-screen play option

- **Question:** Can the game be played full screen?
- **Choice:** A Full screen button in Settings and in the pause menu, where the browser supports it. On iPhone, which has no full-screen mode for web pages, Settings explains Add to Home Screen.
- **Rationale:** Owner request: "can we include an option to play the game full screen as well".
- **Player impact:** Tap Full screen to hide the browser bars; tap again or use the back gesture to leave.
- **Technical impact:** New src/ui/fullscreen.ts and FullScreenButton; uses the Fullscreen API.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi

- **Approval:** Approved: “can we include an option to play the game full screen as well” (2026-10-29)
- **Delivery:** Released (PLAN-039)
- **Review trigger:** Owner tries it on an Android phone and on an iPhone.
- **Reports:** REP-082

DEC-054 · How to play rework and skippable tutorial

- **Question:** How should How to play flow, and can the tutorial be skipped?
- **Choice:** Seven cards in game order: overview with the phases of a game, the seven phases of a day, the four fruits, the rush, evening and night, Market Days and Closing Day, tips. A Skip tutorial button on every first-game tip.
- **Rationale:** Owner request: "i dont like the current flow/and what is included ... make the tutorial skippable please", then "it also needs to outline what phases are there in the game" and "1 (but include details from 4) 5 3 8 9 11 12".
- **Player impact:** A clearer first read of the game, and the tutorial tips can be skipped in one tap (Settings can replay them).
- **Technical impact:** src/ui/HowTo.tsx, src/ui/tutorial.ts, Popovers.tsx and styles.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “"it also needs to outline what phases are there in the game"; "1 (but include details from 4) 5 3 8 9 11 12"” (2026-10-30)
- **Delivery:** Released (PLAN-040)
- **Review trigger:** Owner reads it on the preview and at the first table test.
- **Reports:** REP-083

DEC-055 · Tabletop review fixes

- **Question:** Can everything in the game be done on a tabletop?
- **Choice:** Fix the gaps found in the review: missing guest cards, rules wording, card text, a favour deck that reshuffles only played cards, and new print pieces (stall sheet and screen, paper money, guest cards).
- **Rationale:** Owner request: "ensure that everything can be done on a tabletop"; answered "Yes, fix all".
- **Player impact:** Hotel Chef, Influencer and Gym Club orders are printed; Bad Harvest and Fridge cards say what they do; in long 4-player games a favour card can no longer appear a third time.
- **Technical impact:** drawFavour in src/engine/game.ts; card text in data.ts; RULES.md; the Drive print kit.
- **Data and security:** Rules version stays 6. Only the favour deck after it runs out changes (about 1 game in 6 draws the whole deck).
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Yes, fix all” (2026-10-30)
- **Delivery:** Released (PLAN-041)
- **Review trigger:** Owner reads it on the preview and at the first table test.
- **Reports:** REP-084

DEC-056 · Optional accounts with ID and password

- **Question:** Can players keep their progress across devices?
- **Choice:** Add optional accounts: an ID and a password, no email. Progress (stats, achievements, looks, colours, league stage) syncs and only ever grows. No password recovery. Deleting the account is in Settings.
- **Rationale:** Owner request: "can we add simple account creation, just ID and password, no need email verification or anything"; chose "No recovery".
- **Player impact:** Players can sign up, sign in on another device and get their progress; the game still works with no account.
- **Technical impact:** AccountDO (Durable Object) in server/accounts.ts, PBKDF2-SHA256 passwords, hashed session tokens, lockout after repeated wrong tries; client in src/ui/account.ts and Settings.
- **Data and security:** Stores an ID, a password hash and the progress profile per account. Google Play needs a privacy-policy link and an updated data-safety form (owner).
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “can we add simple account creation, just ID and password” (2026-10-31)
- **Delivery:** Implemented (PLAN-042)
- **Review trigger:** Owner tries accounts and the wardrobe on the live game.
- **Reports:** REP-085

DEC-057 · Achievements, free colours and collectible looks

- **Question:** What do players collect, and what is it for?
- **Choice:** 43 achievements in ten themes; every stall colour is free; 25 collectible looks (roof, trim, counter prop, hats, face, neck, badge) unlocked by achievements, worn by the shopkeeper and shown on the stall. Rivals online see your look. A wardrobe opens from Trophies, the room lobby and the pause menu.
- **Rationale:** Owner request: expand achievements thematically (answered "About 40 in themes"), stall colour free for all, collectibles that build on the stall and are worn by the character, "All in one release", then rivals should see looks and the wardrobe should be easy to reach in a room. Looks are cosmetic: no rule, price or customer number changes.
- **Player impact:** Trophies grouped by theme; Wardrobe screen; online play shows each player's look. Update the app to play online (protocol 3).
- **Technical impact:** Looks catalogue in src/net/looks.ts shared by app and server; server checks look ids and slots only.
- **Data and security:** Rules version stays 6; PROTOCOL_VERSION 3.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “artstyle is awesome i love it” (2026-10-31)
- **Delivery:** Implemented (PLAN-043)
- **Review trigger:** Owner tries accounts and the wardrobe on the live game.
- **Reports:** REP-085

DEC-058 · Balance round 1, readable numbers, privacy page and payout display

- **Question:** Which of the audit's suggested changes go into the game now?
- **Choice:** Build one upgrade per Market Day (as written), a mango costs $1 more, Big Basket's extra piece costs $1 more, computer players take the Handcart first; make small numbers readable, label the rank badge, say 'that day' in event previews, serve the privacy page and link it in Settings, add install and link-preview files and security headers, and add up each customer's payment on screen as its coins land. Not done now: shopkeeper rework, 2-player and Quick match changes, tap queue, set-up screen redesign, print-kit additions.
- **Rationale:** Owner request after REP-086: "do the suggested changes and make it live in the game". The simulation (about 40,000 computer games) measured each rules change in REP-086.
- **Player impact:** Games are tighter (fewer lopsided results), upgrades are a real one-per-Market-Day choice, mango and Big Basket cost a little more, and cash totals climb customer by customer. Rules version 7: update the app to play online, and a game in progress from 1.3.0 can't be resumed.
- **Technical impact:** RULES_VERSION 7; buyCost, buyUpgrade and the computer's upgrade order in the engine; new src/ui/cashShown.ts; public/ manifest, robots and _headers.
- **Data and security:** Daily boards start fresh for rules version 7.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “do the suggested changes and make it live in the game” (2026-11-05)
- **Delivery:** Released (PLAN-044)
- **Review trigger:** Owner plays it, and a week of anonymous stats.
- **Reports:** REP-086

DEC-059 · Balance round 2, screen polish, change password, table kit

- **Question:** Which of the remaining audit items go into the game now?
- **Choice:** Shopkeeper rebalance (Haggler first sale +$1, Big Basket extra piece $2, Nest Egg buys first on day 3, Charmer 4 cards), two stalls get one walk-in, quick matches have no upgrades, a daily twist, taps queued while buying, a sticky Start button and clearer set-up wording, results chart and share, change password, fruit spoiled and unsold stats, a content security policy, and a bigger printed table kit. Not built: Cold Room and Juice Bar redesign (the simulation showed no gain), Table Companion mode, league stars, practice coach, teams, async rooms, offline mode, lockout change.
- **Rationale:** Owner request after the audit status: "ok do everything". Each rules change was measured in a second simulation round (REP-086): every shopkeeper now wins within about 3 points of fair.
- **Player impact:** More even shopkeepers, a more even 2-player game, simpler quick matches, a new daily twist, and several smaller fixes. Rules version 8: update the app to play online, and a game in progress from 1.4.0 can't be resumed.
- **Technical impact:** RULES_VERSION 8; haggler bonus, buyCost, nest egg order, charmer start and 2-player walk-in in the engine; daily twist in src/engine/daily.ts; password action on the account server; quick-match rooms without upgrades; _headers with a content security policy.
- **Data and security:** Daily boards start fresh for rules version 8. Two new anonymous stats: fruit_spoiled and fruit_unsold.
- **Cost:** RM0

- **Owner:** Murshidi
- **Approval:** Approved: “ok do everything” (2026-11-05)
- **Delivery:** Released (PLAN-045)
- **Review trigger:** Owner plays it; a week of the new anonymous stats.
- **Reports:** REP-086

DEC-060 · Printed card backs approved

- **Question:** Is the draft design for the Delivery, Favour and Event card backs approved?
- **Choice:** Approved as drafted (Card_Backs.pdf, REP-079). The backs are now part of the complete play kit in one folder.
- **Rationale:** Owner request: "deploy live game and approve card back".
- **Player impact:** Table players get matching backs for the Delivery, Favour and Event decks.
- **Technical impact:** None. Files on the Drive Print_Kit folder only.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “deploy live game and approve card back” (2026-11-05)
- **Delivery:** Implemented
- **Review trigger:** Owner test print: cut lines, back alignment and sleeves.
- **Reports:** REP-079

DEC-061 · Shopfront home screen, popups and rounder numbers

- **Question:** Which home screen layout, how should New game set-up work, and which font should numbers use?
- **Choice:** Layout 3, the shopfront scene: your shopkeeper at a stall with Dress up (Wardrobe) and a hanging Create account sign, four tiles (New game, Play a friend, Daily market, Market League) and a Continue bar. Option A: the whole New game set-up lives in one popup. Continue and Play a friend also open small menus. Numbers and dollar signs use Fredoka; words keep the pixel font.
- **Rationale:** Owner: "layout 3, option A, yes approve it already", then "Yes, apply it" for the font, then "check this and then approve everything to deploy the game".
- **Player impact:** A friendlier home screen, with the Wardrobe and account on it; set-up is a popup; clearer numbers. No rules change, so online play with 1.5.0 still works.
- **Technical impact:** New src/ui/Home.tsx; App.tsx home and set-up reworked; font alias 'Fruit Pixel' in styles.css; browser test scripts updated.
- **Data and security:** None.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “layout 3, option A, yes approve it already” (2026-11-05)
- **Delivery:** Released (PLAN-046)
- **Review trigger:** Owner plays it on a phone; whether B and C in the pixel font need the same fix.
- **Reports:**

DEC-062 · Rules 9: final customer deck, lorry deck, events and shopkeepers

- **Question:** Which numbers and rules should the economy use after the balance simulations?
- **Choice:** Start cash $15 and buy up to 6 pieces (Handcart 8, Nest Egg $20); customer deck with freshness types (Premium, Everyday, Bargain); ten named lorry cards (two copies each, amounts for 2, 3 and 4 stalls; Mango Growers and Mango Season arrive green); thirteen events with Price Controls replacing Minimum Price and the Health Inspector at the start of the day.
- **Rationale:** Owner supplied the final decks and rules on 2026-11-07 after two simulation rounds. Known trade-offs from the simulations: the moderate lorry deck cuts supply only about 5%, the buy limit of 6 raises blowouts by about 10 points, Premium Display stays weak and Nest Egg is still the weakest shopkeeper.
- **Player impact:** A different economy: more cash and bigger baskets, fussier customers, named lorries, new event numbers. Online play needs the new version (rules version 9); a game in progress from 1.6.0 can't be resumed.
- **Technical impact:** src/engine/data.ts, game.ts and ai.ts; boardModel, table, panel and morning card in the interface; RULES_VERSION 9; new tests/rules9.test.ts.
- **Data and security:** Daily leaderboards for rules 9 start empty (board key includes the rules version).
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “Owner, 2026-11-07: "Final customer deck … Final lorry supply … Event deck … Shopkeeper … deploy end to end per below instructions"” (2026-11-07)
- **Delivery:** Released (PLAN-047)
- **Review trigger:** Owner plays it; if 3 and 4 stall games still feel oversupplied, cut the lorry amounts (simulation: about one third off blueberry, watermelon and apple); test buy limit 4 against 6.
- **Reports:**

DEC-063 · Art direction for the Kickstarter edition: Look E, crate-label world with townsfolk characters

- **Question:** Which visual identity should the Kickstarter-first tabletop edition use, and later the phone game: The Great Fruit Exchange (vintage crate labels), Orchard & Oddballs (quirky personalities), or a controlled hybrid?
- **Choice:** Look E (docs/art/ART_DIRECTION.md, with the corrections of v0.2): vintage crate-label graphics with expressive ordinary townsfolk; flat screen-print illustrations with clean outlines and minimal texture; physical cards prioritise gameplay readability; hidden-information rules as in ART_DIRECTION section 9. Owner's answers to REP-087 section 13: (1) AI concepts are finished by the owner, without commissioning an illustrator; (2) mobile characters will eventually replace the existing pixel characters throughout (not only card views); (3) pilots Benny (broad, bearded, prominent Big Basket) and Sailor Jia Wen (navy clothing, sailor's cap, rolled mango map). Design only: no assets implemented, no game code changed, no artwork replaced, no money spent.
- **Rationale:** Owner brief: Kickstarter first, then the phone game, with a reusable art pipeline. Three AI review passes, each from one viewpoint (tabletop publisher, art-production lead, game identity), scored the hybrid 35-36 of 45 against 22-25 for the Exchange alone and 25-27 for Oddballs alone (REP-087, appendix A); they are a structured opinion, not outside

experts. The hybrid keeps the harbour world, cast rules and palette, avoids the stereotypes of vintage fruit advertising, and keeps the pixel board the owner praised (DEC-057).
- **Player impact:** None until a later implementation plan. Then: illustrated printed cards and box; on the phone, illustrated characters replace the pixel characters throughout.
- **Technical impact:** None now. When implemented: amends DEC-033 and ART-STYLE.md sections 1, 2, 4 and 5 (people become illustrations in print and, later, in the app; colour additions; one display face after its own approval) and the DEC-041 portrait rule. The later app plan must redraw the 11 wearable looks for the illustrated shopkeepers (DEC-057), keep portraits readable at 18-34 px, and keep the performance budget. DEC-042 stands for the app (no expression frames until decided). DEC-060's backs stay until a new back design is approved.
- **Data and security:** None.
- **Cost:** RM0. The owner finishes AI concepts himself; any paid image tool still needs his separate approval.
- **Owner:** Murshidi
- **Approval:** Approved: “Fruit Market — Look E Design Approval. Record the owner's approval of the overall Look E visual direction, subject to the corrections identified in the concept review. Approved: Vintage crate-label graphics with expressive ordinary townsfolk. Flat screen-print illustrations with clean outlines and minimal texture. Benny: broad, bearded, prominent Big Basket. Sailor Jia Wen: navy clothing, sailor's cap, rolled mango map. Physical cards must prioritise gameplay readability. Mobile characters will eventually replace existing pixel characters throughout. AI concepts will be finished by the owner, without commissioning an illustrator. … This approval is for design only. Do not implement assets, modify game code, replace existing artwork or spend money.” (2026-11-07)
- **Delivery:** In progress (PLAN-048)
- **Review trigger:** After the pilot pair is shown next to the current pixel look (DEC-042), or if the owner chooses a different AI or budget policy.
- **Reports:** REP-087

DEC-064 · Shopkeeper and guest names: Uncle Benny, Ah Chen and Kiki; guests unchanged

- **Question:** Which final names do the six shopkeepers and three special guests use (REP-088 sections 5 and 6)?
- **Choice:** Shopkeepers: S1 Rosa (Haggler), S2 Uncle Benny (Big Basket), S3 Ah Chen (Charmer), S4 Saleh (Nest Egg), S5 Raj (Bargain Hunter), S6 Kiki (Tip Jar, was Kiran). All six abilities unchanged. Guests unchanged: The Influencer, The Gym Club, The Hotel Chef. Amends the shopkeeper naming rule of DEC-040 and DEC-041 (plain first names): a shopkeeper may carry a familiar title the owner chose (Uncle, Ah). Design decision only; the 36 customer names are still to come (section 1 of the owner's list).
- **Rationale:** Owner's final list after the naming review (REP-088). 'Ah Chen' reads as a familiar name rather than a bare surname; 'Kiki' replaces Kiran; 'Uncle Benny' fits the approved pilot look (broad, bearded).
- **Player impact:** None until a later implementation plan. Then the shopkeepers show their new names in the app, the print kit and the rulebook.
- **Technical impact:** None now. At implementation: display names in src/engine/data.ts (RIVALS) with the hidden avatar keys unchanged, so saves and online rooms keep working;

tests/cast.test.ts 'shopkeeper names are a plain first name' must allow the approved titles; RULES.md shopkeeper table and daily-market line (Rosa, Benny, Chen); character cards in the print kit.
- **Data and security:** None. Abilities, perks and balance unchanged.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “2. Complete shopkeeper roster. ID Final approved name Existing ability. S1 Rosa Haggler. S2 Uncle Benny Big Basket. S3 Ah Chen Charmer. S4 Saleh Nest Egg. S5 Raj Bargain Hunter. S6 Kiki Tip Jar. All six shopkeeper abilities remain unchanged. 3. Special guests. ID Final approved name. G1 The Influencer. G2 The Gym Club. G3 The Hotel Chef. change to above, appeoved” (2026-11-07)
- **Delivery:** Not started (PLAN-048)
- **Review trigger:** When the customer names (section 1) arrive, or at the implementation plan.
- **Reports:** REP-088

DEC-065 · Customer names: the owner's own playful list (36 cards)

- **Question:** Which final names do the 36 customers use (REP-088)?
- **Choice:** The owner's list, in card order: C1 Cowboy Jack (was John); C2 Bobby Books (was Robert); C3 Dr. Bubbles (was Daniel); C4 Rara Jenab; C5 Zara Zapp (was Aisyah); C6 Ravi Rocket (was Ravi); C7 Coral Mei (was Mei Ling); C8 Postman Pip (was Farid); C9 Madam Siti; C10 Magic Kumar; C11 Rockstar Grace; C12 Professor Pebble (was Hafiz); C13 Inspector Wei (was Wei Jie); C14 Miss Marigold (was Nurul); C15 Dr Priya; C16 Captain Ahmad; C17 Lola Lens (was Sarah); C18 Grandpa Tan; C19 Miss Cha-Cha (was Lina); C20 Sleepy Zul; C21 Anna the Artist (was Anna); C22 Otto Oddjob (was Vijay); C23 Mina Moss (was Hui Min); C24 Firefighter Osman; C25 Marina Wen (was Jia Wen); C26 DJ Rizal; C27 Devi Bloom (was Devi); C28 Marcus Hook (was Marcus); C29 Professor Fatimah; C30 Benny Bell (was Ben); C31 Akira Swift (was Yuki); C32 Hadi Hammer (was Hadi); C33 Baker Lakshmi; C34 Kenji Kaze (was Kenji); C35 Herbalist Mira; C36 Uncle Hock Seng. Style: the owner's own mix of playful nicknames (Bobby Books, Ravi Rocket, Lola Lens) and job titles (Firefighter Osman, Baker Lakshmi), shown as written on one line. Each card keeps its type, wants and prices: the old names stay as hidden IDs. Pilot Sailor Jia Wen is now Marina Wen, with the approved look (navy, sailor's cap, rolled mango map). Assumption: 'Dr. Bubbles' is shown as 'Dr Bubbles' to match 'Dr Priya'. Three names were flagged to the owner for a yes or no (C30 Benny Bell shares 'Benny' with shopkeeper Uncle Benny; C22 'Oddjob' is a famous film character; C8 'Postman Pip' is close to the TV character Postman Pat). Quirks are unchanged until reviewed.
- **Rationale:** Owner's final customer list after the naming review (REP-088). It replaces the 'Role Name' rule of DEC-041 for customers with the owner's playful naming.
- **Player impact:** None until a later implementation plan. Then customers show the new names in the app, the print kit and the rulebook.
- **Technical impact:** None now. At implementation: CUSTOMER_LOOKS titles in src/engine/data.ts with the old names kept as hidden keys (type lists, generous rule, saves, online rooms); RULES.md type lists, example (Hafiz) and generous list rebuilt from the game; print kit; quirks reviewed to fit the new names.
- **Data and security:** None. Types, wants, prices and generous status unchanged.
- **Cost:** RM0

- **Owner:** Murshidi
- **Approval:** Approved: “c1Cowboy Jack C2 Bobby Books C3 Dr. Bubbles C4 Rara Jenab C5 Zara Zapp C6 Ravi Rocket C7 Coral Mei C8 Postman Pip C9 Madam Siti C10 Magic Kumar C11 Rockstar Grace C12 Professor Pebble C13 Inspector Wei C14 Miss Marigold C15 Dr Priya C16 Captain Ahmad C17 Lola Lens C18 Grandpa Tan C19 Miss Cha-Cha C20 Sleepy Zul C21 Anna the Artist C22 Otto Oddjob C23 Mina Moss C24 Firefighter Osman C25 Marina Wen C26 DJ Rizal C27 Devi Bloom C28 Marcus Hook C29 Professor Fatimah C30 Benny Bell C31 Akira Swift C32 Hadi Hammer C33 Baker Lakshmi C34 Kenji Kaze C35 Herbalist Mira C36 Uncle Hock Seng (section 1 of the owner's 'Final approved name' roster; sections 2 and 3 were sent with 'change to above, appeoved')” (2026-11-07)
- **Delivery:** Not started (PLAN-048)
- **Review trigger:** Owner's answer on the three flagged names (C8, C22, C30), a read-aloud test with a crier, and a header-fit test on a printed card.
- **Reports:** REP-088

DEC-066 · Implement the owner's Great Fruit Exchange deck art throughout the game

- **Question:** Which artwork and visual identity does the game use, and may it be implemented now?
- **Choice:** The owner's own deck art (Fruit-Market-Deck-Art-1/2/3.pdf, 'The Great Fruit Exchange'): events use Art 1 'The scene' (Exchange Gazette front pages), shopkeepers use Art 2 'Over the counter' (Stall Signs), customers use Art 3 'At work' (Trading Cards). Implemented throughout: names (DEC-064, DEC-065), portraits, card layouts, the digital game (menus, set-up, character pick, table, events, market, results, buttons and backgrounds) and the tabletop print templates. Fonts and colours follow the PDFs (Alfa Slab One, Abril Fatface, Libre Caslon Text, Oswald, Yellowtail, Caveat; navy ink, gazette paper, red, mustard, green, chalkboard, wood). The art shows old names; the game shows the approved new names. Missing illustrations (guests, favour, lorry and upgrade cards, the face-down walk-in, wearable looks on the new portraits) are listed as gaps, not filled with unrelated art. No gameplay change. Publishing the live game still needs its own approval.
- **Rationale:** Owner instruction with explicit authority to implement immediately.
- **Player impact:** A new look across the app and the printed kit; new character names; no rule, price or balance change.
- **Technical impact:** Art extracted from the owner's PDFs into public/art (app) and print/art (print); new fonts; new palette roles; Face, card and event components; print-deck script; tests updated where they assumed pixel portraits or old names.
- **Data and security:** None. Old names stay as hidden keys; saves and online rooms keep working.
- **Cost:** RM0
- **Owner:** Murshidi
- **Approval:** Approved: “okay now incorporate the designs from the pdf attachhed with over the counter (shopkeeper), scene (events) and at work (customers) as my artstyle/design that i want, incorporate this in the workflow ure on now. Your task is to implement these designs throughout the existing Fruit Market game, ensuring a cohesive experience across the tabletop and digital versions. ... IMPORTANT: You are authorised to implement the approved designs immediately. Do not ask for further approval unless a decision is genuinely necessary to proceed. Start implementation now. The objective is a working, visually updated Fruit

Market game, not another design proposal. (Attachments: Fruit-Market-Deck-Art-1.pdf, -2.pdf, -3.pdf)" (2026-11-07)
- **Delivery:** Released (PLAN-049)
- **Review trigger:** Owner checks the preview on a phone, and a printed test sheet.
- **Reports:**

History (oldest first)

- 2026-10-26 13:15 · REP-001: baseline audit; PLAN-001 proposed; DEC-001 to DEC-005 proposed.
- 2026-10-26 13:20 · Owner: "Approve PLAN-001 only, including its listed tests and documentation". DEC-002 and DEC-003 approved; PLAN-001 approved.
- 2026-10-26 13:25 · PLAN-001 implemented on main (2e4b240). Push started CI and Android only; no live deploy.
- 2026-10-26 13:42 · REP-002: PLAN-001 verified except A4 (owner playtest). DEC-006 recorded, awaiting confirmation.
- 2026-10-26 13:46 · Owner: "Confirm DEC-006, and plan PLAN-002". DEC-006 approved. REP-003: PLAN-002 proposed; DEC-007 to DEC-009 awaiting approval.
- 2026-10-26 13:48 · Owner: "Approve PLAN-002 only, including its listed tests and documentation". PLAN-002, DEC-007, DEC-008 and DEC-009 approved.
- 2026-10-26 13:50 · Correction: the report headers of REP-002 (13:45) and REP-003 (13:50) were rounded estimates; the commit times 13:42 and 13:46 are used above. The reports are left as written.
- 2026-10-26 13:52 · PLAN-002 implemented on workflow/plan-002: records migrated; B1, B2, B3, B5 and B7 PASS; B4 and B6 pending.
- 2026-10-26 14:09 · PLAN-002 verified: B1 to B7 PASS (CI run 40276699547 on a7fa577). DEC-007, DEC-008 and DEC-009 delivered.
- 2026-10-26 14:09 · Known issue found: CI run 40275680793 (docs-only commit 63c3682) failed because the online test's page reload did not show the game within 20 s. Same symptom as the quick-match failure in CI run 1. Pre-existing and intermittent (2 of 5 CI runs); not part of PLAN-002. Proposed as the next package in REP-004.
- 2026-10-26 14:16 · Owner: "PLAN the reconnect fix". REP-005: PLAN-004 proposed; DEC-010 and DEC-011 awaiting approval.
- 2026-10-26 14:18 · Owner: "Approve PLAN-004 only, including its listed tests and documentation". PLAN-004, DEC-010 and DEC-011 approved. (Time approximate: between the REP-005 commit at 14:17 and the first work commit at 14:27.)
- 2026-10-26 14:53 · PLAN-004: root cause reproduced (lost first update after rejoining; 20 of 20 hidden-tab reloads stuck before the fix) and fixed in the client (9d9e911).
- 2026-10-26 15:01 · PLAN-004: R1 to R5 PASS (165 hidden-tab cycles after the fix, 0 failed). R6 waiting for 3 CI runs.
- 2026-10-26 15:34 · REP-006: PLAN-004 verified (R1 to R6 PASS). The fix is on the preview; the live game is unchanged until a release is prepared and deployed.
- 2026-10-26 18:34 · Owner asked how to play with 4 people and requested a plan. Check: 4-person online room works locally (4 joined and played; a 5th refused). REP-007: PLAN-005 proposed; DEC-012 awaiting approval.
- 2026-10-26 18:36 · Owner: "Approve PLAN-005 only, including its listed tests and documentation". PLAN-005 and DEC-012 approved.

- 2026-10-26 18:44 · PLAN-005 implemented: C1 to C5 PASS; C6 waiting for CI.
- 2026-10-26 19:01 · REP-008: PLAN-005 verified (C1 to C6 PASS). On the preview; goes live with the next release.
- 2026-10-26 19:02 · Correction: the REP-008 header time (19:05) was an estimate; the report was completed at 19:01. The report is left as written.
- 2026-10-26 20:26 · Owner: "PREPARE RELEASE". Version set to 1.1.1 (package and Android versionName); release candidate entry added; DEC-013 (publish 1.1.1) awaiting approval. REP-009.
- 2026-10-26 20:27 · CI run 13 (f472186, records-only commit) was cancelled: the browser download took 16 minutes, so the job hit its 20-minute limit. Not a code failure. Limit raised to 35 minutes in the 1.1.1 release commit.
- 2026-10-26 21:47 · Owner asked to apply the Game Concept and Theme Companion. REP-010: theme review (no changes made); DEC-014 proposed; DEC-015 (pillars) and DEC-016 (first playtest) awaiting approval. Open questions: setting identity, renaming Panda Lim, availability of 3 to 5 testers.
- 2026-10-26 22:14 · Owner answers to REP-010: setting C approved (DEC-017); rename Panda Lim approved, keeping the cast coherent (DEC-018); 3 to 5 first-time testers are available. REP-011: creative bible written; PLAN-006 (rename) and DEC-019 awaiting approval. DEC-015 and DEC-016 still await explicit approval.
- 2026-10-26 22:17 · Owner: "Approve PLAN-006 only, including its listed tests and documentation, name: Patches". PLAN-006 and DEC-019 approved; new name Patches.
- 2026-10-26 22:18 · PLAN-006 implemented on workflow/plan-006: N1 to N4 PASS (5 new tests); N5 waiting for CI.
- 2026-10-26 22:48 · REP-012: PLAN-006 verified (N1 to N5 PASS) on workflow/plan-006. Waits off main until 1.1.1 is live; then ships as 1.1.2.
- 2026-10-26 22:56 · Owner: "yes to 1 and 2". DEC-013 (publish 1.1.1) approved; the owner runs Deploy. The owner's own preview playtest result was not reported (recorded as NOT RUN). DEC-015 (pillars) and DEC-016 (first playtest) approved. Recorded on workflow/plan-006 so main stays on the 1.1.1 candidate.
- 2026-10-26 22:57 · Correction: the REP-013 header time (22:58) was an estimate; the report was written at 22:56. The report is left as written.
- 2026-10-26 23:06 · Owner: the game doesn't feel polished; propose changes on the whole UI and UX. REP-014: review of all main screens at 360×640 and 390×844; DEC-020, PLAN-007 (phase 1 fixes) and PLAN-008 (mockups) awaiting approval. PDF export now inlines report images.
- 2026-10-26 23:18 · Owner answers to REP-014: most bothered by clutter, lack of animation and feel, and the art style itself; polished reference: It Takes Two. REP-015: art-direction options; PLAN-008 revised to a style test plus mockups (still awaiting approval); DEC-021 awaiting approval; PLAN-007 suggested to shrink to 3 fixes. PLAN-007 and PLAN-008 not yet approved.
- 2026-10-26 23:50 · Owner: "go with plan b". DEC-021 approved: art direction B (toy-box tabletop). Interpreted also as approval to proceed with PLAN-008 in B only (design-only mockups; the 3-way style test dropped); owner told they can stop it. PLAN-007 and DEC-020 still awaiting a decision.
- 2026-10-26 23:57 · PLAN-008 mockups published (https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB): style sample plus home, buying, pricing, sales reveal and evening in direction B. REP-016. Waiting for the owner's review.

- 2026-10-27 00:12 · Owner on the REP-016 mockups: "yes but align more with the cunning aesthetic given the undercut mechanism of the game". DEC-022 approved (cunning edition of direction B). Mockups revised and republished at the same link; owner's Evening layout edit kept. REP-017. Waiting for the owner's review of the revision.
- 2026-10-27 00:17 · Owner: "its still not enough, maybe balance the artstyle back to pixel art, the ui/ux here looks more polished but lacks the cosiness of how the previous board was set", then "no i meant the version before you tweaked here, that felt cosy". Mockups rebuilt from the pre-cunning version (owner's Evening edit kept) with the game's pixel sprites and font and a light, warm sly touch; dark plum screens dropped. DEC-023 awaiting the owner's confirmation. REP-018.
- 2026-10-27 00:25 · Owner: "revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" and "the one currently feels very menu based, instead of board game". Buying, pricing, reveal and evening mockups rebuilt around one shared market table (customers on the street, rival stalls either side of the market, your stall below), keeping the pixel art and warm frame. DEC-024 awaiting confirmation. REP-019.
- 2026-10-27 01:21 · Owner: "going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible (now it feels to small/ narrow)". DEC-023 (cosy pixel look) and DEC-024 (one shared table) approved. Rival stalls slimmed and the market widened with bigger crates and fruit; republished (version 13). REP-020. DEC-022's dark plum layer is superseded by DEC-023. Next: propose the 1.2.0 build plan.
- 2026-10-27 01:28 · Owner asked for a vertical decay-tracker market (fruits left to right), the decay tracker on your stall, and a delivery lorry that arrives at the start of buying (customers make space) and leaves before selling. DEC-025 approved by that instruction. Buying mockup rebuilt (version 14). REP-021.
- 2026-10-27 01:31 · Owner: "revert back the tracker on player's stall". DEC-025 narrowed: the vertical decay tracker is for the market only; your stall goes back to fruit chips showing nights left. Buying mockup republished (version 15).
- 2026-10-27 01:32 · Owner supplied the delivery lorry art ("use this as the lorry"). Saved as docs/workflow/reports/img/owner-lorry.png; the Buying mockup uses a cut-out of it (background removed, resized). Version 16. Before the build, confirm the picture's origin and that we may use it in the game.
- 2026-10-27 01:35 · Owner: the lorry picture was made with Gemini; asked to trim it so no ground shows below. Ground and background removed (wheels kept); cut-out saved as docs/workflow/reports/img/owner-lorry-cutout.png; Buying mockup republished (version 17). Note for the build: AI-made art may not be protected by copyright, so others could reuse it; check Gemini's terms and Google Play's AI-content rules before publishing it in the game.
- 2026-10-27 01:36 · Owner: "rescale the customers, now it looks disproportionate to the lorry". Street customers shrunk from 54px to 36px on buying, pricing and reveal so they match the lorry (68px tall). Version 18.
- 2026-10-27 01:38 · Owner: "keep the demand supply indicator from the evening's build inside the buying phase". The current game's want / in-shops strip (same meaning and colours as src/ui/table.tsx pulse) added to the Buying mockup above the stalls; it updates as you buy. The street notes it repeated were removed. Version 19.
- 2026-10-27 01:43 · Owner: move the other players off the left and right so the board is the main focus. Rivals now sit in a slim row of seats across the top of the table (portrait, cash, what they hold; sealed board when pricing, flipping board at the reveal, lantern at night). The

market board takes the full width; the want / in-shops chips now sit on top of each fruit column. Applied to buying, pricing, reveal and evening. Version 20.

- 2026-10-27 01:46 · Owner: tap a rival's name in the buying phase to expand their stall tracker and see freshness. Added to the Buying mockup: tapping Foxy Rosa or Patches opens a drop-down decay tracker of their stall (same columns and colours as the market, counts per slot, a one-line summary); tap again or × to close. Shows only what the current game already shows about rival stock (no prices or choices). Version 21.
- 2026-10-27 01:48 · Owner: make the expanded rival tracker show freshness the same way as your stall. The rival drop-down now uses the same fruit chips as your stall (fruit at its freshness, plus nights left) instead of the grid. Version 22.
- 2026-10-27 01:48 · Owner: "sort the fruit from top to bottom aligning at the top". Market columns are now top-aligned: each starts with its newest slot and runs down to not-so-fresh; shorter tracks (blueberry) end early. Version 23.
- 2026-10-27 01:52 · Owner: remove the freshness bar beside the board and just colour-code freshness. Side bar removed; every market slot, empty ones included, is tinted by its freshness (unripe pale green, fresh green, okay gold, not so orange-brown) with a colour key under the board; your stall and rival chips use the same colours. Version 24.
- 2026-10-27 01:58 · Owner: "ok now do this changes", read as: build the mockup look into the game. Under the workflow this needs an approved plan, so PLAN-009 (table and board), PLAN-010 (lorry, outline) and PLAN-011 (pricing, reveal, evening, outline) were proposed; awaiting approval. No game code changed. REP-022.
- 2026-10-27 01:59 · Owner: "yes" to REP-022. Read as approval of PLAN-009 (started) and of folding PLAN-007's text fixes into PLAN-011. The lorry art choice (Gemini picture or pixel lorry) is still open; it only affects PLAN-010.
- 2026-10-27 02:19 · PLAN-009 built and verified on branch workflow/plan-009 (b2b495a): verify and verify:full PASS, new board tests PASS, 9-size screenshot check PASS. Not merged: main holds the 1.1.1 candidate until the owner deploys it. REP-023.
- 2026-10-27 02:20 · Owner: "use the gemini lorry, approve PLAN-011". PLAN-011 approved and started (pricing, reveal and evening on the table). PLAN-010 will use the owner's Gemini lorry picture; owner accepts that AI-made art may not be protected by copyright.
- 2026-10-27 02:51 · PLAN-011 built and verified on workflow/plan-009 (4408cd8): verify and verify:full PASS (leak check 0), screenshots at 3 sizes. PLAN-007 closed as superseded; its text fixes shipped in PLAN-011. Customers walking to the winner was not built (noted in scope). REP-024. Remaining for 1.2.0: PLAN-010 (lorry). Main still waits for the 1.1.1 deploy.
- 2026-10-27 03:26 · Owner: "deploy". Deploy LIVE game run 15 (40385840230) on main 83de99b succeeded, including its live smoke test; tagged v1.1.1. Release 1.1.1 marked published. Then workflow/plan-009 (PLAN-006 Patches rename, PLAN-009, PLAN-011, creative records) fast-forwarded into main so the preview updates; these will ship together as the next release (planned 1.2.0, with PLAN-010).
- 2026-10-27 03:39 · Owner asked for a full audit of how the game is won, lost, progressed and paced, without code changes. REP-025: rules read line by line plus computer-only simulations (balance at 300 games per test and 4 read-only probes). Main findings: league difficulty cliff at stages 5-6, 4-player economy squeezed, snowballing, Juice Press has no measurable value, trait spread 29-42%, hidden seat tie-break, final-day night events do nothing. No decision taken; proposals need plans and approval.
- 2026-10-27 07:39 · Owner: "approve" (reply to the request to approve PLAN-010). PLAN-010 approved and started: delivery lorry with the owner's Gemini art. The league fix from REP-025

is not yet planned.
- 2026-10-27 07:54 · PLAN-010 built and verified (bc2e760): lorry with the owner's Gemini art; verify and verify:full PASS. Merged to main for the preview. All three cosy-table packages (PLAN-009, 010, 011) plus the Patches rename are now on main, ready to become release 1.2.0 when the owner asks.
- 2026-10-27 09:11 · Owner asked for a balance redesign (fruit identity, strategy depth, pacing) with analysis and a patch plan only. REP-027: 813 lab test cells (384,200 computer games) on a separate engine copy, validated on held-out seeds. Proposed PLAN-012 (tighter town, Handcart no longer grows the market, 4-player demand +15%, apple cost 3/2/1, RULES_VERSION 3). Production rules unchanged; awaiting approval.
- 2026-10-27 09:12 · Record note: the PLAN-010 verification (history line of 07:54) is written up in REP-026.
- 2026-10-27 09:58 · Owner: "yes approvrd together with the optional changes and the new layout from the mockup" (reply to the PLAN-012 approval request). PLAN-012 approved and started, widened to the six optional items of REP-027 section 6b and the Home screen from the cosy pixel mockup (the only mockup screen not yet built; interpretation stated to the owner). Optional items still needing values are set by lab test first.
- 2026-10-27 10:47 · PLAN-012 built and verified on workflow/plan-012 (REP-028): rules version 3 with the market rebalance, trait and mango changes, tomorrow's lorry, Juice Press juice tip, retuned Market League and the cosy home screen. verify and verify:full PASS; production equals the lab package on 450 games. Not changed after lab tests failed: Long-mode tuning and Cool Room rework. Skill gap (Hard beats Normal) 60%, below the 65% aim, flagged to the owner. Playtest NOT RUN. Merged to main for the preview; the live game is unchanged.
- 2026-10-27 12:03 · Owner: "edit the UI/UX so that 1. secret price and events pops up to fill the screen (instead of just the bottom half of the screen 2. rework the rivals indicator during secret price, now its not clear still". Explicit instruction, recorded as approval of PLAN-013 (UI only, no rule change); started on workflow/plan-013.
- 2026-10-27 12:21 · PLAN-013 built and verified (REP-029): full-screen pricing and event sheets, 'Your rivals' strip (sealed / deciding / next in line, wanted fruit held), matching seat badges. verify and verify:full PASS (price leaks 0). Merged to main for the preview; live game unchanged.
- 2026-10-27 13:00 · Owner asked to re-analyse fruit pricing, decay, events, traits, upgrades and supply/demand against three guidelines (tabletop-playable, real market feel, fruit playstyle matching payout). REP-030: rules version 3 only partly meets them. Lab sweeps 12-15 (600 held-out games per test): printed numbers reproduce the formula flow; posted prices plus a market pulse plus blueberry/apple price trims give a 12-point plan spread and payouts ordered mango > blueberry > watermelon > apple. Proposed PLAN-014 (awaiting approval). No game code changed.
- 2026-10-27 13:23 · Owner: "approve all the idea, work out the mechanics, and summarise to me game back highlighting the concept and game mechanics as i want to benchmark against similar genre board games and traditional popular board games". PLAN-014 approved including the supporting ideas (queue, Market Days, crates, event redesign). First deliverable: finalised mechanics and a concept summary for benchmarking; code build follows.
- 2026-10-27 13:46 · REP-031: final tabletop mechanics (rules version 4) and a concept and mechanics summary for benchmarking. Lab sweeps 16-18: fruit-specific events (Blueberry Glut, Apple Blight), Good Harvest scaled by players, no Inspector on the final day; Market

Days kept as an optional rule (most even plans but computers can't plan ahead). Open: Shop Bell, Juice Press and Cool Room retune; Long games favour ripening. Also fixed the coach tip covering the Seal button (the cause of the stuck CI run).

- 2026-10-27 14:36 · Owner shared a benchmark review of REP-031 and asked for tweaks plus a favour card that reveals the face-down demand. REP-032: five-beat day (open stalls, bell, rush), prototypes A/B/C, Market Gossip replacing Free Cash. Lab sweep 19: cash catch-up has no exploit (upgrader 33%); uniform freshness improves balance (26-38%); core without traits/favours/upgrades lets ripening dominate (60%). PLAN-015 proposed; PLAN-014 build held until the owner decides.
- 2026-10-27 14:51 · Owner: "approve plan 15". PLAN-015 approved and started on workflow/plan-015; it is built together with the approved PLAN-014 items as rules version 4.
- 2026-10-27 15:37 · PLAN-015 with the PLAN-014 items built as rules version 4 (REP-033): open stalls, the bell, Market Days and Last Call, four specialities, Market Gossip with server-side privacy, day ledger, playtest set-up options, rules text. verify and verify:full PASS (leak check 0). Balance: Hard beats Normal 56%, comebacks 31%, traits 30-38%; Cold Room and Juice Bar weak for computers; little undercutting among computers. Playtest NOT RUN. Merged to main for the preview; live game unchanged.
- 2026-10-27 16:20 · Owner: "approve the speciality price cut". PLAN-016 approved and built (REP-034): Cold Room $4, Juice Bar $3. Same-seed comparison: no measurable gain for computer players (rush Cold Room 12%, Juice Bar 16%); no figure worse beyond the margin. Cause: tight supply means little old fruit or holding value for computers. RULES_VERSION stays 4 (version 4 never published live). Merged to main for the preview.
- 2026-10-27 16:42 · Owner: "approve option 1 and 2". PLAN-017 built (REP-035): farmer cards +1 per listed fruit; Normal/Hard computers use Juice Bar, Cold Room and Market Days. Same seeds: Hard beats Normal 60%, Easy 82%; comebacks 38%. Not fixed: Cold Room and Juice Bar add little (skipping costs nothing). New issue: Big Basket 45%, 2-player blowouts 42%. PLAN-016 CI green (verified). Options for volume and specialities put to the owner.
- 2026-10-27 16:56 · Owner asked for a critical readiness rating for publishing both a physical and a digital version. REP-036: digital 5.5/10, tabletop 3/10. Main risks: rules version 4 never played by people; rules still changing; nightly ageing too fiddly by hand (date-stamped crates proposed, needs approval); balance gaps; missing commercial basics. Recommended path: freeze rules, digital and paper playtests, then balance, art and launch prep. No game change.
- 2026-10-27 17:02 · Owner asked to track balance, retention and data insights. REP-037 proposes PLAN-018: anonymous counters tagged by rules version (set-up, traits, specialities, computer levels, price and spoilage buckets, favours, online health, app version) and a weekly insights report via GitHub Actions. Data handling change: awaiting approval. Three questions: report location, retention 90 or 365 days, STATS_KEY as a GitHub secret.
- 2026-10-27 19:45 · Owner answered the lifecycle trial decision card: "1a 2a 3 tabletop which can be translated into the game at the same time, both with the same mechanics but the digital version to have single player capability". Approved DEC-001, DEC-004, DEC-005 and DEC-014 (previously proposed). Recorded DEC-026 (freeze rules version 4 until the first human playtest) and DEC-027 (one game, two editions with identical mechanics; digital has single-player). PLAN-019 (first human playtest, no game code) proposed, awaiting approval (REP-038). No game change.
- 2026-10-27 19:59 · Owner: "Approve PLAN-019 only." PLAN-019 (first human playtest, no game code) approved; delivery in progress. Approval covers that plan and its listed documentation only. Rules freeze (DEC-026) stays in force. Waiting for the owner's answers

on who the testers are, how nightly ageing is handled in the paper round, and where sessions run, and for the owner to confirm the draft questions and criteria (REP-039). No game change.

- 2026-10-27 20:05 · Owner asked for the paper round to be done with the app. Offered: worked example, play the app as the table, or build a table-helper mode. Owner answered "as it is right now" (taken as: play the current app as the table; no new app mode, no paper kit). PLAN-019 scope amended: the paper round becomes a pass-and-play table round on the current app; acceptance T2 updated. Hand-played ageing, the premise of DEC-027, stays untested and needs a later plan. No game change (REP-040).
- 2026-10-27 20:06 · Owner: "no, i'll get people to play it from the browser link". PLAN-019 scope amended again: no table round; 3 to 5 unfamiliar players play the preview from a link on their own devices and answer a short set of questions. Evidence will be self-reported and unobserved unless the owner joins a call. Hand-played ageing (DEC-027) stays untested. No game change (REP-041).
- 2026-10-27 20:09 · Owner: "The draft questions and criteria are fine. Lock them in as of today." PLAN-019 questions and continue/revise/stop criteria locked (T1 PASS); full text in REP-042. Changing them after results arrive is not allowed without a recorded change made before the results are read. Open point for the owner: how to read "x of 4" if 3 or 5 people test (REP-042). No game change.
- 2026-10-27 20:12 · Owner: "Yes, use the proposed reading for 3 or 5 testers." Confirmed for PLAN-019: with 3 or 5 testers, Continue needs at least three quarters (3 of 3, or 4 of 5); Revise needs at least half (2 of 3, or 3 of 5); anything lower is Stop or park. With 4 testers the locked thresholds apply as written (REP-043). No game change.
- 2026-10-27 21:18 · Owner: "i feel like the questions need to be simpler to answer, and any particular reason we are asking for 1v1? make it 1v3 if possible". Before any tester played: PLAN-019 testers now play one Quick game against three computer shops (four shops in all) instead of one, and the tester questions are replaced by seven simpler ones (four tap-to-answer, three short optional). Locked criteria (REP-042, REP-043) unchanged. DEC-016 said one computer shop; superseded for this test only. No game change (REP-044).
- 2026-10-27 22:18 · Owner passed on informal feedback from one 4-player game (10 rounds, about 30 minutes; setup and tester unknown): supply cannot keep up with demand from mid-game; every player has the same icon; 10 rounds feels long; no undo after tapping a fruit before confirming a buy; cannot edit the game name in the room and the name defaults to "You". Logged in REP-045 as informal feedback, not counted as a PLAN-019 session. Supply and length are rules or balance and stay frozen (DEC-026); the other three are UI backlog items for after the playtest. No game change, no decision.
- 2026-10-27 22:20 · Owner: "the ones i entered here is the feedback, just end the testing". PLAN-019 ended early. No formal session was run and the Tally form had no submissions; the only feedback is the informal 4-player game in REP-045. T1 PASS, T3 PASS, T2 and T4 NOT RUN. Delivery shown as implemented only because the records have no ended-early value; the playtest goal was NOT met and no continue, revise or stop reading exists. DEC-016 (first playtest) is not satisfied. DEC-026 (rules freeze) review trigger reached: owner to say whether the freeze stays or is lifted (REP-046). No game change.
- 2026-10-27 22:24 · Owner: "Lift the rules freeze (DEC-026). Plan fixes for supply, session length and the UI requests." Rules freeze lifted: DEC-026 superseded by DEC-028 (approved). Plans proposed, not approved: PLAN-020 supply, PLAN-021 session length,

PLAN-022 client-only UI requests, PLAN-023 undo a buy and rename in the room. Analysis from code reading only; nothing run or built (REP-047).

- 2026-10-27 22:25 · Correction to REP-047, from reading the code after it was written: an undo button already exists for one-device games and is hidden in online rooms, so PLAN-023 is now "undo in online rooms" (not a new undo). The feedback mentions "the room", so the informal game was probably online with human players (inferred, not confirmed); PLAN-021 timing now includes a 4-human online room. PLAN-020, PLAN-021 and PLAN-023 amended while still proposed (REP-048). No game change.
- 2026-10-27 23:10 · Owner: "do all except 22, for 22 analyse demand as well, feels like theres too much customer towards the final few turns". Read as: PLAN-020 (supply, now also demand), PLAN-021 (session length) and PLAN-023 (online undo and rename) approved; PLAN-022 (icons and blank name) NOT approved, stays proposed. PLAN-020 step 1 is measurement only; steps that change rules, speed or the server wait for the owner choosing an option or approving the design. Repo dependencies installed from the lockfile in the local clone to run the checks (no new dependency). No game change yet (REP-049).
- 2026-10-27 23:16 · PLAN-020 step 1 done (REP-049): in simulation with 4 shops, last-3-day customers rise from 4.9 to 8.9 a day, fruit sold stays near 14 a day, and Market Days (no lorry) leave 15%, 41% and 49% of shop turns facing an empty market. Putting the lorry back on Market Days: shop turns short 35% to 8%, fill rate 30% to 59%. Computer players only. Awaiting the owner: option (A, A+C, A+D or other) and a target before any change. PLAN-021 findings and PLAN-023 design note (Design 1 recommended) are ready for the owner. No game change.
- 2026-10-27 23:29 · Owner: "why is the number in decimal? remember that rules should be able to be implemented on a tabletop elsewhere, else do plan a + c, target is so that the board would not be empty / ppl cant buy anything". Chosen: option A + C for PLAN-020. Target set by the owner: the market should not be empty so that people can always buy something (measured as the share of shop turns that find an empty market). Built on branch workflow/plan-020 as rules version 5, local only: lorry every day, walk-ins players minus 2 every day (+1 on Market Days). Lint, types, 92 unit tests and the build pass; browser tests (verify:full) NOT RUN. Simulation after the change (computer players, 300 games per setting): empty-market turns 0.0% to 1.0% (rules 4: up to 22%). Not on the preview, not pushed, not live (REP-050).
- 2026-10-27 23:29 · Correction: the header of REP-050 and the STATE line say 23:35; the true time of the DEC-029 and PLAN-020 build entry is 23:29. Reports are not rewritten. No game change.
- 2026-10-28 00:28 · Owner: "implement all the fix except for the price ladder, i dont get the problem and your solution". Built on branch workflow/plan-024 (local, not pushed): tie-break by buying order instead of live cash, story and How to play text, tests, and a "Playing on a table" section in RULES.md. Price ladder and colour zones not built. Lint, types, 92 unit tests and the build pass; browser tests NOT RUN; hand-played tabletop test NOT RUN. Still rules version 5 (unreleased) (REP-051).
- 2026-10-28 00:46 · Owner: "make it so that character is selected when you enter the room instead apply all the changes and push it the game". Character cards built on branch workflow/plan-025 (DEC-031): six characters with fixed perks, picked on entering a room or on the set-up screen. Lint, types, 100 unit tests and the build pass; browser tests NOT RUN locally (CI runs them on push). The owner also asked to push: the branch is pushed for CI,

and merged to main (the preview) only if CI passes. The live game is not deployed; that needs the owner to run Deploy LIVE game and type PUBLISH (REP-052).

- 2026-10-28 01:03 · CI on branch workflow/plan-025 (rules 5, tabletop fixes and character cards together) passed in full, including the browser tests: PLAN-020 T3, PLAN-024 T4 and PLAN-025 T5 now PASS. Merging to main, which updates the preview after CI passes there. The live game is not deployed. The new screens are still not seen on a phone (PLAN-025 T4 NOT RUN) (REP-052).
- 2026-10-28 20:42 · Owner: "expand customer card to 36, keep the demand profile and design language in line with the rest of the game" built on local branch workflow/plan-026 (not pushed): 36 customer cards (DEC-032, PLAN-026, REP-053). Then, after the art audit and the three-way look comparison, owner chose "Look C": calm flat scenery with pixel objects, one Farmers Market board shared by the screen and the table (DEC-033). Nothing built for it; PLAN-027 (style guide and palette), PLAN-028 (shared board) and PLAN-029 (animated opening) are proposed for approval (REP-054).
- 2026-10-28 20:46 · Owner: "Board first". PLAN-028 (shared Farmers Market board for the screen and the table) approved. Order of work as the owner chose: the printed A3 prototype first, then the screen. No game code changed in this step (REP-055).
- 2026-10-28 20:49 · PLAN-028 first step done: printable A3 board prototype and token sheets (Board_Prototype_A3, REP-055). Token counts measured in computer-player games: most of one fruit ever in play 27 blueberry, 25 watermelon, 23 apple, 23 mango; one slot up to 14. Not printed on paper, screen board not started, no game code changed.
- 2026-10-28 21:16 · Owner clarified "Board first": decide the board look first, then design the rest of the game back from it (the board is the proof of concept). The earlier reading (approve building the board, print prototype first) was wrong. PLAN-028 returned to proposed and not started; the printable prototype (REP-055) stays as one candidate look. Four board looks with the kit each would hand the rest of the game are ready for the owner to choose (DEC-034, REP-056).
- 2026-10-28 21:16 · Records fix: approvals are never rewritten, so PLAN-028 keeps the owner words "Board first" and is marked superseded. The board work continues as PLAN-030 (proposed), to follow the board look the owner chooses (DEC-034).
- 2026-10-28 21:36 · Owner: "approve everything", straight after the naming audit. Recorded as approval of all names in Naming_Audit including the optional ones (DEC-035) and the rename plan PLAN-031, to be built after the board look is chosen. Not read as approval of the board look (DEC-034) or of PLAN-027, PLAN-029 and PLAN-030, which stay proposed until the owner says so (REP-057).
- 2026-10-28 22:23 · Owner: "do per your suggestion" after the recap of DEC-020, PLAN-022, PLAN-018 and PLAN-003. Done: DEC-020 closed as superseded by DEC-033 (it was never approved); PLAN-022 cut to the blank-name part, approved and built on local branch workflow/plan-022 (not pushed; keyboard on phones NOT RUN); PLAN-003 written up with checks and split into PLAN-032 (free drills) and PLAN-003 (the paid part, no spend until the owner sets a cap), both proposed. PLAN-018 left proposed: approve it when a live release is planned, because the data cannot be filled in later (REP-058).
- 2026-10-28 22:28 · Owner: "approve everything above", after the recap of DEC-034, PLAN-027, PLAN-029 and PLAN-030. Recorded: board look D chosen (DEC-034 approved), and PLAN-027 (style guide and palette), PLAN-029 (animated opening) and PLAN-030 (shared board) approved, to be built in that order. PLAN-027 started: the style guide is drafted from

look D (docs/ART-STYLE.md) for the owner to approve. New proposed plan PLAN-033: restyle the other screens to the board look, which no approved plan covered (REP-059).

- 2026-10-28 22:39 · Owner: "resolve the gap, and approve plan 27". PLAN-033 (restyle the other screens to the board look) approved, to be built after PLAN-027. The style guide is approved. PLAN-027 built on local branch workflow/plan-027 (not pushed): one palette with a lossless colour move, emoji replaced, new size and emoji checks. T3 narrowed to "block new off-list sizes" because fixing the existing ones changes layouts; PLAN-033 takes that over. Phone look and browser tests NOT RUN (REP-060).
- 2026-10-28 22:48 · Owner: "replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady". Uncle Toad was replaced by Mak Jenab (DEC-036, PLAN-034): a lady stallholder with the Nest Egg perk, a headscarf portrait and thrifty lines. Built on local branch workflow/plan-034 (not pushed); phone look NOT RUN (REP-061).
- 2026-10-28 22:54 · Owner: "sorry jenab is a customer". The shopkeeper reading was wrong: DEC-036 and PLAN-034 are closed as superseded (their approval is kept as recorded) and the Uncle Toad change is reverted in code. Jenab is now a customer: Auntie Jenab replaces Chris the Clown with the same order (DEC-037, PLAN-035, implemented on local branch workflow/plan-034, not pushed). The owner also asked which plan covers character theming and design coherence: none did, so PLAN-036 is proposed (REP-062).
- 2026-10-28 23:03 · Owner: "change it to Rara Jenab, design it around the name". Auntie Jenab became Rara Jenab the Cheerleader, designed around "ra-ra" (DEC-038 refines DEC-037; PLAN-035). Owner also said the characters being animals is wrong, customers should all be humans and the whole cast should follow one coherent theme tied to the design style. Then: "approve the above and then work on plan 36" (PLAN-027, PLAN-033 and PLAN-031 were already approved, so nothing new was recorded for them). PLAN-036 approved and started as design work only: the cast audit is done (Cast_Coherence_Audit.html); the owner now decides DEC-039 (cast theme) and DEC-040 (shopkeepers become humans) before any art or code changes (REP-063).
- 2026-10-28 23:14 · Owner: "t1 and proceed with dec040". DEC-039 approved as T1 (townsfolk of the harbour market) and DEC-040 approved (shopkeepers become humans; DEC-018 superseded). First pass built on local branch workflow/plan-034 (not pushed): six human shopkeepers with the perks unchanged, the 18 non-human and costume customers turned into everyday roles, a draft Cast coherence section in the creative bible, and cast checks in tests. Portrait refinement, the bible approval and phone look remain (REP-064).
- 2026-10-28 23:24 · Owner review of the cast: "remove the manner nickname for shopkeepers, remove "cheerleader" from cheerleader Rara, replace the wedding planner, lumberjack, night baker, pony rider". Done on local branch workflow/plan-034 (not pushed): shopkeepers are plain first names (Rosa, Benny, Chen, Saleh, Raj, Kiran); Rara Jenab no longer carries a role word; the wedding planner, lumberjack, night baker and pony rider became Tailor Siti, Fisherman Marcus, Librarian Robert and Photographer Sarah (names and orders unchanged, so demand is unchanged). Side effect: the shopkeepers' plain first names are no longer reserved player names; the old animal-era names stay reserved (REP-065).
- 2026-10-28 23:26 · Owner: "Approve rules": the Cast coherence rules in the creative bible are approved (DEC-041). PLAN-036 T3 and T9 now PASS. The glasses part was recoloured to a brown frame so they read as spectacles (REP-066). Open in PLAN-036: a check that every quirk says what someone does (T5), the screen, printed cards, character pick and gallery

drawing the same portraits (T6, only the sheet checked), phone look (T7) and browser tests (T8). Nothing pushed.

- 2026-10-28 23:44 · Owner: "Push to preview". Branch workflow/plan-034 (36 customers, blank name field, one palette and no emoji, Rara Jenab, the all-human cast and the cast rules) pushed; CI passed in full including the browser tests: PLAN-026 T3, PLAN-022 T4, PLAN-027 T5, PLAN-035 T4 and PLAN-036 T8 now PASS. Merging to main, which updates the preview after CI passes there. The live game is not deployed. Phone look is still NOT RUN for these plans (REP-067).
- 2026-10-29 00:18 · Owner: "ok" to Design 1 for online undo and rename in the lobby only (PLAN-023). Built on local branch workflow/plan-023 (not pushed): the server keeps the state your own buying turn began in and puts you back to it on request (own turn only, refused if stale, repeated or out of turn); everyone sees a message; rename in the lobby only with the usual name cleaning; a new message version so old and new builds never share a room (the rules version is unchanged, so daily boards are not reset). Lint, types, 122 unit tests and the build pass; browser tests and phones NOT RUN (REP-068).
- 2026-10-29 00:32 · CI on branch workflow/plan-023 (run 40724975276) failed in one browser test: the online match test looks for the text "London Friend (you)" in the lobby, and the new rename box put the name inside an input. Fixed: your own name is shown as text again, with a small Rename button that turns it into a box. Everything else in that run passed (quick match, four players, reconnect). Re-running CI (REP-068).
- 2026-10-29 00:33 · Owner: "approve everything except plan 21, for the design related, require my approval on the design, else just run". Recorded as DEC-042: order PLAN-031, PLAN-030, PLAN-033, PLAN-029; the lorry is redrawn flat and smooth; no portrait frames; PLAN-032 and the PLAN-036 quirk check approved; PLAN-021 stays open (time target not set); design work (board, lorry, restyle, opening) is shown to the owner and built only after the design is approved, everything else is just built (REP-069).
- 2026-10-29 00:42 · Rename pass (PLAN-031) built on local branch workflow/plan-031 (not pushed): Wholesale Market, stall, shopkeeper, Tired, piece, Stocking, The rush, Start buying and Start game, Delivery card, Gossip, Tip Jar, perk, stall upgrade, Handcart, regular, special guest, the crier, Closing Day, Daily market, Basic game, Price drift, Fixed prices, with a test that blocks the retired names. Good Riddance was left unrenamed while the owner decides its fate. PLAN-036 quirk check added (T5 PASS). PLAN-032 logic-level restore test and a device checklist added; the real server drills and phone runs are NOT RUN (REP-070).
- 2026-10-29 00:55 · Owner: "okay approve this design": the table screen design is approved (DEC-043). Built on local branch workflow/plan-030 (not pushed; it sits on the rename pass): the Wholesale Market in look D, a flat lorry on a road strip, paper stalls, pill bars, cream mat. The e2e and smoke scripts clicked the old button labels (Open the shops, Open the market); updated to Start game and Start buying. The printed mat in look D and the shared slot data are still to do (REP-071).
- 2026-10-29 01:14 · Owner: "Push to preview". PLAN-023 (online undo and lobby rename) is on the preview: CI passed on the branch and on main. Branch workflow/plan-030 (rename pass, quirk check, free drills, the table screen in look D) pushed; CI passed in full including the browser tests: PLAN-031 T3, PLAN-030 T2 and T6 now PASS. Merging to main, which updates the preview after CI passes there. The live game is not deployed. Phone looks and two-phone undo are still NOT RUN (REP-072).
- 2026-10-29 01:21 · Owner: "cancel 21 and lets work on 42". PLAN-021 (session length) cancelled (DEC-044; its earlier approval stays as recorded and the plan is marked

superseded). Work continues on DEC-042: the order PLAN-031 (done), PLAN-030 (screen done; printed mat and shared slot data next), PLAN-033 and PLAN-029, with design proposals shown to the owner first (REP-073).

- 2026-10-29 01:26 · Continuing DEC-042 (REP-074): the shared slot data is built (trackSlots in the engine, read by the screen and by the printed mat; 4 new tests, 130 in all). Two design proposals are ready for the owner and nothing is built from them yet: the printed mat and tokens in look D (Design_Proposal_Printed_Mat, also as a PDF) and the other screens in look D (Design_Proposal_Screens: home, set-up, morning with a hint, results). The animated opening still needs its design redrawn in the new look before it comes to the owner.
- 2026-10-29 01:34 · Owner: "okay with the design, but include stats like the best moments fir the results". DEC-045 approves both design proposals. PLAN-033 built on local branch workflow/plan-033 (home, set-up, results with stats and best moments, trays, pop-ups in look D); lint, types and 130 tests pass. Not pushed.
- 2026-10-29 02:49 · Owner approved overnight work (DEC-046): push to preview only, never live; no rule changes.
- 2026-10-29 03:12 · Overnight (DEC-046): pop-ups, trays, sheets and desktop margin restyled in look D (REP-075, branch CI green). Balance study: Good Riddance and shopkeeper perks measured, no rule changed (REP-076). Print kit and playtest pack built outside the repo (Print_Kit folder).
- 2026-10-29 03:16 · Overnight (DEC-046): print kit built (mat and tokens, 36 customer cards, delivery, favour, event and shopkeeper cards, playtest pack) and two opening-animation designs drawn for the owner to choose (REP-077). Nothing built from the opening designs.
- 2026-10-29 07:59 · Owner chose opening design 1 (DEC-047). Built on workflow/plan-029: Opening component, played on new game, tap to skip, reduced-motion safe. Lint, types and tests pass; browser tests run in CI.
- 2026-10-29 09:00 · Owner changed the opening (DEC-048): plays on launch, about 2.5 seconds. Supersedes the timing and trigger in DEC-047.
- 2026-10-29 14:10 · Owner approved the bulk session (DEC-049). Work on workflow/plan-037-rulebook-tutorial; QA sweep took contrast findings from 136 to 3; soak of 1500 games clean.
- 2026-10-29 15:30 · Owner answered the decision card: 1C Scout replaces Good Riddance (DEC-050, PLAN-037), 2A perks unchanged, 3A release 1.2.0 now (DEC-051). Release 1.2.0 prepared as a candidate; publishing stays with the owner.
- 2026-10-29 17:30 · Owner chose tiles and sign for the stalls and the bunting roof for the Sunny Pickup lorry (DEC-052). Built as pixel-art props on branch workflow/plan-038-lorry-stalls; included in release candidate 1.2.0.
- 2026-10-29 18:10 · Owner asked for a full-screen option (DEC-053). Built on branch workflow/fullscreen and added to the 1.2.0 candidate.
- 2026-10-29 21:05 · Release 1.2.0 published (Deploy LIVE game on main f5f9390, started by Claude at the owner's explicit request; smoke test passed; tag v1.2.0).
- 2026-10-30 15:30 · Owner asked for a How to play rework with a skippable tutorial (DEC-054), then a tabletop review with fixes (DEC-055). Built on branch feat/how-to-play-rework as release 1.2.1 candidate; publishing stays with the owner.
- 2026-10-30 17:30 · Release 1.2.1 published (Deploy LIVE game on main ed8122f, started by Claude at the owner's explicit request; smoke test passed; tag v1.2.1).
- 2026-11-04 10:00 · Owner asked for accounts, 43 achievements, free colours and looks (DEC-056, DEC-057), then online looks and an easy wardrobe in rooms. Built on branch

feat/accounts-looks as release 1.3.0 candidate; owner said "yes proceed to publish please, i approve".

- 2026-11-05 12:00 · Full audit and balance simulation (REP-086). Owner asked for the suggested changes to be built and made live (DEC-058). Built on branch release/1.4.0 as release 1.4.0 candidate.
- 2026-11-05 14:00 · Release 1.4.0 published (Deploy LIVE game on main 971b500, started by Claude at the owner's explicit request; smoke test passed; tag v1.4.0).
- 2026-11-05 18:00 · Owner asked for everything left in the audit (DEC-059). Built on branch feat/polish-1-5 as release 1.5.0 candidate.
- 2026-11-05 19:00 · Complete play kit built in one folder and one PDF (rulebook, board, cards, tokens, coins, notes, stall sheets, ledger). Numbers on the cards, board and notes now use the rounder font because 5 read as S and 2 as 8 in the pixel font; owner: "yes im okay". Rules and cards unchanged.
- 2026-11-05 20:00 · Release 1.5.0 published (Deploy LIVE game on main b92dc51, started by Claude at the owner's explicit request "deploy live game"; smoke test passed; tag v1.5.0). Owner approved the printed card backs (DEC-060).
- 2026-11-05 22:00 · Owner chose home screen layout 3 and New game option A ("layout 3, option A, yes approve it already"), asked for the rounder number font ("Yes, apply it") and approved deploying everything ("check this and then approve everything to deploy the game"). Built as release 1.6.0, PLAN-046, DEC-061.
- 2026-11-05 23:30 · Release 1.6.0 published (Deploy LIVE game on main 55345c6, started by Claude at the owner's explicit request "deploy live"; smoke test passed; tag v1.6.0). The first build on staging was black because a font rule sat above the palette import; caught on staging before going live, fixed and covered by a test.
- 2026-11-07 03:45 · Owner asked for an art direction and character design workflow for a Kickstarter-first edition (audit first, no assets). Audit done (REP-087): proposed DEC-063 (Look E, crate-label world with townsfolk characters) and PLAN-048 (Character Bible, card templates and a two-character pilot), both awaiting approval. Draft direction in docs/art/ART_DIRECTION.md v0.1. No game, app or print-kit change.
- 2026-11-07 03:58 · Correction: the 2026-11-07 03:45 line was written at about 03:32 and committed at 03:37 (6d24f1a); 03:45 was not read from the clock. Owner's brief, in their words: "Audit the existing Fruit Market art direction and character requirements. Recommend the strongest visual identity for a Kickstarter-first tabletop release with a subsequent mobile adaptation ... Do not generate or implement assets yet. Wait for my approval." Before delivery, an internal review corrected REP-087, ART_DIRECTION.md (now v0.2), DEC-063 and PLAN-048: AI review passes described as such, illustrator cost range fixed, back fruit order and edge rules added for every hidden deck, glimpse and show-through rules, card-template design shown before building, PLAN-048 limited to unpaid work, and DEC-063 no longer pre-answers the owner's three questions.
- 2026-11-07 04:24 · Owner approved Look E for design only (DEC-063, exact words recorded) with three answers: AI concepts finished by the owner with no illustrator; mobile characters will eventually replace the pixel characters throughout; pilots Benny and Sailor Jia Wen. Owner: "before proceeding, i want to rework the character name and naming style, i dont like some names, then can proceed". PLAN-048 rewritten to the owner's design package (naming review first, then Character Bible v1.0 for the pilots, reference sheets and prompts, card mock-up concepts, mobile adaptation specifications) and approved. No concept images or

concept review were found on Drive; 'corrections identified in the concept review' is read as the corrections in ART_DIRECTION v0.2.

- 2026-11-07 04:52 · PLAN-048 step 1: naming review sent to the owner (REP-088): every name with flags, three naming styles (A recommended: Job + first name, tidied), suggested fixes for sound-alike pairs (Ben/Benny, Wei Jie/Jia Wen, Hafiz/Hadi), type-hinting titles and stereotype jokes. Waiting for the owner's choices; no game, data or art change.
- 2026-11-07 11:16 · Owner approved the shopkeeper and guest names (DEC-064): Rosa, Uncle Benny, Ah Chen, Saleh, Raj, Kiki (was Kiran); abilities unchanged; guests unchanged. The owner's message started at section 2, so section 1 (the 36 customers) has not arrived; asked for it. The pilot Benny is now Uncle Benny. Creative bible shopkeeper lines updated; no game, data or art change.
- 2026-11-07 11:22 · Owner sent the 36 customer names (DEC-065, exact list recorded). Pilot customer is now Marina Wen (was Sailor Jia Wen). Flagged to the owner for a yes or no: C30 Benny Bell vs shopkeeper Uncle Benny, C22 Otto Oddjob (famous film character), C8 Postman Pip (close to Postman Pat). Naming done enough to start the design package, which covers the two pilots only. Creative bible customer rule updated.
- 2026-11-07 11:27 · Owner on the three flagged customer names (C30 Benny Bell, C22 Otto Oddjob, C8 Postman Pip): "keep it". All three stay as in DEC-065; the sound-alike and look-alike risks are accepted by the owner and will be watched in the read-aloud test.
- 2026-11-07 12:20 · Owner supplied the Great Fruit Exchange deck art (three PDFs) and authorised implementation throughout the game (DEC-066, exact words recorded): events Art 1, shopkeepers Art 2, customers Art 3. DEC-033 (pixel look) superseded by DEC-066. PLAN-049 started on branch ccr-d9f10c6d-75280w. Baseline before changes: npm run verify PASS (196 tests).
- 2026-11-07 12:36 · Owner: "approve to push it to live" (publication approval for the Great Fruit Exchange release, PLAN-049, given while the work was still being built). Read as approval to publish the finished work once it passes npm run verify, the browser tests and a preview check; nothing is published before then. Release steps: version bump and release candidate entry, merge to main (preview), then Deploy LIVE game.
- 2026-11-07 15:33 · PLAN-049 built and verified on the branch: npm run verify PASS (233 tests), npm run test:e2e PASS (solo, performance budget, online with 0 leaks, quick match, four players, rejoin), all 45 names checked. Release 1.8.0 prepared as a candidate (version bump, player notes). Next, per the owner's "push it after everything passes": merge to main (preview), check the preview, then Deploy LIVE game.
- 2026-11-07 17:00 · Release 1.8.0 published (The Great Fruit Exchange). CI on main df61983 PASS (verify:full) and preview updated with its smoke test PASS; Deploy LIVE game run 23 started by Claude under the owner's "push it after everything passes": verify, release check, deploy, live smoke test PASS; tag v1.8.0. The preview could not be opened from the sandbox (host blocked), so the visual preview check is NOT RUN; the same build passed local visual QA and the browser tests. Print-kit layout fixes (print-only) follow on the branch: four fixes PASS on an independent render check.

REP-001 · Baseline audit and PLAN-001 (Package A)

Game	Fruit Market (`murshidihashim/Fruit-Market`)
Report date	2026-10-26 13:15 (Asia/Kuala_Lumpur)
Build audited	`main @ c915e49` (CI run 2 green, deployed)
Prepared by	Claude (studio lead), from the "Personal workflow for Murshidi", edition 1.2
Type	Planning only. **No game implementation changes were made.**
Approval status	PLAN-001 **awaiting approval**

1 · Baseline: where the game stands

Stage: between *MVP / private alpha* and *browser beta*. The game is complete and publicly reachable, but no unfamiliar players have been observed yet, so player understanding and fun are unproven.

Working and must be preserved

- Rules engine: deterministic, seeded, a single entry point (`applyAction`), rules version 2. 63 unit tests, including 200 simulated games.
- Saves: versioned envelope (`v2` plus rules version); old saves still load.
- Online play is server-authoritative (Cloudflare Worker with Durable Objects):
  - secret prices are redacted; the leak check reads every network message and found 0 leaks
  - reconnect, a 75 s grace period then computer takeover, and quick match
  - per-IP rate limits, a name policy, and a replay-verified daily leaderboard
- Android build (Capacitor) is produced by CI on every push.
- Docs: `RULES.md` , `ONLINE.md` , `PUBLISHING.md` , `docs/ANALYTICS.md` , the studio audit `docs/studio/01–10` (backlog = `10`).

Gaps against the workflow guide

#	Finding	Type	Severity
G1	Every push to `main` deploys straight to the public URL. The guide requires a separate publication approval.	Process defect	High
G2	`main` has **no branch protection** (checked via GitHub: `protected: false`). CI results don't block anything.	Measured	High
G3	No single verification command. There are separate `lint` , `typecheck` , `test` , `build` , `test:e2e` scripts.	Measured	Medium
G4	No `CLAUDE.md` , current-state record, game brief, or owner brief. Each session relies on long chat history.	Measured	Medium
G5	No decision records or PDF register before this report.	Measured	Low
G6	Real-device testing (Android Chrome, iPhone Safari): never done.	NOT RUN	High for release

#	Finding	Type	Severity
G7	No backup or restore for server data (leaderboards, stats). Room crash recovery not tested.	NOT RUN	Medium
G8	Monthly hosting cost unknown (Cloudflare plan not recorded).	Unknown	Medium
G9	The game-screen open time is noisy (217–489 ms locally). Quick match failed once in CI and passed on the rerun.	Measured, pre-existing	Low

2 · Readiness baseline (evidence as of today)

Gate	Check	Result	Evidence and limitation
Build ready	Clean install, lint, typecheck, unit tests, build	PASS	CI run 40266399588 on `c915e49`
Play ready	Automated solo game to results	PASS	`scripts/smoke.cjs` , local and CI
Play ready	Owner completes a match in a preview	NOT RUN	Needs you
Online ready	Two independent clients finish a match; results agree	PASS	`e2e-online.cjs` , local and CI run 2
Online ready	Private state: no secret prices in any message	PASS	40 frames checked, 0 leaks
Online ready	Illegal actions, wrong phase, votes, takeover	PASS	`tests/room.test.ts` (16 tests)
Online ready	Reconnect, abandonment, quick match	PASS	e2e; one CI flake, then a pass
Online ready	Server crash recovery	NOT RUN	Room state is persisted per action; never tested
Release ready	Real Android Chrome and iPhone Safari	NOT RUN	Emulation only
Release ready	Backup and restore, rollback drill	NOT RUN	—
Release ready	Live smoke test after deploy	NOT RUN	The sandbox proxy blocks the live URL; the deploy job itself succeeds
Release ready	Publication approval gate	FAIL	See G1

Release-ready is **not claimed**.

3 · PLAN-001 · Package A: baseline, one verification path, preview and owner brief

Player problem: updates reach players without your approval and without one clear pass/fail check, so a bad change could ship silently. You also have no short guide for trying a build yourself.

Included scope

1. **One verification command.** `npm run verify` runs lint, typecheck, unit tests and build, and stops at the first failure (about 1 minute). `npm run verify:full` adds the browser suite (about 15 minutes). CI calls `verify:full`. Existing scripts are reused; nothing new is installed.
2. **Preview before publication.**
   - Pushes to `main` run checks only.
   - Production deploys happen only when you trigger *Deploy* (a manual GitHub button), which first runs `verify`.
   - Each change also gets a private Cloudflare **preview URL** that doesn't touch the live game. I'll confirm in Cloudflare's current docs that this is free on your plan; if it isn't, the local preview steps are the fallback.
3. **Branch protection instructions.** A 5-step guide for you to turn it on in GitHub settings (I can't change repository settings). It requires the CI check before merging.
4. **Short records** (reusing the studio docs, not duplicating them):
   - `CLAUDE.md`, about 90 lines: rules, actual commands, preservation constraints, pointers
   - `docs/workflow/STATE.md`, about 500 words
   - `docs/workflow/GAME-BRIEF.md`: audience hypothesis, core loop, session length, exclusions
5. **Owner brief.** `docs/workflow/OWNER-BRIEF.md`: how to open a preview, a 10-minute playtest checklist, and what each check means in plain English.

Excluded: any change to rules, balance, art, saves, the server protocol or hosting; paid services; publication; Package B (record automation) and Package C (devices, backups, capacity).

Roles: producer (scope), DevOps and release (deploy gate, CI), QA (verify command), technical lead (`CLAUDE.md`). Self-review only; low risk.

Likely files: `package.json` (scripts), `.github/workflows/{ci,deploy-web}.yml`, new `CLAUDE.md`, new `docs/workflow/{STATE,GAME-BRIEF,OWNER-BRIEF}.md`, `README.md` (pointer).

Acceptance (observable)

- A1. `npm run verify` exits 0 on the current code. A deliberately broken test makes it exit non-zero (shown, then reverted).
- A2. Pushing to `main` does **not** change the live site. Running *Deploy* does, and only after `verify` passes.
- A3. A preview URL (or documented local preview) lets you play the build before you approve publication.
- A4. You can follow `OWNER-BRIEF.md` to open a preview and finish a match, without asking me anything.
- A5. `CLAUDE.md` stays under 120 lines; `STATE.md` stays at or under about 600 words.

Checks: `verify` (PASS and FAIL demonstration), one CI run, one manual deploy dry run (after your separate publication approval, or skipped and marked NOT RUN), and the line and word counts.

Cost and data: RM0 extra. No player data is touched.

Risks: a preview URL feature may need a Cloudflare setting (fallback: local preview). If you forget to press *Deploy*, fixes wait (that's the point).

Rollback: revert the package commit; the deploy workflow returns to deploying on every push.

4 · Packages B and C (outline only; detailed later)

- **PLAN-002 · Package B, records and PDFs:**
  - one canonical record per package (Markdown with an ID header)
  - a deterministic local script that builds the owner brief, changelog, report PDFs and cumulative register, using pandoc and the project's existing Playwright browser (no new dependency)
  - export failures are labelled PDF EXPORT BLOCKED
- **PLAN-003 · Package C, online, capacity and release evidence:**
  - real-device checklist runs (you plus your phones)
  - Durable Object data export and restore drill
  - a room crash-recovery test
  - a small load test within an approved budget, with written workload assumptions (peak concurrent players, 2–4 per room, about 15 min matches)
  - cost record with a dated Cloudflare quote
  - live smoke test from an unrestricted machine (CI)

5 · Decisions

ID	Question	Recommendation	Status	Delivery
DEC-001	Adopt the workflow guide 1.2 for Fruit Market?	Yes; one package at a time	Proposed	Not started
DEC-002	What goes in Package A?	PLAN-001 scope above	Awaiting approval	Not started
DEC-003	When does production deploy?	Only on your manual *Deploy* after `verify`; never on push	Awaiting approval	Not started
DEC-004	PDF tooling	pandoc plus the existing Playwright browser; no installs	Proposed	Used for this report
DEC-005	Budget interpretation	RM500/month portfolio-wide ceiling; current Fruit Market spend unknown (hypothesis: Cloudflare Free, or Workers Paid at about USD 5)	Proposed (assumption)	Not started

Rationale for DEC-003: it is the only way to make "a preview is not a release" true for this repo. Trade-off: one extra tap per release.

6 · Questions for you (max 3)

1. **Devices:** which real phones can you test on (Android model, iPhone model)? This decides Package C.
2. **Hosting cost:** is your Cloudflare account on the Free plan or Workers Paid? Any other paid accounts yet (domain, Google Play developer account)?
3. **Approve PLAN-001** as written (including DEC-003)? Or change it?

7 · Sources

- Repository inspection, `main @ c915e49`, 2026-10-26.
- GitHub branch API (`main protected: false`), 2026-10-26.
- CI run 40266399588 (green), deploy run 40266399518 (success), Android run 40266399555 (success).
- Workflow guide: "Personal workflow for Murshidi", consolidated edition 1.2, 26 Oct 2026.
- Preview-URL and plan-pricing facts to be confirmed against current Cloudflare docs during PLAN-001 (not yet verified).

REP-002 · PLAN-001 (Package A) implemented and verified

Game	Fruit Market
Report date	2026-10-26 13:45 (Asia/Kuala_Lumpur)
Package	PLAN-001, approved by the owner 2026-10-26: "Approve PLAN-001 only, including its listed tests and documentation"
Build	`main @ 2e4b240` ; records on `workflow/plan-001`
Live game	**Unchanged** (still `c915e49`). No publication approval was given and none was performed.
Delivery status	Implemented; verified except A4, which needs the owner

What changed

Added

- `npm run verify` : lint, types, unit tests, build; stops at the first failure.
- `npm run verify:full` : `verify` plus browser tests.
- A **preview** copy of the game, updated automatically after CI passes on `main` , then smoke-tested: https://fruit-market-staging.murshidi-hashim.workers.dev. It is a separate Worker with its own rooms, leaderboard and stats.
- `scripts/smoke-live.mjs` : after a deploy, checks that the page loads and the server runs the expected rules version.
- `CLAUDE.md` (72 lines), `docs/workflow/STATE.md` , `GAME-BRIEF.md` , `OWNER-BRIEF.md` (preview, playtest checklist, publishing, branch-protection steps).

Changed

- The live deploy now runs only through GitHub Actions → *Deploy LIVE game* → type `PUBLISH` . It runs `verify` first and a live smoke test after.
- CI runs `verify:full` . README and ONLINE.md describe the new path.
- Lint now covers the docs export script (it found 12 errors in it, which `verify` caught; now fixed).

Removed: automatic live deploys on every push to `main` .

Player impact: none yet. Players only benefit once you publish, because broken changes can no longer reach them silently.

Acceptance results

#	Criterion	Result	Evidence
A1	`verify` exits 0 on current code; non-zero with a broken	PASS	Local: exit 0 (63 tests, build), then exit 1 with one test deliberately broken ("1 failed, 62 passed"), then exit 0

#	Criterion	Result	Evidence
	test		once restored
A2	Push to `main` does not change the live site	PASS	Push of `2e4b240` started only *CI* (run 40274184197) and *Android build*; no live deploy run
A2	*Deploy* runs only after `verify` passes	PASS (by design) / NOT RUN (live)	`verify` is the step before deploy; not run because publication isn't approved
A3	A preview lets you play before publishing	PASS	Preview job: "Deployed fruit-market-staging", then "SMOKE OK … rules version 2" (version `adf6c304`)
A4	Owner can open the preview and finish a match using the brief	NOT RUN	Needs you (OWNER-BRIEF §1–2)
A5	`CLAUDE.md` under 120 lines; STATE about 500–600 words	PASS	72 lines; STATE about 350 words
—	Full CI (`verify:full`) on `2e4b240`	PASS	Checks job green in 14 min: lint, types, 63 tests, build, solo game, performance budget, online match with a leaver (0 leaks), quick match
—	Smoke script fails on a dead address	PASS	Local: "SMOKE FAILED … fetch failed", exit 1
—	Preview online rooms (Durable Objects) in a real browser	NOT RUN	The smoke test checks the page and API only. Covered by the optional online step in the playtest checklist

Deviation from the plan (needs your confirmation)

PLAN-001 said "a private Cloudflare preview URL; if it isn't free, local preview steps". Cloudflare's docs state that preview and version URLs **are not generated for Workers that implement a Durable Object**, and Fruit Market's server uses Durable Objects for rooms. So I used the next closest option: a second, separate Worker (`fruit-market-staging`). This is recorded as **DEC-006**.

- Cost: no new paid service. It runs within the Cloudflare plan you already use; usage is tiny. Your plan type is still unknown (DEC-005).
- "Private" means unlisted: anyone with the link can open it.
- To undo: delete the `fruit-market-staging` Worker in Cloudflare and remove the `preview` job.

Risks and costs

- Expected extra cost: RM0 (no new paid service). Not yet confirmed against your plan type.
- Until branch protection is on, a direct push can still skip CI, but it can no longer reach players without *Deploy*.
- If CI is red, the preview isn't updated; the live game is unaffected.

Decisions

ID	Decision	Approval status	Delivery
DEC-002	Package A scope = PLAN-001	Approved (owner, 2026-10-26)	Verified (A4 pending)
DEC-003	Live deploys only via manual *Deploy* after `verify`	Approved (owner, 2026-10-26, as part of PLAN-001)	Verified
DEC-006	Preview = separate staging Worker, because per-version URLs don't support Durable Objects	Awaiting confirmation	Implemented

Sources

- Cloudflare docs, Version URLs and Preview URLs (`developers.cloudflare.com/workers/versions-and-deployments/version-urls/` and `/preview-urls/`): not generated for Workers that implement a Durable Object. Retrieved 2026-10-26 via search.
- Wrangler config reference vs the Durable Objects environments page disagree on per-environment migrations; resolved empirically: the staging deploy with top-level migrations succeeded.
- CI run 40274184197 (checks and preview jobs green).

REP-003 · DEC-006 confirmed; PLAN-002 (Package B) proposed

Game	Fruit Market
Report date	2026-10-26 13:50 (Asia/Kuala_Lumpur)
Build	`main @ 63c3682` (live game still `c915e49`)
Type	Planning only. **No game implementation changes were made.**
Approval status	DEC-006 **approved** (owner, 2026-10-26: "Confirm DEC-006"). PLAN-002 **awaiting approval**

1 · DEC-006 confirmed

The preview stays a separate staging Worker (`fruit-market-staging`). Approval status: approved; delivery: verified (REP-002, A3).

2 · Problem Package B solves

Records exist, but they're kept by hand:

- Each decision's status is typed in two or three places (the register, every report that mentions it, STATE). Nothing checks that they agree. REP-002 already needed a manual word-count correction.
- Nothing stops an approval being recorded without your words as evidence, or history being edited.
- There's no player-facing changelog and no version tags, so "what changed in this release" can't be answered from the repo.
- The PDF export works (built in REP-001) but isn't part of any check.

Players never see this directly. It matters because it keeps approvals honest and makes releases traceable when there's money or player data involved later.

3 · PLAN-002 · Package B: canonical records, generated views, checks

Included scope

1. **One canonical source per record** (JSON, no new tools):
   - `docs/workflow/records/decisions.json`: every DEC with question, choice, rationale, impacts, owner, approval evidence (your exact words and date), delivery status, links, review trigger, supersession. A `history` list is only ever appended to.
   - `docs/workflow/records/PLAN-###.json`: one per package with scope, exclusions, acceptance, approval evidence, changes (Added / Changed / Fixed / Removed), checks, build, release state.
   - Reports (`reports/REP-###.md`) stay as hand-written narrative snapshots and are never rewritten.
2. **Generated views**, built by `npm run records` with a deterministic script:

    - `DECISIONS.md`: current decisions and pending approvals first, then the full history.
    - `CHANGELOG.md`: player-facing, grouped by release, with "unreleased" vs "published" state. It is seeded with one summary entry for everything before the workflow, taken from git history.
    - The "Pending approvals" block inside `STATE.md` (between markers; the rest stays hand-written).
    - PDFs: each report, plus the cumulative register (current decisions, then pending approvals, then all reports in order).
3. **Checks**, via `npm run records:check`, added to `npm run verify` so CI runs it. It fails when:
    - an ID is duplicated or skipped, or a report mentions a DEC, PLAN or REP that doesn't exist
    - a decision is "approved" without owner evidence, or a delivery status isn't one of the allowed values
    - a history entry was changed or removed compared with the last commit (append-only)
    - a generated view is out of date with its source
4. **Release tagging**: when you publish, the deploy records the release in the changelog and tags the commit (`v1.1.1` and so on), so every live build can be traced. This only runs when you press *Deploy*.
5. **Migration**: move DEC-001 to DEC-006, PLAN-001 and REP-001 to REP-003 into the new records without losing anything.

Excluded

- Game code, rules, server and hosting.
- Package C (devices, backups, capacity).
- A cross-game studio register (NOT APPLICABLE until there's a second game).
- Installing anything. PDF building stays local because it uses pandoc. If pandoc is missing, the script prints PDF EXPORT BLOCKED and keeps the last good PDFs; it isn't part of CI.

Roles: producer (record design), technical lead (scripts), QA (failure cases), DevOps (tagging). Self-review only; low risk.

Likely files: new `docs/workflow/records/*.json`, new `scripts/records.mjs`, new `CHANGELOG.md`, `docs/workflow/export-pdf.cjs` (reads the records), `package.json` (scripts), `.github/workflows/deploy-web.yml` (tag and changelog step), `CLAUDE.md` and `OWNER-BRIEF.md` (one line each).

Acceptance (observable)

- B1. `npm run records` rebuilds all views. Running it twice in a row changes nothing (no git diff in the Markdown outputs).
- B2. `records:check` fails, with a plain-English message, for each of these deliberate breaks (shown, then reverted):
    - a duplicate ID
    - an unknown DEC referenced in a report
    - "approved" without evidence
    - an edited history line
    - a stale `DECISIONS.md`

- B3. With pandoc made unavailable, the export prints PDF EXPORT BLOCKED, exits non-zero and leaves the previous PDFs untouched.
- B4. The cumulative PDF opens with current decisions and pending approvals, followed by REP-001 to REP-003 in order.
- B5. After migration, every decision, status and approval quote from REP-001 to REP-003 appears in the records.
- B6. `npm run verify` (and CI) runs `records:check` and stays green on the migrated records.
- B7. Tagging is tested on a throwaway branch: a dry run shows the tag and changelog entry it would create. The real tag happens only at your next publication.

Checks: B1–B7, plus `verify`, plus one CI run.

Cost and data: RM0. No player data. `npm run verify` gets about 1 second slower.

Risks:

- Record-keeping overhead: JSON is less friendly to edit by hand. Mitigation: Claude edits the records; you only read the generated Markdown and PDFs.
- The append-only check compares with the last commit, so a deliberate correction needs a new "correction" history entry rather than an edit. That is intended.

Rollback: revert the package commits. The hand-written `DECISIONS.md` and reports remain valid.

4 · Decisions

ID	Question	Recommendation	Status	Delivery
DEC-006	Preview = separate staging Worker	Keep	Approved (owner, 2026-10-26)	Verified
DEC-007	Package B scope = PLAN-002	As above	Awaiting approval	Not started
DEC-008	Canonical record format	JSON records plus Markdown narrative reports; Markdown, changelog and PDF views generated	Awaiting approval (part of PLAN-002)	Not started
DEC-009	Records check in `verify` and CI	Yes, so disagreements block merges once branch protection is on	Awaiting approval (part of PLAN-002)	Not started

Options considered for DEC-008:

- Keep hand-written Markdown only: free, but can't be validated reliably.
- YAML front matter: friendlier, but needs a parser dependency.
- A database: ruled out by the workflow guide.
- **JSON (recommended):** Node reads it without dependencies, and the check can be strict.

5 · Still open from earlier (not blocking PLAN-002)

- Which real phones you can test on (needed for Package C).
- Your Cloudflare plan (Free or Workers Paid) and any other paid accounts (DEC-005).
- Preview playtest (REP-002, A4) and branch protection (owner brief, section 5).

REP-004 · PLAN-002 (Package B) implemented and verified

Game	Fruit Market
Report date	2026-10-26 14:10 (Asia/Kuala_Lumpur)
Package	PLAN-002, approved by the owner 2026-10-26: "Approve PLAN-002 only, including its listed tests and documentation"
Build	`main @ a7fa577` (records), plus this report
Live game	**Unchanged** (still `c915e49`, version 1.1.0). No publication.
Delivery status	Verified (B1–B7 PASS)

What changed

Added

- `docs/workflow/records/`, the single source for every record:
  - `decisions.json`: all decisions, plus a history that can only grow
  - `PLAN-001.json` to `PLAN-003.json`
  - `releases.json`
- `npm run records`: rebuilds `DECISIONS.md`, `CHANGELOG.md`, the pending-approvals block in `STATE.md`, and every PDF.
- `npm run records:check` (now the first step of `verify`, so CI runs it). It rejects:
  - broken or duplicate IDs
  - references to things that don't exist
  - an approval without your words and a date
  - an edited history line or a rewritten approval
  - out-of-date views
- `CHANGELOG.md`, in plain player language. It is seeded with version 1.1.0 (what's live now).
- Release step in *Deploy LIVE game*: it refuses to run without release notes for the new version, and after a successful deploy it tags the commit (for example `v1.1.1`).

Changed

- The PDF export is all-or-nothing. On failure it prints PDF EXPORT BLOCKED and keeps the previous PDFs.
- CI fetches the full history so the records check can compare with the previous commit.

Removed

- The hand-edited `DECISIONS.md` (it is now generated).

Player impact: none directly. The release process is now traceable: every live build will have a version tag and changelog notes.

Acceptance results

#	Criterion	Result	Evidence
B1	Rebuilding twice changes nothing	PASS	Checksums of all three generated files identical after a second run
B2	Check fails on 5 deliberate breaks	PASS	Duplicate DEC-008; REP-003 citing a nonexistent DEC; DEC-001 "approved" with no evidence; history line 1 edited; stale register. Each exits 1 with a plain message. All reverted
B3	No pandoc → PDF EXPORT BLOCKED, old PDFs kept	PASS	Exit 1; all 4 PDFs byte-identical; no temporary files left
B4	Cumulative PDF order	PASS	Pending approvals, then current decisions, then REP-001, REP-002, REP-003
B5	Migration lost nothing	PASS	Every row of the old register matches. DEC-004 "In use" maps to the allowed value "implemented"; DEC-007 to DEC-009 changed only because of today's approval (in history)
B6	`verify` and CI run the check, green	PASS	CI run 40276699547: "RECORDS OK … history checked against 63c3682", full suite green, preview updated
B7	Tagging dry run	PASS	Throwaway branch: refuses while 1.1.0 is published; with a 1.1.1 candidate, "would tag v1.1.1" plus the notes. Branch deleted, no tag created

Known issue found (not part of this package)

The game screen sometimes doesn't appear after rejoining an online room. The automated tests caught it in 2 of the last 5 CI runs, with no game-code change between them:

- Run 1: after quick match, the second player's game didn't show within 15 s.
- Run 4 (a docs-only commit): after reloading the page mid-match, the game didn't come back within 20 s.

Every other run passed, including locally (more than 8 runs). Real players could hit this when their phone reloads the page or their connection drops.

This is pre-existing and intermittent; the cause isn't known yet. I recommend it as the next package, before Package C: reproduce it by repeating the reconnect test many times, find the cause, fix it, and prove the fix with repeated runs.

Risks and costs

- RM0. `verify` is about 1 second slower.
- The first live release under the new process needs "PREPARE RELEASE" first, because Deploy refuses without release notes. That is intended.
- Until the reconnect issue is fixed, CI can occasionally fail for reasons unrelated to the change being tested. Read red runs before assuming the latest change caused them.

Decisions

ID	Decision	Approval	Delivery
DEC-007	Package B scope = PLAN-002	Approved (owner, 2026-10-26)	Verified
DEC-008	JSON records plus narrative reports; generated views	Approved (owner, 2026-10-26)	Verified
DEC-009	Records check in `verify` and CI	Approved (owner, 2026-10-26)	Verified

Still pending your decision: DEC-001 (adopt the workflow), DEC-004 (PDF tooling), DEC-005 (budget assumption) and PLAN-003 (outline). See the register's "Pending approvals".

REP-005 · PLAN-004 proposed: fix the rejoin problem in online play

Game	Fruit Market
Report date	2026-10-26 14:20 (Asia/Kuala_Lumpur)
Build	`main @ 935a287` (live game still 1.1.0, `c915e49`)
Type	Planning only. **No game implementation changes were made.** Read-only code inspection only.
Approval status	PLAN-004 **awaiting approval** (DEC-010, DEC-011)

1 · The problem

Sometimes, after a player rejoins an online room, the game screen doesn't come back:

CI run	Situation	Waited	Result
Run 1 (`da2eea9`)	Second player joins via Quick match	15 s	Game screen never appeared
Run 4 (`63c3682`, docs only)	Player A reloads the page mid-match	20 s	Game screen never appeared
Runs 2, 3 and 5, plus 8+ local runs	Same steps	—	Passed (usually about 1 s)

There was no game-code change between the failing and passing runs, so the problem is timing-dependent. Real players can trigger it: phones reload pages when memory is low, and mobile connections drop. The tests saved **no evidence** when it happened: the local server's log was discarded, and the online test takes no screenshot on failure.

2 · What the code shows (read-only)

How rejoining works today:

1. The phone keeps a seat token per room.
2. On reload, the online screen sees `?room=CODE` plus the token and reconnects.
3. The server's `hello` matches the token to the seat and sends the room state.
4. The game screen appears once the state says the match has started.

Hypotheses, none confirmed yet:

#	Hypothesis	Where	Fits run 1	Fits run 4	Evidence so far
H1	In a quick-match lobby, a seat is deleted when its connection closes. If the waiting player's connection	`server/room.ts closed()`	Yes	No	Plausible from code

#	Hypothesis	Where	Fits run 1	Fits run 4	Evidence so far
	blips just as the stranger joins, the match never starts				
H2	The old connection's "closed" event reaches the server *after* the new `hello`, wrongly marking the seat offline or freeing it	`server/room.ts` `closed()` / `hello()`	Yes	Yes	Plausible; order depends on server timing
H3	The server answers the rejoin with a non-fatal error and the screen stays on "Connecting…" with no retry	`server/room.ts`, `src/ui/Online.tsx`	Maybe	Yes	No error is shown or logged in the test
H4	The client drops or ignores the first room state after reconnecting (version check or listener timing)	`src/net/client.ts`, `useGameController.ts`	Maybe	Yes	Code looks right, but it isn't tested under slow timing
H5	Only the test environment: an overloaded CI machine is slower than the timeout	Test scripts	Yes	Yes	Normal time is about 1 s, so 15–20 s would be extreme. Not assumed without data

Ruled out: a seat being lost when the server sleeps. Seats are stored on each connection in a way that survives the server sleeping (`serializeAttachment`).

3 · PLAN-004 · Reconnect fix (evidence first)

Player problem: a player who reloads or loses connection can be stuck on a waiting screen instead of returning to their match.

Included scope (small steps)

1. **Evidence.** When the online or quick-match test fails, it saves:
    - both players' screens, page text and browser console
    - the room messages each player received
    - the local server log, kept as a CI artifact for 7 days
2. **Reproduce.** A new stress test, `npm run test:reconnect`, repeats three kinds of cycle and times how long the game screen takes to return:
    - reload mid-match
    - a connection drop (the browser goes offline, then online)
    - quick-match pairing

    It runs normally and with the CPU slowed 4× (like a cheap phone). It can be run locally and from a manual CI workflow (DEC-011). I'll run it in batches (target 300 cycles) and record the failure rate.
3. **Isolate.** Compare failing and passing cycles to confirm or reject H1–H5.

4. **Fix the measured cause** with the smallest change, plus a regression test that fails before the fix and passes after.
    - If the cause is in seat or token handling (who owns a seat), a fresh, separate review checks the change, because that's where security matters.
    - If the cause is only test timing (H5), the fix goes in the tests, but only with stress-test data showing real rejoins work.
5. **Prove.** The stress test and CI must stay green repeatedly (acceptance below).

Excluded

- Rules, balance, `RULES_VERSION` and saves.
- Message-format changes that old app versions can't understand (if one is needed, I stop and replan).
- Publishing (that needs "PREPARE RELEASE" and your Deploy).
- Package C.
- Making tests "pass" by just waiting longer or retrying.

Roles: network/backend engineer and QA (lead), technical lead, plus security review if seat or token logic changes.

Likely files:

- `scripts/e2e-online.cjs`, `scripts/e2e-quick.cjs`, `scripts/e2e.mjs` (diagnostics, server log)
- new `scripts/e2e-reconnect.cjs`
- new `.github/workflows/reconnect-stress.yml` (manual)
- `package.json`
- then, depending on the cause: `server/room.ts` or `src/net/client.ts` or `src/ui/Online.tsx`, plus a test in `tests/room.test.ts`

Acceptance (observable)

- R1. `npm run test:reconnect` prints time-to-game for every cycle, the failures, and the 95th percentile.
- R2. The root cause is named with evidence from a failing cycle (a log or message excerpt). If it isn't reproduced after at least 300 cycles, I report **NOT REPRODUCED** with the conditions tried, and stop rather than guess.
- R3. If reproduced: the failure rate before the fix is recorded, and after the fix at least 100 slowed-down cycles have 0 failures.
- R4. A deliberately forced failure in the online test produces the diagnostics files and the server log artifact.
- R5. A regression test fails on the old code and passes on the fixed code.
- R6. `verify:full` is green on 3 consecutive CI runs; the leak check stays at 0; all unit tests stay green; no rules or save changes.

Checks: R1–R6, `verify`, the manual stress workflow, and CI.

Cost and data: RM0. GitHub Actions is free for public repositories. Diagnostics contain only test players and expire after 7 days.

Risks

- It may not reproduce outside CI. Mitigation: the slowed-down mode and the CI workflow, and a hard stop at 300 cycles with a report.
- A fix in seat handling could affect reconnect security. Mitigation: a regression test, the existing room tests, and a separate review.
- Time: likely 1–2 sessions. I'll report at the end of step 3 if the cause turns out larger than a small fix.

Rollback: revert the package commits. Players are unaffected until a release is published.

How you'll see the result: once the fix is on the preview, reload the page mid-match on your phone, or switch to airplane mode for 5 seconds and back. The match should return within a few seconds.

4 · Decisions

ID	Question	Recommendation	Status
DEC-010	How to fix the rejoin problem	Evidence and reproduction first, then fix only the measured cause (PLAN-004)	Awaiting approval
DEC-011	Run the stress test on every push?	No: on demand (manual workflow and local runs)	Awaiting approval

5 · Still open

- Pending decisions from earlier: DEC-001, DEC-004, DEC-005, and PLAN-003 (outline only).
- Your preview playtest, branch protection, your phones and your Cloudflare plan.

REP-006 · PLAN-004 done: the online "stuck on Connecting..." bug is found and fixed

Game	Fruit Market
Report date	2026-10-26 15:25 (Asia/Kuala_Lumpur)
Package	PLAN-004, approved by the owner 2026-10-26: "Approve PLAN-004 only, including its listed tests and documentation"
Build	`main @ d7e5d24` (fix in `9d9e911`)
Live game	**Unchanged** (1.1.0). The fix is on the preview; players get it only after "PREPARE RELEASE" and your Deploy
Delivery status	Verified (R1–R6 PASS)

The bug, in plain words

After a player reloaded the page or rejoined an online match, the game sometimes stayed on **"Connecting to room ..."**, even though the match was there and the server had already sent it.

Why: the screen started listening for room updates a moment *after* it first appeared. The connection was already open by then. If the server answered in that gap, the answer was missed, and the screen waited for a next message that might never come (for example, when everyone was waiting on that player).

The gap is tiny on a fast computer, so it usually worked. On a slow phone, or when the app is in the background (browsers pause drawing for hidden tabs), the gap is long and the bug happens often. The game screen had the same weakness for updates arriving just as it opened.

Fix: two small changes in the app (`src/ui/Online.tsx`, `src/ui/game/useGameController.ts`). Right after starting to listen, each screen checks what has already arrived.

- No server change, no message-format change, no rules or save change.
- Old and new app versions stay compatible.

Evidence

Check	Result
Normal timing, old code (local)	270 cycles (reloads, network drops, quick matches; normal and 4× CPU slowdown), 0 failed. The bug is too rare to see this way
Background-tab mode, **old code**	**20 of 20 reloads failed**; every one stuck on "Connecting to room" with the full game already received (`room v2 status=playing`)
Background-tab mode, **fixed code** (local)	65 cycles, 0 failed

Check	Result
Background tab plus 4× slowdown, **fixed code** (local)	100 cycles, 0 failed (reload p95 0.59 s, drop p95 0.63 s, quick match p95 1.3 s)
Same on a CI machine (where the original failures happened)	110 cycles, 0 failed (Reconnect stress run 40283316566)

Acceptance results

#	Criterion	Result	Evidence
R1	Stress test with per-cycle timings	PASS	`npm run test:reconnect` ; modes: slowdown, background tab, preview target
R2	Root cause named with evidence	PASS	Lost first update after (re)joining; see the old-code evidence above
R3	Before rate recorded; at least 100 slowed cycles with 0 failures after	PASS	20/20 failed before; 100/100 passed after (plus 175 more)
R4	Forced failure leaves evidence	PASS	Page text, room messages, console, both screenshots and the server log were saved
R5	Regression test fails before, passes after	PASS	New suite step "rejoin from a background tab"
R6	3 consecutive green CI runs; leak check 0; unit tests green; no rule or save changes	PASS	CI runs 7 (push), 8 and 9 (manual) on `d7e5d24` : all green, each with leaks 0 and 63 unit tests; no rules or save files changed

Side observations (no action taken)

- **The local test server never finishes closing a socket when the page closes it.** Cloudflare's docs say production replies automatically for this project's settings, and the real app never waits on a socket it closed itself, so no player impact was found. If it ever matters, the documented safe fix is one line in the server's close handler.
- **Playwright's offline mode doesn't break open connections**, so the stress test cuts them with a small local relay instead. A "silent" dead connection (no signal at all) isn't simulated yet. That would be part of Package C.

What changed

Added

- `npm run test:reconnect` and a manual *Reconnect stress* workflow (local or preview, optional slowdown and background-tab mode).
- Failure evidence for all online browser tests.
- The regression step in the browser suite.

Fixed (player-facing, on the preview, not yet live)

- After reloading or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.

Risks and costs

- RM0.
- The browser suite is about 1 minute longer (the regression step).
- The fix affects all online screens. Mitigation: the full suite, including the secret-price leak check, stayed green, and 545 stress cycles ran in total.
- Remaining NOT RUN: a real phone in a real background tab. Your preview playtest covers that (see below).

How to try it (preview)

1. Open https://fruit-market-staging.murshidi-hashim.workers.dev on two phones (or one phone and a computer).
2. Start an online game (Play a friend → Create room → open the link on the other device).
3. During the match, on one phone: switch to another app for 10 seconds, come back, then pull down to reload.
4. Expected: the match comes back within a few seconds both times, never stuck on "Connecting to room...".

REP-007 · PLAN-005 proposed: make 4-player games easy to find

Game	Fruit Market
Report date	2026-10-26 18:36 (Asia/Kuala_Lumpur)
Build	`main @ cc27848` (live game still 1.1.0)
Type	Planning only. **No game implementation changes were made.** One read-only test was run locally.
Approval status	PLAN-005 **awaiting approval** (DEC-012)

1 · What you noticed

You couldn't see an option for 4 human players. 4-player games are supported, but the screens don't say so.

Online room (Play a friend), as it looks today on a 360 px phone:

- The heading says **"Invite a friend"** (singular).
- There is one empty row: **"Waiting for a friend to join"**, which disappears once one person joins.
- The count "1/4" is in small grey text next to "Players".

Same phone (New game): 4 people means choosing **3** rival shops and switching each from Computer to Friend. Nothing on the screen explains that.

Quick match is always 2 strangers, by design. Nothing points groups to Play a friend instead.

2 · Check that 4 players really work (read-only, local server)

- 4 browsers joined one room by the same link. The lobby showed "Players 4/4". All 4 pressed ready, the host started, and **all 4 reached the game** (each saw 3 rivals).
- A **5th** person was turned away: "Couldn't join · This room is full."
- Limitation: tested on one machine, not 4 real phones. The automated test below makes this a permanent check.

3 · PLAN-005 · 4-player clarity

Player problem: groups of 3 or 4 think the game is for 2 and give up, or play with computers instead.

Included scope (screen text and layout only)

Screen	Today	Proposed
Online lobby, invite box	"Invite a friend"	**"Invite friends (up to 3)"**
Online lobby, player list	"Players 1/4" (small, grey), one row "Waiting for a friend to join"	**"Players 1 of 4"** in normal text, plus one row per free seat: "Open seat · send the link"
Online lobby, when full	(nothing)	"Room full · start when everyone is ready"
Home, under "Play a friend"	(nothing)	Small line: "2 to 4 players online"
Quick match screen	(nothing)	"Quick match is always 2 players. For 3 or 4, use Play a friend."
New game, under "Rival shops"	(nothing)	"Up to 4 shops. Pick Friend to pass this phone around." plus a live total, for example "4 shops: you, 2 friends, 1 computer"

Computer shops added in the lobby take seats, so the open-seat rows and the count include them. For example: 2 people + 1 computer = "Players 3 of 4", one open seat.

Also included: a new automated browser test where 4 people join one room and all reach the game, and a 5th is refused. It runs in CI on every change.

Excluded

- Rules, balance, the 4-player limit and quick-match size.
- Any server or message change.
- Renaming the buttons people tap to get there ("Play a friend", "New game"), so tests and habits keep working.
- Publishing.

Roles: UX designer (wording), gameplay engineer, QA. Self-review; low risk.

Likely files: `src/ui/Online.tsx` (lobby), `src/ui/App.tsx` (home, new game), `src/ui/styles.css`, new `scripts/e2e-four.cjs`, `scripts/e2e.mjs`.

Acceptance (observable)

- C1. The lobby shows "Players N of 4" and one open-seat row per free seat. Both update when people join or leave and when computer shops are added.
- C2. With 4 people there are no open seats and the lobby says the room is full.
- C3. The new browser test passes: 4 people join one room, all see the game, and a 5th is refused.
- C4. The new lines fit on a 360 px wide phone without overflow (screenshots of each screen in the report).
- C5. Store screenshots change only where intended (home and lobby text), checked with the visual-diff tool.
- C6. `verify:full` is green in CI; no rules, server or save changes.

Cost and data: RM0. No data changes.

Risks: longer text could wrap awkwardly on small phones; C4 checks this. Wording is a matter of taste, so you can change any line before or after.

Rollback: revert the package commit.

After it's built: open the preview, tap Play a friend → Create room, and check that the lobby clearly invites up to 3 friends.

4 · Decision

ID	Question	Recommendation	Status
DEC-012	How to make 4-player play easy to find	Screen text and open-seat rows plus a 4-person test (PLAN-005); no rule or server change	Awaiting approval

5 · Note on order

The reconnect fix (PLAN-004) is on the preview and not live yet. PLAN-005 can be built first and both can go out together in 1.1.1, which saves you one Deploy.

REP-008 · PLAN-005 done: 4-player games are easy to find

Game	Fruit Market
Report date	2026-10-26 19:05 (Asia/Kuala_Lumpur)
Package	PLAN-005, approved by the owner 2026-10-26: "Approve PLAN-005 only, including its listed tests and documentation"
Build	`main @ 04b72b5`
Live game	**Unchanged** (1.1.0). On the preview; goes live with the next release
Delivery status	Verified (C1–C6 PASS)

What changed (screen text and layout only)

Screen	Before	After
Online lobby, invite box	"Invite a friend"	"Invite friends (up to 3)". Fewer if computer shops are added, for example "(up to 2)"
Online lobby, player list	"Players 1/4" in small grey text, one row "Waiting for a friend to join"	"Players 1 of 4", plus one "Open seat · send the link" row per free seat
Online lobby, when full	Nothing	"Room full · start when everyone is ready"
Online screen	"Play a stranger..." / "send to a friend"	Quick match: "Always 2 players; for 3 or 4, start a private game below." Private game: "send to up to 3 friends... 2 to 4 players."
Home	"Play a friend"	Same button, with a small line "2 to 4 players online"
New game	Nothing explained 4 players	"Up to 4 shops. Pick Friend to pass this phone around." plus a live total, for example "4 shops: you, 2 friends, 1 computer"

Wording note: the quick-match hint says "start a private game below" instead of the planned "use Play a friend", because both options are on the same screen. The meaning is the same.

No rules, server or save changes. The buttons you tap keep their names.

Acceptance results

#	Criterion	Result	Evidence
C1	Lobby count and open seats update with joins and computer shops	PASS	New test: 1 of 4 with 3 open; 2 people + 1 computer = "3 of 4" with 1 open and "(up to 2)"

#	Criterion	Result	Evidence
C2	Full room shows no open seats and "Room full"	PASS	New test: "Players 4 of 4", 0 open, "Room full"
C3	4 people play; a 5th is refused	PASS	New test: 5th sees "This room is full"; all 4 reach the game, each seeing 3 rivals
C4	Fits a 360 px phone	PASS	Home, online and new-game screens: page exactly 360 px wide, no sideways scrolling
C5	Store screenshots change only where intended	PASS	Home 7.3% different (the button's new second line moves the rows below it). All 6 in-game screens 0.00%
C6	Full CI green; no rule, server or save changes	PASS	CI run 11 on `04b72b5` green, including the new four-player step and the preview update; diff touches only screen code, styles, tests and docs

Process note: my first `verify` run failed because I had updated a record without regenerating the views. The records check caught it, as designed (PLAN-002), and the run passed after `npm run records`.

Risks and costs

- RM0.
- The store screenshot of the home screen (`store/screenshots`) should be retaken before the next Play listing update. It still shows the old button.

How to see it

On the preview (https://fruit-market-staging.murshidi-hashim.workers.dev):

1. The home screen shows "2 to 4 players online" under Play a friend.
2. Tap Play a friend → Create room. The lobby says "Invite friends (up to 3)" and shows three open seats.
3. Tap New game → 3 rival shops → switch two to Friend. It reads "4 shops: you, 2 friends, 1 computer".

Next

Both fixes (reconnect and 4-player clarity) are on the preview. After your preview playtest, say "PREPARE RELEASE" to send them to players as 1.1.1.

REP-009 · Release 1.1.1 prepared: readiness and how to publish

Game	Fruit Market
Report date	2026-10-26 20:30 (Asia/Kuala_Lumpur)
Request	Owner: "PREPARE RELEASE"
Candidate	Version **1.1.1**: the `main` commit that adds this report (the Deploy button tags exactly that commit `v1.1.1`)
Live now	1.1.0 (`c915e49`)
Approval status	DEC-013 (publish 1.1.1) **awaiting your approval**. Nothing has been published

1 · What players get (changelog for 1.1.1)

- **Fixed:** after reloading the page or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.
- **Clearer group play:** the online lobby shows how many of the 4 seats are taken and which are still open, and the menus explain that 2 to 4 people can play together, online or by passing one phone.
- **Behind the scenes:** updates now reach players only after a preview and your approval, with automatic checks before and after.

Included packages: PLAN-001, PLAN-002, PLAN-004 and PLAN-005.

2 · Size of the change since 1.1.0

- **Game code:** 4 files, about 40 lines (`src/ui/Online.tsx` , `src/ui/App.tsx` , `src/ui/game/useGameController.ts` , `src/ui/styles.css`).
- **Server code:** unchanged. **Server settings for the live game:** unchanged (only a separate preview section was added).
- **Rules, balance, rules version (2) and saves:** unchanged. Old and new app versions can play together.
- **Version number:** 1.1.0 → 1.1.1 (shown in Settings; also the Android version name).
- **CI setting:** job time limit raised from 20 to 35 minutes (see section 3). No effect on players.

3 · Readiness gates

Gate	Check	Result	Evidence and limitation
Build ready	Clean install, records, lint, types, 63 unit tests, build	PASS	Latest green CI: run 12 on `f319548` (same game code as the candidate). Run 13 on `f472186` was **cancelled**: its browser download took 16 minutes (usually about 20 seconds), so it hit the job's 20-minute limit. The limit is now 35 minutes. The candidate's own CI run must be green before you press Deploy

Gate	Check	Result	Evidence and limitation
Play ready	Computer-played solo game; speed budget	PASS	Browser suite in CI
Play ready	**You finish a match on the preview**	**NOT RUN**	Needs you, about 10–15 minutes (owner brief, section 2)
Online ready	Two players, secret prices never leak, a player leaves	PASS	Leak check 0 on every CI run
Online ready	Rejoin and reconnect	PASS	545 stress cycles after the fix, 0 failures; regression test in CI
Online ready	4 players in one room; a 5th refused	PASS	New test in CI
Online ready	Server crash recovery	NOT RUN	Package C. Server code is unchanged since 1.1.0, so this release adds no new risk here
Release ready	Real phones (Android Chrome, iPhone Safari)	NOT RUN	Emulated phones only; Package C
Release ready	Data backup and restore	NOT RUN	No data format change in this release; Package C
Release ready	Live smoke test after deploy	Automatic	The Deploy workflow checks the live site and its rules version, then tags v1.1.1
Release ready	Rollback path	Ready	See section 5

Release-ready is not claimed, because required checks are NOT RUN. The workflow guide lets you choose between:

- **A (recommended):** do the 15-minute preview playtest first. If it passes, publish. Real phones, backups and crash recovery stay as documented gaps for Package C. That's reasonable here because 1.1.0 is already public and this release changes no server code or data.
- **B:** publish now and explicitly accept the playtest as NOT RUN (a "scoped exception"). Not recommended: it skips the only human check.

4 · How to publish (after you decide)

1. Make sure CI is green for the candidate. I'll confirm in my reply.
2. GitHub → **Actions** → **Deploy LIVE game** → **Run workflow** → type `PUBLISH` → **Run workflow**.
3. It runs the checks again, confirms the release notes for 1.1.1 exist and that `v1.1.1` isn't already used, publishes, smoke-tests the live site, and tags the release.
4. Tell me when it's green. I'll mark 1.1.1 as published (date, commit, tag) and check the live site's version.

If anything else lands on `main` before you press Deploy, ask me to re-check, because the Deploy publishes whatever `main` holds.

5 · If something goes wrong after publishing

- **The Deploy run fails:** the old version keeps running. Nothing to undo. Tell me.
- **A problem is found after a successful deploy:**
  - Fastest: Cloudflare dashboard → Workers → `fruit-market` → Deployments → roll back to the previous deployment (takes effect in seconds; no data changes in this release, so it's safe).
  - Standard: I prepare 1.1.2 with the change reverted; you press Deploy (about 30 minutes including checks).

6 · Costs and data

RM0 change. No new services. No player data or storage format changes.

7 · Decision

ID	Question	Recommendation	Status
DEC-013	Publish 1.1.1 to players?	Yes, after your preview playtest passes (option A)	Awaiting approval

To approve, reply for example: **"Playtest passed. Approve DEC-013, publish 1.1.1."** Then press Deploy. To choose option B, say so explicitly, for example "Approve DEC-013 with the playtest NOT RUN exception".

REP-010 · Theme review: does Fruit Market's theme support how it plays?

Game	Fruit Market
Report date	2026-10-26 21:47 (Asia/Kuala_Lumpur)
Request	Apply the "Game Concept and Theme Companion" (edition 1.0) to this game
Build reviewed	`main @ 83de99b` (release 1.1.1 candidate); content in `src/engine/data.ts` and `RULES.md`
Type	Creative review only. **No game, rule or content changes were made.**
Approval status	DEC-014 proposed; DEC-015 and DEC-016 **awaiting your choice**

The direction is already chosen and the game is live, so this follows the companion's path for an existing game: check what works, review coherence, propose the smallest useful test. No new concepts.

1 · Creative brief (what is confirmed, what is assumed)

Item	Status	Value
Player fantasy	Confirmed (game brief)	Run a fruit stall; outsell rival shops before your stock goes bad
Core interaction	Confirmed (rules)	Buy from the farmers, set secret prices, reveal, age overnight
Players and session	Confirmed	2–4 shops; Quick 8 / Standard 10 / Long 14 days; 10–20 min (assumption, unmeasured)
Platform	Confirmed	Phone browser and Android; touch first
Audience	**Assumption**	Casual players 12+, families sharing a phone, friends online
Desired feeling	**Assumption**	Light, cheeky tension: "will my price win?" plus "sell it before it rots"
Setting and identity	**Open**	See section 4

2 · Does the theme show up in what players do?

Repeated action	How the theme expresses it	Fit
Buying at the farmers market; poorest buys first	A morning market where small shops get first pick	**Strong**. The catch-up rule feels like market goodwill
Fruit ages each night; older fruit sells for less; spoiled fruit is binned	The freshness track and "Not so" fruit	**Strong**. This *is* the fantasy; every choice is a race against rot
Secret prices, all revealed at once; cheapest sells first	"Showdown" reveal, customer by customer	**Mixed**. Exciting and fair, but real stalls post prices openly. The reveal needs a stall-side reason (presentation idea in section 5)

Repeated action	How the theme expresses it	Fit
Customers with orders and price caps	24 named characters with quirks	**Strong** for orders and caps. The quirks are cosmetic (they never change play), which is fine but should stay labelled cosmetic
Events (Heatwave, Health Inspector, Lorry Breakdown…)	Each bends the market in a believable way	**Strong**. Heatwave aging fruit and the Inspector fining old fruit are the best theme-mechanic matches
Upgrades (Handcart, Cool Room, Juice Press, Signboard…)	Things a stall owner would actually buy	**Strong**
Favour cards	Rewards for serving a whole order yourself	**Mixed**. "Free Cash" and "Good Riddance" read as abstract game cards rather than market favours
Rival shops (animal traders)	Six characters with reaction lines	**Mixed**. Fun, but personality and play style don't match: lines are cosmetic, and how cleverly they price comes from difficulty, not character

Overall the core loop expresses the fantasy well. The weak spots are presentation: why prices are secret, a few abstract card names, and cast identity. None needs a rule change.

3 · Proposed design pillars (for your approval: DEC-015)

Pillar	Include when…	Exclude when…
1. Fresh fruit, fast money: every decision is a race against spoilage	It makes freshness visible or raises the cost of waiting	It lets fruit sit forever at no cost, or hides freshness
2. Outprice rivals in secret: commit a price blind, then see the reveal	It keeps prices hidden until the reveal and makes the reveal explain *why*	It shows rivals' prices early, or adds open haggling or chat
3. A cheerful town market: playful characters, short sessions, readable on a phone	Jokes target behaviour (a vampire who hates garlic); games stay under about 20 min	Mean or stereotype-based jokes, grind, pay-to-win, long text

4 · Coherence issues found (review before expanding)

1. **Mixed setting identity** *(owner decision, see question 1)*:
    - Customer names are Malaysian and multicultural (Aisyah, Mei Ling, Kumar, Wei Jie), and one rival is "Uncle Toad".
    - The palette is Mediterranean (terracotta, harbour teal, olive).
    - Prices are in **$**, and the fruits are blueberry, watermelon, apple and mango (two are imports in Malaysia).

    It reads as "anywhere town". That's fine, but it isn't a choice anyone made deliberately.

2. **Possible ethnic association:** "Panda Lim" pairs a Chinese surname with a panda. Most other names pair freely with professions or fantasy roles (Astronaut Aisyah, Ninja Farid), so this one stands out. Recommend renaming it (cosmetic only). "Kumar the Genie" is milder; worth a second look.

3. **Why are prices secret?** The best moment in the game has no in-world reason. That's a presentation fix, not a rule change (section 5).

4. **Repeated jokes:** each rival has 2–3 lines per situation, and a 10-day game has many reveals, so lines probably repeat. Hypothesis; check in the playtest.

5. **Icon style:** some emoji still sit beside the pixel art (🔒 Prices locked, 📊, 🙋/🤖, 🕵). Already in backlog B-33.

5 · Representative day (actual rules)

Day 4 of 10, Standard game. You and Foxy Rosa (computer). You have $14; she has $19.

1. **Morning:** the event is 🔥 **Heatwave**: watermelons sell for $4 more, but all fruit ages an extra step tonight. Tomorrow's event, shown early: 🕵 Health Inspector.
2. **Buy:** you're poorer, so you **buy first**. Fresh watermelon costs $3. You take 3 and leave the cheap blueberries: in a heatwave they'd rot by tomorrow. *Choice: profit now vs. stock that survives tonight.*
3. **Customers:** Cowboy John wants 2 watermelons and pays up to $6 + 4 = * *10** for fresh ones.
4. **Secret price:** you offer 2 at **$9**; Rosa secretly offers 2 at $8.
5. **Showdown:** prices flip; the cheapest sells first. Rosa's $8 fills John's order, and you sell nothing. Rosa quips "A dollar cheaper. Sorry, not sorry!" *Feedback:* your offer is tagged "$1 too high".
6. **Evening:** you build the ❄ **Cool Room** ($7). Your fruit now ages only every other night, and heatwaves can't touch it. *Choice: spend on protection or keep cash for tomorrow.*
7. **Night:** heatwave aging hits Rosa's shelf. Your Cool Room saves your 3 watermelons for tomorrow's customers, and tomorrow's Inspector will fine Rosa for her old fruit.

Theme-only idea for pillar 2 (not approved, no rule change): present secret pricing as *chalkboards*. Each stall chalks its price face-down, the market bell rings, and the boards flip. Same rule, with a market reason.

6 · Mini creative bible (draft: samples only, per the companion)

- **World and tone:** a sunny town market where anyone might shop: cowboys, astronauts, a vampire after sunset. Light, cheeky, never mean. Stakes: a good day's takings, never ruin.
- **Sample character:** *Grandpa Tan* (cosmetic). Silhouette: cardigan, cane, paper bag. Desire: a fair price. Flaw: remembers when apples cost ten cents. Sample reaction when you sell at full price: "Ten dollars? In my day..."
- **Sample event:** *Health Inspector*. The clipboard arrives at night; every Not-so fruit costs $2. This teaches "sell old fruit first" without a tutorial line.
- **Key screen:** the reveal. Face-down boards flip highest to lowest, the winner's coins fly to their stall, and the losing offer is tagged "$1 too high". It works without sound and without colour alone: text plus movement.
- **Do:** jokes about what characters *do*; short lines; a mix of names and roles across cultures. **Avoid:** pairing a person's ethnicity with an animal or cultural cliché; jokes longer than one line; new lore that doesn't change a decision.

7 · Riskiest creative hypothesis and the smallest test (DEC-016)

Hypothesis: first-time players understand *why* they won or lost a sale at the reveal. That is pillar 2, and it's what makes the game feel fair rather than random.

Format: a playtest with the existing game (preview or live 1.1.1). **No code needed.** It can't tell us retention, market demand or online-server behaviour.

Playtest card

Testers	3–5 people who have never played; no coaching
Setup	Their own phone. New game → Quick (8 days) → 1 computer shop (Normal)
Tasks	1) Start a game. 2) Buy fruit. 3) Set your prices. 4) After the first sales, say why you sold or didn't. 5) Finish the game
Watch for	Time to first price; whether they mention "cheapest sells first" or freshness; whether they read the event card; where they hesitate
Ask after	"What were you trying to do?" "Why did you lose that customer?" "Would you play again?" (note if they start another game unprompted)
Provisional criteria (you set them; these are suggestions, not benchmarks)	**Continue** if at least 3 of 4 can explain a lost sale by day 3. **Revise** (reveal presentation, section 5) if 2 or fewer can
Limits	Small sample; friends are kind; not proof of fun or retention

I'll record what happens (observations separate from interpretations) in the next report.

8 · Questions for you (max 3)

1. **Where is this market?**
   - (A) "Anywhere town": keep as is. *Recommended for now:* costs nothing and doesn't affect the biggest risk.
   - (B) A Malaysian pasar: RM prices, local touches; the fruit artwork could change later; much bigger art scope.
   - (C) A deliberate mix ("a market where the whole world shops"): keep the names, lean into variety.
2. **Rename Panda Lim** (and review Kumar the Genie)? Recommended: yes, a cosmetic change in a later small package.
3. **Can you find 3–5 people who've never played** for the playtest card above?

9 · Decisions

ID	Question	Recommendation	Status
DEC-014	Use the Concept and Theme Companion for Fruit Market's creative work, alongside the development workflow	Yes; one shared record system	Proposed
DEC-015	Adopt the three design pillars (section 3)?	Yes	Awaiting approval

ID	Question	Recommendation	Status
DEC-016	First creative test = the reveal-comprehension playtest (section 7)	Yes; no code	Awaiting approval

No conflict with approved decisions: release 1.1.1 (DEC-013) is unaffected, and nothing here changes rules, saves or multiplayer.

REP-011 · Setting chosen; panda rename planned (PLAN-006)

Game	Fruit Market
Report date	2026-10-26 22:16 (Asia/Kuala_Lumpur)
Owner input	Answers to REP-010's three questions
Type	Creative decisions and planning. **No game changes were made.**
Approval status	DEC-017 and DEC-018 **approved**; DEC-019 (PLAN-006) **awaiting approval**; DEC-015 and DEC-016 still awaiting approval

1 · Your answers, recorded

Question	Your answer	Recorded as
Where is this market?	"C: a deliberate 'the whole world shops here' mix"	DEC-017, approved
Rename Panda Lim?	"sure, but dont run away from the general theme coherence"	DEC-018, approved
3–5 first-time testers?	"yes"	Noted. This confirms testers are available; it isn't approval of the playtest itself (DEC-016)

2 · What option C means in practice

Written into `docs/workflow/CREATIVE-BIBLE.md` (short, one page):

- **A sunny harbour market where the whole world comes to shop.** The seaside colours you already have become the harbour; a port that trades with everywhere explains the mix of fruit.
- **Customers** keep their current mix of names and roles, freely combined, never tying a culture to a cliché.
- **Money** stays plain market dollars ($), the same for every visitor.
- **No game changes are needed for C.** It turns what's already there into a deliberate choice and sets the rules for future content.

3 · The rename, keeping the cast coherent

The rivals are animal traders with friendly nicknames: Foxy Rosa (fox), Big Benny (bear), Uncle Toad, Rocky (raccoon), Pinky (pig). The panda's lines are calm and sleepy ("Bamboo-zled you!", "very zen", "Nap time, I guess"). The new name should fit that pattern and keep the lines working.

Option	Why
Patches (recommended)	Describes the panda (black-and-white patches), like Pinky and Rocky; no real-world link
Bamboo Bea	Keeps the bamboo joke; alliterative like Big Benny

Option	Why
Sleepy Sid	Matches the napping lines

Kumar the Genie, reviewed: keep. Genie is a fantasy role like Ninja, Wizard or Astronaut; the name and role don't form a national cliché.

Checked in the code (read-only): the panda's reaction lines are found by its picture (🐼), not its name. So saved games and games in progress keep working, and they just show "Panda Lim" until they end. The panda is also one of the three daily-challenge rivals; names don't affect the score check, and PLAN-006 tests that.

4 · PLAN-006 · Rename the panda rival (cosmetic)

Scope

- Change the display name in the rival list (`src/engine/data.ts`) to your chosen name. Picture, lines, difficulty and play stay the same.
- Update the daily-challenge sentence in `RULES.md` .
- Keep "Panda Lim" reserved, so no player can take the old rival name.
- Add tests: an old save naming "Panda Lim" still gets the panda's lines; daily-challenge verification gives the same score; both names are refused as player names.

Excluded: other renames, art, lines, rules, AI or balance; renaming inside saved or in-progress games.

Acceptance

- N1. New solo games, daily challenges and online rooms show the new name.
- N2. An old save loads, still says "Panda Lim", and the panda's lines still appear.
- N3. The daily-challenge score check is unchanged for the same moves.
- N4. Both names are refused as player names.
- N5. `verify:full` is green in CI; no rule, save-format or server-logic changes.

When: built on a separate branch now and released as **1.1.2 after 1.1.1 is live**, so the 1.1.1 candidate you're testing doesn't change.

Cost: RM0. **Risk:** very low (one name plus tests). **Rollback:** revert the commit.

5 · Decisions

ID	Decision	Status
DEC-017	Setting: a market where the whole world shops	Approved (owner, 2026-10-26)
DEC-018	Rename Panda Lim, keeping the cast coherent	Approved (owner, 2026-10-26)
DEC-019	Rename scope = PLAN-006, shipped in 1.1.2	Awaiting approval
DEC-015	Three design pillars	Awaiting approval (from REP-010)
DEC-016	Reveal-comprehension playtest with your 3–5 testers	Awaiting approval (from REP-010)

REP-012 · PLAN-006 done: the panda rival is now Patches

Game	Fruit Market
Report date	2026-10-26 22:48 (Asia/Kuala_Lumpur)
Package	PLAN-006, approved by the owner 2026-10-26: "Approve PLAN-006 only, including its listed tests and documentation, name: Patches"
Build	Branch `workflow/plan-006 @ f14245a` (not on `main` yet)
Live game	**Unchanged** (1.1.0). `main` is still the 1.1.1 candidate
Delivery status	Verified (N1–N5 PASS); ships in 1.1.2, after 1.1.1

What changed

- The panda trader is now called **Patches** in new solo games, daily challenges and online rooms. Same picture (🐼), same calm, sleepy lines ("Bamboo-zled you!"), same difficulty and play.
- "Panda Lim" stays blocked as a player name, so nobody can pose as the old rival.
- The daily-challenge sentence in `RULES.md` names Patches.
- Five new tests (`tests/rename.test.ts`). Game code change: one line, plus the blocked-names list.

Acceptance results

#	Criterion	Result	Evidence
N1	New games show the new name	PASS	Rival list and daily lineup contain Patches, not Panda Lim; online rooms draw from the same list
N2	Old saves load, keep "Panda Lim", still get the panda's lines	PASS	Test saves and reloads such a game unchanged; the panda's undercut line is still chosen
N3	Daily-challenge score check unchanged	PASS	The same daily played with either name ends with identical cash and sales for every shop
N4	Both names refused for players	PASS	"Patches", "Panda Lim" and "panda lim" are reserved
N5	Full CI green; no rule, save-format or server-logic change	PASS	CI run 15 on `f14245a` green (68 unit tests plus the full browser suite)

How it reaches players

`main` holds the 1.1.1 candidate, and Deploy publishes whatever `main` holds, so this waits on its branch. After 1.1.1 is live: I merge it, you say "PREPARE RELEASE" for 1.1.2, then you press Deploy. Its player note: "The panda trader is now called Patches."

Risks and costs

- RM0, very low risk.
- **Observation (not part of this package):** the CI machines' browser download took 11–16 minutes in the last 4 runs (it used to take about 20 seconds). The 35-minute limit absorbs it, but caching the download would cut each CI run by about 10 minutes. That would be a small separate package if you want it.

REP-013 · Approvals recorded: publish 1.1.1, pillars, first playtest

Game	Fruit Market
Report date	2026-10-26 22:58 (Asia/Kuala_Lumpur)
Owner input	"yes to 1 and 2" (reply to REP-012)
Type	Decisions recorded. **No game changes.** Nothing has been published yet

1 · What you approved

ID	Decision	Status
DEC-013	Publish 1.1.1 (reconnect fix + 4-player clarity)	**Approved**. Not yet deployed: you press the button
DEC-015	The three design pillars	**Approved**. Now marked approved in the creative bible
DEC-016	First playtest with 3–5 first-time players	**Approved**. Ready to run with the kit below

One gap, recorded honestly: you didn't say whether you tried the preview yourself. It's recorded as NOT RUN. It's your call; a 5-minute look at https://fruit-market-staging.murshidi-hashim.workers.dev before deploying is still worth it (Settings should show version 1.1.1).

2 · Publish 1.1.1 now (about 3 minutes, plus about 15 minutes of automatic checks)

1. Open GitHub → **Fruit-Market** → **Actions** → **Deploy LIVE game**.
2. Tap **Run workflow**, type `PUBLISH`, then tap **Run workflow**.
3. Wait for the green tick. It checks the code, publishes, tests the live site and tags `v1.1.1`.
4. Tell me "deployed". I'll check the live version, mark 1.1.1 published, then bring in the Patches rename and prepare 1.1.2.

`main` is unchanged since the candidate (`83de99b`). These records are on a separate branch until you've deployed.

3 · Playtest kit (DEC-016)

Goal: find out whether first-time players understand *why* they won or lost a sale.

Before: use the live game after 1.1.1 is published (or the preview). Each tester uses their own phone. Don't explain the rules. You only watch and take notes.

Say only this: "This is a game about running a fruit stall. Please think aloud while you play. I can't help, but I'll ask a few questions at the end."

Tasks: New game → Quick (8 days) → 1 computer shop. Play to the end.

Note for each tester (copy this per person):

```
Tester:            Age range:        Plays phone games? (often / sometimes / rarely)
Minutes to first price set:
After day 1–3 sales, in their words: why did they sell or not sell?
Did they mention "cheapest sells first" or fresher fruit? (yes / no)
Did they read the event card? (yes / no / unsure)
Where did they hesitate or get stuck?
Finished the game? (yes / no)   Started another game without being asked? (yes / no)
"What were you trying to do?"
"Why did you lose that customer?"
"Would you play again?"
Anything that made them laugh or frown:
```

Decision rule (provisional, you set it): continue if at least 3 of 4 testers can explain a lost sale by day 3; **revise the reveal** (the chalkboard idea in REP-010) if 2 or fewer can.

Send me: the notes, as typed text or photos. I'll record what you *saw* separately from what we *think* it means, and propose the smallest next step.

4 · Records

DEC-013, DEC-015 and DEC-016 approved, with your words, in the register. Still pending from earlier: DEC-001, DEC-004, DEC-005, PLAN-003 (outline).

REP-014 · UI and UX polish: what feels unfinished, and a plan for the whole game

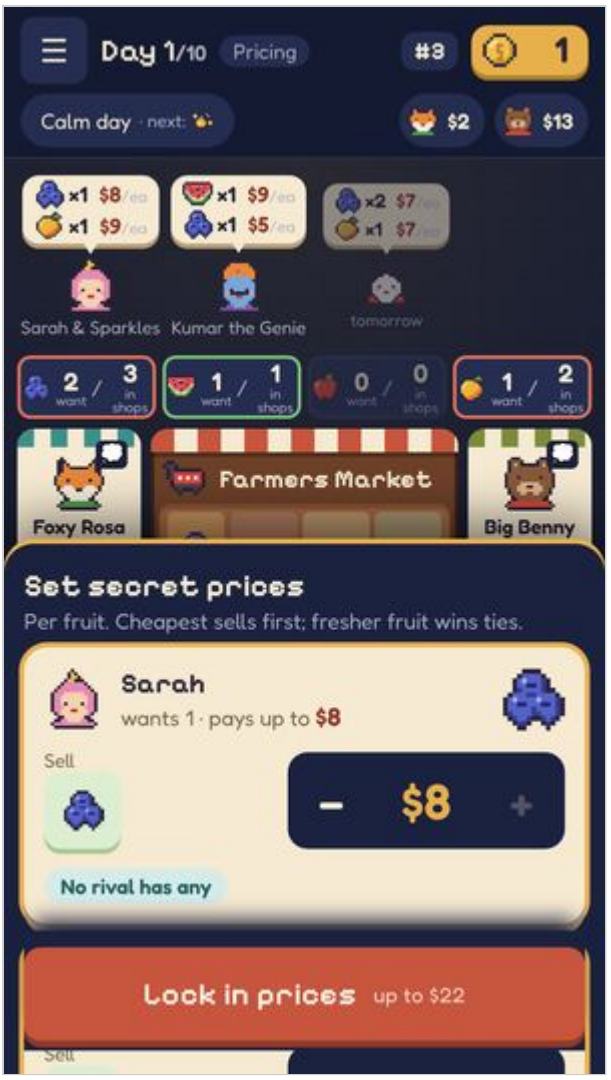
Game	Fruit Market
Report date	2026-10-26 23:06 (Asia/Kuala_Lumpur)
Owner input	"the look and feel of the game doesnt feel polished" and "the whole UI ... propose changes on the whole UI UX"
Build reviewed	1.1.1 candidate (`main @ 83de99b`), phone sizes 360×640 and 390×844
Type	Review and proposal. **No game changes were made.**
Approval status	DEC-020, PLAN-007 and PLAN-008 **awaiting approval**

1 · What I looked at

Every main screen at two common phone sizes: home, trait pick, buying, pricing, sales reveal, evening upgrades, results, trophies, online lobby and new game. The same checks the browser tests use: whether anything is cut off, overlaps or wastes space.


Day 1/10
Buying
#3
1
Calm day · next:
$10
$13
×1 $8
×1 $9
Sarah & Sparkles
walk-in
1+ / 3 want / in shops
? / 1 want / in shops
? / 0 want / in shops
1+ / 0 want / in shops
Foxy Rosa
Haggler
10
Farmers Market
Big Benny
Charmer
13
spoils
fresh
You
YOU
Big Basket
4/8
4/5
1
Your move
Tap a fruit in the market · 4/5 bought
Done buying


Day 1/10
Sales
+$8
#1
23
Calm day · next:
$11
$13
×2 $7
×1 $7
Sarah & Sparkles
Kumar the Genie
tomorrow
Foxy Rosa
Haggler
11
SHOWDOWN 1/3
cheapest sells first
Sarah & Sparkles
×1
pays up to $8 each
"Lovely!"
You
offers 1
SOLD
$8
No rivals! 1 blueberry sold at full price.
Big Benny
Charmer
13
You
YOU
Big Basket
1/8
23
Free Fruit
Skip
Selling... tap to speed up





2 · Why it doesn't feel polished: four causes

A. Things that look broken (defects, easy to fix)

1. **Sales reveal:** on shorter phones the explanation is cut off at the bottom ("...sold at full price" disappears behind the bar), and the card covers your own stall.
2. **The "+$8" money pop** flies off the top of the screen.
3. **Grammar:** "**You offers** 1" on the reveal card.
4. **Pricing panel:** the "Lock in prices" button floats over the next customer's card, which shows underneath it.
5. **Evening upgrades:** the Build buttons sit at different heights; card text is long and uneven.
6. **Trait picker:** the title wraps awkwardly and the panel covers your stall.
7. **Mixed icon styles:** emoji (🃏 favour count, ⏩ Skip, 💭, 📊, 🔒, 🧑/🤖) beside the pixel art.
8. **Placeholders that look unfinished:** the grey "walk-in" and "tomorrow" ghosts and the dashed "?" bubble.

9. **Trophies:** achievements have no icons, and locked and unlocked look the same (all grey text).

B. Too much on screen at once (layout)

- Before the table you see: the top bar, a second row (event, rival cash), customers, and a 4-box demand strip ("1+ want / 3 in shops"). That's four bands of small information.
- **Numbers shown twice:** your cash (top right and your stall) and rivals' cash (top chips and their stalls).
- **Rival stalls** take about 40% of the width to show mostly "0" rows, which squeezes the farmers market and makes its price labels ($1 $2 $3) tiny.
- **Your stall** is a big grid of empty pastel boxes with tiny "−$2" labels. Hard to read at a glance.

C. No single focus per moment Every phase shows the whole table at the same size. When you're pricing, the customers matter most; when you're buying, the market does. Right now nothing steps forward, so your eye has nowhere to land.

D. The visual system isn't consistent

- Two fonts are used without a clear rule: the pixel font appears in titles, numbers and small labels alike.
- Cards come in several styles: cream paper, navy chips, outlined boxes.
- Colours carry meaning (red outline, green outline, teal), but it isn't explained or applied consistently.
- Feedback and motion are uneven: some actions animate, others just switch.



3 · Proposal: three phases, smallest risk first

Phase 1 · Fix what looks broken (PLAN-007, build now, low risk)

All nine items in **A**, presentation only:

- the reveal card fits every phone and stops covering your stall
- the money pop stays on screen
- the grammar and other copy are corrected
- the pricing button sits in a solid footer
- upgrade cards are aligned and their text shortened
- the trait picker fits
- the remaining emoji become pixel icons (backlog B-33)
- the walk-in and tomorrow placeholders get proper art
- trophies get icons and a clear locked look

Ships with the Patches rename as **1.1.2**.

Phase 2 · Calm the table (PLAN-008 mockups first, then a build plan)

- **One compact top bar:** menu · day · phase · today's event icon · your cash and rank. Rivals' cash appears only on their stalls.
- **Slim rival stalls:** face, name, cash, and only the fruit they actually hold. This frees width so the farmers market and its prices get bigger.
- **Plain-language demand:** replace "1+ want / 3 in shops" with icons and words on the customer bubbles, for example "2 shops selling: price war".
- **Your stall:** show your fruit as items with "nights left", not a grid of empty boxes.
- **One focus per phase:** buying enlarges the market; pricing brings customers forward and dims the rest; the reveal gets the full stage; the evening shows only the upgrades.

Phase 3 · Make it feel finished (after Phase 2 is approved)

- **A small design system:** one type scale (pixel font for titles and big numbers only), one card style, spacing on a fixed grid, and colour roles (green good, red risk, gold money, teal info).
- **Motion and feedback:** phase transitions, a press effect on every button, and the **chalkboard reveal** from REP-010 (price boards flip). The reveal waits for the playtest result, because it's pillar 2.
- **Sound paired with the key moments** (backlog B-34), plus the harbour-market identity (DEC-017) on the home screen: a harbour backdrop, crates, chalkboards.

4 · Order of work (recommended)

1. **Now:** publish 1.1.1 and run the first playtest (DEC-016) on the *current* look. That gives a baseline, so we can show whether the new UI really makes the game clearer.
2. **Now, in parallel:** build Phase 1 (PLAN-007). Low risk; goes out as 1.1.2.
3. **Then:** PLAN-008, interactive mockups of the buy, price, reveal and evening screens plus home, as a page you can open on your phone, with two options where there's a real choice (for example, the rival stalls). **No game code.** You pick.
4. **Then:** a build plan for the chosen design (Phases 2 and 3), released as **1.2.0**, followed by a second playtest compared with the baseline.

5 · PLAN-007 acceptance (Phase 1)

- P1. Every item in **A** has before-and-after screenshots at 360×640 and 390×844.
- P2. Nothing is cut off or overlapping, and nothing scrolls sideways, on those sizes (automated check on each phase screen).
- P3. The visual-diff tool shows changes only on the screens being fixed.
- P4. The speed budget still passes (no long tasks over 200 ms; page size within budget).
- P5. `verify:full` is green; no rule, save or server changes.

PLAN-008 acceptance (mockups): an interactive page you can open on your phone, covering 5 screens, with clear options where a choice is needed, and every idea mapped to a problem above. No game code.

6 · Costs and risks

- RM0. No new tools or services.
- **Risk:** Phase 2 changes how the table is laid out, which could confuse returning players. Mitigation: mockups first, then the second playtest.
- **Risk:** polish can expand forever. Mitigation: each phase has a fixed list and its own approval.

7 · Questions for you (max 3)

1. **What bothers you most?**

   - (a) cluttered screens
   - (b) things that look broken
   - (c) it lacks animation or "feel"
   - (d) the art style itself

   (This sets the Phase 2–3 priorities. If it's (d), that's a bigger art decision.)

2. **Is there a game whose look you'd call "polished"?** A name or a screenshot helps me aim the mockups.

3. **Approve PLAN-007 (Phase 1 fixes) and PLAN-008 (mockups)?**

8 · Decisions

ID	Question	Recommendation	Status
DEC-020	How to polish the UI and UX	Three phases; fix defects first; mockups before any layout change; a playtest before and after	Awaiting approval
PLAN-007	Phase 1: fix what looks broken	Build now; ship in 1.1.2	Awaiting approval
PLAN-008	Interactive mockups for Phases 2–3	Design only, no game code	Awaiting approval

REP-015 · Art direction: what "It Takes Two" can teach a 2D phone game

Game	Fruit Market
Report date	2026-10-26 23:18 (Asia/Kuala_Lumpur)
Owner input	Bothers most: (a) clutter, (c) lack of animation and feel, (d) the art style itself. Polished reference: *It Takes Two*
Type	Creative direction options and a revised plan. **No game changes.**
Approval status	DEC-021 and the revised PLAN-008 **awaiting approval**; PLAN-007 still awaiting approval

1 · What this changes

REP-014 started with fixing broken-looking details. Your answer puts the priority elsewhere: **the art style, the feel, and the clutter**. That's a bigger and more important decision, so it should be designed and tested before anything is built.

2 · What makes *It Takes Two* feel polished (the qualities we can borrow)

It Takes Two (Hazelight, 2021) is a 3D console game, so we can't copy its look or technology, and we must not reuse any of its art or characters. Its polish comes from qualities that translate to 2D:

Quality	In *It Takes Two*	What it could mean for Fruit Market
Toy-like, touchable materials	The world is shrunk to toy size: wood, felt, clay, fabric	Stalls of painted wood, fruit like shiny clay or wooden toys, cloth awnings, chalkboard price tags
Chunky, rounded shapes	Soft silhouettes, big heads, readable at a glance	Rounder cards and buttons, bigger and simpler characters, fewer thin lines
Warm light and depth	Soft shadows, glow, layers of foreground and background	Gentle drop shadows, a warm late-afternoon harbour backdrop, depth between the table and the cards
Everything reacts	Squash and stretch, bounces, particles on every action	Fruit hops into your basket, coins bounce, buttons squish, the price reveal *flips*
Expressive characters	Faces and poses show feeling all the time	Customers smile, frown or shrug at the reveal; rivals react with a pose, not just a text line
One thing to look at	The camera always frames the important moment	Each phase zooms into what matters (market, customers, reveal) and dims the rest

3 · Three art directions (choose one)

	A · Polished pixel	**B · Toy-box tabletop (recommended)**	**C · 3D toy diorama**
Idea	Keep pixel art; make it consistent and animated	A smooth 2D illustrated style: toy-like, rounded, warm, with soft shadows and textures	Real 3D: a miniature market you look down on

	A · Polished pixel	B · Toy-box tabletop (recommended)	C · 3D toy diorama
Fixes your (d) art style?	**Weakly**: same look, just cleaner	**Yes**: a clearly new, warmer, toy-like look	Yes, and closest to the reference
Fixes (c) feel	Partly	Yes: built around motion from the start	Yes
Phone speed	Excellent	Good (vector art; we keep the speed budget)	Risky on cheap phones; heavier download
Scope	Small	Medium-large: redraw fruits, ~30 characters, rivals, ~40 icons, UI	Very large: a new rendering engine and 3D models
Who makes the art	Claude, in code (as now)	Claude, in code as vector art (consistent, revisable), **or** a hired illustrator later	3D modelling is beyond what Claude can do well; it needs an artist (paid)
Cost	RM0	RM0 if Claude draws it; an illustrator would be a separate paid decision	Paid artist plus large engineering
Risk	Doesn't satisfy (d)	Art quality by Claude is good but not AAA; judge it on the style test before committing	Highest cost, slowest; could stall the game

Recommendation: B, proven by a small style test first. It addresses all three of your concerns, stays fast on phones, costs nothing if I draw it, and can be judged before any real work. If the test looks too plain, the next step would be to brief a paid illustrator. That would be a separate, priced decision under your RM500/month ceiling.

4 · Revised PLAN-008 · Style test plus layout mockups (design only, no game code)

1. **Style test:** the *same* buying screen drawn three ways (A, B, C as a flat render), plus one fruit at its four freshness stages, one customer and one rival in direction B. It's a page you open on your phone.
2. **Layout mockups in the chosen direction** for buy, price, reveal, evening and home, with:
   - the clutter fixes from REP-014 (one top bar, slim rival stalls, demand in plain words)
   - one focus per phase
   - motion notes (what moves, when) and a short animated example of the reveal flip
3. **Your choice:** you pick the direction and the layout options. Only then do I propose the build plan(s).

Acceptance: M1, a phone-viewable page with the style test (3 directions on one screen) and the 5 mockup screens. M2, every idea is linked to a problem in REP-014 or a quality in section 2. M3, no game code changed. All art is original; reference games are inspiration only.

5 · Where PLAN-007 (the 9 broken-looking fixes) fits now

You didn't pick (b), and a new art style would redraw most of those screens anyway. Two options:

- **Keep it small (recommended):** do only the fixes that survive any art change. That's the grammar ("You offers"), the money pop going off-screen, and the sales text being cut off. Ship them in 1.1.2 with Patches; park the rest until the new style.
- Or drop PLAN-007 entirely and fold everything into the new design.

6 · Order of work

1. Publish 1.1.1 (your Deploy) and run the first playtest on the current look (baseline).
2. PLAN-008 style test, then you choose a direction.
3. Layout mockups in that direction, then you choose.
4. A build plan for the new look and feel (likely several packages), released as **1.2.0**, then a second playtest.

7 · Questions (max 3)

1. **Approve the revised PLAN-008** (style test plus mockups, no game code)?
2. **PLAN-007:** keep it small (3 fixes in 1.1.2) or drop it?
3. **Is a paid illustrator possible later,** if the B style test isn't good enough? A yes or no now helps me plan; any spending would still need its own quote and approval.

8 · Decisions

ID	Question	Recommendation	Status
DEC-021	Art direction for the polish	Explore A/B/C in a style test; recommend B (toy-box tabletop)	Awaiting approval
PLAN-008 (revised)	Style test plus layout mockups, design only	Approve	Awaiting approval
PLAN-007	Broken-looking fixes	Reduce to 3 fixes that survive any art change	Awaiting your choice

REP-016 · Toy-box mockups ready for your review (PLAN-008)

Game	Fruit Market
Report date	2026-10-26 23:57 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only), approved: "go with plan b"
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (private to you)
Type	Design mockups. **No game code changed.** The live game is unchanged
Delivery status	Implemented; waiting for your review and choices

What's on the page (phone-sized, tap to try)

Screen	What to notice	Problem it answers
Style sample	Fruit at four freshness stages (shiny, then spotted, then droopy); Grandpa Tan and Patches as rounded toy figures; wood, chalkboard, coin and a pressable button	Art style (d); "freshness you can see" (pillar 1)
Home	Harbour backdrop with bobbing boats, a market sign "Where the whole world shops", big pressable buttons, "2 to 4 players online"	Harbour identity (DEC-017); clutter (a)
Buying	One top bar; customers with plain-language demand ("3 shops sell apples"); slim rival chips with only what they hold; big crates (tap to buy: coins drop, the basket count bounces, fruit hops onto your stall); fruit shows "nights left"	Clutter (a): no repeated cash, no "1+ want / 3 in shops"; feel (c)
Pricing	One customer in focus, the table dimmed; chalkboard price with big – and +; a hint about rivals in words; a solid "Lock in prices" footer	No focus (REP-014 C); floating button defect
Sales reveal	Ring the bell: boards flip highest to lowest, the cheapest wins, coins fly to the winner, " too high" on yours, plus a plain explanation and the rival's quip	Pillar 2 and the reveal-comprehension test; feel (c)
Evening	Dusk sky, three equal upgrade crates with aligned buttons, a "BUILT" stamp, a favour card, "Lights out"	Misaligned upgrade buttons; feel (c)

Limits of the mockups

- They're **mockups, not the game**: the numbers are examples, and only the parts needed to show the idea are interactive.
- Characters other than Grandpa Tan, Patches, Foxy Rosa and Cowboy John are not drawn yet; the build would redraw all of them in this style.
- I didn't add sound; that comes with the build.
- The style is original. *It Takes Two* informed the qualities only.

What I need from you

1. **Overall:** does this look and feel like the direction you want? (yes / closer to … / no)

2. **Anything you love or dislike** on a specific screen (a comment on the page works too).
3. If yes, I'll propose the **build plan** to turn the real game into this style. It's several packages: art and the shared style first, then each screen. It would be released as 1.2.0, after the baseline playtest.

Acceptance (PLAN-008)

#	Check	Result	Evidence
M1	Phone-viewable interactive page with the style sample and 5 screens	PASS	Published at the link above: 6 phone screens plus a shared fruit part. Your review is the real check
M2	Every idea linked to a REP-014 problem or REP-015 quality	PASS	Table above
M3	No game code changed	PASS	No changes under `src/` or `server/`

REP-017 · Mockups revised: the cunning edition (PLAN-008, DEC-022)

Game	Fruit Market
Report date	2026-10-27 00:12 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only); direction DEC-021, tone DEC-022
Owner feedback	"yes but align more with the cunning aesthetic given the undercut mechanism of the game"
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 9, private to you)
Type	Design mockups. **No game code changed.** The live game is unchanged
Delivery status	Implemented; waiting for your review

The idea in one line

The market is sunny by day, but **pricing is a secret game**. When something is secret, the screen turns **plum and gold**, rivals look sly, and the reveal plays like a showdown. Cheeky, never mean.

What changed, screen by screen

Screen	Cunning touches
Style sample	Plum header; Foxy Rosa as the sly mascot (raised brow, wink, smirk: "Psst... I'll go a dollar lower."); a wax-sealed price board; the dotted "Psst..." whisper bubble; palette row: market colours by day, plum and gold for secrets
Home	New tagline "Undercut. Outfox. Sell it fresh." (kept "Where the whole world shops"); Foxy Rosa peeks out from behind the sign every few seconds; online buttons in plum; "Outfox a stranger"
Buying	A whisper: "Psst... Foxy Rosa is sitting on 2 apples. Who blinks first?"; sly rival faces (Foxy Rosa narrowed eyes, Patches half-lidded)
Pricing	Plum top bar ("top secret"); "Seal your prices"; "Only you see this" over your chalkboard; the rival hint is now a whisper from Foxy Rosa; a "Rivals are scheming..." row with their sealed boards and shifty eyes; footer "Seal my prices" with a wax seal
Sales reveal	Dark stage with a curtain and spotlight; the winning board glows gold; headline "Outfoxed by $1!" with Foxy Rosa's quip and face
Evening	"Plotting time"; the favour card is "Up your sleeve"; a whisper: "the rivals are plotting tonight too". Your layout edit (favour card lower) is kept

Fit with the approved pillars

- **Pillar 2, outprice rivals in secret:** now the strongest visual idea.
- **Pillar 3, a cheerful town market:** kept. Rivals tease and wink; nobody is cruel; daytime stays sunny. If playtesters read the rivals as mean, we soften them (DEC-022 review trigger).
- **Privacy rule:** whispers only repeat what the game already shows (who holds what). They never reveal a rival's secret price or choice.

Limits

- Mockups, not the game; numbers are examples. Characters other than Foxy Rosa, Patches, Grandpa Tan and Cowboy John are not drawn yet. No sound yet.
- Plum screens need a contrast check in the build (text sizes and colours were chosen for it, not measured).

Acceptance (PLAN-008, added check)

#	Check	Result	Evidence
M4	Owner feedback applied; owner's own page edit kept	PASS	Version 9 published; Evening spacer kept
M3	No game code changed	PASS	Changes only in docs/workflow

What I need from you

1. Is this the right amount of cunning? (more / about right / less)
2. If about right, I'll propose the **build plan** for 1.2.0 (art and shared style first, then each screen).

REP-018 · Mockups rebuilt: the cosy pixel edition (PLAN-008, DEC-023)

Game	Fruit Market
Report date	2026-10-27 00:17 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only); look decision DEC-023 (awaiting your confirmation)
Owner feedback	"its still not enough, maybe balance the artstyle back to pixel art, the ui/ux here looks more polished but lacks the cosiness of how the previous board was set" · "no i meant the version before you tweaked here, that felt cosy"
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 10, private to you)
Type	Design mockups. **No game code changed.** The live game is unchanged
Delivery status	Implemented; waiting for your review

The idea in one line

Pixel heart, smooth frame. The first toy-box mockups (the cosy ones) are back, but the characters, fruit and icons are now the game's own pixel art, and headings use the game's pixel font.

What changed from the cunning edition

- **Cosy restored:** the first toy-box layout and warm colours (cream, teal, wood, terracotta). Your Evening layout edit is kept. The dark plum screens, curtain and spotlight are gone.
- **Pixel art:** the fruit at four freshness stages, Grandpa Tan, Cowboy John, Foxy Rosa, Patches, you, the upgrades, bell, coin and favour card all use the game's own sprites (built from `src/ui/pixel.tsx`). Foxy Rosa and Patches get sly half-closed eyes.
- **Light sly touch, in warm colours:** dotted "Psst..." whispers from Foxy Rosa (buying, pricing), "Seal your prices" and "only you see this", "Outfoxed by $1!" on the reveal, "...and every price is a secret" on the home sign, with Foxy Rosa peeking out from behind it.

Why this is cheaper to build

The full vector redraw (DEC-021) is no longer needed. The build reuses and tidies the existing sprites; most of the work is the new frame and layout.

Limits

- Mockups, not the game; numbers are examples. No sound yet.
- The sly eyes for Foxy Rosa and Patches are mockup variants; the build would add them as sprite variants.

Acceptance (PLAN-008, added check)

#	Check	Result	Evidence
M5	Cosy pixel revision; your page edit kept	PASS	Version 10 published; Evening spacer kept
M3	No game code changed	PASS	Changes only in docs/workflow

What I need from you

1. Does this feel cosy again, and is the pixel balance right? (yes / more pixel / less pixel)
2. If yes, I'll record DEC-023 as approved and propose the 1.2.0 build plan.

REP-019 · Mockups rebuilt as a board game: one shared market table (PLAN-008, DEC-024)

Game	Fruit Market
Report date	2026-10-27 00:25 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only); layout decision DEC-024 (awaiting your confirmation)
Owner feedback	"revamp the current UI/UX to match the cosiness of the previous UI/UX board layout" · "the one currently feels very menu based, instead of board game"
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (same link, version 12, private to you)
Type	Design mockups. **No game code changed.** The live game is unchanged
Delivery status	Implemented; waiting for your review

The idea in one line

You sit at the market table. Every game screen is the same board, like the current game, now with the cleaner, pixel-art, warm look.

The table

- **Street (top):** customers queue as pixel figures with a small tag ("apple ×2 · up to $10") and one plain hint ("3 shops sell apples").
- **Middle:** Foxy Rosa's stall, the Farmers market, Patches' stall, each under a striped awning. Rival stalls show only what matters: cash and what they hold.
- **Bottom:** your stall, with your fruit and how many nights each has left.

Each phase on the same table

Phase	What happens on the board
Buying	Tap crates in the market; fruit hops onto your stall. Foxy Rosa whispers.
Pricing	Tap a customer on the street to serve them. Rival stalls show sealed boards ("sealed..."). Your stall becomes your price board, which only you see. "Seal my prices".
Reveal	Ring the bell: the boards flip **on the stalls** (Patches $10, you $9, Foxy Rosa $8). Foxy Rosa's stall glows, coins fly to it, Grandpa Tan walks over, then "Outfoxed by $1!".
Evening	Night falls over the same table: lanterns on the rival stalls ("plotting..."), the market covered, and the upgrades at your stall. Favour card kept low, as in your earlier edit.

Limits

- Mockups, not the game; numbers are examples. Home and the style sample are unchanged apart from a subtitle.

- NOT RUN: I did not view the page myself; the four screens' scripts were checked to load.

Acceptance (PLAN-008, added check)

#	Check	Result	Evidence
M6	Board-game layout on one shared table	PASS	Version 12 published
M3	No game code changed	PASS	Changes only in docs/workflow

What I need from you

1. Does this feel like a cosy board game now? (yes / closer to …)
2. If yes, I'll record DEC-023 and DEC-024 as approved and propose the 1.2.0 build plan.

REP-020 · Look approved; market made bigger (PLAN-008, DEC-023, DEC-024)

Game	Fruit Market
Report date	2026-10-27 01:21 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only)
Owner feedback	"going back to your mockup, i like it now, but tweak the layout so that the fruit market is more visible (now it feels to small/ narrow)"
Decisions	DEC-023 (cosy pixel look) and DEC-024 (one shared table) approved
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (version 13, private to you)
Type	Design mockups. **No game code changed**

What changed

- Rival stalls are slimmer (80 to 64 px wide); the Farmers market grows from about 206 to 234 px.
- Crates are taller (58 to 74 px) with bigger pixel fruit (28 to 42 px) and bigger prices.
- The market title is larger; the street strip is a little shorter to make room.
- The same slimmer stalls are used on pricing, reveal and evening, so every screen matches.

Validation

Check	Result	Evidence
Market wider, crates and fruit bigger	PASS	Version 13 published
Screen scripts load	PASS	All four parse
Viewed on a phone by me	NOT RUN	Please check the Buying screen
No game code changed	PASS	Changes only in docs/workflow

Next

Propose the 1.2.0 build plan for this look (separate packages, each needing your approval).

REP-021 · Vertical decay board and the delivery lorry (PLAN-008, DEC-025)

Game	Fruit Market
Report date	2026-10-27 01:28 (Asia/Kuala_Lumpur)
Package	PLAN-008 (design only)
Decision	DEC-025, approved by your instruction
Mockups	https://claude.ai/artifact/YQUpnHeNNWkAUqFhKRz9VB (version 14, Buying screen, private to you)
Type	Design mockups. **No game code or rules changed**

What's on the Buying screen now

- **Market as a decay tracker, vertical:** four columns left to right (blueberry, watermelon, apple, mango). New and ripening fruit at the top, spoiling at the bottom, with a colour bar (green, gold, brown). Every slot shows its price; empty slots stay as dashed outlines, like today's game. The tracks match the real rules: blueberry 3 slots, watermelon 5 + 1 unripe, apple 7, mango 4 + 2 unripe.
- **Your stall uses the same tracker** as four small strips; fruit you buy pops into its slot.
- **The lorry:** when buying starts it drives in along the street, Grandpa Tan and Cowboy John shuffle left to make room, and the crates drop one by one onto the board. Press **Done buying** and the lorry drives off, the customers spread back out, and the button changes to "Lorry's gone · set prices →". A small ↻ **replay lorry** button lets you watch it again.

Validation

Check	Result	Evidence
Slots and prices follow the game's tracks	PASS	Matches src/engine/data.ts
Buying moves fruit to the right slot on your stall	PASS	Logic test of the screen
Lorry leaves on Done buying	PASS	Logic test of the screen
Viewed on a phone by me	NOT RUN	Please check the animation
No game code changed	PASS	Changes only in docs/workflow

Notes for the build

- The lorry must be quick (about 2 seconds) and skippable with a tap, so it never slows play.
- The other screens keep the shared table; only Buying shows the market.

REP-022 · Build plan for the cosy table look (PLAN-009 to PLAN-011)

Game	Fruit Market
Report date	2026-10-27 01:58 (Asia/Kuala_Lumpur)
Owner request	"ok now do this changes" (read as: build the mockup look into the game)
Type	Planning only. **No game code changed**
Approval status	PLAN-009 awaiting approval; PLAN-010 and PLAN-011 are outlines

Good news first

The current game already has rival seats, a rival pop-up, a market and the want / in-shops strip. So the build mostly **restyles and rearranges** what exists; rules, saves and the server stay as they are.

Three packages, one at a time

Plan	What you get	Size
PLAN-009	Rivals seated across the top; tap a name to see their freshness chips; full-width market board (fruit columns, top-aligned, colour-coded, key, want / in-shops on each column); your stall as freshness chips; smaller customers	Largest; first
PLAN-010	The delivery lorry: drives in, customers step aside, crates land; leaves before selling; skippable	Small
PLAN-011	Pricing, reveal and evening on the table (sealed boards, flips in the seats, lanterns); folds in PLAN-007's text fixes	Medium

Release as **1.2.0** once all three are done and checked on the preview.

Checks for PLAN-009 (all must PASS)

verify and verify:full; board slots and prices tested against the rules; column chips tested against today's numbers; screenshots with 2, 3 and 4 players at small phone, normal phone and desktop with no clipping; secret-price leak check stays 0; old saves load.

Risks and costs

- RM0. No new dependencies.
- **Small phones (360×640):** the 7-row board is tall. I'll make the rows shorter there; if it still doesn't fit, the table scrolls rather than squashing.
- **Release order:** `main` is holding the 1.1.1 candidate until you press Deploy. I'll build on the work branch, and nothing reaches the preview until 1.1.1 is live.
- **Lorry art (PLAN-010):** made with Gemini. Google doesn't claim ownership of it, but AI-made art may not be protected by copyright, so others could copy it. The alternative is a pixel lorry

drawn in the game's own sprite style, which would match the other art.

Decisions needed

1. Approve PLAN-009?
2. Lorry: your Gemini picture, or a pixel lorry in the game's sprite style?
3. PLAN-007 (old small fixes): fold its text fixes into PLAN-011 and close it?

REP-023 · PLAN-009 built: seated rivals and the new market board

Game	Fruit Market
Report date	2026-10-27 02:19 (Asia/Kuala_Lumpur)
Package	PLAN-009, approved ("yes" to REP-022)
Branch	`workflow/plan-009` , commit b2b495a. **Not on the preview or the live game yet**
Delivery status	Verified

What changed in the game

- **Rivals sit across the table** in one row: portrait, name, cash and what they hold. Tap a seat and their stall drops down as **freshness chips**, the same as your own.
- **The market is the centre:** full width, one column per fruit, newest at the top, every slot coloured by freshness, with a colour key. Each column starts with its **want / in shops** chip.
- **Your stall** shows chips such as "×4 ripens in 2" or "3 nights"; a red "!" marks fruit on its last day.
- Tutorial and how-to text updated for the new board. Rules, saves, server and online play are unchanged.

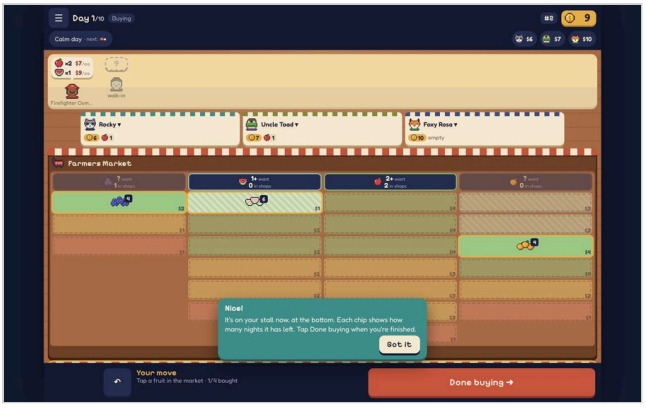
Screenshots


Day 1/10
Buying
#3
8
Calm day · next:
$10
$10
$7
×1 $10/ea
×1 $7/ea
Princess Siti
walk-in
Big Benny ▾
10 empty
Rocky ▾
10 empty
Patches ▾
7 1
Farmers Market
? want 1 in shops
1+ want 1 in shops
? want 0 in shops
1+ want 0 in shops
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying


Day 1/10
Buying
#3
8
Calm day · next:
$10
$10
$7
×1 $10/ea
×1 $7/ea
Princess Siti
walk-in
Big Benny ▴
10 empty
Rocky ▾
10 empty
Patches ▾
7 1
Big Benny's stall
Local Favourite
10
Nothing on the shelf yet.
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying


Day 1/10
Buying
#3
2
Calm day · next:
$16
$10
Princess Siti
walk-in
Patches ▾
16 empty
Uncle Toad ▾
10 empty
Farmers Market
? want 0 in shops
1+ want 0 in shops
? want 0 in shops
1+ want 0 in shops
unripe
fresh
okay
not so
You
YOU
Local Favourite
4/8
4/4
2
×4 ripens in 2
Your move
Tap a fruit in the market · 4/4 bought
Done buying


Day 1/10
Buying
#4
8
Calm day · next:
$10
$10
$10
Dr Priya
walk-in
Uncle Toa...
10 empty
Rocky ▾
10 empty
Pinky ▾
10 empty
Farmers Market
1+ want 1 in shops
? want 0 in shops
2+ want 0 in shops
? want 0 in shops
Nice!
It's on your stall now, at the bottom. Each chip shows how many nights it has left. Tap Done buying when you're finished.
Got it
Your move
Tap a fruit in the market · 1/4 bought
Done buying



Validation

#	Check	Result
B1	Quick checks (lint, types, 73 unit tests, build)	PASS
B2	Full browser tests (solo, speed budget, online with a leaver, quick match, four players, reconnect)	PASS
B3	Board slots and prices match the rules (30 games)	PASS
B4	Want / in-shops numbers unchanged (30 games)	PASS
B5	2, 3 and 4 players on small phone, phone and desktop: no sideways scroll or errors	PASS
B6	Rival drop-down shows only public stock; secret-price leak check 0	PASS
B7	Rules, saves, server untouched	PASS

Notes

- On small phones the table scrolls to reach your stall (as planned). Long names shorten ("Uncle Toa…") with three rivals on the smallest phones.
- Pricing, reveal and evening still use today's panels; that is PLAN-011. The lorry is PLAN-010.

Next

1. Press **Deploy LIVE game** for 1.1.1 and tell me "deployed". Then I can merge this to `main`, which puts it on the **preview** for you to try.
2. Lorry art choice for PLAN-010: your Gemini picture, or a pixel lorry in the game's style.

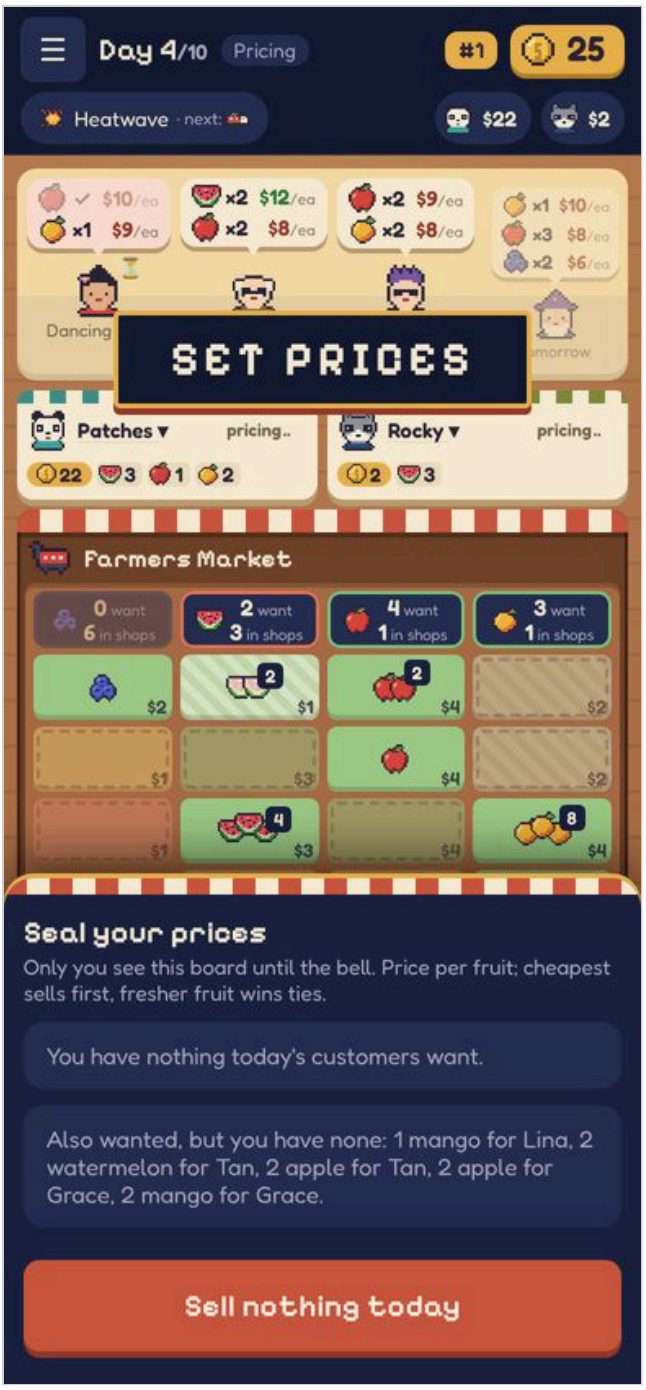
REP-024 · PLAN-011 built: pricing, reveal and evening on the table

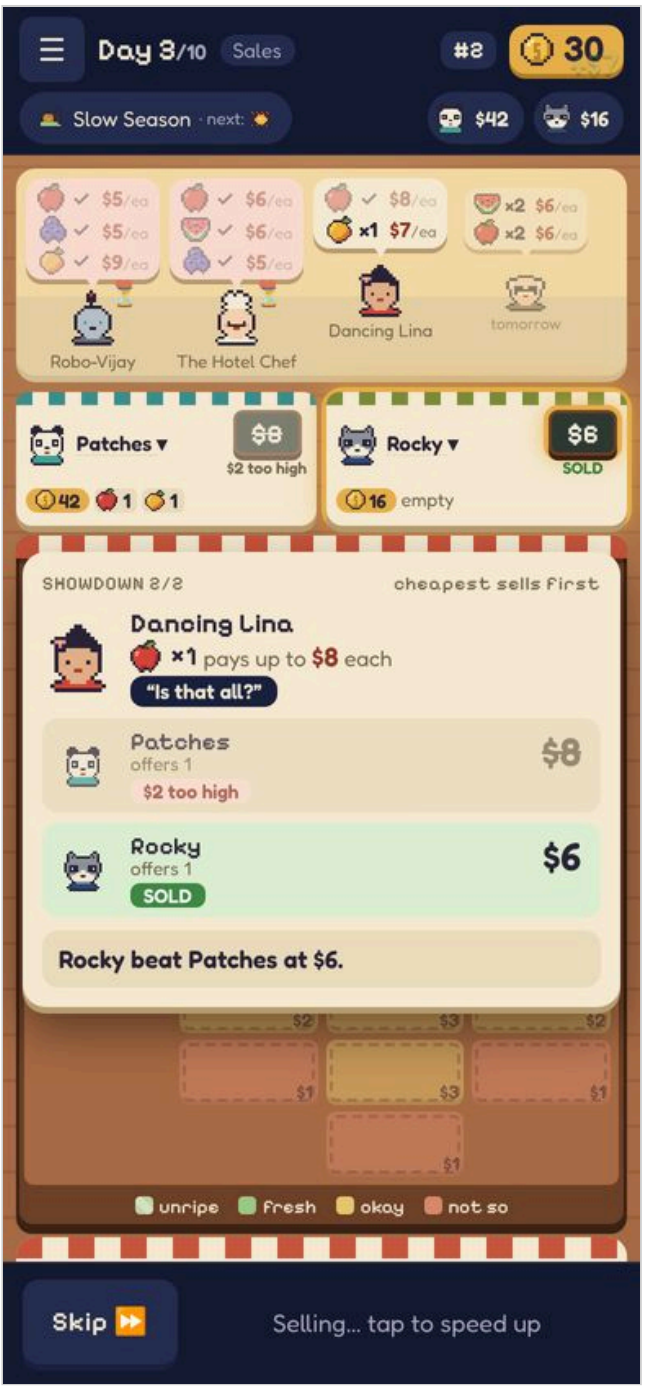
Game	Fruit Market
Report date	2026-10-27 02:51 (Asia/Kuala_Lumpur)
Package	PLAN-011, approved ("use the gemini lorry, approve PLAN-011")
Branch	`workflow/plan-009` , commit 4408cd8. **Not on the preview or the live game yet**
Delivery status	Verified

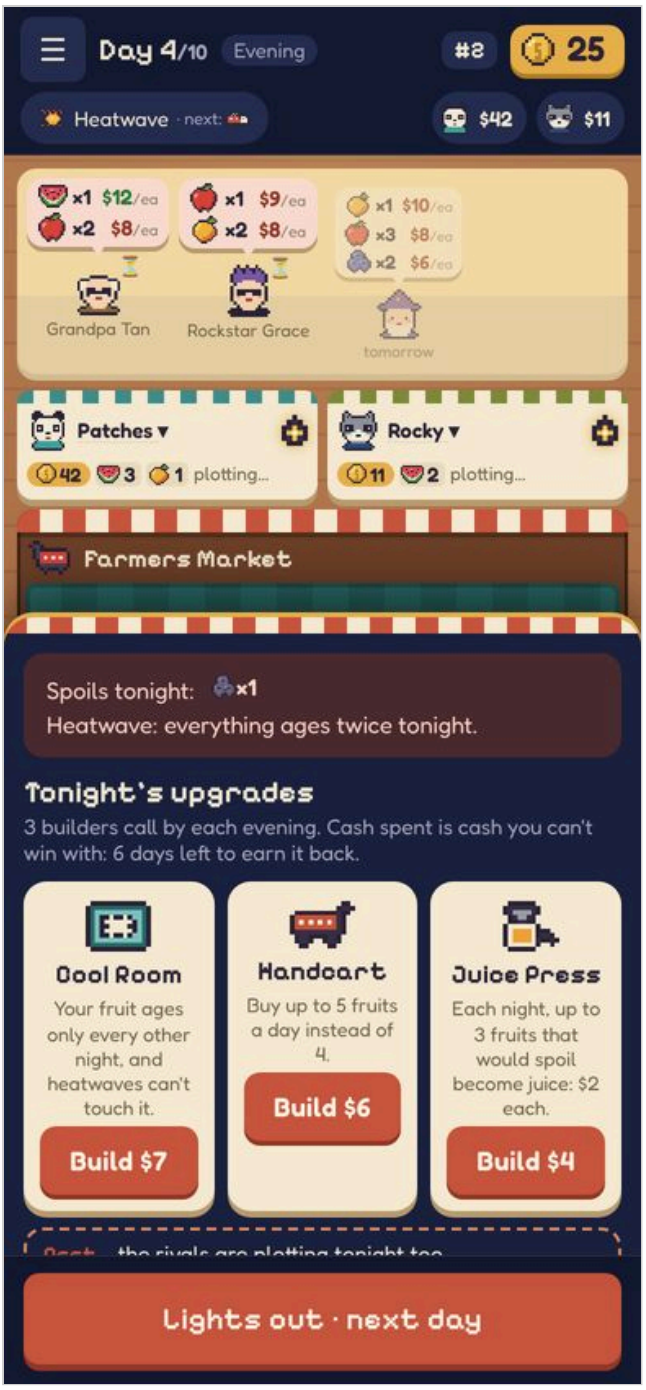
What changed in the game

- **Pricing:** each rival's seat shows "pricing...", then a sealed board once they're done. Nobody sees a price. Your panel says **Seal your prices**, with a "Psst..." read of the competition, and the button says **Seal my prices**.
- **The reveal happens on the table:** the boards flip open on each seat and on your stall, highest price first. The winner's seat glows, and tags like **SOLD** or **$1 too high** sit under the boards. If a rival undercuts you, the card says **Outfoxed!**
- **Evening:** lanterns and "plotting..." on the rival seats, and the market crates are covered for the night. Favour cards sit under **Up your sleeve**, with a "Psst... the rivals are plotting tonight too."
- **Old small fixes (PLAN-007, now closed):** "You offer" instead of "You offers", money pop-ups stay on screen, and the showdown card scrolls instead of being cut off.

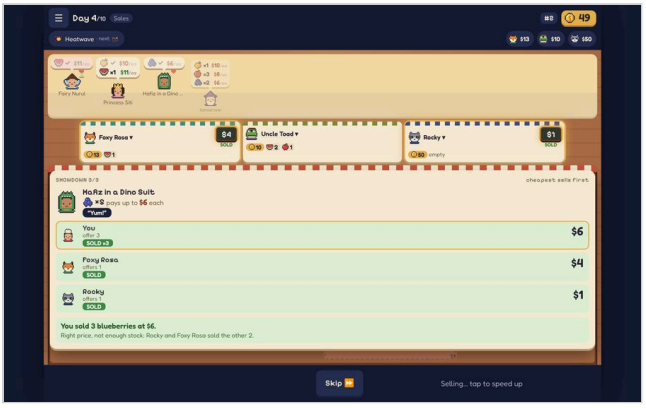
Screenshots


Day 4/10
Pricing
#1
25
Heatwave · next:
$22
$2
SET PRICES
Patches ▾
pricing..
Rocky ▾
pricing..
Farmers Market
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
You have nothing today's customers want.
Also wanted, but you have none: 1 mango for Lina, 2 watermelon for Tan, 2 apple for Tan, 2 apple for Grace, 2 mango for Grace.
Sell nothing today


Day 3/10
Sales
#2
30
Slow Season · next:
$42
$16
Robo-Vijay
The Hotel Chef
Dancing Lina
tomorrow
Patches ▾
$8
$2 too high
Rocky ▾
$6
SOLD
empty
SHOWDOWN 2/2
cheapest sells first
Dancing Lina
×1 pays up to $8 each
"Is that all?"
Patches
offers 1
$2 too high
$8
Rocky
offers 1
SOLD
$6
Rocky beat Patches at $6.
unripe
fresh
okay
not so
Skip
Selling... tap to speed up


Day 4/10
Evening
#8
25
Heatwave · next:
$42
$11
×1 $12/ea
×2 $8/ea
×1 $9/ea
×2 $8/ea
×1 $10/ea
×3 $8/ea
×2 $6/ea
Grandpa Tan
Rockstar Grace
tomorrow
Patches ▾
42 3 1 plotting...
Rocky ▾
11 2 plotting...
Farmers Market
Spoils tonight: ×1
Heatwave: everything ages twice tonight.
Tonight's upgrades
3 builders call by each evening. Cash spent is cash you can't win with: 6 days left to earn it back.
Cool Room
Your fruit ages only every other night, and heatwaves can't touch it.
Build $7
Handcart
Buy up to 5 fruits a day instead of 4.
Build $6
Juice Press
Each night, up to 3 fruits that would spoil become juice: $2 each.
Build $4
Lights out · next day


Day 4/10
Sales
+$9
#8
35
Minimum Price · next
$48
$10
$31
$9/ea
×2 $8/ea
$10/ea
×2 $9/ea
$6/ea
Rockstar Grace
Dancing Lina
Hafiz in a Dino ...
Hui Min th
Patches ▾
48 4 1
$9
SOLD
Foxy Rosa...
10 1
$10
$1 too high
Uncle Toa...
31 empty
SHOWDOWN 3/3
cheapest sells first
Hui Min the Elf
×1 pays up to $10 each
"A bargain!"
Foxy Rosa
offers 1
$1 too high
$10
Patches
offers 1
SOLD
$9
Skip
Selling... tap to speed up



Validation

#	Check	Result
T1	Full browser tests incl. secret-price leak check 0; screenshots at 3 sizes, no sideways scroll	PASS
T2	Quick checks (73 tests, lint, types, build)	PASS
T3	Rules, saves, server and online protocol untouched	PASS
T4	PLAN-007 fixes	PASS

Not done

- The customer walking over to the winner (from the mockup) is not built. Fruit and coins already fly between the stall and the customer. It can follow later if you want it.

Next

1. **Deploy 1.1.1** (Actions → Deploy LIVE game → `PUBLISH`) and tell me "deployed". Then I merge the new look to `main`, and it appears on the **preview** for you to play.
2. Approve **PLAN-010** (the lorry with your Gemini picture). It's the last piece before 1.2.0.

REP-025 · Balance and pacing audit: how Fruit Market is won, lost and paced

Game	Fruit Market (rules version 2, `main` at 15f358a)
Report date	2026-10-27 03:36 (Asia/Kuala_Lumpur)
Type	Audit only. **No code changed.** Planning and documentation work, not a game change
Method	Read the rules engine line by line (`src/engine/game.ts` , `data.ts` , `ai.ts` , `daily.ts` , `rng.ts`), the mode set-up (`src/ui/App.tsx` , `src/ui/progress.ts` , `server/room.ts` , `server/meta.ts` , `src/net/protocol.ts`) and the scoring screens; then measured with computer-only simulations (`npm run balance` at 300 games per test, plus 4 extra read-only probes, 300–400 games each)
Approval status	Not applicable (findings and proposals; any change needs a plan and your approval)

Caution on the numbers. Every measured figure comes from computer players playing each other. People price, buy and take risks differently, so treat win rates as the shape of the game, not as human results. Human results need the playtest (DEC-016) and the telemetry proposed in section 6.

A. How the game is won and lost (overview)

- **You win by having the most cash when the last day's sales end.** Nothing else counts: unsold fruit, upgrades and favour cards are worth $0 at the end (`ranking()` in `src/engine/game.ts`).
- **Nobody is ever knocked out.** There is no bankruptcy, no health, no timer and no early finish. Every game lasts exactly its length: 8, 10 or 14 days. "Losing" just means not finishing first.
- **Ties:** most cash wins; if level, most fruit sold; if still level, the **seat order decides silently** (seat 0, which is you in solo, daily and league games). The results screen then wrongly says it was "decided on fruit sold".
- **Where money comes from:** buying fruit cheaply at the farmers market and selling it to customers through **secret, sealed prices** where the **cheapest offer sells first**. Freshness limits the top price you may ask and breaks price ties. Everything else (upgrades, traits, favour cards, events) moves the margins.
- **What really decides games** (measured): pricing skill (Hard beats Normal 84% of the time head to head), buying only what customers want, the Signboard and Cool Room upgrades, and the event deck. Turn order is a deliberate catch-up: the poorest shop buys first.
- **Long-term goals outside a single game:** the 6-stage Market League (win to advance, unlimited retries), the daily challenge (score = final cash on a shared leaderboard), and 12 achievements.

B. Win / loss condition table

#	Outcome	Exact check (code)	When it's evaluated	Notes and problems
1	**Game ends**	`finishReveal()`: `if (s.day >= s.config.days) phase = 'over'`	After the final day's sales reveal	No evening on the last day: no upgrades, no ageing, no night fines
2	**Win (1st place)**	`ranking()`: sort by `cash` desc, then `stats.sold` desc	Phase `over`	Third tie-break is the array's original order (stable sort) = seat order. **Hidden outcome**
3	2nd–4th place ("loss")	Same ranking	Phase `over`	No other penalty
4	Dead heat message	`Results.tsx`: `margin === 0` → "A dead heat, decided on fruit sold!"	Results screen	**Wrong text** when sold counts are also equal (then seat order decided). Measured: 0.33% of 3–4-player games
5	No defeat / elimination	none exists	—	Cash can't go below 0 (buys need cash; the inspector fine is capped at your cash)
6	"Broke" state (soft loss)	Not in code; arises when cash < $1 and no ripe stock	Any morning	You can still play but can't buy (every fruit costs ≥ $1). Only favour cards (Free Cash, Free Fruit) can rescue you. Measured: 0.3–0.6% of computer players in 3-player games, 2.2% in 4-player Hard games, typically from day 4–6
7	Online leaver	Server: grace period, then a computer plays the seat (`aiActFor`)	Live	The match always finishes; the winner is still decided by cash
8	Local quit	Leaving a local game	Any time	No result; only `quit_day` telemetry
9	Daily challenge score	`server/meta.ts`: board sorted by `cash`, then `sold`, then submit time; the server replays your moves to verify	On submit	Your "win" against the 3 rivals is separate from your leaderboard rank
10	League stage cleared	`recordLeague()`: `ranking(g)[0] === me` and you're on that stage	Results screen	Lose = replay the same stage (new shuffle). Clearing stage 6 unlocks the League Champion achievement and the Royal colour
11	Achievements (optional)	`checkAchievements()` in `src/ui/progress.ts`	Results screen	`rich` ($200) reached by 0–1.8% of computer players and effectively impossible in 4-player Hard games (best seen $185). `comeback` needs being last on day 5

Order inside a day (it matters for every outcome above): morning (event revealed; delivery added) → buying in turn order → customers arrive → everyone seals prices → sales resolved customer by customer (yesterday's waiting customers first, then today's) → evening upgrades (not on the last day) → night: inspector fine, then everything ages, then the next day starts.

Conflicting or unreachable outcomes found

- On the **final day**, Health Inspector and Heatwave's extra ageing do nothing (they act at night, and there is no last night). Priority can't be played (no evening). These cards still appear as the final day's event.
- The **seat-order tie-break** favours seat 0 (you in solo, daily and league games, the room host online).
- `rich` ($200) is practically unreachable in 4-player games.

C. Balance values (grouped by system, with code references)

Legend for "Type": **C** = constant in `src/engine/data.ts` (one place, easy to tune); **H** = hard-coded number inside logic; **F** = formula; **T** = also written in player-facing text (must be edited too). ⚠ marks a value that needs edits in several places. ★ marks the highest-impact controls.

C1. Run length, start and end

Value	Current	Changes	Affects	Where	Type
Game length ⚠	8 / 10 / 14 days (default 10); league 8 or 10; daily 10	Chosen at set-up	Everything; final cash roughly scales with days (3-player Hard: $61 / $81 / $127)	`App.tsx` `setDays`, options list; `server/room.ts` allowed list; `src/net/protocol.ts` `RoomConfig`, `DEFAULT_CONFIG`; `progress.ts` `LEAGUE`; `daily.ts`	H ×5, T (`RULES.md`, `GAME-BRIEF.md`)
Starting cash ⚠	$10	Nest Egg makes it $16	Day 1–2 buying power, upgrade timing	`START_CASH` (data.ts); `applyTrait()` `START_CASH + 6`	C, H, T (`RULES.md`, `HowTo.tsx` "$10", trait text "$16 instead of $10")
Winner rule ★	most cash → most sold → seat order	Never	The goal	`ranking()` game.ts	H
Final-day stock value	$0	Never	End-of-game buying	implicit	—

C2. Supply: the farmers market

Value	Current	Changes	Affects	Where	Type
Delivery size ★	each card fruit: `max(1, round(base × scale))`; fruits not on the card: base 2	Daily	How much fruit exists; price pressure	`startDay()`; `FARMER_CARDS` (20 cards, 3 fruits each, 4–6 per fruit)	C + F
Supply scale ★	`buyCapacity × (1 + 0.5 ×`	Rises over the game;	≈100% of total buy capacity	`supplyScale()` game.ts; 17.15 = `FARMER_AVG_QTY`	F with H 1, 0.5

Value	Current	Changes	Affects	Where	Type
	`progress) ÷ 17.15`	rises when anyone builds a Handcart or has Big Basket	on day 1 → 150% on the last day		
Unripe deliveries	9 of 20 farmer cards; mango (2 nights) or watermelon (1 night); only if "Unripe fruit" is on	Per card	The best margins in the game (unripe mango $2 → sells as fresh)	`FARMER_CARDS[].unripe`, `FRUIT.*.unripeSlots`	C
Market fruit ages	1 step a night (+1 Heatwave)	Nightly	Cheap old fruit available every day	`nightfall()`	H
Buy cost by freshness	blueberry 2/1/1 (fresh/okay/not so), unripe 1; watermelon 3/2/1, 1; apple 4/3/1, 1; mango 4/2/1, unripe 2; floor $1	Events ±$1	Margins	`FRUIT.*.cost`, `marketCost()`	C, T (`RULES.md` table)
Buy limit ⚠	4 a day; 5 with Handcart; +1 Big Basket	Upgrade or trait	Throughput, **and everyone's supply and demand** (see finding E7)	`BASE_BUY`, `CART_BUY`, `buyLimit()`	C, T (upgrade text, `RULES.md`, `HowTo.tsx`)
Shelf size ⚠	8; 14 with Bigger Shelf	Upgrade	Rarely binding	`BASE_SHELF`, `BIG_SHELF`	C, T

C3. Demand: customers

Value	Current	Changes	Affects	Where	Type
Demand target ★	`buyCapacity × (0.40 + 0.45 × progress)` units/day, split over (1 + walk-ins) customers	Daily	How much can be sold at all; price wars	`demandScale()` game.ts	F with H 0.4, 0.45
Card size	24 customers, 3–6 fruit each (avg 4.375), 1–3 fruit types	Deck reshuffles when empty	Order sizes	`RAW_CUSTOMERS`; `CARD_AVG_QTY`	C
Order rounding	`max(1, round(qty × scale))`	Per draw	Small early games get more demand than the target (every want is at least 1)	`drawCustomer()`	H

Value	Current	Changes	Affects	Where	Type
Walk-ins ⚠	1 a day; 2 from the halfway day (day 6 of 10, day 5 of 8, day 8 of 14); events ±1	At halfway	More, smaller orders → more showdowns in the second half	`baseWalkIns()` (threshold 0.5)	H
Forecast customer	1, known a day early	Daily	Planning	`customersArrive()`	H
Waiting customers	Unserved new customers wait 1 more day and are served first; then leave (no penalty)	Daily	Carry-over demand	`finishReveal()`, `resolve()`	H
Customer top price ★	card `max` + fruit shift (apple −1, mango +2), + event; weighted averages: blueberry $6.5, watermelon $7.4, apple $7.7, mango $9.1	Events	Revenue ceiling	`RAW_CUSTOMERS[].max`, `PRICE_SHIFT`, `effectiveMax()`	C
Event extra customers	Influencer, Gym Club, Hotel Chef	With their event	One-day demand spike	`EVENT_CARDS[].extra`	C (not price-shifted, unlike normal cards)

C4. Fruit life and selling

Value	Current	Changes	Affects	Where	Type
Life tracks ★	blueberry 3 slots (not so, okay, fresh); watermelon 5 (+1 unripe); apple 7; mango 4 (+2 unripe)	Fruit moves one slot a night	Spoilage risk	`FRUIT.*.ripe`, `unripeSlots`	C, T
Freshness discount ★	blueberry −4/−2/0; watermelon −3/−1/−1/0/0; apple −2/−1/−1/−1/0/0/0; mango −3/−1/0/0	With age	Top price allowed per fruit	`FRUIT.*.drop`, `capAt()`	C, T
Price floor	$1; Minimum Price event: watermelon 5, apple 5, mango 6, blueberry 4	Event	Price wars	`minPrice()`	H, C
Who sells first	lowest price → fresher fruit → tie edge (Favoured Trade card, or Haggler v richer shop) → buying order (poorest first)	Every sale	All income	`resolve()` sort	H
Customers pay	your sealed price (not their top price)	—	Undercutting	`resolve()`	H
Signboard tip ⚠	+$1 per fruit sold	Upgrade	Income	`resolve()`; display repeated in `animator.tsx`	H ×2, T

C5. Turn order and catch-up

Value	Current	Where	Type
Buying order ★	Day 1 random; then **poorest first** (ties round the table); Priority card jumps the queue	`startDay()`	H
Tie-break by buying order	Poorer shop wins full price-and-freshness ties	`resolve()`	H

C6. Events (13 cards, one a day from day 2, tomorrow's shown a day early; reshuffled when used)

Event	Effect	Measured income vs same-day average (2p / 4p)
Festival	+$3 all prices, +1 walk-in	+11.1 / +8.1
Superfood	+$3 blueberry, Gym Club wants 4 at $8	+10.6 / +4.5
Mango Craze	+$3 mango, Influencer wants 3 at $11	+10.4 / +4.2 (largest gap between shops: 25–35)
Hotel Order	Hotel Chef orders 7	+6.6 / +1.0
Bad Harvest	delivery ×0.5, +$2 prices	+1.7 / +2.6
Heatwave	+$4 watermelon, everything ages 2 steps tonight	−1.1 / −0.1
Minimum Price	price floors	−2.3 / +1.1
Good Harvest	+3 each fruit, costs −$1	−3.6 / −2.0
Health Inspector	$2 fine per not-so fruit at night (capped by cash)	−3.9 / −2.0
Price Hike	costs +$1	−4.0 / −2.3
Lorry Breakdown	no delivery	−6.0 / −2.9
Rainy Day	no walk-in, costs −$1	−8.0 / −4.8
Slow Season	−$2 all prices, no walk-in	**−12.0** / −7.3

Where: `EVENT_CARDS` (data.ts, C), applied in `startDay()`, `marketCost()`, `priceMod()`, `customersArrive()`, `nightfall()`; text repeated in `RULES.md`.

C7. Upgrades (each evening each shop is offered 3 it hasn't built; it may build any of those it can afford)

Upgrade	Cost	Effect (code)	Measured: AI that never builds it (3p, control 37% win)	Payback estimate
Signboard ★	$8	+$1 per fruit sold	**26.3%** (biggest loss)	~2.5 days (2p, 3.2 sales/day) to ~3.2 days (4p, 2.5/day)

Upgrade	Cost	Effect (code)	Measured: AI that never builds it (3p, control 37% win)	Payback estimate
Cool Room	$7	ages only on even-numbered nights (`s.day % 2 === 1` skips); immune to Heatwave	32.3%	Indirect: keeps fruit fresh (higher top price, wins freshness ties)
Bigger Shelf	$3	8 → 14 shelf	29.7%	Only when hoarding
Handcart	$6	4 → 5 buys a day	32.0%	Low: shops are limited by customers, not by buying
Shop Bell	$4	2 favour cards now; hand 3 → 4	34.3%	~2 cards × ~$2–3 value
Juice Press	$4	up to 3 spoiled fruit a night → $2 each	**37.3% (no loss at all)**	Never for careful play: computers spoil 0.1–0.7 fruit per game

Rushing any single upgrade on the first evening loses (13–26% win). Building all six loses badly (–$87 versus rivals). No upgrades at all: 21% (vs 37% for the computer's normal choices). Where: `UPGRADES` (C, T), effects in `buyLimit`, `shelfSize`, `handLimit`, `resolve`, `ageShop`, `buyUpgrade` (H). The Juice numbers (3, $2) are hard-coded in `ageShop()`.

C8. Merchant traits (pick 1 of 3 on day 1; computers take a random one)

Trait	Effect	Where	Measured win rate (3p, fair 33.3%)
Bargain Hunter	first fruit each day $1 cheaper (min $1)	`buyCost()`	**42.0%** (+$7.7 vs rivals)
Charmer	3 favour cards at start; hand +1	`applyTrait()`, `handLimit()`	39.0%
Nest Egg	start with $16	`applyTrait()`	38.0%
Local Favourite	+$3 per customer you complete alone (no daily limit)	`resolve()`	35.3%
Haggler	wins tied prices against richer shops	`tieEdge()`	30.7%
Big Basket	+1 buy a day	`buyLimit()`	**29.0%**

C9. Favour cards (12-card deck; earned when you alone complete a customer; at most 1 per shop per day; hand limit 3)

Free Cash +$3 · Free Fruit (any market fruit, free, ignores the buy limit) · Priority (buy first tomorrow) · Good Riddance (sell all ripe stock back for $1 each; it returns to the market) · Favoured Trade (win tied prices today) · Fridge (all your ripe fruit one step fresher). Where: `FAVOURS`, `FAVOUR_DECK` (C), `playFavour()` (H: the $3, the $1), `resolve()` (earning rule).

C10. Computer difficulty

Knob	Easy	Normal	Hard	Where
Buy if value − cost ≥	0.3 (+ random ±2)	0.8	0.3	`pickPurchase()`
Pricing	rival's cap minus 0–1 (random)	best response, then −0 or −1	best response, then −1; reads tomorrow's event	`planBids()`, `HARD` constant
Keeps unsold value	—	30%	10%	`planBids()`
Upgrades	skips 60% of nights	list order with day limits, keeps ≥ $6	same	`pickUpgrade()`
Measured head to head (2p)	—	beats Easy 72%	beats Normal 84%, Easy 91%	`npm run balance`

C11. Modes

Mode	Set-up	Where
Daily	You vs Normal, **Hard**, Normal; 10 days; same decks for everyone	`daily.ts`
League	1: 1 Easy (8 days) · 2: 2 Easy (8) · 3: 1 Normal · 4: 2 Normal · 5: Hard + Normal · 6: 3 Hard (all 10 days)	`LEAGUE` in `progress.ts`
Online	2–4 people, optional computers, 8/10/14 days, toggles	`server/room.ts`

Versioning rule: any change to C1–C10 that alters results needs `RULES_VERSION` (`src/net/protocol.ts`) raised. That **discards games saved in progress** (`store.ts`), keeps old and new apps out of each other's online rooms, and separates daily scores. League and mode set-ups (C11) can change without it.

D. Progression and pacing map

Units: money in $, time in game days. Computer-only measurements, 10-day games unless stated.

Early game: days 1–3 ("set-up")

- Day 1 is thin: one forecast customer plus one walk-in, orders scaled to 40% of buying capacity. Average income per shop on day 1: **$6.3 (2p), $4.9 (4p)**, against $10 starting cash. Most shops spend nearly all their cash on day 1.
- Supply is about **2.5× new demand on day 1** (formula: supply 100% vs demand 40% of capacity). Unbought market fruit ages and becomes cheap, so not-so-fresh fruit at $1 is almost always available.
- First upgrades: computers buy the Signboard on day ~2.6, Handcart ~2.5–3.1, Cool Room ~2.8, Bell ~2.8–3.4. The AI only builds when it keeps ≥ $6 afterwards, so the $8 Signboard needs $14 in hand.
- Traits make the biggest difference here: Nest Egg's +$6 is about one full day's income at this point.

Mid game: days 4–6

- Income climbs: 2p **22–27*aday* * *,4*p* * *13–16 a day**.
- Walk-ins double at the halfway day (day 6 of 10), so there are more but smaller orders, more showdowns and longer reveals.
- Juice Press is usually built here (day ~5) and pays nothing back for careful players.
- **Who's ahead on day 5 usually wins:** 66% (2p), 54% (3p), 47% (4p), against fair chances of 50/33/25%.

Late game: days 7–10

- Income plateaus: 2p **28–30*aday* * *,4*p* * *16–19 a day** (demand reaches 85% of capacity; supply 150%).
- Upgrades stop (the computer needs 3–6 days left for each). Last day: no evening; unsold fruit is worth $0.
- Computers still buy a full basket (~4) on the last day to sell the same day.

Whole run

Measure	2 players	3 players	4 players	Source
Average final cash (Hard mirror)	$131	$81	$56	balance, probes
Final cash range (10th–90th percentile)	$92–170	$49–117	$24–91	probes
Median winning margin	$27	$17	$15	balance
Close games (≤ $3)	8%	10%	12%	balance
Blowouts (≥ $20)	63%	45%	41%	balance
Lead changes per game	2.7	3.6	4.2	balance
Comeback from last on day 5	33%	19%	9%	balance
Sales at ≤ $2 ("price war")	6%	—	16%	balance
Sales near the customer's top price	51%	—	20%	balance
Fruit spoiled per shop per game	0.1	—	0.7	balance

Profit per fruit bought (3p): mango $2.59 > blueberry $2.21 > watermelon $1.85 > apple $1.58. In 4-player games every fruit falls to $1.1–1.8.

Real time per run: not measured. From the animation code, each showdown takes about 2.3–3.2 seconds (0.55× with "Fast"); people's thinking time dominates and isn't recorded. Assuming 30–90 seconds per day, a 10-day game takes about 5–15 minutes. Telemetry should measure this (section 6).

League difficulty curve (computer standing in for you, 300 games per stage):

Stage	Easy-level player	Normal-level player	Hard-level player
1 Village Square (1 Easy, 8 days)	47%	65%	87%
2 Harbour Steps (2 Easy, 8 days)	27%	67%	84%
3 Old Town (1 Normal)	29%	54%	84%
4 Spice Lane (2 Normal)	7%	38%	80%

Stage	Easy-level player	Normal-level player	Hard-level player
5 Grand Bazaar (Hard + Normal)	5%	**11%** (~9 tries)	54%
6 Royal Market (3 Hard, 4 players)	5%	**5%** (~20 tries)	28%
Daily challenge (Normal, Hard, Normal)	2%	**8%**	41%

Downtime, spikes and snowballing

- Downtime: day 1 (little to sell) and long showdown sequences in the late game.
- **Difficulty spike:** league stages 4 → 5 → 6 for an average player (38% → 11% → 5%).
- **Snowball:** the day-5 leader wins well above fair, and blowouts are common. The poorest-first buying rule does help: the old rotating order had 62% blowouts in 3-player games against 45% now. But being poor enough to buy first on 4 or more days still means winning only 15.5% of the time.
- **Unrecoverable:** a shop with less than $1 and no ripe fruit can't buy again without a lucky favour card.
- **Player count changes the game more than any setting:** 4-player games earn less than half what 2-player games earn.

E. Prioritised findings and proposed adjustments

Confirmed = shown by the code or by measurement. **Hypothesis** = needs testing or human playtest.

#	Finding	Evidence	Player impact	Where to change	Proposal and side effects
E1	**League cliff at stages 5–6** (confirmed)	Normal-level player: 38% → 11% → 5%	Average players stall for 10–20 tries; the "League Champion" goal feels out of reach	`LEAGUE` in `src/ui/progress.ts` (no rules change, no `RULES_VERSION` bump)	Add a step between (e.g. Hard + Normal + Normal), or make stage 6 three players. Test until each step loses no more than ~10–15 points for a Normal-level player. Side effect: easier champion achievement
E2	**4-player economy is squeezed** (confirmed)	Final cash $56 vs $131; 16% of sales at ≤ $2; only 20% near top price	More price wars and fewer good days; `rich` achievement out of reach; Royal Market feels grim	`demandScale()` and `supplyScale()` coefficients in `game.ts`; could depend on the number of shops	Hypothesis: lower supply growth (the 0.5) or raise demand (0.40/0.45) for 3–4 shops. Tune **together**: the gap between supply and demand drives prices. Side effect: changes daily scores (always 4 players), so needs `RULES_VERSION`

#	Finding	Evidence	Player impact	Where to change	Proposal and side effects
E3	**Snowballing and blowouts** (confirmed; how much is too much is a hypothesis)	Day-5 leader wins 54% (3p); blowouts 41–63%; close games 8–12%	Many games are decided by mid-game	`startDay()` order, `resolve()` tie rules, the event deck	Options to test one at a time: tie advantage for the poorest beyond full ties; smaller late-game event swings. Side effect: skill matters less
E4	**Juice Press is a dead choice** (confirmed)	AI without it wins as often (37.3% vs 37%); spoilage < 1 fruit per game	Wastes $4 and an evening for new players who pick it	`UPGRADES.juicer` and `ageShop()` (3 and $2 hard-coded)	Rework so it helps careful players too, e.g. also juices unsold fruit on the last day, or more per fruit. Needs testing so it doesn't become a farm (I tested "buy $1 dying fruit to juice": it **loses**, 17% win)
E5	**Trait spread 29–42%** (confirmed at 300 games; ±3 points noise)	Bargain Hunter 42%, Big Basket 29%, Haggler 30.7%	Trait pick quietly decides some games	`TRAITS` text + `buyCost()`, `buyLimit()`, `tieEdge()`	Haggler almost never fires because freshness breaks ties first (code order). Make it apply before freshness, or give it a small extra. Big Basket is weak because buying capacity isn't the limit: try +1 shelf space or a $1-off second fruit. Trim Bargain Hunter only if playtests agree. Test each trait **independently**
E6	**Handcart and Big Basket feed everyone** (confirmed in code)	`buyCapacity()` adds every shop's limit into demand and supply	Your Handcart also gives rivals more customers and fruit	`buyCapacity()`, `demandScale()`, `supplyScale()`	Decide if intended (the comment says it keeps the market tight). If not, base the scales on the shop count instead. Side effect: Handcart becomes stronger
E7	**Hidden seat tie-break and wrong text** (confirmed)	`ranking()` stable sort; Results text	0.3% of games; seat 0 wins dead heats	`ranking()`, `Results.tsx`	Add an explicit third rule (e.g. fewer fruit spoiled, or a shared win) and correct the text. Low effort
E8	**Final-day events that do nothing** (confirmed)	Inspector and Heatwave ageing act at night; no last night	A "dramatic" final event that changes nothing	`startDay()` event draw	Don't draw night-only events for the last day, or apply them at the final sales. Low effort
E9	**No way back from broke**	0.3–2.2% of computer players;	A player can sit out the rest	`canBuy()`, `startDay()`	Hypothesis: a tiny safety net (e.g. $1

#	Finding	Evidence	Player impact	Where to change	Proposal and side effects
	(confirmed, rare)	humans may be higher	of a game		morning credit when cash and ripe stock are both 0). Measure humans first (E-telemetry)
E10	**Slow Season and Rainy Day are big shared losses** (confirmed size; fun is a hypothesis)	−12/−8 a day in 2p	Can feel like a wasted day	`EVENT_CARDS`	Soften (e.g. Slow Season −1*instead of* −2) if playtesters dislike them
E11	**Daily challenge is hard for an average player** (confirmed for the computer stand-in)	Normal-level wins 8%; Hard rival in the line-up	Daily "wins" are rare, but leaderboard rank still rewards	`daily.ts` rival levels	Optional: Normal, Normal, Hard on alternate days. Changing it changes daily seeds' outcomes; do it at a date boundary
E12	**Rounding inflates small orders** (confirmed, minor)	`max(1, round(qty × scale))` at scale 0.46–0.7	2-player early days have slightly more demand than intended	`drawCustomer()`	Leave unless tuning E2
E13	**Numbers duplicated in text** (confirmed)	Section C ⚠ items	A balance change can leave the rules, how-to or upgrade text wrong	`RULES.md`, `HowTo.tsx`, `UPGRADES` / `TRAITS` text, `animator.tsx` tip	When tuning, update together; better: build those texts from the constants
—	Checked, **not** a problem	"Juice farming" (17% win, loses); "buy only the cheapest old fruit" (0% win); last-day buying (sold the same day, warned "Unsold = $0")	—	—	No change

F. Missing information and playtest questions

Not available from the code or simulations

- How people actually price (do they undercut by $1 like the Hard computer, or price high?).
- Real play time per day and per game, and where people stop (only `quit_day` exists).
- Human win rates per difficulty, trait and mode; how often humans go broke.

- Your target experience: how often should an average player win against Normal? Is a 4-player game meant to feel tighter? Should the League be finishable by most players?

Playtest questions (fit the DEC-016 kit)

1. After the first game: did you understand why you won or lost a sale? (freshness and ties)
2. Did any day feel pointless (day 1, Slow Season, a broke stall)?
3. Which upgrade did you build first and why? Did the Juice Press feel useful?
4. Which trait did you pick, and did it feel strong?
5. Did the game feel decided before the end? From which day?
6. How long did it take, and was that the right length?

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6. Tuning and measurement plan

Order of testing (expected impact first)

1. **League steps (E1)**: no rules change, quick to simulate and ship.
2. **Player-count economy (E2)**: supply and demand coefficients, tuned **together** (their ratio matters), checked in 2, 3 and 4 players.
3. **Dead or weak choices (E4, E5, E6)**: Juice Press, Haggler, Big Basket, each **independently**, one change per test run.
4. **Snowballing (E3)** after the economy, because price wars and margins change it.
5. **Small fixes (E7, E8)**: tie-break and final-day events. Safe; ship with the next rules change.
6. **Difficulty and daily line-up (E11)** last, once the economy is settled.

Tune together: demand 0.40/0.45 with supply 1.0/0.5; customer top prices with fruit costs (margin); Signboard tip with upgrade cost. **Tune alone:** each trait, each upgrade, each event, the league line-up.

Telemetry to add (anonymous counters, in line with `docs/ANALYTICS.md`; adding events changes data handling, so it needs its own approved plan):

- Game result by mode, number of players, length and rival levels (win/lose, final rank, winning margin bucket).
- Trait picked × win; upgrades built (which, which day) × win.
- Fruit spoiled and unsold at the end (buckets); fruit sold; share of sales at the customer's top price.
- Rank on day 5 × final rank (snowball and comeback).
- Game duration in minutes (bucket) and minutes per day.
- Times a player was broke (cash < $1, no ripe stock).
- League: attempts per stage and clears; daily: rank against the three rivals.

Small test matrix | Axis | Values | |---|---| | Players | 2, 3, 4 | | Rivals | all Easy, all Normal, mixed, all Hard | | Length | 8, 10, 14 days | | Strategy for the test seat | computer default; no upgrades; Signboard first; each trait fixed; cautious buyer (only forecast demand) | | Toggles | events on/off; unripe on/off |

Run each cell 300–400 games with `npm run balance` (plus the probes used here), compare against this report's baseline, then confirm the top 2–3 changes with 3–5 human playtesters before any release.

Appendix: probes used for this report (read-only, not committed)

- `curves` : supply and demand per day from `demandScale()` / `supplyScale()` for 2–4 players.
- `probes` : juice farming; broke shops; ties; last-day buying; final cash spread.
- `ladder` : league stage and daily win rates for Easy-, Normal- and Hard-level stand-ins (300 games each).
- `bottom` : "buy only the oldest, cheapest fruit" strategy (400 games).

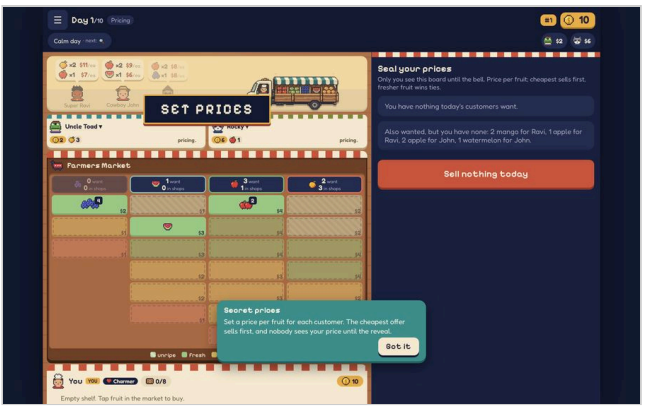
REP-026 · PLAN-010 built: the farmer's lorry

Game	Fruit Market
Report date	2026-10-27 07:54 (Asia/Kuala_Lumpur)
Package	PLAN-010, approved ("approve")
Build	commit bc2e760, merged to `main` for the preview
Delivery status	Verified

What changed

- Each morning your Gemini lorry drives into the street and parks; the customers step aside.
- The delivered fruit flies from the lorry into its own slot on the market board.
- When buying ends and pricing starts, the lorry drives off. On "Lorry Breakdown" days it doesn't come.
- It's quick (about 1 second to arrive) and never blocks play. Tapping speeds the unloading; reduced-motion turns the driving off.





Validation

Check	Result
Quick checks (73 tests, lint, types, build)	PASS
Full browser tests incl. speed budget and secret-price leak check 0	PASS
Phone and desktop screenshots, no sideways scroll or errors	PASS
Rules, saves, online play untouched	PASS

Next

All of the new look (PLAN-009, PLAN-010, PLAN-011) and the Patches rename are on the preview. When you're happy, say **PREPARE RELEASE** and I'll set up 1.2.0 for you to publish.

REP-027 · Balance redesign: fruit identity, strategy depth and pacing (proposal, no rule change)

Game	Fruit Market (rules version 2, `main` at 66bc299)
Report date	2026-10-27 09:04 (Asia/Kuala_Lumpur)
Type	Analysis, experiments and a patch plan. **Production rules unchanged.** All variants ran in a separate lab copy of the engine
Builds on	REP-025 (balance and pacing audit)
Proposed plan	PLAN-012 "Market rebalance: tighter town, fair Handcart, better apples" (awaiting approval)
Approval status	Awaiting approval (PLAN-012). This report approves nothing

How to read the numbers. Three kinds of statement appear below, and they are labelled:

- **[Code]**: a fact read from the source code.
- **[Sim]**: a measured result from computer players. The ± figures are 95% intervals. With 600 games per cell, a win rate is good to about ±4 points.
- **[Hyp]**: a hypothesis about how people will feel. Only the playtest (section 7) can confirm it.

Computer players are not people. Win rates show the shape of the game, not human results.

---

1. Diagnosis: what the strategic game is today

The core loop (code). Each day you buy up to 4 fruits (5 with a Handcart), then set sealed prices per customer. The cheapest offer sells first, and freshness caps your price and breaks ties. The goal is the most cash at the end.

What decides games (sim, current rules, 3 players, 600 games per plan on held-out seeds):

1. **Two plans dominate.**
   - **Undercut**: always price $1 under the expected rival. It wins 43% (fair share is 33%).
   - **Salvage**: dump old stock cheaply and keep buying fresh. It wins 51%.
   - In 4 players it is worse: 45% and 54% (fair share 25%).
   - **Premium pricing** (hold out for top price) wins only 13% in 3 players and 6% in 4 players.
   - In short, cutting the price is almost always right. That is the opposite of an interesting sealed-bid game.
2. **The market is loose early, so price wars pay.**
   - Customer demand starts at 40% of what the shops can buy, while farmers bring 100%.
   - Shops are therefore fighting over too few customers with too much fruit.
   - 11% of 3-player sales and 16% of 4-player sales happen at $2 or less ("price-war share").

3. **The Handcart (and Big Basket) grows the market for everyone.** [Code] Demand and deliveries are sized from the sum of every shop's buy limit (`buyCapacity()` in `src/engine/game.ts`). So when one shop buys a Handcart, the town gets more customers and more fruit for all shops. That rewards whoever floods the market and blurs what the upgrade is for.
4. **Apples are a trap.**
   - Apples cost $4 fresh but are priced $1 below other fruit (`PRICE_SHIFT` in `data.ts`).
   - [Sim] Net profit per apple bought is $1.64, the lowest of the four fruits (mango is $2.58).
   - 25% of okay-age apples are still unsold at the end.
   - The "safe, long-keeping" identity does not pay, because almost nothing is held long enough for keeping to matter.
5. **Fruit is barely held, so decay rarely bites.**
   - [Sim] The average fruit sells 0.6 days after purchase, and 87–93% of fresh fruit sells the same or next day.
   - The only real "hold" decision is green mango (held 2.6 days, net $3.33 each) and green watermelon.
   - Spoilage is 1.2 fruits per shop per game in 3 players.
6. **Early leaders snowball.**
   - [Sim] The mid-game leader wins 57% in 3 players (fair is 33%).
   - Only 15% of 3-player games and 8% of 4-player games are won by whoever was last at mid-game.
   - 45% of games are blowouts.
7. **4 players is squeezed.** [Sim] Average final cash is $130 in 2 players, $80 in 3 and $56 in 4. Four-player shops barely grow, and price wars are most common there.

Summary. The tension the game wants is "freshness and timing against price". Today it collapses into "price lowest, buy fresh, repeat". Fruit identities exist on paper, but only mango's shows up in play.

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2. Fruit identity and decay

2a. Before and after (exact values)

Decay is unchanged in the recommendation. The reason is in 2c.

Fruit	Track, oldest → freshest (price drop $)	Cost: fresh / okay / not so / unripe (now → proposed)	Price shift on cards	Unripe nights	Intended identity
🫐 Blueberry	−4 · −2 · 0 (unchanged)	2/1/1/– (unchanged)	0	none	Cheap, fast, sell today
🍉 Watermelon	−3 · −1 · −1 · 0 · 0 (unchanged)	3/2/1/1 (unchanged)	0	1	Steady middle; small ripening bet
🍎 Apple	−2 · −1 · −1 · −1 · 0 · 0 · 0 (unchanged)	**4/3/1/1 → 3/2/1/1**	−1 (unchanged)	none	The keeper: safe margin any day, sells late

Fruit	Track, oldest → freshest (price drop $)	Cost: fresh / okay / not so / unripe (now → proposed)	Price shift on cards	Unripe nights	Intended identity
🥭 Mango	−3 · −1 · 0 · 0 (unchanged)	4/2/1/2 (unchanged)	+2	2	Premium; buy green and wait

Measured fruit economics (sim, 3 players, held-out seeds; "net" = sale price minus cost, per fruit bought):

Fruit	Net now	Net with package	Share of fruit bought, now → package	Note
Blueberry	$2.14	$3.43	26% → 24%	Still the quick-profit fruit
Watermelon	$1.90	$2.55	22% → 24%	Green watermelon nets $3.57
Apple	**$1.64 (last)**	**$2.69**	22% → 26%	Moves from trap to solid choice
Mango	$2.58	$4.26	30% → 25%	Green mango nets $4.07, held 2.9 days

All fruits gain under the package because the tighter market raises prices (section 3). What matters is the order: apple goes from clearly worst to level with watermelon.

2b. Specific investigations

Question	Test	Result	Verdict
Is apple too expensive?	Apple cost 3/2/1 (A1); apple price +$1 (A2); both (A3)	A1 and A3 both lift apple to mid-table. A3 adds cash and widens the Hard-vs-Normal gap slightly. A1 is the smaller change	**A1 recommended**
Should apples keep value even longer?	Apple drop −1 · −1 · 0…	No measurable change to any plan (all within ±3 points)	Rejected: no effect
Is blueberry too punishing?	Blueberry drop −3 · −1 · 0	No measurable change	Rejected: no effect
Is green mango too strong?	Green mango $2 → $3	Premium rises (27%) but Reserve becomes too strong (45%). The Hard-vs-Normal gap shrinks to 57% (3 players), so skill matters less	Optional experiment, not recommended now
Should unripe fruit stop arriving when it can't ripen before the end?	Unripe cutoff	No measurable change	Rejected
Does unsold fruit at the end need a salvage value?	$1 per fruit at the end; juice at the end	No measurable change	Rejected

2c. Why decay is not the lever

[Sim] Shops sell almost everything within a day, so changing decay curves moves nothing. Decay only matters if holding is a real choice. Holding becomes a real choice when the market is tight and

has gaps: some days your fruit is scarce and worth keeping, other days it is everywhere.

The package creates that through market size, not through faster rot. That honours "do not simply increase decay everywhere or force a spoilage quota". Spoilage actually falls (1.2 → 0.3 per shop) because shops buy more selectively.

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3. A deeper strategy layer

3a. The plans tested

Each plan is the Hard computer with one strategic lean layered on top (lab "styles").

Plan	Lean
Undercut	Always price $1 under the best rival guess
Salvage	Dump old stock cheaply, restock fresh
Premium	Keep prices near the top; accept unsold fruit
Reserve	Keep a cash cushion; buy only clear profits
Ripen	Prefer green mango and watermelon
Adaptive	Learn each rival's discount from past reveals
Turnover	Buy the cheapest stock and sell it fast
Event	Lean hard into event cards (festival, heatwave and so on)

3b. Matchup table: one plan against two (or three) standard Hard computers

Held-out seeds 100001–100600 (600 games per cell, about ±4 points). Fair share is 33% in 3 players and 25% in 4 players.

Plan	3p now	3p package	4p now	4p package
Undercut	**43**	25	**45**	21
Salvage	**51**	32	**54**	29
Premium	13	25	6	16
Reserve	30	36	–	–
Ripen	27	31	26	23
Adaptive (reads rivals)	36	34	–	–
Turnover	18	21	–	–
Event	32	30	–	–
Spread (best – worst)	**38 points**	**15 points**	**48 points**	**13 points**

Four different plans at one table (4 players, fair share 25%):

	Turnover	Reserve	Ripen	Event
Now	17	19	29	34
Package	16	33	32	20

Reading this. [Sim] Under the package no plan is a must-pick, and several different plans sit within a few points of fair. Turnover stays weakest. [Hyp] That is acceptable, because it is the "no thinking" plan. Reading rivals (Adaptive) stays strong, which keeps the sealed-price duel rewarding.

3c. Sealed-price analysis

- [Code] The cheapest offer sells first. Ties go to the fresher fruit, then Favoured Trade or Haggler, then buying order.
- [Sim] Price-war share (sales at $2 or less) falls from 11.4% to 7.0% in 3 players and from 16.3% to 7.6% in 4 players.
- [Sim] Sales within $1 of the customer's top price rise from 30% to 55%.
- **Meaning.** With fewer spare fruits around, the question changes from "how low must I go?" to "will they dare to go high, and is my fruit fresh enough to win a tie?". [Hyp] That is the more interesting guess.
- **Adaptive still wins 34%.** Reading rivals still beats autopilot.

3d. Handcart and Big Basket: the shared-market effect

- [Code] Today, any shop's Handcart or Big Basket raises everyone's demand and supply.
- **Proposed:** size the town from the base 4 fruits per shop, so upgrades help only their owner.
- [Sim] This one change does most of the work against undercutting (P1 test: Undercut 32 → 24, Reserve 32 → 39, comeback from last 23% → 28%).
- [Sim] Handcart is still bought in 99% of games, so it keeps its value.
- **Trade-off:** average cash falls about 12%. The demand start is raised (0.55 → 0.60) to give some back.

3e. Traits (supporting evidence; changes optional)

2,000 games per player count, current rules (development seeds) → package without the 0.60 demand start (held-out seeds).

Trait	3p now	3p package	4p now	4p package
Haggler	28.6	28.6	24.0	22.9
Big Basket	29.6	33.3	22.4	23.2
Bargain Hunter	38.4	36.8	29.9	29.5
Charmer	36.1	35.3	26.1	27.4
Local Favourite	30.0	31.5	18.4	20.5
Nest Egg	37.4	34.6	29.2	26.3
Spread	10 points	8 points	11.5 points	9 points

Optional trait tweak (Haggler wins ties before freshness; Local Favourite tip $3 → $4; Nest Egg $16 → $15):

- 3 players: spread 6 points (30.5–36.2).
- 4 players: 21.4–29.4.
- A small gain, kept as an optional experiment.
- Bargain Hunter needing a $3+ fruit and Big Basket also adding 3 shelf spaces: no measurable change.

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4. Pacing

4a. Standard game (10 days), held-out seeds

Measure	3p now	3p package	4p now	4p package
Average final cash	$80	$105	$56	$97
Day 1 income	$5.9	$8.8	$5.0	$9.2
Blowouts (winner far ahead)	45.5%	40.0%	42.5%	30.0%
Mid-game leader wins	56.8%	38.7%	47.8%	34.8%
Last at mid-game but wins	15.3%	28.8%	8.2%	18.3%
Lead changes per game	3.6	4.3	4.3	4.7
Close games	10.8%	15.2%	14.5%	15.8%

In 2 players, final cash goes from $130 to $146. A 4-player game is no longer the poor relation ($97 against $56); the 15% extra 4-player demand is what fixes this.

4b. Quick (8 days) and Long (14 days), 3 players

"Now" uses development seeds; "package" uses held-out seeds.

Measure	8d now	8d package	14d now	14d package
Undercut	45	29	40	18
Salvage	52	39	43	23
Premium	10.5	21	16.5	35
Reserve	–	34	–	41
4-plan table T/R/Ri/E	22/13/26/38	20/30/24/26	15/25/36/24	**10/45/35/10**
Comeback from last	19%	28%	17%	30%

- **Quick (8 days) becomes the most balanced length.**
- **Long (14 days) swings towards patient plans** (Reserve and Ripen); Turnover and Event fall to 10%.
- [Hyp] Patience winning a long game may feel right, but it is a watch item. Long-mode tuning is listed as an optional follow-up rather than mixed into this package.

4c. Provisional success criteria (computer-only, 3 and 4 players, held-out seeds)

#	Criterion	Target	Package result
S1	Against standard Hard computers, no single plan wins more than 1.25 × fair share (42% in 3p, 31% in 4p)	≤ 42% / ≤ 31%	36% / 29%: **PASS**
S2	At least 4 of 8 plans within ±7 points of fair (3p)	≥ 4	5: **PASS**
S3	Hard beats Normal head to head (skill still matters)	≥ 65%	66%: **PASS (narrow)**
S4	Comeback from last at mid-game	≥ 20% (3p), ≥ 15% (4p)	28.8% / 18.3%: **PASS**
S5	4-player cash at least 60% of 2-player cash	≥ 60%	66%: **PASS**
S6	Price-war share	≤ 10%	7.0% / 7.6%: **PASS**
S7	Long mode: no plan above 42% in the 4-plan table	≤ 42%	45% (Reserve): **FAIL**, watch item
S8	Human enjoyment: clearer choices, "I could have won" feeling	Playtest	**NOT RUN** (needs section 7)

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5. Test method and evidence

- **Engine copy with switches.**
  - The lab is a copy of the rules engine with switches for each proposed change.
  - With every switch off it reproduces production exactly (checked on 450 games, identical results).
  - The production code was not touched.
- **Paired seeds and seat rotation.** Every variant replays the same seeds. Seats rotate each game, so every plan plays every seat equally often.
- **Many opponent types.**
  - The plans in 3a, plus Normal and Easy computers.
  - Crude single-rule bots, which all lose under both rule sets (0–3% wins), so they are not reported.
- **Sweeps.**
  - Eight runs (a baseline, a plan check and six sweeps): 813 test cells and 384,200 simulated games.
  - Round 1: demand × supply grid.
  - Rounds 2–3: market candidates, apple and mango costs, delivery forecast, end-of-game salvage.
  - Round 4: traits, Handcart basis, broke credit, earlier walk-ins, 4-player demand, Slow Season, unripe cutoff.
  - Round 5: decay variants, trait sample of 2,000 games, 8 and 14 days.
  - Round 6: final candidates on held-out seeds.
- **Validation.**

    - Seeds 1–600 were used to develop the package.
    - Seeds 100001–100600 (and 100001–102000 for traits) were kept back for the final check.
    - The current-rules numbers on held-out seeds match the development numbers within the error margin (Undercut 46 vs 43, Salvage 54 vs 51), so the results are not seed-specific.

- **Market League check** (300 games per stage, a computer standing in for you):

Stand-in skill	Village	Harbour	Old Town	Spice Lane	Grand Bazaar	Royal Market	Daily win
Normal, now	65	67	54	38	11	5	8
Normal, package	54	41	49	36	19	14	15
Hard, now	87	84	84	80	54	28	41
Hard, package	75	58	65	62	45	24	33

  The late-stage cliff softens, but early stages get harder because Easy and Normal computers do relatively better in a tight market. **Risk:** new players may find Harbour Steps harder. The league fix already suggested in REP-025 should follow this package, and the playtest checks it.

- **Rejected or no measurable effect:**
    - credit for broke shops;
    - earlier second walk-in;
    - Slow Season at −$1;
    - Bargain Hunter minimum cost;
    - Big Basket shelf bonus;
    - unripe cutoff;
    - end-of-game salvage or juice;
    - apple and blueberry decay changes.

- **Optional, not in the package:**
    - **Showing tomorrow's farmer delivery.** Undercut 28%, Turnover 14%, Hard-vs-Normal gap up to 72% in 2 players. It is an information feature with UI cost.
    - **Green mango at $3.**

- **Coverage gaps (NOT RUN):**
    - human play;
    - online rooms under new rules (needs verify:full after the patch);
    - Juice Press and Cool Room rework (Juice Press stays near-useless: 0.24 juiced per game);
    - favour-card balance;
    - every event card individually;
    - 2-player plan matchups (only mirror and skill tests);
    - Easy-level opponents in the plan tests.

- **Reproducibility.**
    - The lab, experiment lists and raw results are archived as `docs/workflow/reports/img/REP-027-lab.tar.gz`, with a README inside.

---

6. Recommendation: one cohesive package (PLAN-012)

6a. Essential changes

For each change: what decision it improves, why it improves that decision, and what it could break.

#	Change	Decision improved	Why	What could break
E1	**Tighter town.** Demand starts at 60% of base capacity instead of 40% (growth unchanged at +45%). Deliveries are 80% + 20% growth instead of 100% + 50%	"How low do I price?" becomes "how high dare I go?"	Fewer spare fruits, so cutting is no longer automatic; premium and reserve become real plans	Higher cash totals make the achievement thresholds easier; new players see fewer bargains at the farmers market
E2	**Town size uses the base 4 fruits per shop.** Handcart and Big Basket no longer grow the market for everyone	"Should I buy a Handcart?" is about my own capacity, not flooding the town	Removes a hidden subsidy to market-flooding play; the main reason undercutting fell	Cash about 12% lower than E1 alone (offset by E1's 0.60 start); Handcart is slightly less exciting
E3	**4-player towns have 15% more demand**	Whether to fight or share in big games	4-player shops were starved ($56 average)	If over-tuned, 4 players becomes easier than 3 (now $97 against $105: fine)
E4	**Apple cost 4/3/1 → 3/2/1**	"Which fruit fills my last slot?"	Apple becomes a real safe choice instead of a trap ($1.64 → $2.69 net per fruit)	Apple is now equal to watermelon on cost; identity rests on keeping a week, which the text must say
E5	**RULES_VERSION 2 → 3** and compatibility work	–	Required by the rules-change policy	Old saves and old clients are cut off (by design, with clear messages)

6b. Optional experiments (not in PLAN-012)

1. Trait tweak (Haggler before freshness, Favourite $4, Nest Egg $15).
2. Green mango $3.
3. Show tomorrow's delivery.
4. Long-mode tuning (demand growth for 14 days).
5. Juice Press and Cool Room rework.
6. League stage rebalance (follows REP-025).

6c. Exact patch plan

File	Change
`src/engine/data.ts`	Add `export const MARKET = { demandStart: 0.6, demandGrowth: 0.45, supplyStart: 0.8, supplyGrowth: 0.2, fourPlayerDemand: 1.15 }`. Apple `cost: { fresh: 3, okay: 2, notso: 1, unripe: 1 }`. Apple `theme`: "Keeps a week and barely loses value. Never the best price, never a loss." Update the header comment (apple no longer "costs the most")
`src/engine/game.ts` (lines 179–209)	New `marketCapacity(s) = s.players.length * BASE_BUY` used by `demandScale` and `supplyScale` instead of `buyCapacity`. `demandScale`: `marketCapacity(s) * (MARKET.demandStart + MARKET.demandGrowth * progress) * (n === 4 ? MARKET.fourPlayerDemand : 1)`. `supplyScale`: `marketCapacity(s) * (MARKET.supplyStart + MARKET.supplyGrowth * progress) / FARMER_AVG_QTY`. Remove the unused `buyCapacity` if nothing else uses it. Fix the stale comment at line 179 ("3 fruit each a day")
`src/engine/ai.ts`	No code change. The AI reads `demandScale`, `FRUIT` costs and prices at run time (lab parity confirmed). Re-measure Easy, Normal and Hard gaps after the patch
`src/net/protocol.ts`	`RULES_VERSION = 3`. Old clients get the existing "please update" message from `server/room.ts` (line 203) and `server/meta.ts` (line 70)
`src/ui/store.ts`	No code change: line 66 already discards a game in progress with "from an older version of the rules". Release notes say so
`server/worker.ts` (lines 107 and 110)	Daily board key `daily:\${date}` → `daily:\${date}:r\${RULES_VERSION}`, so release-day scores under old and new rules never share a board. Replays already use server rules (`verifyDaily`)
`src/engine/daily.ts`	No change (same seeds; the outcomes simply follow the new rules)
`RULES.md`	Fruit table (apple $3/$2/$1, new character line). "The town grows": orders and deliveries are sized for 4 fruits per shop, whatever upgrades or traits you have; 4-player towns have 15% more customers. Handcart row: "Buy up to 5 fruits a day instead of 4 (the town doesn't grow with it)." Fix the existing wrong price-cap example (it says okay apples lose $2 and not-so lose $4; the real figures are $1 and $2). Balance notes refreshed
`src/ui/HowTo.tsx`, `src/ui/tutorial.ts`	No apple cost or demand numbers found; re-read after the patch. Fruit lines come from `FRUIT.theme`
`src/engine/data.ts` `UPGRADES.cart.text` / `TRAITS.basket.text`	Add "(the town doesn't grow with it)" to the Handcart; Big Basket unchanged
`src/ui/progress.ts`	Keep "Fruit Tycoon" ($200) and "Market Day" ($40/day). They become easier; earned badges stay. Owner choice listed in section 8
`tests/`	New tests: demand and supply are the same with or without a Handcart; demand rises 15% at 4 players; apple cost values. Update any snapshots
Balance records	Re-run `npm run balance` and the plan checks on the patched engine; attach to the PLAN-012 verification report

Release: this would ship as a minor version (1.2.0 or 1.3.0) with player notes, for example "Markets are livelier: customers come sooner, and big undercuts pay less. Apples are cheaper. Games in progress from the old version can't be resumed."

7. Human playtest plan

- **Who.** 4–6 people. Mix newcomers and people who have played 3+ games. At least one session with 4 players on phones (pass-and-play or online).
- **Format.**
  - Each tester plays current rules (live game) and the package (a preview copy built from a branch), in alternating order.
  - Each set is 2 games: one Standard and one Quick or Long.
  - Testers are not told which version is which.
- **Questions after each game** (1–5 scale, plus one sentence):
  1. Did you have more than one sensible way to play?
  2. Did prices feel like a guessing duel or a race to the bottom?
  3. Did each fruit feel different? Which one did you avoid, and why?
  4. Did you feel you could still win in the last third?
  5. Was the game the right length?
  6. Was anything confusing?
- **Watch for:**
  - the price testers set relative to the top price;
  - whether anyone buys a Handcart and says why;
  - whether apples get bought;
  - league stage 2 difficulty for newcomers.
- **Pass mark.**
  - The package scores at least as well on questions 1, 2 and 4 for most testers.
  - No tester finds it more confusing.
  - Any FAIL goes back to the lab.
- **No telemetry.** Notes are written by hand. No new data collection, as instructed.

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8. Decisions for the owner

1. **Approve PLAN-012** (E1–E5 as above), or ask for changes.
2. Achievement thresholds: keep $200 and $40 (recommended, simplest), or raise them in the same plan.
3. Run the human playtest before or after building the preview copy (recommended: build the package on a branch preview, then playtest, then publish).

REP-028 · PLAN-012 built: market rebalance, optional tweaks and the cosy home screen

Game	Fruit Market, rules version 3 (branch `workflow/plan-012`)
Report date	2026-10-27 10:35 (Asia/Kuala_Lumpur)
Type	Implementation and verification of PLAN-012 (approved: "yes approvrd together with the optional changes and the new layout from the mockup")
Builds on	REP-027 (balance proposal)
Approval status	Approved (PLAN-012). Publishing the live game still needs its own approval

How to read the numbers.

- **[Code]** marks a fact about the code.
- **[Sim]** marks a computer-player result: 600 games per test, about ±4 points, on seeds 100001–100600, which were never used for tuning.
- **[Hyp]** marks a guess about people. Only the playtest can confirm it.

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1. What was built

Item	Change	Where
E1	Customers start at 60% of the town's buying power (was 40%), growing +45%. Farmers bring 80% + 20% growth (was 100% + 50%)	`MARKET` in `src/engine/data.ts`; `demandScale`, `supplyScale` in `src/engine/game.ts`
E2	The town is sized for 4 fruits per shop. Handcart and Big Basket help only their owner	`marketCapacity()` in `game.ts`; Handcart text
E3	4-shop towns have 15% more customers	`MARKET.fourPlayerDemand`
E4	Apples cost $3 / $2 / $1 (were $4 / $3 / $1)	`FRUIT.apple`
E5	`RULES_VERSION` 3; daily boards keyed by rules version (`daily:<date>:r3`), so release-day scores never mix	`src/net/protocol.ts`, `dailyBoardKey()` in `server/meta.ts`, `server/worker.ts`
O1	Haggler wins ties against richer shops before freshness is compared. Local Favourite tip $4 (was $3). Nest Egg starts with $15 (was $16)	`hagglerEdge()` in `game.ts`; `TRAITS`, `NEST_EGG_BONUS`, `LOCAL_FAVOURITE_TIP`
O2	Green mango costs $3 (was $2)	`FRUIT.mango`
O3	**Tomorrow's lorry**: tomorrow's farmer card is drawn a day early and shown to everyone as plenty or few per fruit. Computer players use it to hold scarce fruit	`nextFarmer` in `game.ts`; `NextLorry` in `src/ui/table.tsx` (market and morning card); `src/engine/ai.ts`
O5	**Juice Press**: each older fruit sold (okay or not so) earns $1 extra as juice; the nightly juicing of spoiling fruit stays. The sale pop-up includes it	`JUICE_TIP`; `resolve()` in `game.ts`; `src/ui/game/animator.tsx`

Item	Change	Where
O5	**Cool Room**: unchanged (see section 3)	–
O4	**Long (14-day) games**: unchanged (see section 3)	–
O6	**Market League** stages retuned (section 2d)	`LEAGUE` in `src/ui/progress.ts`
L1	**Home screen in the cosy pixel layout** from the mockup: harbour scene with bobbing boats, bunting, the "Fruit Market" sign with Foxy Rosa peeking out, fruit on the pier, then New game, Play a friend / Quick match, Daily challenge and Market League cards, and How to play / Trophies / Settings. Motion stops when reduced motion is on	`src/ui/App.tsx` , `src/ui/styles.css`
Text	Rules page (fruit table, the town grows, tomorrow's lorry, ties, Handcart, Juice Press, traits, balance notes). The wrong apple price-cap example is fixed. How-to mentions the lorry. Sale explanations name the Haggler rule	`RULES.md` , `src/ui/HowTo.tsx` , `src/engine/story.ts`

Saves and online play.

- A game in progress from the old version shows the existing "older version of the rules" notice and is cleared (`src/ui/store.ts` , unchanged).
- Old app versions are refused by online rooms and the daily board with the existing "please update" message.

---

2. Evidence

2a. The game code plays exactly like the tested package

[Sim] Production engine against the lab with the package switched on: **450 games** (2–4 players, 8/10/14 days, mixed levels). **0 differences in final cash.**

2b. Strategy matchups, 3 players (fair share 33%)

One plan against two Hard computers.

Plan	Before (rules 2)	After (rules 3)
Undercut	43	27
Dump old stock (Salvage)	51	36
Premium prices	13	28
Reserve (keep cash)	30	36
Ripen green fruit	27	34
Read rivals (Adaptive)	36	37
Turnover	18	23
Events	32	29
Spread	**38 points**	**14 points**

In 4 players (fair share 25%): Undercut 45 → 20, Salvage 54 → 28, Premium 6 → 20.

Four plans at one 4-player table:

	Turnover	Reserve	Ripen	Events
Before	17	19	29	34
After	16	32	33	20

2c. Pacing and the success criteria from REP-027

#	Criterion	Before	After	Result
S1	No plan above 42% (3p) / 31% (4p)	51% / 54%	37% / 28%	**PASS**
S2	At least 4 of 8 plans within ±7 of fair (3p)	3	7	**PASS**
S3	Hard beats Normal in 2 players at least 65% of the time	80%	**60%**	**FAIL** (see 3)
S4	Comeback from last at half-time, 3p at least 20% / 4p at least 15%	15.3% / 8.2%	25.8% / 19.2%	**PASS**
S5	4-player cash at least 60% of 2-player cash	43% ($56 / $130)	67% ($100 / $150)	**PASS**
S6	Price-war sales at most 10%	11.4% / 16.3%	5.8% / 6.7%	**PASS**
S7	Long mode: no plan above 42% at a 4-plan table	–	42% (Reserve); Reserve alone wins 50% against Hard	**FAIL** (see 3)
S8	Human enjoyment	–	–	**NOT RUN** (playtest)

Other measures, 3 players:

- Blowouts: 45.5% → 36.3%.
- Mid-game leader wins: 56.8% → 43.5%.
- Lead changes per game: 3.6 → 4.3.
- Average cash: 2 players $130 → $150, 3 players $80 → $108, 4 players $56 → $100.

Quick (8-day) games are the most even length. At the 4-plan table the plans score 23 / 28 / 25 / 24. Salvage reaches 40% against Hard computers, still under the 42% limit.

2d. Traits, upgrades and the league

Traits (2,000 games per player count):

- 3 players: from 29.9% (Charmer) to 37.7% (Bargain Hunter), an 8-point spread (was 10).
- 4 players: from 23.3% to 28.5%, a 5-point spread (was 11.5).

Upgrades. A Hard computer building only one upgrade, 3 players. Building nothing wins 10.8%.

Upgrade	Win %
Signboard	20.7
Cool Room	15.8
Handcart	14.7
Shop Bell	13.5
Juice Press	12.8 (was below nothing: 9.7)
Bigger Shelf	12.2

Every upgrade is now worth more than none.

Market League (300 games per stage, a computer standing in for you):

Stand-in	Village	Harbour	Old Town	Spice Lane	Grand Bazaar	Royal Market
Normal, rules 2 and old league	65	67	54	38	11	5
Normal, rules 3 and new league	64	64	52	41	33	20
Hard, rules 3 and new league	76	76	64	58	48	35

New stages:

1. Village Square: 1 Easy rival, 8 days.
2. Harbour Steps: 1 Easy rival, 8 days.
3. Old Town: 1 Normal rival.
4. Spice Lane: 2 Easy rivals.
5. Grand Bazaar: 2 Normal rivals.
6. Royal Market: 1 Hard and 2 Normal rivals.

Stages 3–6 are 10 days. The old final stage (three Hard rivals) was a wall.

2e. Checks

Check	Result	Evidence
`npm run verify` (records, lint, types, 81 unit tests, build)	PASS	exit 0
`npm run verify:full` (browser tests, online leak check, reconnect, quick match, four players)	PASS	exit 0; see section 5
Home screen at 390×844, 360×640 and 1280×800	PASS	no sideways scroll, no page errors; screenshots checked
Tomorrow's lorry visible on the morning card and under the market	PASS	screenshot at 390×844

---

3. What did not work, and what I did instead

- **Long games (O4).** I tried 10 different levers in the lab:
    - demand growth, demand start, supply start, supply growth;

    - when the second walk-in arrives;
    - the Cool Room price;
    - and others.

  None stopped patient plans (Reserve and Ripen) winning most 14-day games; Reserve stays at 39–52%. Long mode therefore keeps the standard rules. [Hyp] Patience winning a long game may feel right to players. The playtest should check it.

- **Cool Room (O5).**

    - A cheaper Cool Room ($4–5) adds little.
    - A Cool Room that stops ageing completely makes the Reserve plan too strong (45–49%).
    - Under the new market the current Cool Room is already worth buying (15.8%, against 10.8% for nothing), so it stays unchanged.

- **Juice Press (O5).** Four juice designs were tested. Juicing any leftover fruit was worse, because it turns good fruit into juice. The +$1 "juice tip" on older fruit was the only design that helped without boosting any plan.

- **Skill gap (S3).** Hard beats Normal 60% of the time (target 65%). Most of the drop comes from green mango at $3: without it the figure is about 70%, but Ripen and Turnover weaken.

    - Mango is on the preview as approved.
    - Changing it back to $2 is a one-line change. Decide after playtesting.
    - [Hyp] A smaller gap makes the game friendlier to new players and less punishing against Hard.

---

4. Risks

- Fewer spare fruits on Day 1: the market looks sparser. [Hyp] This may feel tight to new players.
- The $200 "Fruit Tycoon" and $40 "Market Day" achievements become easier (thresholds kept, as recommended).
- League progress is kept by stage number, so a player midway through keeps their stage under the new line-up.

5. Browser tests (`npm run verify:full`, exit 0)

- **Solo game:** results screen reached, no page errors.
- **Performance budget:** BUDGET OK.
- **Online match:**
    - 40 frames checked during pricing, 0 leaks of secret prices;
    - a refresh rejoined on Day 2;
    - both phones showed the same final cash;
    - rematch started.
- **Abandoned match:** a computer took over and the other player finished alone.

- **Quick match:** both players reached the game.
- **Four players:** lobby counts OK, a 5th person refused, all 4 in the game.
- **Reconnect stress:** 5 cycles, 0 failed (median 255 ms after a reload, 538 ms after a drop).
- **Old apps:** turned away by online rooms (`tests/room.test.ts`) and by the daily board (`tests/meta.test.ts`).

Lab files (engine copy, sweeps 7–11, parity check against the game code) are archived in `docs/workflow/reports/img/REP-028-lab.tar.gz` .

REP-029 · PLAN-013: full-screen pricing and event sheets, clearer rivals while pricing

Game	Fruit Market, rules version 3 (branch `workflow/plan-013`)
Report date	2026-10-27 12:25 (Asia/Kuala_Lumpur)
Type	UI only. No rule, balance, server or protocol change
Owner request	"edit the UI/UX so that 1. secret price and events pops up to fill the screen (instead of just the bottom half of the screen 2. rework the rivals indicator during secret price, now its not clear still"
Approval status	Approved (PLAN-013). Publishing needs its own approval

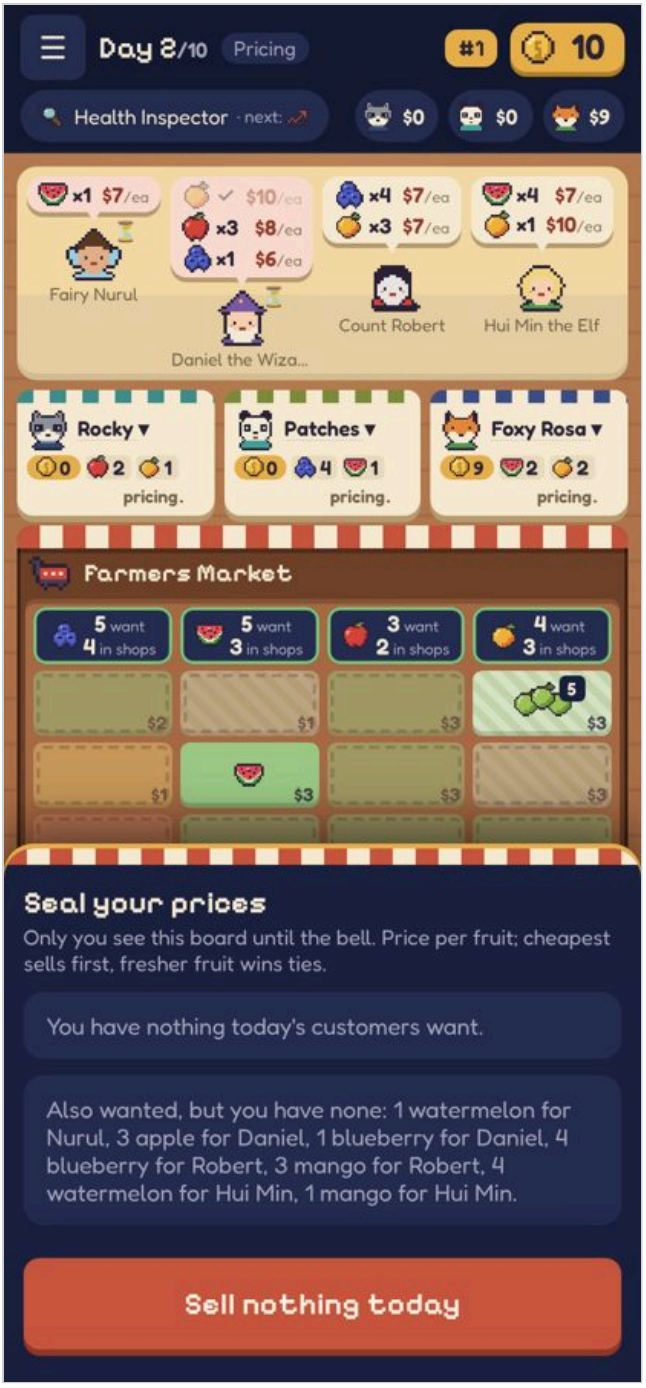
What changed

1. **Pricing fills the screen.** On phones the "Seal your prices" sheet now covers everything below the top bar. It was the bottom half before. Desktop keeps its side panel.
2. **Events fill the screen.**
    - The morning card (day, standing, today's event, tomorrow's event, tomorrow's lorry) opens as a full-screen sheet. "Open the market" stays at the bottom.
    - Tapping the event chip in the top bar opens the same full-screen view, with "Back to the market".
3. **"Your rivals" strip at the top of the pricing sheet.** Each rival gets one card showing:
    - their portrait and name;
    - a clear status: **sealed** (green, wax seal), **deciding** (amber, a person still choosing) or **next in line** (amber, a computer, which always prices after the people at the table);
    - the fruit they hold that today's customers want, or **sits out** if they have none.

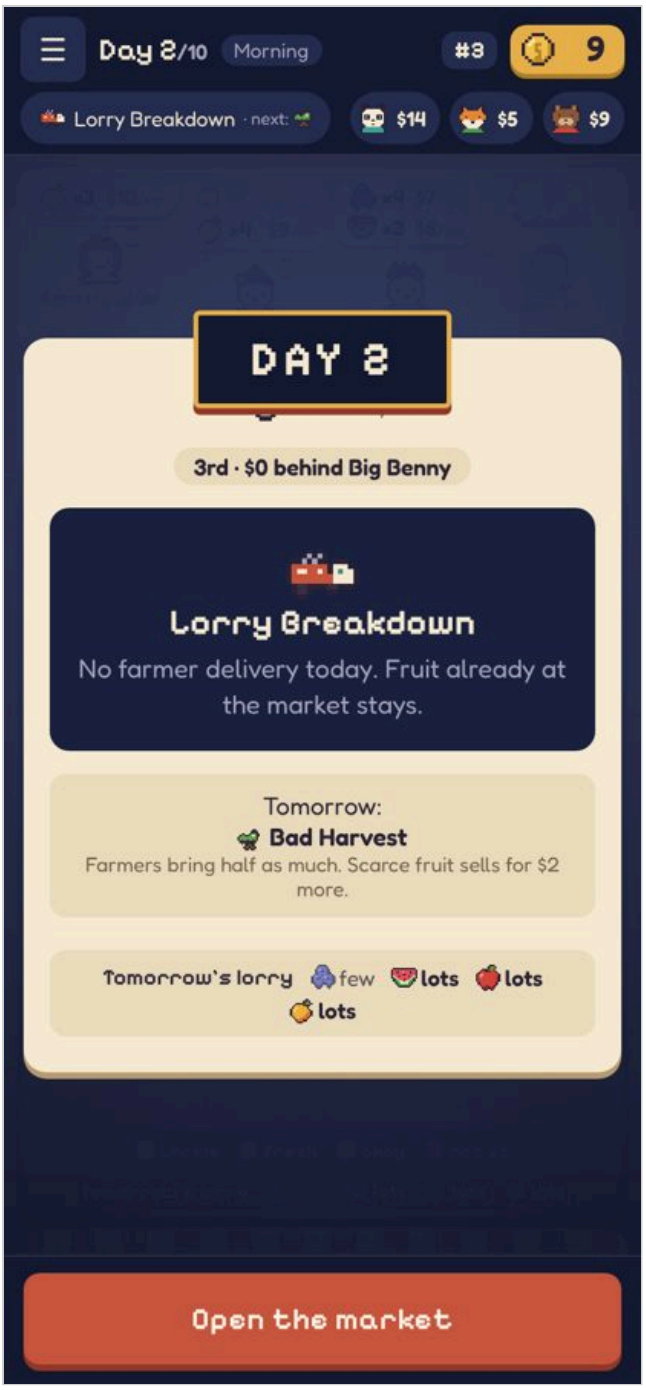
    A summary line reads, for example, "1 of 3 sealed · nobody sees a price until the bell".
4. **Seat badges** use the same words and colours: a larger amber "deciding" or "next in line" pill, and a "sealed" label under the wax seal.

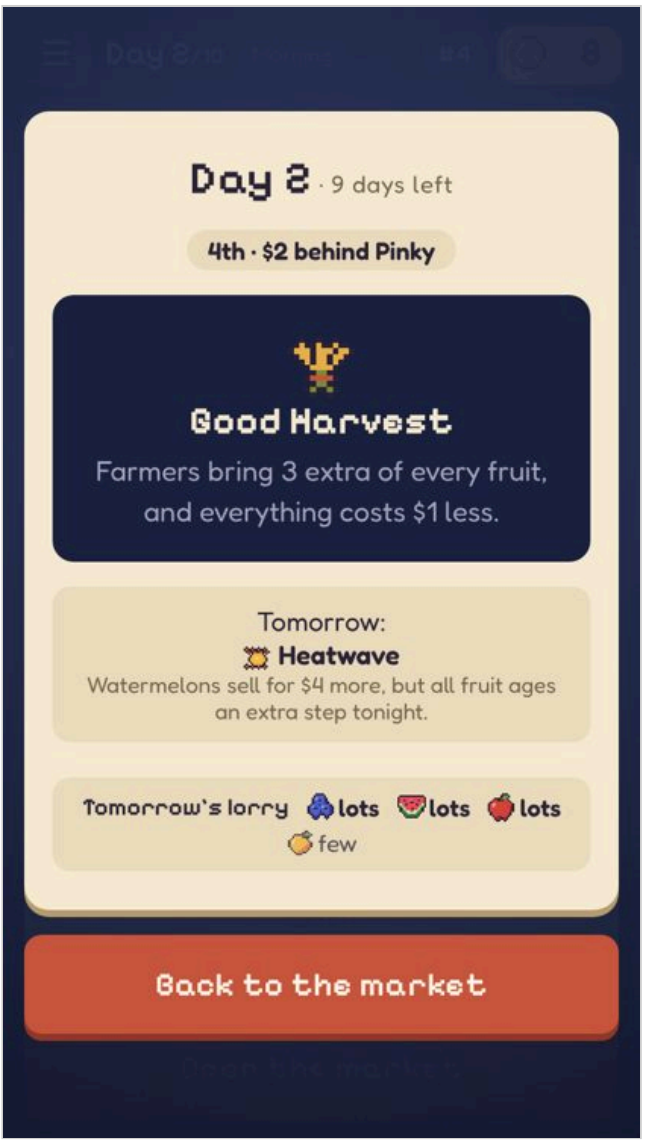
Only public facts are shown: whether a shop has sealed, and the stock that is already visible on its stall. No price is shown before the bell.

Before and after (390×844)


Day 8/10
Pricing
#1
10
Health Inspector · next:
$0
$0
$9
×1 $7/ea
Fairy Nurul
$10/ea
×3 $8/ea
×1 $6/ea
Daniel the Wiza...
×4 $7/ea
×3 $7/ea
Count Robert
×4 $7/ea
×1 $10/ea
Hui Min the Elf
Rocky ▾
0 2 1
pricing.
Patches ▾
0 4 1
pricing.
Foxy Rosa ▾
9 2 2
pricing.
Farmers Market
5 want 4 in shops
5 want 3 in shops
3 want 2 in shops
4 want 3 in shops
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
You have nothing today's customers want.
Also wanted, but you have none: 1 watermelon for Nurul, 3 apple for Daniel, 1 blueberry for Daniel, 4 blueberry for Robert, 3 mango for Robert, 4 watermelon for Hui Min, 1 mango for Hui Min.
Sell nothing today


Day 8/10
Pricing
#1
6
Heatwave · next:
$0
$0
$0
Seal your prices
Only you see this board until the bell. Price per fruit; cheapest sells first, fresher fruit wins ties.
Your rivals
0 of 3 sealed · nobody sees a price until the bell
Patches
next in line
2 2 2
Pinky
next in line
2 1
Uncle Toad
next in line
1 2
Vijay
wants 3 · pays up to $7
Sell
$4
Psst... 5 rival fruits chasing 4
≤$7 ≤$5 ≤$7
Chris
wants 1 · pays up to $7
Sell
$7
Psst... 5 rival fruits chasing 4
≤$7 ≤$5 ≤$7
Also wanted, but you have none: 4 mango for Lina, 4
Seal my prices up to $12





Checks

Check	Result	Evidence
`npm run verify`	PASS	exit 0: records, lint, types, 81 unit tests, build
`npm run verify:full`	PASS	exit 0: solo game, BUDGET OK, online 40 pricing frames with 0 price leaks, computer takeover, quick match, four players, reconnect 5 cycles 0 failed
Screens at 390×844, 360×640, 1280×800	PASS	no sideways scroll, no page errors

Code

- `src/ui/Game.tsx` : the morning card moves into a full-screen sheet, the pricing tray gets the full-screen class, and seat boards record computer rivals.
- `src/ui/panel.tsx` : new `RivalWatch` strip.
- `src/ui/game/panels/Popovers.tsx` : full-screen event view.
- `src/ui/table.tsx` : seat badges.
- `src/ui/styles.css` : PLAN-013 block.

REP-030 · Rules re-analysis for a tabletop-ready market game (proposal, no rule change)

Game	Fruit Market, rules version 3 (`main` at 416f381)
Report date	2026-10-27 12:59 (Asia/Kuala_Lumpur)
Type	Analysis, lab experiments and a proposal. **Game code unchanged**
Owner request	Re-analyse fruit pricing (lorry and customer), decay, events, traits, power-ups and supply/demand control, against three guidelines: (1) playable on a physical table, (2) a real market feel with inventory, forecasting, strategy and undercutting, balanced all round, (3) each fruit's playstyle in line with its payout
Proposed plan	PLAN-014 "Tabletop edition rules" (awaiting approval)

Labels. [Code] = how the game works now. **[Sim]** = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, for the playtest.

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1. Verdict against the three guidelines (rules version 3)

Guideline	Today	Main gap
1. Playable on a table	**Partly**	Deliveries and order sizes come from formulas (player count × game progress, rounded); 4-player ×1.15; up to ~10 secret prices per player per day; sliding every fruit one step each night is slow by hand
2. Market feel	**Partly**	Undercutting and forecasting work; prices never react to shortage or glut; fruit is sold within half a day (85–90% bought fresh, 0.3 spoiled per shop per game), so inventory and decay barely matter; shared-loss events (Slow Season, Rainy Day) are luck with no decision
3. Fruit playstyle = payout	**No**	Blueberries earn the best return on cost of all four ($1.72 → $5.61) though they are meant to be the cheap volume fruit; apples (safest) earn more than watermelons

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2. How the rules work now [Code]

2a. Fruit: lorry price, customer price, decay

Fruit	Farmers-market cost fresh / okay / not so / green	Customer top price (24 cards, after shift)	Price drop by age, oldest → freshest	Life on the shelf
Blueberry	2 / 1 / 1 / –	5–8 (shift 0)	−4 · −2 · 0	3 nights
Watermelon	3 / 2 / 1 / 1	6–9 (0)	−3 · −1 · −1 · 0 · 0	5 nights (+1 green)

Fruit	Farmers-market cost fresh / okay / not so / green	Customer top price (24 cards, after shift)	Price drop by age, oldest → freshest	Life on the shelf
Apple	3 / 2 / 1 / –	6–10 (−1)	−2 · −1 · −1 · −1 · 0 · 0 · 0	7 nights
Mango	4 / 2 / 1 / 3	7–11 (+2)	−3 · −1 · 0 · 0	4 nights (+2 green)

- The farmers market cost is fixed per freshness stage; events change it ±1.
- The customer's price is the card value plus the fruit shift plus the event, minus the age drop.
- Fruit at the farmers market ages too, and every night all fruit moves one step.

2b. Supply and demand control (the tabletop problem)

- **Deliveries:** one farmer card a day lists three fruits (4–6 each); the fourth fruit comes as 2. Each number is then multiplied by a formula: 4 per shop × (0.8 + 0.2 × game progress) ÷ 17, rounded. Tomorrow's card is shown a day early.
- **Customers:** one expected customer, known a day early, plus 1 walk-in (2 from halfway), drawn face down. Every order size is multiplied by a formula: 4 per shop × (0.6 + 0.45 × progress), × 1.15 with 4 players, ÷ cards, rounded.
- **Selling:** every shop sets a secret price and quantity for every customer and fruit. Then:
  - cheapest sells first;
  - ties go to a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.

2c. What events, traits and upgrades do to outcomes

Events (from REP-025, income vs a normal day, 2p / 4p):

- Big gains: Festival +11 / +8, Superfood +11 / +5, Mango Craze +10 / +4 (the largest gap between shops).
- Neutral: Heatwave, Minimum Price.
- Losses: Slow Season **−12** / −7, Rainy Day −8 / −5, Lorry Breakdown −6 / −3.
- Most events hit every shop equally. They reward planning only when they are announced a day early, which they are.

Traits [Sim, rules 3]:

- 3 players: 30–37%, a 7-point spread (fair 33%).
- 4 players: 23–29%.

Close enough.

Upgrades [Sim]: building only one upgrade, against 10.8% for building none:

Upgrade	Win %
Signboard	20.7
Cool Room	15.8
Handcart	14.7

Upgrade	Win %
Shop Bell	13.5
Juice Press	12.8
Bigger Shelf	12.2

All are worth something; the Signboard is the clear best.

Favour cards: small one-off boosts. Fine on a table.

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3. Proposed "Tabletop edition" (rules version 4)

3a. Core changes (tested)

#	Change	Replaces	Physical form	What decision it improves
T1	**Printed deliveries.** Each farmer card shows 3-player amounts: its three fruits at 2–4 (today's number −2), the fourth fruit at 1. With 2 players, one less of each listed fruit (minimum 1); with 4 players, one more of every fruit	The supply formula	Numbers on the cards plus one line in the rules	None directly; makes the game runnable by hand with almost the same flow
T2	**Printed orders and a walk-in rule.** Customer cards are used as printed. Walk-ins: players − 2 in the first half of the game, players − 1 in the second half	The demand formula and ×1.15	The deck and one line in the rules	Same as T1
T3	**One posted price per fruit.** Each shop secretly sets one price per fruit on a 4-slider price board behind its screen, and puts the fruit it will sell in its "for sale" crate. On reveal, customers (waiting ones first) buy from the cheapest shop whose fruit is fresh enough for the price	~10 secret prices per customer and fruit	A price board and a screen	The core duel becomes "what will they post for mangoes today?", which reads like a real market stall and is quick on a table
T4	**Market pulse.** Each fruit has a marker on a track from −$1 to +$2. After sales: if any customer still wants that fruit, the marker goes up $1 (shortage). If at least players + 1 ripe ones are left at the farmers market, it goes down $1 (glut). Otherwise it drifts one step back to 0. Customers pay the marker on top of their card price	Fixed prices	One cube per fruit on a small track	Prices respond to supply and demand. Spotting tomorrow's shortage pays, and flooding a fruit is punished
F1	**Fruit payouts lined up with risk.** Blueberry customers pay $1 less; apple customers $1 less again (apple shift −2)	Card shifts	Printed on the customer cards	See 3c

3b. Results [Sim, 3 players unless noted]

Measure	Rules 3 now	T1+T2	+ T3 + T4	+ F1 (proposed)
Undercut	27	22	29	28

Measure	Rules 3 now	T1+T2	+ T3 + T4	+ F1 (proposed)
Dump old stock	36	32	32	28
Premium prices	28	30	31	33
Reserve	36	38	33	34
Ripen	34	35	39	38
Read rivals	37	35	38	35
Turnover	23	24	24	26
Events	29	30	24	28
Spread	14	16	15	**12**
4 players: Undercut / Salvage / Premium	20 / 28 / 20	20 / 30 / 22	21 / 28 / 21	23 / 29 / 20
Hard beats Normal (2p)	60	59	63	62
Comeback from last at half-time (3p / 4p)	25.8 / 19.2	28 / 19	32.3 / 19.2	30.2 / 18.0
Price-war sales (3p / 4p)	5.8 / 6.7	5.0 / 7.1	3.9 / 4.2	5.9 / 7.0
Average cash (2p / 3p / 4p)	150 / 108 / 100	137 / 116 / 94	146 / 133 / 116	135 / 122 / 103
4-plan table (Turnover / Reserve / Ripen / Events)	16/32/33/20	16/40/26/19	18/29/35/18	15/29/38/19

Daily flow, formula against printed numbers (fruit delivered / units customers want, first half → second half):

Players	Formula (now)	Printed numbers
3	9.7 → 11.0 / 8.4 → 12.1	10.1 → 10.1 / 8.7 → 13.1
4	13.7 → 15.3 / 12.8 → 17.3	14.1 → 14.1 / 13.1 → 17.4
2	6.8 → 8.0 / 5.5 → 8.1	7.1 → 7.1 / 4.4 → 8.8

The printed rules reproduce the market closely.

Traits under the proposal: 31–37% in 3 players.

Caution. The computers' posted price is a rough stand-in: they take their lowest planned price for each fruit. People will price more cleverly, so T3 must be playtested.

3c. Fruit identities and payout [Sim, proposal]

Fruit	Target playstyle	Cost (fresh)	Average sale	Net per fruit bought	Risk
🥭 Mango	**Speculator:** buy green for $3, wait 2 nights, sell big on Mango Craze or shortage days	$3.61	$9.15	**$5.15**	Highest: 2-night wait, spoils fast once ripe, few delivered

Fruit	Target playstyle	Cost (fresh)	Average sale	Net per fruit bought	Risk
🫐 Blueberry	**Quick flipper:** cheap and plentiful; sell today or lose $2 a night	$1.72	$5.58	**$3.51**	High: 3-night life, steepest decay
🍉 Watermelon	**Steady bulk seller:** big orders, slow decay, a 1-night green bet	$2.51	$6.20	**$3.15**	Medium
🍎 Apple	**The keeper:** lasts a week, sells any day, the safe stock for a Lorry Breakdown or a shortage	$2.79	$6.08	**$3.00**	Lowest

Now the more a fruit can hurt you, the more it pays: mango > blueberry > watermelon > apple. Under rules 3 apples paid more than watermelons, and blueberries gave the best return on cost.

3d. Supporting proposals (not simulated: they need people, or computer players that plan ahead)

1. **The queue: forecasting the crowd.**
    - Tomorrow's walk-ins are drawn tonight and placed face down.
    - The card backs show which fruits each customer wants, but not how many or at what price.
    - With the face-up expected customer and tomorrow's lorry, players can forecast demand: "two walk-ins want apples tomorrow, and the lorry brings few".
    - Physically: printed card backs. Digitally: a row of silhouettes with fruit icons.
2. **Market Days.** Every third day is a busy fair day: +2 walk-ins, and the lorry doesn't come.
    - Shops must stock up the day before, which is where shelf space, decay and apples matter.
    - [Sim] Comebacks rise to 34% and blowouts fall to 31%.
    - But the computers don't plan ahead, and skill matters less in their games (Hard beats Normal 52%).
    - [Hyp] This may be the strongest inventory mechanic, so it should go to a paper playtest before any decision.
3. **Crates instead of sliding.** On a table, fruit goes into a crate stamped with the day it arrived. Its age (today minus that day) gives its freshness from a printed chart. Nothing needs sliding each night, and the digital game already works like this underneath.
4. **Events with decisions.**
    - Keep the events that are announced a day early and reward preparation: Lorry Breakdown, Mango Craze, Superfood, Heatwave, Bad Harvest, Hotel Order.
    - Replace the two pure shared losses (Slow Season, Rainy Day) with single-fruit shocks such as "Blueberry glut: +2 blueberries per player, blueberry customers pay −$1" or "Apple shortage: no apples delivered".
    - Scale Good Harvest by players: +1 of each fruit per 2 players, rounded up.
    - Never draw a night-only event (Inspector, Heatwave ageing) for the final day.
    - Needs a simulation round before numbers are fixed.
5. **Traits and upgrades:** no changes needed for the table; all are printed cards.
    - Juice Press stays the weakest upgrade (12.8%). [Hyp] It gains value with Market Days and stocking ahead.

3e. What it would mean for the app

- Rules version 4: old saves cleared; old apps refused online; separate daily boards (all already handled).
- **Pricing screen:** four fruit price sliders plus "how many to put out", instead of one card per customer. It is simpler and fits the full-screen sheet from PLAN-013.
- **Board:** a market pulse marker on each fruit column, and walk-in silhouettes in the street (if 3d-1 is approved).
- **Computer players:** must learn posted pricing properly before release. Today's stand-in is crude.
- **Rules page and how-to:** rewritten. The digital game and a paper version share one rulebook.

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4. Risks

- **Posted prices remove per-customer pricing.** A shop can no longer charge a rich customer more than a poor one. [Hyp] Simpler and more market-like, but less fine control for experts. Hard beats Normal 62%, still under the 65% aim.
- **Market pulse adds one track and a rule after sales.** [Hyp] It is easy on a table, but it is a new idea for players.
- **The computers can't judge** Market Days, the queue or event redesigns, so those depend on playtests.
- **It is a large change:** a new pricing screen, computer pricing work, and a rules rewrite.

5. Playtest plan

1. **Paper prototype:** print the farmer and customer cards with the new numbers, use coins and tokens, play 3 evenings with 3–4 people. Test T1–T4, the queue and Market Days. Time each day; target 6 minutes or less on a table.
2. **Digital preview:** after building, the same questions as REP-027 section 7, plus:
    - "Did the posted prices feel like a market?"
    - "Did you ever stock up ahead?"

6. Recommendation

Order	Item	Status
1	Approve **PLAN-014 core: T1 + T2 + T3 + T4 + F1** (tested above), with computer posted-price pricing and the new pricing screen	Awaiting approval
2	Paper playtest of the queue, Market Days and crates, then decide	Proposal
3	Event redesign (3d-4) after a simulation round	Proposal

Lab files (engine copy, sweeps 12–15) are archived in `docs/workflow/reports/img/REP-030-lab.tar.gz`.

REP-031 · Fruit Market: game concept and mechanics (Tabletop edition, rules version 4)

Game	Fruit Market. Design: Murshidi bin Md Hashim
Report date	2026-10-27 13:46 (Asia/Kuala_Lumpur)
Type	Final mechanics (worked out and lab-tested) and a concept summary for benchmarking against board games
Plan	PLAN-014, approved: "approve all the idea, work out the mechanics, and summarise to me game back highlighting the concept and game mechanics as i want to benchmark against similar genre board games and traditional popular board games"
Status	Design final for the build. The app still runs rules version 3 until PLAN-014 is built

Labels. [Sim] = computer-player result: 600 games per test on seeds never used for tuning, about ±4 points. **[Hyp]** = a guess about people, to be checked in the paper playtest.

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1. The game on one page

Pitch. You run a fruit stall in a busy harbour market. Each day you buy fruit from the farmers' lorry, guess what the crowd will want and what your rivals will charge, then post your prices in secret. The cheapest stall makes the sale. But fruit goes off every night, so stock you can't sell is money rotting on your shelf. After 8, 10 or 14 days, the richest stall wins.

Players	2–4 (solo against computer shops in the app)
Length	Quick 8, Standard 10 or Long 14 days. App: about 15–20 minutes. Table: about 45–60 minutes **[Hyp]**
Age / weight	10+. Light-to-medium strategy **[Hyp]**: simple turns, with depth in reading the market and rivals
Win condition	Most cash at the end. Unsold fruit and upgrades are worth nothing. Ties go to the shop that sold more fruit
Elimination	None. Everyone plays every day
Luck vs skill	Luck from the order of the farmer, customer and event decks, softened by a one-day look ahead. Skill in buying, timing, pricing and reading rivals. [Sim] Hard computer beats Normal 62%

The four pillars (your brief).

Pillar	How the game delivers it
A real market	Supply comes off a lorry; demand comes from customers with budgets. One posted price per fruit; the cheapest stall sells first. The market pulse moves prices up after a shortage and down after a glut
Inventory and decay	Every fruit ages one step a night and sells for less as it ages; the oldest spoils. Shelves are limited. Fruits rot at different speeds

Pillar	How the game delivers it
Demand forecasting	The crowd queues a day ahead: the expected customer is face up, and the walk-ins are face down with their wanted fruits printed on the back. Tomorrow's event and tomorrow's lorry are also shown a day early
Strategy and undercutting	Sealed simultaneous pricing creates the "do I go $1 lower?" duel. Eight different plans are viable (section 4)

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2. Components (tabletop)

Component	Count	Purpose
Fruit tokens	4 kinds (about 40 each)	Blueberry, watermelon, apple, mango
Market board	1	Farmers market (crates by arrival day), customer queue, market pulse track, day track
Shop mats	4	Shelf (8 spaces), crates by arrival day, "for sale" crate
Price boards and screens	4	One slider per fruit ($1–$14), hidden until the reveal
Farmer cards	20	Tomorrow's lorry: 3 fruits in quantity plus 1 of the fourth; some mark one fruit as arriving green
Customer cards	24	Front: name, wanted fruits, quantity, top price for fresh fruit. Back: fruit icons only
Event cards	13	One a day from Day 2, each announced a day early
Favour cards	12 (6 kinds × 2)	One-off bonuses earned by serving a whole order alone
Upgrade cards	6	Evening draft
Trait cards	6	Pick 1 of 3 at the start
Coins, cubes, markers		Cash; one market pulse cube per fruit; day marker

Crates instead of sliding. Fruit sits in a crate marked with the day it arrived. Its age (today minus that day) gives its freshness from the chart on the mat. Nothing needs moving each night, and the app shows the same thing as a decay track.

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3. A day, step by step

1. **Morning**
    - Reveal today's event (announced yesterday) and turn up tomorrow's.
    - **The lorry arrives:** put today's farmer card's fruit into the farmers market. Green fruit goes to the green crate.
    - Turn up **tomorrow's lorry** card.
2. **Buying**
    - **Poorest shop first** (ties go round the table).

    - Each shop buys up to **4 fruits** (5 with a Handcart), paying the cost for that freshness.
    - Shelf limit 8 (14 with a Bigger Shelf).
3. **The crowd**
    - Customers still waiting from yesterday stand at the front.
    - Today's expected customer and today's walk-ins are turned face up.
    - Draw **tomorrow's queue**: 1 expected customer face up, and the walk-ins face down (backs show their fruits).
4. **Pricing (secret)**
    - Behind your screen, set **one price per fruit** on your price board.
    - Put the fruit you are willing to sell in your "for sale" crate.
    - Everyone reveals at once.
5. **Sales**
    - Customers are served in queue order. For each fruit a customer wants, they buy from the **cheapest stall**.
    - Your price must not exceed what they pay for your fruit's freshness: their top price + market pulse + today's event − the age drop.
    - You hand over your oldest fruit that is fresh enough.
    - **Ties:** a Haggler facing a richer shop, then fresher fruit, then Favoured Trade, then turn order.
    - Paid on the spot.
    - Serve a customer's whole order by yourself to draw a **favour card** (one a day).
    - An unserved new customer waits until tomorrow; a waiting customer then leaves.
6. **Market pulse** (for each fruit)
    - **Shortage:** a customer still wants it, so the cube goes up $1 (to +$2 at most).
    - **Glut:** at least players + 1 ripe ones are left at the farmers market, so the cube goes down $1 (to −$1 at least).
    - Otherwise the cube moves one step back towards 0.
7. **Evening**
    - Upgrade draft: each shop sees 3 upgrades and may build any it can afford.
    - Evening favour cards can be played.
8. **Night**
    - All fruit ages one step, and fruit past its last step spoils.
    - The Juice Press turns up to 3 spoiling fruits into $2 each.
    - The Cool Room ages your fruit only every other night.
    - Night events apply (Health Inspector fines, Heatwave ageing).

Supply and demand on a table (no formulas).

- **Lorry:** the card's three fruits arrive at the printed number (2–4); the fourth fruit arrives as 1.
    - 2 players: one less of each listed fruit (minimum 1).
    - 4 players: one more of every fruit.
- **Walk-ins:** players − 2 in the first half of the game, players − 1 in the second half.
    - 2 players: 0 → 1.
    - 3 players: 1 → 2.
    - 4 players: 2 → 3.
- [Sim] These printed rules reproduce the old formula flow almost exactly. In 3 players, about 10 fruits a day are delivered, and customers want about 9 units a day early and 13 late.

Optional rule: Market Days (off by default, for the paper playtest).

- Every third day is a fair: +1 walk-in, and **no lorry**.
- Tomorrow's lorry shows "Market Day", so shops must stock up the day before.
- See section 6 for why it is optional.

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4. Fruit: price, decay and playstyle

	🫐 Blueberry	🍉 Watermelon	🍎 Apple	🥭 Mango
Playstyle	Quick flipper	Steady bulk seller	The keeper	Speculator
Lorry cost fresh / okay / not so / green	$2 / $1 / $1 / –	$3 / $2 / $1 / $1	$3 / $2 / $1 / –	$4 / $2 / $1 / $3
Customer top price (fresh)	$4–7	$6–9	$5–9	$7–11
Price drop by age, oldest → freshest	−4 · −2 · 0	−3 · −1 · −1 · 0 · 0	−2 · −1 · −1 · −1 · 0 · 0 · 0	−3 · −1 · 0 · 0
Shelf life	3 nights	5 nights (+1 green)	7 nights	4 nights (+2 green)
Risk	High: steepest decay	Medium	Lowest	Highest: wait to ripen, fast spoil, few delivered
[Sim] Net profit per fruit bought	$3.51	$3.15	$3.00	$5.15
Best moment	Sell the day you buy	Big orders, Heatwave	Stock for a Lorry Breakdown or shortage	Buy green before a Mango Craze or a shortage

The more a fruit can hurt you, the more it pays.

---

5. Events, traits, upgrades and favours

Events (13 cards; one a day from Day 2, each shown a day early):

Event	Effect
Festival	Customers pay +$3, plus an extra walk-in
Blueberry Glut *(new, replaces Slow Season)*	+1 blueberry per player delivered; blueberry customers pay −$1
Heatwave	Watermelons +$4; all fruit ages an extra step tonight
Apple Blight *(new, replaces Rainy Day)*	No apples delivered; apple customers pay +$1
Mango Craze	Mangoes +$3; the Influencer wants 3
Superfood Trend	Blueberries +$3; the Gym Club wants 4
Good Harvest *(scaled)*	+1 of every fruit per 2 players (rounded up); lorry prices −$1

Event	Effect
Bad Harvest	Half delivery (rounded up); customers pay +$2
Price Hike	Lorry prices +$1
Minimum Price	Price floors per fruit
Health Inspector	$2 fine per not-so fruit tonight. Never the final day's event (draw again)
Lorry Breakdown	No delivery today
Hotel Order	The Hotel Chef orders 3 apples, 2 watermelons and 2 blueberries

Traits (pick 1 of 3 at the start, public):

Trait	Effect
Haggler	Wins ties against richer shops, even over fresher fruit
Big Basket	+1 fruit a day
Bargain Hunter	First fruit each day $1 cheaper
Charmer	Start with 3 favour cards; hold 4
Local Favourite	+$4 per customer you serve completely alone
Nest Egg	Start with $15 instead of $10

[Sim] 31–38% wins each in 3 players.

Upgrades (evening draft, 3 offered, built once):

Upgrade	Cost	Effect
Handcart	$6	Buy 5 a day
Signboard	$8	+$1 per fruit sold
Cool Room	$7	Ages every other night; Heatwave-proof
Bigger Shelf	$3	14 spaces
Juice Press	$4	+$1 per older fruit sold, and up to 3 spoiling fruits a night become $2 each
Shop Bell	$4	Draw 2 favour cards; hold 4

Favour cards (2 of each): Free Cash ($3), Free Fruit, Good Riddance (sell ripe stock back at $1 each), Fridge (all fruit one step fresher), Favoured Trade (win ties today), Priority (buy first tomorrow).

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6. Balance evidence (computer players)

Measure	Rules 3 (app now)	Rules 4 (this design)
Win share of 8 plans, 3 players (fair 33%)	23–37%	25–42% (Ripen at the top)
Win share, 4 players (fair 25%)	20–28%	18–25%

Measure	Rules 3 (app now)	Rules 4 (this design)
Hard beats Normal (2 players)	60%	56–62%
Comeback from last at half-time (3p / 4p)	26% / 19%	28–32% / 18–19%
Price-war sales at $2 or less (3p)	6%	5%
Quick (8 days): 4-plan table	–	19 / 30 / 30 / 21 (even)
Long (14 days): 4-plan table	Reserve 42%	Ripen 49% (watch item)
Traits (3 players)	30–37%	31–38%

Market Days (optional rule):

- Every third day with +1 walk-in gives the most even plans of anything tested (24–35%).
- It also gives 38% comebacks and spoilage halves, because shops stock ahead.
- But Hard and Normal computers become even (50%), because the computers don't plan ahead.
- [Hyp] People who plan ahead may make it the most skilful mode. The paper playtest decides.

Open items for the build:

- Shop Bell, Juice Press and Cool Room are worth little on their own under posted prices (12.8–14.5%, against 13.5% for no upgrade). They need retuning.
- Long games favour the ripening plan.
- The computer players need proper posted-price logic.

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7. For your benchmark: mechanism profile

Mechanism names follow the usual board-game catalogue terms, so you can compare like for like.

Mechanism	In Fruit Market	Weight in this game
Simultaneous action selection / sealed pricing	Secret price board, revealed together	Core
Lowest-price-wins competition (a reverse auction)	Cheapest stall serves the customer	Core
Dynamic market pricing	Market pulse after shortages and gluts; lorry prices by freshness	Core
Perishable inventory / resource decay	Fruit ages nightly, loses value, spoils	Core
Demand forecasting / hidden information	Face-down queue with fruit-icon backs; tomorrow's lorry and event shown	Core
Open market buying	Shared farmers market, first come, first served	Medium
Catch-up turn order	Poorest buys first	Medium
Variable player powers	Traits	Light
Card drafting	Evening upgrade draft (3 offered)	Light

Mechanism	In Fruit Market	Weight in this game
Event deck with look ahead	One event a day, announced a day early	Medium
Contract fulfilment	Serve a whole order alone for a favour	Light
Engine building	Upgrades that pay each day	Light

Starting points for comparison (well-known games sharing a mechanism; please verify details in your own benchmark):

Game	Shared idea	Where Fruit Market differs
Power Grid	A resource market whose prices rise as supply runs out; the trailing player buys first	Fruit also rots, and selling prices are set by the players in secret
Food Chain Magnate	Customers buy from the cheapest restaurant; price-setting competition	Much lighter; demand is forecast from a visible queue rather than built by marketing
For Sale / Modern Art	Blind or sealed bidding duels	Bidding is for customers each day, and goods spoil
Monopoly (traditional)	Win by cash; rents and luck	No elimination, no property; every day is a fresh price duel; catch-up turn order
Catan (traditional)	Variable supply each turn	Supply is shown a day ahead and bought, not rolled; no trading between players
Sushi Go / 7 Wonders	Simultaneous play and drafting	Drafting is a side system (upgrades); the core is pricing and stock

Possible distinctive hooks for your benchmark to confirm: perishable stock with a different decay speed per good; one posted price per good settled by a simultaneous reveal; and a self-adjusting price track driven by unmet demand.

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8. Next steps

1. **Paper playtest:**
    - print the cards with these numbers;
    - 3 sessions of 3–4 players, with Market Days on in one;
    - time each day.
2. **Build PLAN-014 in the app** (rules version 4):
    - new pricing screen (four sliders);
    - market pulse on the board;
    - queue silhouettes;
    - printed supply and demand;
    - event changes;
    - Market Days as an option in the game set-up;
    - computer posted pricing;
    - upgrade retune.
3. **Benchmark** using section 7, then feed changes back before the build is finalised.

Lab files (sweeps 16–18) are archived in `docs/workflow/reports/img/REP-031-lab.tar.gz`.

REP-032 · Tweaks from the benchmark review: a readable price duel, three prototypes and the Market Gossip card

Game	Fruit Market (app on rules version 3; design target rules version 4, REP-031)
Report date	2026-10-27 14:34 (Asia/Kuala_Lumpur)
Type	Design proposal with lab checks. **Game code unchanged**
Input	The owner's benchmark review of REP-031 (Stockpile, Food Chain Magnate, Modern Art, For Sale, Catan, Monopoly) and the request: "suggest tweaks based on the above, dont take current game turn as the final form, feel free to tweak accordingly, introduce one favour cards which allows player to see the demand for the face down card"
Proposed plan	PLAN-015 (awaiting approval). It reshapes PLAN-014 before that is built

Labels. [Sim] = computer players: 600 games per test on unseen seeds, about ±4 points. **[Hyp]** = a guess about people, to test at the table.

Correction to REP-031. Food Chain Magnate does not simply give customers to the cheapest restaurant. Whether a restaurant can serve the order, price plus distance, and tie-breakers all matter. Thank you for the check.

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1. The direction

The review's main point holds: **make the price duel exciting and easy to read; don't add machinery.** The hook stays:

> **"How much can I charge before my rival, or the rot, gets me?"**

Every tweak below either:

- removes bookkeeping;
- makes the reveal clearer;
- gives a real alternative to "$1 lower"; or
- gives the session an arc.

The new favour card is the **only** new information mechanism, as the review advised.

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2. A new day: five beats instead of eight

Beat	What happens	Why
1. Dawn: the "Tomorrow" strip	The lorry unloads today's fruit. One strip shows everything known about tomorrow: the lorry card, the event, the expected customer (face up) and the walk-ins (face down, backs show their fruit)	All forecasting in one place, so there are no separate things to check
2. Market	Buy up to 4 fruits, poorest shop first	Unchanged
3. Open stalls *(new)*	Each shop puts the fruit it will sell today into its "for sale" crate, **face up**. Quantities are public before anyone prices	Makes the alternative to undercutting visible: "He only has two apples, so I can charge more and serve the rest"
4. The bell	Prices are set secretly on the price board, then **everyone reveals at once**. This is the centrepiece	One clear dramatic moment a day
5. The rush	Customers come in queue order; each fruit goes from the cheapest stall, and leftover demand **spills to the next stall**. Every sale is shown as customer → stall. Then night: fruit ages	Players see why they sold or didn't

Removed from the core:

- the market pulse track, now an optional advanced rule;
- evening admin on most days (upgrades are bought only on Market Day evenings, see section 4);
- the Haggler tie rule. Ties become **fresher fruit, then poorer shop**, which uses one catch-up idea, not three.

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3. Prototype A: the core market (readability)

Tweak	Change	Evidence
A1 **Uniform freshness**	Every fruit sells at full price when fresh, −$1 when okay, −$2 when not so. Fruits differ only in **how many nights** they stay in each stage. One rule replaces four price-drop tables	[Sim] Plans win 26–38% (was 25–42%); Hard beats Normal 56% (same); comebacks 32% (was 28%)
A2 **Street prices (option)**	Each fruit has one printed street price ($5 blueberry, $7 watermelon, $6 apple, $9 mango). Customer cards show only what and how many, plus a "generous" +$2 icon on a few cards	[Sim] Balanced (plans 25–43%), but skill matters a little less (Hard beats Normal 53%). Keep as the beginner option
A3 **Catch-up stays "poorest buys first"**	Keep cash order; don't switch to net worth	[Sim] Adversarial test: a shop that buys every upgrade wins 33% (fair), so there's no exploit. Ordering by net worth makes hoarding cash viable (13% → 32%) but makes upgrades optional, and it's harder to count by hand
A4 **Simpler ties**	Fresher fruit, then poorer shop	[Hyp] Fewer exceptions to remember

Prototype A without traits, favours and upgrades [Sim]:

- Games are close (17% blowouts).
- **Ripening green fruit dominates** (60% wins), and Hard beats Normal only 52%.
- The layers we're removing currently keep that plan in check. If people find it the same, Prototype B must fix it.

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4. Prototype B: different businesses (upgrades that change decisions)

There are four "stall specialities", one per plan. Each changes *what you do*, not just how much you earn.

Speciality	Cost	Effect	Decision it changes	Replaces
Cold Room	$5	Your fruit ages only every other night; Heatwave-proof	Hold stock for a shortage or Market Day	Cool Room ($7)
Juice Bar	$4	+$1 for each okay or not-so fruit sold; up to 3 spoiling fruits a night become $2 each	Buy cheap old fruit at the farmers market on purpose	Juice Press
Barrow	$6	Buy 5 a day and hold 12	Volume and turnover	Handcart + Bigger Shelf
Premium Display	$8	Your **fresh** fruit may be priced $1 above a customer's normal limit	Stay expensive, and sell to whoever is left after the cheap stalls run out	Signboard

- Removed: **Shop Bell** (the Market Gossip card takes over the information role).
- **When:** upgrades are drafted only on the evening after each Market Day (see section 5): three build moments a game, not ten.
- **Traits** stay as starting stall capabilities.
- [Hyp] The upgrades are designed to change decisions, so measure them by whether they change purchasing, holding and pricing, not only by win rate. The computer players need teaching before the lab can rate them.

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5. Prototype C: market rhythm (the arc)

Day (10-day game)	Rhythm
1–2	**Opening:** small crowds; learn the rivals
3	**Market Day:** +1 walk-in, **no lorry**; then the evening upgrade draft
4–5	Ordinary trading; demand grows
6	**Market Day**, then the draft
7–8	Second-half crowds (more walk-ins)
9	**Market Day**, then the last draft
10	**Last Call:** customers pay +$1; afterwards the jam maker buys leftover fruit at $1 each

Market Days are printed on the day track, so everyone knows they're coming. In 8-day games they fall on days 3 and 6; in 14-day games on 3, 6, 9 and 12.

- **Why:**
  - preparation, holding stock for a day with no lorry, becomes a planned, satisfying choice;
  - the jam maker softens the final-day dump without making stock valuable.
- [Sim] Market Days every third day:
  - plans become the most even of anything tested (24–35%);
  - comebacks rise to 38%;
  - spoilage halves.
- **But** Hard and Normal computers become even (50%), because the computers don't plan ahead.
- [Hyp] Following the review: lower spoilage is not automatically better. Test whether preparing creates hard choices, or just an obvious stock-up.
- A $1 jam-maker salvage was tested earlier: no balance change.

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6. The new favour card: **Market Gossip** 👂

> **Market Gossip** *(evening)*. Secretly look at the front of every face-down walk-in card in tomorrow's queue: who they are, what they want, how many, and their top price. Don't show anyone. Everyone sees that you played it.

- **Replaces Free Cash** (2 copies; the deck stays at 12). Free Cash is the only favour that is pure money with no decision.
- **Why it fits the review:**
  - it is the single, contained piece of private information that the review suggested;
  - rivals know you know *something*, and your buying gives clues ("why is she buying so many mangoes?");
  - that adds Stockpile-style reading without forecasts, secret objectives and trading all at once.
- **How it's earned:** like any favour, by serving a customer's whole order by yourself. It rewards the sales that build rivalry.
- **App:**
  - the walk-in silhouettes turn face up **only on your screen**, with a small 👂 badge on your seat for rivals;
  - online, the server must send the card fronts to that player only, the same privacy rule as secret prices, with a leak check added to the browser tests.
- **Needs:** the face-down queue (REP-031). It works whether or not the street-price option is used; with street prices it reveals quantities and "generous" icons.
- [Hyp] Test it on its own, after A–C, so its effect can be told apart (as the review advised).

Favour deck after the change (2 of each):

- Market Gossip *(new)*
- Free Fruit
- Good Riddance
- Fridge

- Favoured Trade
- Priority

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7. App feedback: understand why you won or lost

Feature	Status today	Tweak
Which customer bought from which stall	Showdown cards per line	One **"rush" view**: customers walk to the stalls they bought from
Why a sale was missed	Tags: "$1 too high", "lost tie: fresher won", "Right price, not enough stock"	Keep; add "too old for that price" (freshness)
Day's money in, money out and waste	Day income only	**Day ledger:** sales, purchases, spoilage and upgrades, with net for the day
Reveal pacing	Fast, skippable	A short pause after the bell before the rush resolves
Forecast and rival stock	Lorry strip, rival drop-downs, rivals strip while pricing	All in the one Tomorrow strip, plus the public "for sale" crates
Playtest modes	Toggles for events, upgrades, unripe	Add **"Core market" (Prototype A)** and **"Market Days"** toggles in the set-up, so playtests can compare versions

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8. Test plan (the review's targets)

Step	Version	Question	Measure
1	A (core) on paper and in the app	Is the core fun on its own?	Teach in ~5 minutes; new players finish unaided; players can explain their best and worst decision
2	B (specialities)	Does specialising improve rivalry and replay?	Did upgrades change what people bought, held or priced? Did a "stay expensive" win happen?
3	C (rhythm), on the better of A and B	Does preparation improve the arc?	Hard choices before Market Days; the final two days still have real decisions
4	+ Market Gossip	Does one private clue add reading?	Did anyone change their play after seeing a rival use Gossip?
Always	Same group repeatedly, plus fresh groups	Repetition and dominant plans; teaching problems	Replay requests; time per game; the reveals people remember

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9. What changes against the approved PLAN-014

PLAN-014 item	Proposal
Printed supply and demand, posted prices, crates, queue, fruit payouts	**Keep**
Market pulse in the core	Move to an **optional advanced rule**
Market Days optional	Make the rhythm **standard in Prototype C** (fixed days on the track)
Event redesign	Keep
Upgrades unchanged (retune)	Replace with the **four specialities**, drafted on Market Day evenings
Tie rules	Simplify to fresher, then poorer
Free Cash	Replace with **Market Gossip**

Lab files: `docs/workflow/reports/img/REP-032-lab.tar.gz` .

REP-033 · Rules version 4 built: open stalls, the bell, Market Days, specialities and Market Gossip

Game	Fruit Market (app now on rules version 4, "tabletop edition")
Report date	2026-10-27 15:37 (Asia/Kuala_Lumpur)
Type	Build and validation report
Plans	PLAN-015 (approved: "approve plan 15") built together with the approved PLAN-014 items
Branch	`workflow/plan-015` , merged to `main` for the preview. **Live game unchanged**

Labels. [Sim] = computer players, 300 games per test, about ±5 points. **[Hyp]** = a guess about people.

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1. What was built

The day (five beats).

1. **Dawn:** the lorry unloads; the Tomorrow strip shows the next lorry, event, expected customer and the face-down walk-ins (fruit icons on the backs).
2. **Market:** buying, poorest shop first.
3. **Open stalls (new):** each shop secretly fills its for-sale crate; all crates show together.
4. **The bell:** one secret price per fruit, revealed together, with a short pause before the rush.
5. **The rush:** customers buy from the cheapest stall, leftover demand spills to the next; then evening and night.

Prototype A (core market).

- Uniform freshness: −$1 okay, −$2 not so, for every fruit.
- Ties: Favoured Trade, then Haggler, then fresher fruit, then poorer shop, then turn order.
- Cash catch-up kept ("poorest buys first").
- Options in the set-up: **Core market** (no traits, favours or specialities), **Street prices** (beginner) and **Market pulse** (advanced).

Prototype B (four specialities), drafted on the evenings after days 3, 6, 9 and 12: Cold Room $5, Juice Bar $4, Barrow $6, Premium Display $8. Shop Bell, Signboard, Bigger Shelf and Handcart are gone.

Prototype C (rhythm). Market Days on days 3, 6, 9 and 12 (+1 walk-in, no lorry); Last Call on the final day (+$1, then the jam maker buys leftovers at $1). Set-up toggle "Market Days", on by default.

Market Gossip replaces Free Cash (2 copies, evening): secretly see tomorrow's walk-ins. The fronts appear only on your screen, with a 👂 badge.

PLAN-014 items.

- Printed farmer cards (2–4 per fruit for 3 shops; 2 shops −1, 4 shops +1; missing fruit 1).
- Walk-ins: players − 2, then players − 1 in the second half. Customer cards are no longer scaled.
- Price shifts: blueberry −1, apple −2, mango +2.
- Events: Blueberry Glut and Apple Blight replace Slow Season and Rainy Day; Good Harvest +1 per 2 shops; Bad Harvest rounds up; no Health Inspector on the final day.
- `RULES_VERSION` 4. Older apps are asked to update; daily scores are kept separately per rules version.

App feedback.

- Stall screen and price screen: each shows rivals' crates, which customers can pay your price, and a hint.
- Day ledger: sales, bought, other (fines, favours), net, and fruit spoiling tonight; jam maker money on the results.
- Missed-sale reason "too old for $X". Bell pause before the rush. New tutorial tips.
- `RULES.md` and the in-app How to play rewritten.

One rule detail added for privacy. The evening is now held whenever anyone has a favour card. Before, it was held only when someone had an *evening* card, which told rivals what that player held. No balance effect.

2. Privacy (online)

Hidden until everyone commits	How it is protected	Check
Crates during Open stalls	Server keeps them aside; every view shows all crates empty	Unit test + browser test: PASS
Prices during the bell	Same	Unit test + browser test: PASS (0 leaks in 94 frames)
Face-down walk-ins	Each player's view shows only the fruit, no name, amount or price, unless that player heard the gossip	Unit test (fronts only to the gossiper) + browser test frame check: PASS
Rivals' favour cards	Only the number of cards is sent	Existing test: PASS

3. Validation

Check	Result	Evidence
`npm run verify` (records, lint, types, 92 unit tests, build)	PASS	Local run, commit after this report
`npm run verify:full` (solo game, performance budget, online match with leaver, quick match, 4 players, reconnect)	PASS	Exit 0; "leaks: 0"; "E2E OK"
New screens on a phone-sized browser	PASS	Stall, price, Market Day notice, evening, day ledger checked by screenshot; no page errors

Check	Result	Evidence
Paper or preview playtest (PLAN-015 T3)	NOT RUN	Needs the owner

4. Balance of the production build [Sim]

Measure	Result	Target or earlier figure
Hard beats Normal (2 players)	56%	REP-032: 56% (earlier aim 65%)
Hard beats Easy (2 players)	63%	—
Normal beats Easy (2 players)	55%	—
Comeback from last at half-time (3 players)	31%	REP-032: 32%
Day-5 leader wins (3 players)	34%	lower is more open
Blowouts, margin $20+ (2 / 3 / 4 players)	34% / 27% / 13%	watch 2-player games
Traits win (3 players, fair 33%)	30–38% (Big Basket highest)	REP-028: 30–38%
Old "first buyer rotates" rule for comparison	comebacks 8%, blowouts 58%	catch-up still essential

Specialities (3 players, fair 33%).

- Rushing **Barrow** wins 29% and **Premium Display** 27%; skipping either costs a lot (23%, 26%).
- Rushing **Cold Room** wins only 13%, **Juice Bar** 17%. The computers buy Juice Bar rarely (0.14 per game).
- Never building anything wins 19%, so specialities are worth having.
- This matches REP-032's warning: Cold Room and Juice Bar change *what you do*, and the computers don't play those plans well yet, so these numbers undersell them. [Hyp] Check in the playtest whether people find them useful.

Market feel.

- 90% of computer sales are within $1 of the customer's top price, and deep price wars are rare (0.1% of sales). Spoilage is near zero.
- Supply is tight enough that the computers rarely need to undercut. [Hyp] People may still undercut; watch for it in the playtest. If not, the next lever is +1 supply per listed fruit (needs a new plan).
- Market Days are lean days (income dips on days 3, 6 and 9), as designed.

5. Risks and open items

1. **Cold Room and Juice Bar are weak for computers.** Possible retune: Cold Room $4, Juice Bar $3. Not changed without your approval.
2. **Little undercutting among computers.** The hook ("how low dare I go?") needs a human check.
3. **2-player blowouts at 34%.** Watch in playtests.
4. **Hard vs Normal 56%,** below the old 65% aim. The computers are fair practice, not experts.

5. **Online settings:** online rooms use the defaults (Market Days on, no Core, Street or Pulse options). The playtest toggles are for games on one phone.

6. Next steps

1. Preview playtest in the order of REP-032 section 8: Core market first, then specialities, then Market Days, then Market Gossip.
2. Decide on the speciality retune (section 5, item 1).
3. PREPARE RELEASE 1.2.0 when you want rules version 4 on the live game (separate approval to publish).

REP-034 · Speciality price cut: Cold Room $4, Juice Bar $3

Game	Fruit Market (rules version 4)
Report date	2026-10-27 16:20 (Asia/Kuala_Lumpur)
Type	Balance change with a same-seed comparison
Plan	PLAN-016, approved: "approve the speciality price cut"
Branch	`workflow/plan-016`, merged to `main` for the preview. **Live game unchanged**

Labels. [Sim] = computer players, 300 games per test, the same seeds as REP-033, about ±5 points.

1. Change

- Cold Room: $5 → **$4**.
- Juice Bar: $4 → **$3**.
- Updated in the engine and `RULES.md`. The screens read the cost from the engine.
- `RULES_VERSION` stays 4. Version 4 has never been published to the live game, so no live player or saved score is affected. The preview's version-4 daily scores are test data.

2. Same-seed comparison [Sim]

Measure (3 players unless noted)	Before (REP-033)	After
Rush Cold Room	13%	12%
Rush Juice Bar	17%	16%
Rush Barrow / Premium Display	29% / 27%	27% / 25%
Never build a speciality	19%	16%
Juice Bar built per player (2 players)	0.14	0.06
Hard beats Normal (2 players)	56%	56%
Hard beats Easy (2 players)	63%	66%
Traits (range)	30–38%	31–40% (Big Basket 40%)
Comeback from last at half-time	31%	28%
Close games (10 days)	21%	21%

All differences are within the margin.

3. Finding

The price was not the problem. Cheaper Cold Room and Juice Bar did not help the computers win with them.

- Supply is tight in rules version 4. The computers sell almost everything while it's fresh, and spoilage is near zero (REP-033). So the Juice Bar has little old fruit to work with, and holding stock in a Cold Room rarely pays.
- These two specialities reward plans the computers don't play: buying cheap old fruit on purpose, and holding stock for a Market Day.
- [Hyp] People may use them better. The cheaper prices make them easier to try in the playtest.

4. Validation

Check	Result	Evidence
Same-seed balance comparison (PLAN-016 T1)	PASS	No figure worse by more than the margin (section 2)
`npm run verify`	PASS	Local run before merge
CI `verify:full` on `main`	Reported after the push	GitHub Actions CI

5. Options (need your decision; nothing changed yet)

1. **Keep as is and playtest.** Recommended: see whether people use Cold Room and Juice Bar before more tuning.
2. Teach the computers to plan for Market Days and to buy cheap old fruit (computer-player work, no rule change).
3. Give Juice Bar more to work with, for example +1 supply per listed fruit (a balance change; needs a plan).

REP-035 · More fruit on the farmer cards, and computers that use specialities

Game	Fruit Market (rules version 4)
Report date	2026-10-27 16:42 (Asia/Kuala_Lumpur)
Type	Balance change and computer-player change, with same-seed comparisons
Plan	PLAN-017, approved: "approve option 1 and 2" (REP-034 section 5)
Branch	`workflow/plan-017`, merged to `main` for the preview. **Live game unchanged**

Labels. [Sim] = computer players, 300 games per test, the same seeds as REP-033/034, about ±5 points. **[Hyp]** = a guess about people.

1. Changes

Option 2: more fruit (rule change). Every fruit listed on a farmer card brings 1 more. Printed values are now 3–5 (were 2–4). The unlisted fruit still comes as 1; the 2-shop and 4-shop adjustments are unchanged. `RULES.md` updated.

Option 1: smarter computers (no rule change). Normal and Hard computers now:

- with a **Juice Bar**, count the $1 tip on older fruit and the $2 juice floor, so cheap old fruit is worth buying;
- with a **Cold Room**, value stock as lasting about twice as long, and hold back on the nights their fruit doesn't age;
- stock up harder the day before a **Market Day**;
- pick the speciality that suits them (old fruit about → Juice Bar; Market Days to come → Cold Room), so plans vary across the table.

Easy computers are unchanged.

2. Results [Sim]

Measure (3 players unless noted)	Before (REP-034)	More fruit only	Both changes
Hard beats Normal (2 players)	56%	60%	**60%**
Hard beats Easy (2 players)	66%	75%	**82%**
Normal beats Easy (2 players)	56%	—	**72%**
Comeback from last at half-time	28%	37%	**38%**
Day-5 leader wins	32%	35%	31%
Blowouts (2 / 3 / 4 players)	36 / 27 / 14%	38 / 23 / 18%	42 / 23 / 15%
Big Basket trait (fair 33%)	40%	48%	**45%**
Other traits	31–34%	26–35%	27–33%

Measure (3 players unless noted)	Before (REP-034)	More fruit only	Both changes
Skip Cold Room / skip Juice Bar	32% / 35%	36% / 32%	33% / 34%
Skip Barrow	21%	21%	**16%**
Juice Bar built per player (2 players)	0.06	0.30	0.35
Sales within $1 of the top price	90%	86%	86%
Average final cash (2 players, 10 days)	$125	$162	$162

3. What this means

Good.

- Skill now matters clearly: Hard beats Normal 60% (the earlier aim was 62–65%), and Easy is a real beginner level.
- Comebacks rose to 38%, and day-5 leaders win less often. Games stay open longer.
- Customers are served more often (46% of wanted fruit, was 31%).

Not fixed: Cold Room and Juice Bar.

- The REP-033 "rush" test was misleading. A shop that rushes one speciality also skips the Barrow, and skipping the Barrow alone drops a shop to 16%.
- The fairer measure is "skip it": skipping Cold Room or Juice Bar still makes no difference (33–34%). These two are harmless but add little. Even with the smarter computers, almost no fruit spoils, so the Juice Bar has little to do.

New problem: volume wins.

- With more fruit, buying more each day is the strongest plan: Big Basket wins 45% and the Barrow is close to essential.
- 2-player blowouts rose to 42%.

4. Validation

Check	Result	Evidence
Same-seed comparisons (section 2)	PASS (run and recorded)	3 runs of the balance report, 300 games per test
`npm run verify`	PASS	Local run before merge
CI `verify:full` on `main`	Reported after the push	GitHub Actions CI
PLAN-016 CI on `main`	PASS	CI run 26 green, preview updated

5. Options (need your decision; nothing changed yet)

1. **Rein in volume.** Big Basket becomes "buy 1 extra fruit on Market Days", or the Barrow becomes "hold 12" only (no extra buy). Recommended: test both in the lab and bring you the numbers first.

2. **Give the Juice Bar and Cold Room a sharper job.** For example, Juice Bar: "buy not-so fruit at the market for $0".
3. **Playtest first** and see whether people find the same.

REP-036 · Readiness review: Fruit Market as a tabletop game and as a mobile/browser game

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 16:56 (Asia/Kuala_Lumpur)
Type	Critical readiness assessment. No game change
Request	"rate the readiness of the game ... rate it out 10, be real and critical, rate as both a tabletop and a mobile/browser games, i want to publish both version, physical copy and digital"

How to read the scores. 10 = ready to sell next to good commercial games of this type. 7 = solid, needs polish. 5 = works, but a paying customer would notice gaps. 3 or less = not started or a serious blocker. Scores are my judgement from the code, the test results and the records. **No unfamiliar player has played rules version 4 yet**, and that caps every score about fun or balance.

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1. Headline

Version	Score	One line
Digital (mobile and browser)	**5.5 / 10**	Technically strong and well tested, but the rules changed heavily today, have never been played by people, and the presentation is still small-studio prototype level
Tabletop (physical copy)	**3 / 10**	A good idea on paper, but no printed prototype, no table playtest, and one rule (nightly ageing) that is too fiddly to move by hand

Neither version is ready to publish as a product today. The digital one is much closer.

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2. Digital version

Angle	Score	What's good	What's holding it back
Code quality and cleanup	**7.5**	Strict TypeScript; 92 unit tests plus browser tests in CI; lint clean; only 7 small dependencies; game rules kept separate from the screens and deterministic (seeded)	`game.ts` (953 lines) and `App.tsx` (422) are large; four rule rewrites in two days left leftovers (sprites and names for removed upgrades, old experiments in the balance script, "market pulse" also used as the name of the demand bar); 3 type escapes for CSS variables
Network and server	**7**	Server decides everything; hidden choices stay on the server (leak checks at 0); reconnect, grace period and computer takeover are tested; rate limits and name rules	No backup of leaderboard or stats data; capacity never load-tested; Cloudflare plan unconfirmed; branch protection unconfirmed; crash recovery NOT RUN

Angle	Score	What's good	What's holding it back
Gameplay balance	**4.5**	Catch-up rule works (comebacks 38%); skill ladder now clear (Hard beats Normal 60%)	Computer-only evidence; Big Basket 45% (fair 33%); Cold Room and Juice Bar add nothing; 2-player blowouts 42%; rules changed three times today
Core fun (the hook)	**6** (unproven)	The "open stalls, then the bell" duel is a clear, dramatic moment; freshness gives every fruit a role	Computers rarely undercut (86% of sales near the top price), so the hook may feel flat against them. Only people can confirm the fun
Single-player richness	**6.5**	3 computer levels with personalities; tutorial; daily challenge; Market League; 25 achievements; cosmetics; playtest options	Little long-term goal after the league; 24 customers and 13 events can repeat within a few games; no story or campaign; computers don't bluff or show a style you can read
Multiplayer	**6**	Online rooms with invite links, quick match, pass-and-play, 2–4 players, rematch, takeover when someone leaves	Quick match needs many players online at once (it will often be empty at launch); no friends list, accounts, chat or emotes; no async (play-by-turn) mode, which suits phones
UI and UX	**6**	Phone-first; full-screen decisions; clear stall and price screens with hints; day ledger; colour-blind patterns	Very dense: a price card shows stock, buyer caps, rivals and hints at once; tutorial and How to play were rewritten today and not tested; never tested with new players
Art style	**5**	One consistent cosy pixel style; characters with costumes; day/night table	Sprites are drawn as small text maps: charming but programmer-art level beside commercial pixel games; no key art, logo or store art; little character animation
Theme and identity	**6**	A harbour market with Malaysian names is a distinctive setting	Tone is mixed (Princess, Zombie, Cowboy customers next to a real market); the name "Fruit Market" is generic: hard to search for and hard to trademark
Audio	**3.5**	Sound effects and haptics exist; music can be turned off	Sounds and the one 16-note music loop are generated tones; no recorded audio
Release and store readiness	**4**	Android build in CI; privacy policy; anonymous stats with opt-out; release process with tags	No real-phone testing; no iOS build; store screenshots out of date; no monetisation decision (paid, ads or free); no data backups
Accessibility and localisation	**5**	Colour-blind patterns; large tap targets checked	English only (Malay would suit the theme and market); no text-size option; heavy reliance on small numbers

Digital overall: 5.5 / 10.

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3. Tabletop version

Angle	Score	What's good	What's holding it back
Rules completeness	**5**	Full rules written (`RULES.md`), designed to work without an app	Written as a reference, not a rulebook: no set-up diagram, no worked example turn, no component list, no quick-reference card
Upkeep at the table	**2.5**	Simultaneous crates and prices keep downtime low	**Ageing by hand is the biggest problem.** Every night, every fruit in 4 shops plus the market moves one space: roughly 40–80 cube moves a night, about 10 times a game. Also: waiting customers with part-filled orders, Market Day and Last Call tracking, a five-step tie order
Balance for people	**3.5**	The app's computer tests give a starting point	No paper playtest has ever happened; computer results don't show how people bluff, read and undercut
Components and production	**1.5**	Card counts are known (20 farmer, 24 customer, 13 event, 12 favour, 6 trait, 4 speciality)	No prototype; no card or board art; no price dials or screens designed; no manufacturing quote, retail price or box size
Table presence and theme	**4.5**	A colourful fruit market would look great on a table	Nothing designed yet for the physical look
Play time and player count	**4**	2–4 players; 8/10/14-day lengths	Real play time unknown (my guess is 60–90 minutes for 10 days, which is long for this weight); 2 players is the weakest count (42% blowouts)

Tabletop overall: 3 / 10.

The ageing fix to test first. Replace "move every fruit each night" with **date-stamped crates**. When you buy a fruit, place it on the day track at the day it goes off (for example, an apple bought on day 3 goes on day 10). Nothing moves at night; at dawn you clear the crate for today, and freshness is read from how many days are left. This keeps the same rule and removes almost all upkeep. It needs your approval as a rule change.

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4. The biggest risks, in order

1. **Nobody has played rules version 4.** Every balance and fun score here is a guess until real people play it.
2. **The rules keep moving.** Three changes today. A game needs a frozen rule set to playtest, print and sell.
3. **Tabletop upkeep (nightly ageing)** would put off physical players.
4. **Balance gaps:** Big Basket, dead specialities, 2-player blowouts.
5. **Commercial basics are missing:** name or trademark check, store art, iOS, monetisation, a backup plan.
6. **Online population:** quick match will be empty at launch without an async mode or friends invites.

---

5. Recommended path to publish both

Step	What	Why
1	**Freeze the rules** for a playtest round; fix only bugs	Stable target for testing
2	**Digital playtest:** 5 or more new players on the preview; watch, don't help	The only way to learn whether the bell is fun
3	**Paper prototype** with date-stamped crates; play at least 5 times	Proves the tabletop works and gives balance data from people
4	**Balance pass** from both playtests (Big Basket, specialities, 2 players)	Fix what people actually hit
5	**Name, art and audio direction:** trademark search, key art, card frames, recorded sound	Needed for the store page and the box
6	**Digital launch prep:** real-phone tests, backups, monetisation decision, store assets, then a soft launch	Release-ready evidence
7	**Tabletop production prep:** rulebook with examples, component list, print-and-play version, manufacturing quote	Before any crowdfunding or print run

A sensible order is **digital first**. It is the cheapest way to test the rules and gather players, and the app can later teach the physical game.

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6. Evidence used

- Code: 2,097 lines of rules, 3,542 lines of screens, 994 lines of server, 1,408 lines of tests (`main` at `e7e5132`).
- Checks: `verify` and `verify:full` PASS (REP-033 to REP-035); online leak checks 0.
- Balance: REP-033 to REP-035, 300 computer games per test.
- Records: `STATE.md` open items (backups, branch protection, real phones, playtests NOT RUN).

REP-037 · Proposal: data insights for balance, retention and online health

Game	Fruit Market
Report date	2026-10-27 17:02 (Asia/Kuala_Lumpur)
Type	Plan proposal (data handling change). **Nothing built yet**
Request	"can we track this please, for balance changes, user retention etc, generally data insights"
Proposed plan	PLAN-018 (awaiting approval)

1. What we have today

- Anonymous daily totals, no IDs, opt-out in Settings, kept 90 days (`docs/ANALYTICS.md`):
  - app opens, installs per week and return days (retention);
  - games started and finished, days reached and quit days;
  - wins and losses, and final cash buckets;
  - tutorial steps, online room creation and errors.
- **Gaps:**
  - nothing about balance (traits, specialities, computer levels, prices, spoilage);
  - counts don't say which rules version a game used;
  - nobody reads them regularly;
  - this session can't read them (no key, server not reachable from here).

2. Proposal

a. New anonymous counters (same privacy rules: totals only, no IDs, opt-out respected)

Question it answers	Counters (each tagged with the rules version)
Which set-ups do people play?	length, players, computer levels, options (Core, Street prices, Market Days, pulse)
Is a trait too strong?	your trait, and whether you won with it
Are specialities worth building?	specialities built (and on which day), and wins with each
Are computer levels right?	wins and losses against Easy, Normal and Hard
Is the price duel working?	per game: share of sales near the customer's top price (bucket); fruit spoiled (bucket); final winning margin (bucket)
Which favour cards matter?	favour cards played
Is online healthy?	matches finished vs abandoned; computer takeovers; quick match found vs gave up, and wait time (bucket)

Question it answers	Counters (each tagged with the rules version)
Did a release change retention?	app version on each app start

b. Weekly insights report

- A GitHub Action reads the live and preview stats with your `STATS_KEY` (stored as a GitHub secret), once a week and on demand.
- It builds a one-page report:
  - return rates on day 1, 7 and 30;
  - first-game and tutorial funnel, and where people quit;
  - win rates by trait, speciality and computer level, each with its sample size;
  - balance compared with the computer-player simulations;
  - errors.
- Each balance change is then judged on real players by rules version, not only on simulations.

c. Paperwork

- `docs/ANALYTICS.md` and `docs/privacy.html` list every new counter.
- Tests prove nothing identifying can be sent and that opt-out sends nothing.

3. Cost and risk

- **Money:** no new service or dependency. More counter writes on Cloudflare, still small (batched). Whether that stays free depends on your Cloudflare plan, which is still unconfirmed.
- **Privacy:** unchanged in kind (anonymous totals). Small groups can still be revealing, so the report hides any number under 5.
- **Limit:** counters can show *what* happens, not *why* or whether it's fun. Watched playtests are still needed.
- **Live game:** the counters only reach live players after the next release (separate publish approval).

4. Questions for you

1. **Where should the weekly report go?** (a) A private GitHub Actions download, recommended; (b) saved into the repo's docs folder, visible to anyone with repo access.
2. **How long to keep the data?** 90 days today. For trends across releases, 365 days is better (still anonymous).
3. **Will you add `STATS_KEY` as a GitHub secret?** I'll give step-by-step instructions; without it, the report can't run automatically.

REP-038 · Owner decisions after the lifecycle trial: freeze, approvals, one game in two editions

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 19:45 (Asia/Kuala_Lumpur)
Type	Decision record and plan proposal. **No game change**
Request	"1a 2a 3 tabletop which can be translated into the game at the same time, both with the same mechanics but the digital version to have single player capability"
Proposed plan	PLAN-019 (awaiting approval)

STATUS

Stage: complete digital game; no unfamiliar-player evidence for rules version 4. Gate view (from the trial): G0 PASS, G1 CONCERNS, G2 CONCERNS, G3 NOT ASSESSED, G4 FAIL, G5 NOT ASSESSED, G6 not started. Branch `workflow/decisions-020`, records only. Nothing is pushed.

SUMMARY

You chose to stop changing the rules and test with people first, approved the four open process decisions, and set the direction: one game, two editions, same mechanics. My recommendation is to approve PLAN-019, a playtest with no game code.

PLAN OR WORK COMPLETED

Recorded as approved (your words in each record):

- DEC-001 adopt the workflow; DEC-004 PDF method; DEC-005 budget read as RM500 per month across all games; DEC-014 use the Concept and Theme Companion. If you disagree with any after reading it, say so; approvals are never rewritten, a correction is a new line.
- **DEC-026** freeze rules version 4 until the first human playtest is done. Allowed during the freeze: fixes for bugs that stop play, lose data or break security, plus docs and playtest materials.
- **DEC-027** tabletop and digital are developed together with identical mechanics. The tabletop rules are the master. The digital game adds single-player capability (computer shops; League and daily challenge as modes) but no digital-only mechanics. Interpretation to confirm: single-player already exists; the requirement is that it follows the tabletop rules.

Proposed, not approved: PLAN-019 First human playtest (no game code): 3 to 5 unfamiliar players on one digital Quick game each, one paper round of the tabletop rules, a report per session, then your continue/revise/stop decision.

What DEC-027 changes in practice

- A rule is only good if it works by hand. Nightly ageing is already flagged as too fiddly (REP-036); the date-stamped crates idea is still undecided.
- Digital-only extras such as bluffing computers, accounts or cosmetics stay modes or presentation, never rules.
- The paper round is now evidence for both editions.

VALIDATION

Check	Result	Evidence
Records check	PASS	`RECORDS OK: 27 decisions, 19 packages, 2 releases; history checked against HEAD`. Run with a temporary Windows-safe copy of `scripts/records.mjs`: the repo script fails on Windows paths (`C:\C:\...`). It works on Linux, so CI is unaffected. The temporary copy was deleted; the repo script is unchanged.
Game checks (`verify`, `verify:full`)	NOT RUN	No game code changed
Playtest	NOT RUN	PLAN-019 not approved

CHANGELOG

Added: none for players. Changed: decision records and one plan record. Player impact: none. State: proposed, unreleased.

RISKS AND COSTS

- Physical-edition costs are unknown: printing, components, manufacturing, shipping, storage. A dated quote and a separate approval are needed before any spend.
- Open questions for you: print-and-play first or a manufactured box; who the strangers will be; how nightly ageing is handled in the paper round.
- A hand-playable rule set may limit what the digital game can do. Review if a rule that makes the game fun cannot be done by hand.
- Playtesters are hard to find; 3 to 5 sessions give early signals only, not proof of demand.
- Cost of this report: RM0.

NEXT ACTION OR APPROVAL

One decision: approve PLAN-019, revise it, or reject it. After that: write the hypotheses and criteria before any session, then recruit and run.

DOCUMENT RECORD

Report REP-038. Decisions: DEC-001, DEC-004, DEC-005, DEC-014 approved; DEC-026, DEC-027 approved; PLAN-019 proposed. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** the export needs pandoc and the Playwright package, and neither is installed on this Windows machine

(`Cannot find module 'playwright'` ; pandoc not found). The source report and the regenerated register are saved. Recovery: install the repo dependencies (`npm ci`) and pandoc, then run `npm run records` . Installing needs your approval. No earlier PDF was replaced.

REP-039 · PLAN-019 approved: first human playtest

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 19:59 (Asia/Kuala_Lumpur)
Type	Approval record. **No game change**
Request	"Approve PLAN-019 only."
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 is approved and in progress. Rules version 4 stays frozen (DEC-026). No testing has happened yet. Branch `workflow/decisions-020`, records only, not pushed.

SUMMARY

You approved the first human playtest: 3 to 5 unfamiliar players on the digital game plus one paper round of the tabletop rules. The approval covers PLAN-019 and its listed documentation only. Nothing is approved for game changes, spending or publishing.

PLAN OR WORK COMPLETED

Done: approval recorded in `PLAN-019.json` with your exact words; history line added. Next, all documentation or playtest material:

- You answer three questions: who the testers are, how nightly ageing works in the paper round (recommended: let testers try the simplest way and record what happens), and where sessions run.
- You confirm or edit the draft questions and the continue/revise/stop criteria (in the plain-English plan). They are dated and fixed before the first session (T1).
- I prepare the printable session sheet and a paper kit from `RULES.md`.

VALIDATION

Check	Result	Evidence
T1 questions and criteria fixed before first session	NOT RUN	Awaiting your confirmation
T2 sessions saved with tester count and build	NOT RUN	No session yet
T3 no change to game files or RULES.md during testing	NOT RUN	Check at the end
T4 owner decision recorded	NOT RUN	After the sessions
Records check	See DOCUMENT RECORD	

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

Cost: RM0 expected. Finding strangers is the slow part. 3 to 5 players show early confusion, not demand. Printing a paper kit, if you want one, needs a quote first.

NEXT ACTION OR APPROVAL

Your three answers and your confirmation of the draft criteria. No further approval is needed for the playtest itself.

DOCUMENT RECORD

Report REP-039. PLAN-019 approved (evidence: "Approve PLAN-019 only.", 2026-10-27), delivery in progress. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine. Recovery: install the repo dependencies and pandoc, then run `npm run records`. Installing needs your approval.

REP-040 · PLAN-019 amended: the "paper round" is a table round on the current app

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 20:05 (Asia/Kuala_Lumpur)
Type	Plan scope amendment. **No game change**
Request	"do my poc for this with the app, that would be the 'paper round'" ; answer to the options put to the owner: "as it is right now"
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 stays approved and in progress. Scope changed by the owner's instruction. Rules version 4 frozen (DEC-026). No testing yet.

SUMMARY

The owner wants the paper round done with the app, as it is now. I read "as it is right now" as: play the current app as the table, with no new app mode and no paper kit. PLAN-019's paper round is replaced by one pass-and-play table round with 2 to 4 people on the current app.

PLAN OR WORK COMPLETED

- PLAN-019 scope item and acceptance T2 amended in `PLAN-019.json`; history line added.
- Options considered: (a) worked example from the engine; (b) play the app as the table (taken); (c) build a table-helper mode, which is a code change and would need its own plan.
- Plain-English plan updated; a one-page sheet for the app-as-table round added.

VALIDATION

Check	Result	Evidence
Records check	See DOCUMENT RECORD	
T1 to T4	NOT RUN	No sessions yet

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

- **What this no longer tests:** whether the tabletop rules work by hand. The app moves the fruit for people, so nightly ageing by hand, the premise of DEC-027, stays unproven. It needs a later plan once the digital sessions show the game is worth taking further.
- A table round on one device shows how groups read the reveal and the bell; it does not show how long physical set-up takes.
- Cost: RM0.

NEXT ACTION OR APPROVAL

No new approval needed: the owner's instruction amends the scope. Still needed from the owner: who the testers are, where sessions run, and confirmation of the draft questions and criteria.

DOCUMENT RECORD

Report REP-040. PLAN-019 approved (unchanged evidence), scope amended 2026-10-27 20:05. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-041 · PLAN-019 amended again: testers play from the browser link

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 20:06 (Asia/Kuala_Lumpur)
Type	Plan scope amendment. **No game change**
Request	"no, i'll get people to play it from the browser link"
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 stays approved and in progress. Rules version 4 frozen (DEC-026). No testing yet.

SUMMARY

The owner will send people a browser link; they play on their own devices. The earlier table round (REP-040) is dropped. Testers use the preview copy, which runs rules version 4.

PLAN OR WORK COMPLETED

- PLAN-019 scope and acceptance T2 amended in `PLAN-019.json`; history line added.
- A message to send testers, with the link and a short set of questions, is prepared as a document for the owner to send. Claude does not contact anyone.

VALIDATION

Check	Result	Evidence
Records check	See DOCUMENT RECORD	
T1 to T4	NOT RUN	No sessions yet

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

- **Unobserved play.** Without watching, confusion is only visible through what testers write or say. Self-reported answers are weaker evidence. Option: join a short call or ask for a screen recording from willing testers.
- **Preview, not live.** Testers must use the preview address (rules version 4). The live game is still 1.1.1 and would test the wrong rules. The preview is not advertised; sharing the link is the owner's choice, and it has its own rooms and stats.

- **Hand-played ageing and the tabletop edition (DEC-027) remain untested**; they need a later plan.
- Cost: RM0.

NEXT ACTION OR APPROVAL

No new approval needed. The owner confirms the draft questions and criteria, then sends the link and questions. The sessions are counted when answers come back.

DOCUMENT RECORD

Report REP-041. PLAN-019 approved (unchanged evidence), scope amended 2026-10-27 20:06. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-042 · PLAN-019: questions and criteria locked

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 20:09 (Asia/Kuala_Lumpur)
Type	Test criteria locked before testing. **No game change**
Request	"The draft questions and criteria are fine. Lock them in as of today."
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 in progress. Acceptance T1 passed: the questions and criteria are fixed before any tester has the link. No tester has played yet. Rules version 4 frozen (DEC-026).

SUMMARY

These are the exact questions and thresholds the results will be judged against. They are project targets set by the owner, not industry benchmarks. 3 to 5 testers show early confusion; they do not prove retention or demand.

PLAN OR WORK COMPLETED

Criteria (locked 2026-10-27 20:09)

Question	Continue if	Revise if	Stop or park if
Can a new player say why they lost a sale by day 3? (DEC-016)	At least 3 of 4 can	2 of 4 can	0 or 1 of 4 can
Do they finish their first Quick game without help?	At least 3 of 4 finish	Most finish but ask for help	Most give up
Would they play again without being asked?	At least half say yes or start another game	Mixed	Almost nobody

Tester questions (locked)

1. In one sentence: what is this game about?
2. Think about day 3. A customer didn't buy from you. Why do you think that happened?
3. When your fruit got old, what happened to it and its price?
4. What was the most confusing moment? Which screen or day was it?
5. What did you enjoy most? What annoyed you most?
6. Did you finish the game? If you stopped, on which day and why?
7. Would you play again? Why or why not?
8. Anything that looked broken or too small to read? (Phone model and browser, if known)

Testers play the preview from a browser link, one Quick game against one computer shop, without help.

Rule for changes: criteria and questions are not changed once results are read. A change is recorded before any result is read, with the reason.

Open point for the owner (not yet decided): the thresholds say "x of 4". If 3 or 5 people test, this is the proposed reading: Continue needs at least three quarters (3 of 3, or 4 of 5); Revise needs at least half (2 of 3, or 3 of 5); anything lower is Stop or park. The owner confirms or changes this before the results are read.

VALIDATION

Check	Result	Evidence
T1 questions and criteria written and dated before the first tester	PASS	This report, 2026-10-27 20:09; no tester has been sent the link
T2 sessions saved	NOT RUN	No sessions yet
T3 no change to game files or RULES.md	NOT RUN	Check at the end
T4 owner decision recorded	NOT RUN	After the sessions

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

Unobserved remote play is weaker evidence than watching. Self-reported answers can miss confusion. A small sample cannot show demand. Hand-played ageing (DEC-027) is untested. Cost: RM0.

NEXT ACTION OR APPROVAL

The owner sends the link and questions to 3 to 5 unfamiliar players and passes the answers back, one tester at a time, with no names. The owner confirms the "x of 4" reading before results are read.

DOCUMENT RECORD

Report REP-042. PLAN-019 approved (unchanged evidence); T1 PASS. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-043 · PLAN-019: how to read the criteria with 3 or 5 testers

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 20:12 (Asia/Kuala_Lumpur)
Type	Test criteria clarified before testing. **No game change**
Request	"Yes, use the proposed reading for 3 or 5 testers."
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 in progress. No tester has played yet. Rules version 4 frozen (DEC-026). Locked criteria are in REP-042.

SUMMARY

The owner confirmed how the "x of 4" thresholds apply when 3 or 5 people test. This was decided before any result.

PLAN OR WORK COMPLETED

Testers	Continue if	Revise if	Stop or park if
3	3 of 3	2 of 3	0 or 1 of 3
4	3 of 4	2 of 4	0 or 1 of 4 (as locked in REP-042)
5	4 of 5	3 of 5	0, 1 or 2 of 5

Applies to the two count-based questions (explain a lost sale by day 3; finish the first Quick game). The "would they play again" question keeps its wording: at least half say yes or start another game / mixed / almost nobody.

VALIDATION

Check	Result	Evidence
T1 questions and criteria fixed before testing	PASS	REP-042; this report, before any tester
T2 to T4	NOT RUN	No sessions yet

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

Same as REP-042: unobserved remote play is weaker evidence; a small sample shows confusion, not demand; hand-played ageing (DEC-027) is untested. Cost: RM0.

NEXT ACTION OR APPROVAL

The owner sends the preview link and the questions to 3 to 5 unfamiliar players and passes each tester's answers back, one at a time, with no names.

DOCUMENT RECORD

Report REP-043. PLAN-019 approved (unchanged evidence). Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-044 · PLAN-019: simpler tester questions and a 1-vs-3 game

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 21:18 (Asia/Kuala_Lumpur)
Type	Test design change before any tester played. **No game change**
Request	"i feel like the questions need to be simpler to answer, and any particular reason we are asking for 1v1? make it 1v3 if possible"
Plan	PLAN-019 (approved, in progress)

STATUS

PLAN-019 in progress. No tester has played or answered yet, so this change is made before any result (the rule from REP-042). Rules version 4 frozen (DEC-026). The locked criteria (REP-042, REP-043) are unchanged.

SUMMARY

Testers now play one Quick game against **three** computer shops (four shops in all) and answer **seven** simpler questions: four tap-to-answer and three short optional ones.

PLAN OR WORK COMPLETED

Why it was 1 vs 1. DEC-016 (2026-10-26) set "one computer shop" to keep a first game short and the cause of a lost sale easy to read. It was a test-design choice, not a rule. The app supports 1 to 3 rival shops (new-game screen, `src/ui/App.tsx`), each set to Computer or Friend.

Trade-off of 1 vs 3 (the owner chose 1 vs 3). More like the intended 3 to 4 player game and the tabletop edition; more demand pressure and more to read on screen, so a lost sale may be harder to explain. Four shops also change supply (farmer cards add 1 of each fruit) and walk-ins per day.

New tester questions (live at the form link)

1. Did you finish the game? Yes, on my own / Yes, but I needed help / No, I stopped early
2. Day 3: a customer didn't buy from you. Do you know why? Yes, I knew exactly why / I had a rough idea / No idea
3. (optional) If you knew or had an idea, what was the reason?
4. How easy was it to understand what to do? Very easy / Okay / Confusing
5. Would you play again? Yes / Maybe / No
6. (optional) What was the most confusing moment, if any? (the day or screen)
7. (optional) Anything broken or hard to read? Your phone and browser, if you know.

How they map to the locked criteria

- Explain a lost sale by day 3: question 2 says Yes or rough idea AND the reason in question 3 is a correct one (judged against what the game shows). "No idea", or a wrong reason, counts as cannot explain.
- Finish the first game without help: question 1, "Yes, on my own".
- Play again: question 5 (at least half "Yes", with "Maybe" not counted as yes).
- Questions 4, 6 and 7 are supporting evidence only.

Setup shown to testers: New game, Rival shops 3 (all Computer), Quick, Open the shops. Other settings left as they are.

VALIDATION

Check	Result	Evidence
New form live and readable	PASS	Public link read after publishing; seven questions, four required
1 vs 3 possible in the app	PASS (by code)	`src/ui/App.tsx` new-game screen allows 1 to 3 rivals; not tried on screen
T1 criteria unchanged	PASS	REP-042, REP-043
T2 to T4	NOT RUN	No sessions yet

CHANGELOG

Added: none for players. Changed: test design and tester form only. State: unreleased.

RISKS AND COSTS

Tap answers are easier but give less detail. The reason question (3) carries the evidence for the main criterion and is optional, so some "Yes" answers will have no reason and will not count as explained. Three computers may undercut little (REP-036), which could make the game feel easier. Unobserved remote play remains weaker evidence than watching. Hand-played ageing (DEC-027) is untested. Cost: RM0 (Tally free plan, fair use unverified).

NEXT ACTION OR APPROVAL

The owner sends the preview link and the form link to 3 to 5 unfamiliar players and passes the answers back, one tester at a time, with no names.

DOCUMENT RECORD

Report REP-044. PLAN-019 approved (unchanged evidence). Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-045 · Informal feedback from a 4-player game

Game	Fruit Market (build not stated)
Report date	2026-10-27 22:18 (Asia/Kuala_Lumpur)
Type	Feedback log. **No decision, no game change**
Request	"log this feedback"
Plan	PLAN-019 in progress; this is **not** counted as a PLAN-019 session

STATUS

PLAN-019 in progress; rules version 4 frozen (DEC-026). No game change.

SUMMARY

One 4-player game produced five points: one is rules or balance evidence (supply), one is about session length, and three are UI requests. Nothing here is built or decided. Rules and balance stay frozen; UI items wait until the playtest is finished (PLAN-019 T3 says no change to game files between the first and last session).

PLAN OR WORK COMPLETED

Feedback as received (verbatim, no name)

- "supply is so weird, it feels like mid-game onwards, supply cant keep up with demand and players ended up with nothing to buy"
- "player Icon is the same for everyone, vary this character/icon choice"
- "10 rounds feels long (took around 30mins)"
- "no undo button after clicking a fruit on the board before confirming buy"
- "cant edit game name in the room, and dont default name to "you" make it blank and pop up the keyboard"

What we do not know: who played (unfamiliar or not), how many were human, whether it was online or pass-and-play, which build (preview or live), the exact settings, or the day supply first ran short. The game keeps no per-game log, and the server was not readable from here, so nothing could be checked.

Sorted

#	Point	Kind	Under the freeze	Suggested follow-up
1	Supply cannot keep up with demand from mid-game; players left with nothing to buy	Rules or balance evidence	Frozen. Evidence only	Ask for screenshots of the day ledger and the day supply ran short. Compare with the 4-player supply rule (4 shops add 1 of each fruit to farmer cards, RULES.md) and the PLAN-017 farmer-card change. Decide after the playtest.

#	Point	Kind	Under the freeze	Suggested follow-up
2	10 rounds took about 30 minutes	Session length	Frozen if it means changing the rules; the tester guidance is not frozen	Brief assumed 10 to 20 minutes (GAME-BRIEF, hypothesis). Time a Quick (8 day) 4-shop game. Check the tester message wording.
3	Same icon for everyone	UI	Backlog, after the playtest	Owner to say what variety is wanted (icon, character, colour)
4	No undo after tapping a fruit before confirming a buy	UI	Backlog, after the playtest	Small UI plan; check it does not change what a legal action is
5	Cannot edit game name in the room; name defaults to "You"; wants a blank field that opens the keyboard	UI	Backlog, after the playtest	Small UI plan; online room name handling needs a look at `server/room.ts` before scoping

VALIDATION

Check	Result	Evidence
Feedback recorded without a name or personal data	PASS	This report
Any item verified in the game	NOT RUN	No build, settings or log available
Counts toward PLAN-019 T2	No	Setup differs (4 players, 10 rounds); tester status unknown

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

One game is one data point, and it may have been played by someone who knows the game. The supply complaint is the one worth chasing, but it needs the ledger, the settings and the build before anyone treats it as a balance finding. Cost: RM0.

NEXT ACTION OR APPROVAL

Optional, from the owner: screenshots of the end-of-game results and the day ledgers, the settings used, and the link used. No approval is needed for this report. UI backlog items need a plan and approval after the playtest.

DOCUMENT RECORD

Report REP-045. No decision record. Register: `docs/workflow/DECISIONS.md` (history line added). **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-046 · PLAN-019 ended early by the owner

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 22:20 (Asia/Kuala_Lumpur)
Type	Plan closure record. **No game change**
Request	"the ones i entered here is the feedback, just end the testing"
Plan	PLAN-019 (approved; ended early)

STATUS

PLAN-019 ended early at the owner's instruction. The playtest goal was **not met**. Rules version 4 remains unplayed by any recorded unfamiliar tester. Branch `workflow/decisions-020`, records only; one commit is local and not pushed.

SUMMARY

No formal session was run. The Tally form had no submissions when checked. The only feedback is the informal 4-player game (REP-045), which does not match the plan's setup. So there is no continue, revise or stop reading for the game, and the criteria locked in REP-042 and REP-043 were never applied.

PLAN OR WORK COMPLETED

Check	Result	Evidence
T1 questions and criteria locked before testing	PASS	REP-042, REP-043
T2 3 to 5 remote sessions with saved reports	NOT RUN	None run; no form submissions
T3 no change to game files	PASS	git diff of src, server, RULES.md and package.json between origin/main and this branch is empty
T4 owner continue, revise or stop decision	NOT RUN	Owner ended the plan; no game-level decision made

What the evidence does and does not give. One informal game suggests, with low confidence, three things to look at: supply running short from mid-game in a 4-player game, session length (about 30 minutes for 10 rounds), and four small UI requests (REP-045). It does not show whether new players understand the game, finish it, or want to play again.

Record keeping. The records have no "ended early" delivery value, so delivery is shown as implemented. That wording does not mean the goal was met. This gap is noted for the guides.

VALIDATION

Check	Result	Evidence
Records check	See DOCUMENT RECORD	Run after this report
Game checks	NOT RUN	No code changed
G1 core proof (gate view)	Still CONCERNS	No unfamiliar-player evidence for rules version 4

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

Without human evidence, any further rule or balance work rests on computer-player simulations alone, which was the original problem. The supply complaint and the length complaint are single-source. The Tally form is still published and anyone with its link can still answer; it can be unpublished or deleted by the owner. Cost: RM0.

NEXT ACTION OR APPROVAL

One decision for the owner: DEC-026 (rules freeze) was to last until the first playtest round was recorded. That round did not happen. Keep the freeze, or lift it so that supply, session length and the UI requests can be planned? Either way, any change needs its own plan and approval. DEC-016 (first playtest) stays not started.

DOCUMENT RECORD

Report REP-046. PLAN-019 approval unchanged (owner, 2026-10-27); acceptance T1 PASS, T3 PASS, T2 and T4 NOT RUN. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-047 · Rules freeze lifted; four fix plans proposed

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 22:24 (Asia/Kuala_Lumpur)
Type	Decision record and plan proposals. **Nothing built or run**
Request	"Lift the rules freeze (DEC-026). Plan fixes for supply, session length and the UI requests."
Proposed plans	PLAN-020, PLAN-021, PLAN-022, PLAN-023 (all awaiting approval)

STATUS

Freeze lifted: DEC-026 superseded by DEC-028 (approved). No plan is approved, so no game code may change yet. Analysis is from reading the code and rules only. No simulation or build was run, because that needs the repo's dependencies installed, which needs your approval. Branch `workflow/decisions-020`, records only, not pushed.

SUMMARY

Four small plans, each answering one point from the informal feedback (REP-045). Recommended order: PLAN-022 first (client-only, low risk), then PLAN-020 step 1 (measure supply), then PLAN-021, then PLAN-023.

PLAN OR WORK COMPLETED

Supply (PLAN-020). Evidence from code, not from play:

- No lorry comes on a Market Day (days 3, 6, 9, 12; `game.ts`, `RULES.md`). In a 10-day game that is 3 days with no new fruit.
- With 4 shops, every fruit gets +1 (`printedSupply`). A card brings about 17 fruits (three listed fruits at 3 to 5 plus 1 each, and the fourth fruit at 1, all +1). Four shops can each buy up to 4 fruits a day, so 16.
- Fruit also spoils overnight, so leftovers do not carry the market through a no-lorry day.
- This fits the complaint but is a hypothesis. The first step is to measure it by simulation: per day and per fruit, how much is on the market when each shop buys.
- Options for step 2, to choose after measuring: a smaller delivery on Market Days, or more supply for 4 shops. Either must work by hand with cards (DEC-027), so a simple printed number is preferred over a new card rule.
- Needs your target first: the largest share of shop turns that may find nothing to buy.

Session length (PLAN-021).

- The new-game screen defaults to 10 days (Standard); Quick is 8 (`App.tsx`). The 30-minute game used 10 rounds with 4 players.
- First steps with no rule change: default to Quick for 3 or 4 shops, and make waiting skippable or faster (lorry, reveal, computer turns), without hiding information.

- Time a Quick 4-shop game with a stopwatch before and after. Proposed target: 20 minutes or less (yours to set).
- Shortening the number of days is a rules change and stays out of this plan.

UI requests, client only (PLAN-022).

- Same icon: the default avatar is the first one unless a player picks another in the online lobby (`Online.tsx`). Proposed: each seat gets the first unused icon.
- Name: it falls back to "You" (`store.ts`). Proposed: blank field, focused, keyboard opens, with a fallback if left empty.

Undo and rename in the room (PLAN-023).

- Tapping a fruit sends the buy at once (`useGameController.ts`). Undo is a new legal action; online it must be checked by the server, tested for duplicates, stale actions and reconnect, and must not leak hidden information.
- Renaming inside a room needs a server message. Step 1 is a design note for your approval; no code until then.

VALIDATION

Check	Result	Evidence
Supply cause confirmed	NOT RUN	Code reading only; no simulation or play data
Session length measured	NOT RUN	One informal report (about 30 minutes); no stopwatch test
UI items checked on screen	NOT RUN	Code reading only
Records check	See DOCUMENT RECORD	

CHANGELOG

Added: none for players. Changed: records only. State: unreleased and proposed.

RISKS AND COSTS

- **No human playtest** stands behind any rule change. Simulations use computer players, which was the original problem (REP-036). The owner accepted this when lifting the freeze (DEC-028).
- The supply fix changes the rules version. The tabletop must be updated with it (DEC-027).
- Undo online is the riskiest item because it touches server authority.
- Running the simulations needs `npm ci` in the local clone: installs the repo's own dependencies from the lockfile. Needs your approval. PDF export also needs pandoc.
- Cost: RM0 expected.

NEXT ACTION OR APPROVAL

Decide which plans to approve, one at a time. My recommendation: approve PLAN-022 now, and approve PLAN-020 step 1 together with permission to run `npm ci` in the local clone. Set the supply target and the time target before the changes are chosen.

DOCUMENT RECORD

Report REP-047. DEC-028 approved (evidence: "Lift the rules freeze (DEC-026). Plan fixes for supply, session length and the UI requests."); DEC-026 superseded; PLAN-020 to PLAN-023 proposed. Register: `docs/workflow/DECISIONS.md` . **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-048 · Correction to REP-047: undo already exists offline

Game	Fruit Market
Report date	2026-10-27 22:30 (Asia/Kuala_Lumpur)
Type	Correction. **No game change**
Corrects	REP-047 (reports are not rewritten; this is the correction)
Plans	PLAN-021, PLAN-023 amended while still proposed

STATUS

No plan approved. No game change. Branch `workflow/decisions-020`, records only, not pushed.

SUMMARY

REP-047 said tapping a fruit commits a buy and that undo would be a new action. Reading the code afterwards shows an undo button already exists for one-device games and is hidden in online rooms. So the request is really "undo in online rooms". The feedback also mentions "the room", so the informal game was probably an online game with human players (inferred, not confirmed).

PLAN OR WORK COMPLETED

- Code: `ActionBar.tsx` shows the undo button (↶) only when the game is not online and a purchase has been made; `turn.ts undoPoint` keeps the state where the current person's buying turn began.
- PLAN-023 retitled "Undo a buy in online rooms, and rename inside the room". One-device undo stays unchanged.
- PLAN-021 timing now covers both a 1-human and 3-computer game and a 4-human online room, because waiting for people adds time.
- Feedback in REP-045 is now read as probably online: it explains the same default icon (the online lobby default), the name field in the room, and the extra length.

VALIDATION

Check	Result	Evidence
Undo hidden online	PASS (by code)	`ActionBar.tsx` line 20: shown only when not online
Feedback game was online	NOT RUN	Inferred from the word "room" only
Records check	See DOCUMENT RECORD	

CHANGELOG

Added: none for players. Changed: records only. State: unreleased and proposed.

RISKS AND COSTS

If the feedback game was not online, the undo point may be a different problem (for example, the button was disabled or not noticed). Confirm with the tester or on screen before building. Cost: RM0.

NEXT ACTION OR APPROVAL

No new approval. Same recommendation as REP-047: approve PLAN-022 first, then PLAN-020 step 1.

DOCUMENT RECORD

Report REP-048. No decision record. Register: `docs/workflow/DECISIONS.md` (history correction line added). **PDF EXPORT BLOCKED:** pandoc and the Playwright package are not installed on this machine; recovery: install the repo dependencies and pandoc, then run `npm run records` (needs owner approval).

REP-049 · Approvals, and PLAN-020 step 1: supply and customer demand measured

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 23:20 (Asia/Kuala_Lumpur)
Type	Approval record and measurement report. **No game change**
Request	"do all except 22, for 22 analyse demand as well, feels like theres too much customer towards the final few turns"
Plans	PLAN-020, PLAN-021, PLAN-023 approved and in progress; PLAN-022 not approved

STATUS

PLAN-020 step 1 (measurement) done. Steps that change rules, speed or the server wait for your choice. The game files are unchanged (checked against git). Branch `workflow/decisions-020`, records only, not pushed.

SUMMARY

Both your impressions are confirmed in simulation: supply runs short and customers pile up late in the game. The biggest single cause is the Market Days with no lorry (days 3, 6, 9). Putting the lorry back on Market Days is the biggest single fix in the tests. Results come from computer players only.

PLAN OR WORK COMPLETED

How I read your message. "all except 22" = approve PLAN-020, 021 and 023, not 022. "for 22 analyse demand" = PLAN-020, which now also analyses demand. If that is wrong, say so.

Method. I installed the repo's dependencies from the lockfile in the local clone (no new dependency) and ran simulated games through the real engine: 300 games per setting for the baseline, 200 for each what-if, all computer shops at Normal, Standard 10 days with events, specialities and unripe fruit on. Per day, the script recorded the fruit on the market when each shop's buying turn starts, the customers, the fruit they wanted and the fruit sold. The what-ifs were temporary edits to a copy of the rules in the working folder, reverted afterwards; nothing was committed. Script and raw results are saved in the owner's Guides folder, `supply_demand_measurement`.

Baseline (today's rules, 10 days)

Shops	First 3 days: customers/day	Last 3 days: customers/day	Fill rate first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
2	2.1	4.9	57% / 36%	4% / 32%	0% / 10%

Shops	First 3 days: customers/day	Last 3 days: customers/day	Fill rate first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
3	3.5	6.9	52% / 32%	9% / 36%	3% / 17%
4	4.9	8.9	52% / 30%	12% / 35%	5% / 22%

(Fill rate = fruit sold divided by fruit the customers wanted.)

What this shows (4 shops)

- **Demand nearly doubles.** Customers per day go from 4.9 to 8.9; the fruit they want goes from about 19 to about 35.
- **Fruit sold per day stays flat** at about 14 from day 4 on. Supply does not grow with demand.
- **Market Days starve the market.** On days 3, 6 and 9 shops find an empty market on 15%, 41% and 49% of their turns (units on the market at a shop's turn start fall to 6.0, 3.5 and 2.7, against 13 to 18 on other days). Fill rate on day 9 is 12%.
- **Why demand grows:** one extra walk-in in the second half, one extra on each Market Day, and unserved customers coming back the next day. About 7 to 8 customers a day go unserved late in the game.
- **Quick (8 days, 4 shops):** the same pattern: last 3 days 8.7 customers a day, 33% fill, 33% of turns short, 21% empty.

What-ifs (4 shops, 10 days, last 3 days)

Variant	Customers/day	Fruit wanted/day	Fill rate	Turns short	Turns with an empty market	Empty on days 3 / 6 / 9
Today	8.9	35.5	30%	35%	22%	15% / 41% / 49%
A: lorry also comes on Market Days	7.7	28.4	59%	8%	2%	1% / 2% / 2%
D: +1 of each fruit in the second half	8.7	34.0	37%	25%	17%	15% / 41% / 39%
C: no extra walk-in in the second half	6.8	25.8	41%	31%	17%	15% / 37% / 38%
A + D	7.1	26.4	71%	2%	0%	1% / 2% / 0%
A + C	5.4	20.0	75%	2%	0%	1% / 2% / 0%

Reading the what-ifs

- **A alone fixes most of the supply problem** and also lowers late demand (7.7 vs 8.9), because fewer customers are left waiting. It is the simplest to play by hand: a farmer card every day.
- **D and C each help only a little** because they leave the Market Day starvation in place.
- **A + C** makes late-game demand level with early-game (5.4 vs 5.0 customers a day) and gives the highest fill rate; it removes one extra customer per day in the second half, a printed number.

- **Untested idea:** a half delivery on Market Days (keeps some scarcity). It needs an extra rule, so I did not test it.
- **Trade-off for A:** Market Days lose their "nothing arrives today" tension and keep only the extra walk-in and Last Call. That is a design call for you.

VALIDATION

Check	Result	Evidence
PLAN-020 T1 measurement report	PASS	This report; 2, 3 and 4 shops, 300 games each; what-ifs 200 games
PLAN-020 T2 owner target set	NOT RUN	Awaiting the owner
Game files unchanged	PASS	`git status` shows no change to src, server or RULES.md; temporary edits reverted (line endings only, restored with git)
Human evidence	NOT RUN	Computer players only; all Normal; no humans
`npm run verify`	NOT RUN	No game change to verify

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

- **Computer players only.** They rarely undercut and may buy differently from people, so absolute numbers may not match human games. The direction is clear; the size is uncertain.
- **One configuration.** All Normal computers, events on. Easy or Hard computers, or the daily challenge and League settings, were not measured.
- Any change bumps the rules version and must be mirrored on the tabletop (DEC-027). A and C are simple printed rules; D needs a "from day 6" note on the table.
- Cost: RM0. The dependency install is on the C: drive, outside Drive.

NEXT ACTION OR APPROVAL

Decisions for you (PLAN-020 step 2):

1. Which option: A (recommended first), A + C, A + D, or something else such as a half delivery.
2. Your target before building, for example: "no more than 10% of shop turns find fewer fruit than their buy limit, in any 3-day stretch, with 4 shops". A meets it at 8%. Then step 3 (build, bump the rules version, update RULES.md and tests) can start.

PLAN-021 (session length): read-only findings so far, no build yet. Computer players wait 380 ms per action (150 ms with the existing Fast setting), which adds up to roughly 10 to 15 seconds a day with 3 computers; human thinking time is the bigger part. I propose: default length Quick for 3 to 4

shops, turn the existing fast setting on by default, then time a game with a stopwatch before and after. Needs your time target (proposed 20 minutes or less).

PLAN-023 (online undo and rename): design note ready for your approval (Design 1 recommended: server-side undo of the current turn; rename in the lobby only).

DOCUMENT RECORD

Report REP-049. Plans PLAN-020, PLAN-021 and PLAN-023 approved (evidence: "do all except 22, for 22 analyse demand as well, feels like theres too much customer towards the final few turns"), in progress; PLAN-022 proposed, not approved. PLAN-020 T1 PASS. Register: `docs/workflow/DECISIONS.md` . **PDF EXPORT BLOCKED:** pandoc is not installed on this machine (the repo's dependencies now are); recovery: install pandoc, then run `npm run records` (needs owner approval).

REP-050 · PLAN-020 built: rules version 5 (lorry every day, no second-half walk-in)

Game	Fruit Market (rules version 5 on local branch `workflow/plan-020`; preview and live unchanged)
Report date	2026-10-27 23:35 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change, local only, not pushed, not on the preview**
Request	"why is the number in decimal? remember that rules should be able to be implemented on a tabletop elsewhere, else do plan a + c, target is so that the board would not be empty / ppl cant buy anything"
Plan	PLAN-020 (approved; implemented; one check NOT RUN)

STATUS

Rules version 5 is built on local branch `workflow/plan-020`, two commits on top of the records branch. Not pushed, so CI has not run and the preview is unchanged. Lint, types, 92 unit tests and the build pass. The browser tests (`verify:full`) did not run.

SUMMARY

Option A + C is in: the lorry comes every day, and walk-ins are players minus 2 every day (plus 1 on a Market Day). In simulation the market is no longer empty (0.0% to 1.0% of shop turns, down from up to 22%), and late-game demand is level with early-game. Both rules are printed numbers, playable by hand, so the tabletop master (`RULES.md`) is updated to match.

PLAN OR WORK COMPLETED

How I read the message. "else do plan a + c" = go with A + C. "target is so that the board would not be empty" = the target is that shops can always buy something, measured as the share of shop turns that find an empty market. "Why the number in decimal" is answered below.

Why the numbers had decimals. They are averages over hundreds of simulated games. A single game always has whole numbers (for example 5 customers on a day); the average of 300 games is 5.3. The percentages are shares of shop turns.

Changes

- `src/engine/game.ts`: the lorry card is drawn every day; walk-ins are players minus 2, plus 1 on a Market Day, plus any event.
- `src/engine/ai.ts`: computer shops no longer treat a Market Day as a no-lorry shortage (one fixed weight for scarce fruit).
- `src/net/protocol.ts`: rules version 5 (old saves will not resume; daily boards are kept per version; online players need the same version).
- Screens: the lorry shows on Market Days, the strip says "Market Day: one more walk-in", and How to play, set-up and hint text are updated.

- `RULES.md` (tabletop master) updated: lorry every day, walk-ins every day, version 5, balance note.
- Tests updated: walk-ins, Market Day delivery, demand staying level, version 5.

Tabletop check (DEC-027). The farmer card is flipped every day, as it already was on other days. Walk-ins are a printed number: players minus 2. Nothing new to move or count.

Results after the change (computer players, all Normal, same seeds 1 to 300)

Shops	Customers/day, first 3 / last 3 days	Fruit sold vs wanted, first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
2	2.1 / 2.2	63% / 86%	0.5% / 0.1%	0.1% / 0.0%
3	3.5 / 3.7	60% / 79%	1.4% / 0.6%	0.3% / 0.0%
4	4.9 / 5.3	62% / 75%	1.5% / 2.4%	0.6% / 0.2%
4 (Quick, 8 days)	4.9 / 5.5	62% / 73%	1.5% / 4.2%	0.6% / 1.0%

Before the change (4 shops, 10 days): last 3 days had customers 8.9, fill 30%, turns short 35%, empty-market turns 22%. The highest empty-market rate on any single day after the change is 1.6% (4 shops, day 7).

VALIDATION

Check	Result	Evidence
Owner target: market not empty	PASS (simulation)	0.0% to 1.0% of shop turns find an empty market; computer players only
Lint	PASS	`npx eslint .` no output
Type checks (app and server)	PASS	`tsc --noEmit` and `tsc -p server --noEmit`
Unit tests	PASS	92 of 92 (`npx vitest run`)
Build	PASS	`npm run build`
Records check	PASS	Run with a patched copy; the repo script fails on Windows paths
Browser tests (`verify:full`)	NOT RUN	Needs Playwright browsers; about 15 minutes
Human playtest of rules 5	NOT RUN	None; computer players only
Tabletop paper test	NOT RUN	Not done; hand-played ageing is still unproven

CHANGELOG

Added: none. Changed: lorry every day; no extra walk-in in the second half; computer shops stop treating Market Days as a shortage; text updated; rules version 5. Removed: the no-lorry Market

Day. Player impact: the market no longer runs empty mid-game and the last days are no longer flooded. State: implemented, unreleased, local only.

RISKS AND COSTS

- **2-shop games are now quiet:** about 2 customers a day all game (they were 4.9 in the last days before). The market is well supplied (fill 86%), so shops sell most of what they put out but earn less per day. Worth a look in a human game; a 2-shop tweak could come later.
- **Market Days lost their tension.** They are now only "one more walk-in" plus the speciality draft afterwards.
- **Computer players only.** Humans may buy and price differently.
- Not pushed, so CI has not checked it and the preview does not show it yet.
- Old saves cannot be resumed and the daily leaderboard restarts for version 5; online rooms need matching versions.
- Cost: RM0.

NEXT ACTION OR APPROVAL

The owner decides whether to push `workflow/plan-020` to GitHub (CI runs; main stays untouched), and later whether to merge it to `main` (the preview updates after CI passes). Also open: PLAN-021 time target, PLAN-023 design choice, and whether to approve PLAN-022.

DOCUMENT RECORD

Report REP-050. DEC-029 approved (owner words recorded); PLAN-020 implemented, T1 and T2 PASS, T3 NOT RUN (browser tests), T4 NOT RUN. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc is not installed on this machine (the repo's dependencies now are); recovery: install pandoc, then run `npm run records` (needs owner approval).

REP-051 · PLAN-024: tabletop fixes built (tie-break and table procedures)

Game	Fruit Market (rules version 5 on local branch `workflow/plan-024`; preview and live unchanged)
Report date	2026-10-28 00:28 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change, local only, not pushed**
Request	"implement all the fix except for the price ladder, i dont get the problem and your solution"
Plan	PLAN-024 (approved; implemented; two checks NOT RUN)

STATUS

Built on local branch `workflow/plan-024`, on top of the PLAN-020 branch. Not pushed, so CI has not run. Lint, types, 92 unit tests and the build pass. Browser tests and a hand-played tabletop test did not run. Still rules version 5, because version 5 is not released.

SUMMARY

One rule line changed: a price tie, after Favoured Trade, Haggler and fresher fruit, now goes to whoever bought earlier today instead of the poorer shop by cash at that moment. The other fixes are table procedures, written into `RULES.md` as "Playing on a table". The price ladder and the colour zones that depend on it were left out, as asked.

PLAN OR WORK COMPLETED

How I read the request. "implement all the fix except for the price ladder" = build the tie-break and write the table procedures into the rules, without the ladder (fix 1) or the colour zones (fix 5, which sit on the ladder). A reader announcing which freshness qualifies at each price replaces the zones. If you wanted printed art, that is a separate step.

In the code and the app

- `src/engine/game.ts`: the poorer-shop step is gone from the tie-break; the order is now Favoured Trade, Haggler, fresher fruit, today's buying order.
- `src/engine/story.ts` and `src/ui/HowTo.tsx`: the explanations say "bought earlier today".
- `tests/rules4.test.ts`: the tie test now checks that the earlier buyer wins whatever the cash is.

In the rules (`RULES.md`, the tabletop master): the tie line, and a new "Playing on a table" section: customer order, the reader (announces the real top price and which freshness qualifies, and pays from the bank), order circles, the sole-supplier cube, the dawn line-up, evening for card holders only, day chips with age cards for blueberry, watermelon, apple and mango (with Cold Room, Fridge and Heatwave handled), and a waiting pile.

Why the tie-break is safe. Same 300 games with and without the change, computer players only: average final cash per shop 4 shops $146.9 to $146.3, 2 shops $145.9 to $145.9. Individual games differ because one different tie changes later choices; that does not show a balance change.

VALIDATION

Check	Result	Evidence
Lint	PASS	`npx eslint .`
Type checks	PASS	app and server
Unit tests	PASS	92 of 92
Build	PASS	`npm run build`
Records check	PASS	Run with a patched copy of the records script (the repo script fails on Windows paths)
Browser tests (`verify:full`)	NOT RUN	Needs Playwright browsers
Hand-played tabletop test	NOT RUN	Day chips are the least proven part

CHANGELOG

Changed: a price tie after fresher fruit goes to whoever bought earlier today. Added: the table procedures in the rules. Player impact: tiny; ties after freshness no longer favour the poorer shop. State: implemented, unreleased, local only.

RISKS AND COSTS

- The day-chip method moves complexity from moving pieces to reading ages; it needs a hand-played test.
- Not pushed, so CI has not checked it. Cost: RM0.

NEXT ACTION OR APPROVAL

The owner decides whether to push `workflow/plan-024`, and later whether to merge. A printable kit (customer cards with circles, age cards, mats, chips) and a hand-played test would be a new plan.

DOCUMENT RECORD

Report REP-051. DEC-030 approved; PLAN-024 implemented, T1 to T3 PASS, T4 and T5 NOT RUN. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc is not installed on this machine; recovery: install pandoc, then run `npm run records` (needs owner approval).

REP-052 · PLAN-025: character cards built

Game	Fruit Market (rules version 5; preview and live unchanged until merged and deployed)
Report date	2026-10-28 00:46 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change**
Request	"make it so that character is selected when you enter the room instead apply all the changes and push it the game"
Plan	PLAN-025 (approved; implemented; two checks NOT RUN)

STATUS

Built on branch `workflow/plan-025`, on top of the PLAN-020 and PLAN-024 branches. Lint, types, 100 unit tests and the build pass. Browser tests did not run locally (they need Playwright browsers); CI runs them on push. The new screens were not looked at on a phone.

SUMMARY

Each shop now picks one of six characters, each with one fixed perk, when it enters an online room or on the set-up screen. No two shops can be the same character, so every seat has a different icon. Computer shops take the characters people did not pick. The first-morning "deal 3, keep 1" trait pick is gone for characters.

PLAN OR WORK COMPLETED

How I read the request. "character is selected when you enter the room instead" = pick a character on entering a room (and on the local set-up screen), replacing the dealt traits. "apply all the changes and push it" = commit everything, push the branch so CI runs, and merge to `main` (the preview) if CI passes. I do not deploy the live game: that needs the owner to run Deploy LIVE game and type PUBLISH.

Characters and perks: Foxy Rosa Haggler, Big Benny Big Basket, Rocky Bargain Hunter, Patches Charmer, Pinky Local Favourite, Uncle Toad Nest Egg.

Code: `data.ts` (character map and list), `game.ts` (a character's perk applies at the start), `protocol.ts` and `client.ts` (a `character` message), `server/room.ts` (unique characters on joining, switching in the lobby, computer shops take the rest), a new `CharacterPick` screen part used in the room entry, the lobby and the set-up screen, and the set-up logic in `App.tsx`. `RULES.md` has a Characters section; How to play text updated.

Unchanged: the daily challenge and league still let the person pick a trait; their computer rivals now carry their animal's perk.

VALIDATION

Check	Result	Evidence
Lint, types	PASS	`eslint`, `tsc` for app and server
Unit tests	PASS	100 of 100 (8 new: 5 character, 3 room)
Build	PASS	`npm run build`
Records check	PASS	Run with a patched copy of the records script
New screens on a phone	NOT RUN	Not looked at
Browser tests (`verify:full`)	See CI	Run by GitHub after the push

CHANGELOG

Added: character cards. Changed: pick a character on entering a room or the set-up screen; computer shops take the rest. Fixed: every player no longer starts with the same icon. Removed: the first-morning trait pick for characters. Player impact: a new pick step before the game. State: implemented, unreleased.

RISKS AND COSTS

- The room entry and set-up screens are longer (six character cards each; a friend seat adds six more). Not checked for look or tap size on a phone.
- A taken character on entering a room is resolved by the server giving a free one; the player can change it in the lobby.
- Old and new clients cannot meet: rules version 5 is unreleased, so no live client is affected.
- Cost: RM0.

NEXT ACTION OR APPROVAL

The owner looks at the preview once CI passes and the preview updates, then decides on the live deploy (Deploy LIVE game, type PUBLISH).

DOCUMENT RECORD

Report REP-052. DEC-031 approved; PLAN-025 implemented, T1 to T3 PASS, T4 and T5 NOT RUN. Register: `docs/workflow/DECISIONS.md`. **PDF EXPORT BLOCKED:** pandoc is not installed on this machine; recovery: install pandoc, then run `npm run records` (needs owner approval).

REP-053 · PLAN-026: customer deck grows to 36 cards

Game	Fruit Market (rules version 5, unreleased; local branch `workflow/plan-026`, not pushed)
Report date	2026-10-28 20:42 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change**
Request	"expand customer card to 36, keep the demand profile and design language in line with the rest of the game, regenerate the Character Sheet together with the demand per card, i want to see how it looks on the physical copy"
Plan	PLAN-026 (approved; implemented; four checks NOT RUN)

STATUS

Built on local branch `workflow/plan-026` (commit 7a4473d). Lint, types, 100 unit tests and the build pass. Browser tests, a look on a phone and a print of the cards did not run. Not pushed.

SUMMARY

Twelve new customers join the original 24. They match the old cards: 4.3 pieces and $31.8 per card on average against 4.4 and $31.9, the same fruit shares and the same mix of card sizes and fruits per card. The original 24 are untouched. The character sheet (Characters_Gallery.html) has all 36, a demand table, and printable card fronts and backs.

CHANGES

`data.ts` (twelve cards, their titles and quirks), `pixel.tsx` (twelve portraits from the shared face template and parts), `RULES.md` (36 cards), one test made independent of the dealt traits (the new deck order changed which traits the seed dealt). Card backs show the fruits wanted, as before; a simulation of blank backs against fruit backs (REP-054 appendix files in card_backs) found no change in cash or spoilage and a slightly narrower gap between winner and last with fruit backs.

NOT DONE OR NOT VERIFIED

Browser tests; the portraits on a phone; a print of the cards.

REP-054 · Art direction: Look C chosen; three plans proposed

Game	Fruit Market (no game change)
Report date	2026-10-28 20:42 (Asia/Kuala_Lumpur)
Type	Decision and planning. **No game change**
Request	Art style audit, a board that looks the same in the game and on a real table, an animated market opening, then a comparison of a single coherent look against the owner's flat harbour picture. Owner answer: "Look C"
Plans	PLAN-027, PLAN-028, PLAN-029 (all proposed)

STATUS

Art direction decided (DEC-033). Nothing built. Three plans wait for approval.

SUMMARY

The art audit (Art_Style_Audit) found the game mixes pixel sprites, flat vector, CSS shapes and one smoothed raster lorry with no shared rule. The owner compared pixel everywhere (A), flat vector everywhere (B) and calm flat scenery with pixel objects (C), and chose C. The Farmers Market board will be drawn once and used for the screen and the printed table mat; league stages can later be skins of it. A concept prototype of an animated market opening exists.

PLANS, IN ORDER

1. PLAN-027 style guide and one palette (small).
2. PLAN-028 the shared board, Village Square skin only (medium; touches the market, animation anchors and the printed set).
3. PLAN-029 animated opening (small to medium; shares sprites with PLAN-028).

DECISIONS STILL OPEN

The lorry (redraw as flat scenery or keep the picture), the printed board size (A3 landscape is the working assumption; needs a print), the table token size, and whether to do the other league skins later.

FILES

Guides_v2/trial_FruitMarket: Art_Style_Audit, Art_Style_Reference.html, Market_Opening_Prototype.html, Look_Comparison.html, Characters_Gallery.html, art_reference/ and supply_demand_measurement/card_backs/.

REP-055 · PLAN-028 first step: printable board prototype

Game	Fruit Market (no game change)
Report date	2026-10-28 20:49 (Asia/Kuala_Lumpur)
Type	Plan step. **No game change**
Request	"Board first" (approve PLAN-028; printed A3 prototype before the screen)
Plan	PLAN-028 (approved; in progress; print and table checks NOT RUN)

STATUS

A printable prototype exists: one A3 landscape mat and three A4 pages (two token sheets and a "what to test" page). A PDF render shows 4 pages at the right sizes (A3 landscape then A4 portrait). It has not been printed on paper. The screen market is not started.

SUMMARY

The mat is the Look C board: flat, smooth frame, awning, sign and slots, with the pixel fruit inside. Four tracks, one per fruit (3, 6, 7 and 6 slots, newest at the top). Every slot has the printed price from the game's cost table, a freshness band, a stage word, a "nights left" label and a faint fruit; gold slots are where the lorry delivers. The footer has a lorry, spots for the farmer and event cards (lying down) and a legend. Fruit tokens are 20 mm, 30 of each fruit.

EVIDENCE: HOW MANY TOKENS

Computer-player games (200 per row, events on, 10 days). Most of one fruit ever in play (market and all shelves): 2 shops blueberry 14, watermelon 21, apple 17, mango 18; 3 shops 20, 23, 20, 22; 4 shops 27, 25, 23, 23. One market slot held up to 14, so tokens stack. Thirty of each fruit covers the highest figure with a margin. People may hold more than computers do; the playtest will tell.

THINGS TO DECIDE OR TEST

1. **Night handling:** the prototype slides each pile down a slot at night. The current table rules use a fruit clock and day chips instead (nothing moves). Test which is quicker with 4 shops. This changes the "Playing on a table" text if the board wins.
2. **Stage on tokens:** the token always shows the fresh picture and the slot gives the stage; on the screen the picture dulls. Decide if table tokens should change.
3. **Shelves:** each shop's own stock also needs a track (smaller). Not on this board yet.
4. **Print:** A3 at 100%, background graphics on; slot 22 mm; the apple column (7 slots) fits with a 7 mm margin.
5. **Fonts:** the file loads Pixelify Sans and Fredoka online; print with internet on.

FILES

Guides_v2/trial_FruitMarket: Board_Prototype_A3.html and Board_Prototype_A3.pdf; generator in art_reference/make_board_prototype.ts.

REP-056 · Board look options (decide the look first)

Game	Fruit Market (no game change)
Report date	2026-10-28 21:16 (Asia/Kuala_Lumpur)
Type	Design options for decision. **No game change**
Request	"i want to decide on the board look first, then design the game back because that the proof of concept"
Decision	DEC-034 (proposed); plan PLAN-030 (proposed)

STATUS

Correction: "Board first" was read as approval to build the board, and PLAN-028 was recorded as approved with a printed prototype made (REP-055). The owner meant: choose the board's look first, then design the rest of the game back from it. PLAN-028 is marked superseded (its recorded approval is kept as it was) and the work continues as PLAN-030, proposed. Nothing is built; the printable prototype remains as one candidate.

SUMMARY

Four looks for the same board (same tracks, slots, prices and pixel fruit), each with the kit it would give the rest of the game (buttons, cash chip, customer order, event card, portrait): A stall front, B chalkboard, C crate rack, D soft tabletop mat. File: Board_Look_Options.html.

RECOMMENDATION

D: calmest, lightest to print, cheapest to build and easiest to read on a small screen. Its weak point is less market character, which the home scene, street and lorry can carry. A is the safe familiar choice, C the most charming and most work, B the best fit with the dark screen UI.

NEXT, ONCE CHOSEN

Write the board's palette, corner, line and button rules as the style guide (PLAN-027 becomes "derived from the chosen board"), then redraw the home scene, street and lorry in that language, then build the screen and table versions (PLAN-030).

REP-057 · Naming adopted

Game	Fruit Market (no game change)
Report date	2026-10-28 21:36 (Asia/Kuala_Lumpur)
Type	Decision. **No game change**
Request	"approve everything", after the naming audit
Decision and plan	DEC-035 and PLAN-031 (approved; not started)

STATUS

All names in Naming_Audit are adopted, including the optional ones. Nothing is renamed yet: the rename is planned for after the board look is chosen.

NOT READ AS APPROVAL OF

The board look (DEC-034), PLAN-027 (style guide), PLAN-029 (animated opening) and PLAN-030 (shared board). They stay proposed. If "everything" was meant to include them, say so and they will be recorded.

REP-058 · Recap items done: DEC-020 closed, PLAN-022 built, PLAN-003 split

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-022, not pushed)
Report date	2026-10-28 22:23 (Asia/Kuala_Lumpur)
Type	Plan implementation and records. **Game change** (name field only)
Request	"do per your suggestion"
Plans	PLAN-022 (approved; implemented; keyboard on phones and browser tests NOT RUN), PLAN-032 and PLAN-003 (proposed)

STATUS

PLAN-022 is built: lint, types, 105 unit tests and the build pass. It is on local branch workflow/plan-022 (on top of workflow/plan-026), not pushed. The keyboard behaviour was not tried on a phone.

WHAT CHANGED

The name field on the set-up screen and the online entry starts blank and takes focus. If left blank the game uses "You" (one device) or "Player" (online) but does not save that as the player's chosen name. Older saves that stored those fallbacks start blank. Code: store.ts, App.tsx, Online.tsx, tests/name.test.ts.

RECORDS

DEC-020 closed as superseded by DEC-033. PLAN-003 split: PLAN-032 (free drills) and PLAN-003 (paid load test and cost record, no spend until you set a cap; the Cloudflare plan is still unknown). PLAN-018 left proposed on purpose.

NOT DONE OR NOT VERIFIED

Keyboard on Android Chrome and iPhone Safari; browser tests (CI); nothing pushed.

REP-059 · Board look D chosen; art plans approved; style guide drafted

Game	Fruit Market (no game change)
Report date	2026-10-28 22:28 (Asia/Kuala_Lumpur)
Type	Decision and plan step. **No game change**
Request	"approve everything above"
Decision and plans	DEC-034 (approved: D), PLAN-027, PLAN-029, PLAN-030 (approved; PLAN-027 in progress), PLAN-033 (proposed)

STATUS

Board look D is chosen. The style guide for it is drafted (docs/ART-STYLE.md) and waits for your approval of its content before any code changes. Nothing in the game changed.

ORDER OF WORK

1. PLAN-027: guide, one palette, whole-number sprite sizes, no emoji.
2. PLAN-029: animated opening.
3. PLAN-030: shared board (a printed mat is tried on a table before the screen market is rebuilt).

A GAP

No approved plan restyles the other screens to the new look (buttons, chips, cards, panels, the dark information layer). Without it the game is half old and half new. PLAN-033 is proposed to cover it.

OPEN

The lorry style, the printed board size, token stage and table nights (PLAN-030 decides them).

REP-060 · PLAN-027 built: one palette, no emoji, size checks

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-027, not pushed)
Report date	2026-10-28 22:39 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (small visible change: replaced emoji)
Request	"resolve the gap, and approve plan 27"
Plans	PLAN-027 (approved; implemented; phone look and browser tests NOT RUN), PLAN-033 (approved; not started)

STATUS

Built on local branch workflow/plan-027 on top of workflow/plan-022. Lint, types, 111 unit tests and the build pass. Browser tests and a look on a phone did not run. Not pushed.

WHAT CHANGED

- **One palette.** src/ui/palette.ts names every colour the interface and sprites use (197). styles.css reads them as variables from the generated palette.css. The move is lossless: putting the palette values back gives the old files exactly. The old colours are kept as c_xxxxxx entries; PLAN-033 merges them into the guide roles. A test fails on any colour written outside the palette.
- **No emoji.** Seat buttons (Computer, Friend), the waiting, special, heard and happy marks on customers, the Favoured Trade card and several labels now use text or pixel icons (new hourglass and ear icons). A few plain glyphs stay (the close, tick, arrow and menu marks). A test enforces this.
- **Sizes.** The allowed sprite sizes are defined and a test blocks any new off-list size. The 29 existing off-list sizes stay until PLAN-033, because changing a size changes the layout.
- **Guide.** docs/ART-STYLE.md approved and updated.

NOT DONE OR NOT VERIFIED

Browser tests (CI); the six replaced marks on a phone; fixing the existing sprite sizes (PLAN-033 T5); merging the older colours (PLAN-033 T6).

REP-061 · Mak Jenab replaces Uncle Toad

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 22:48 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change**
Request	"replace one of the character with Jenab (add any suitable attributes for her name) jenab should be a lady"
Plan	PLAN-034 (approved; implemented; phone look and browser tests NOT RUN)

STATUS

Built on local branch workflow/plan-034 (on top of workflow/plan-027). Lint, types and the affected tests pass. Not pushed.

WHAT CHANGED

Uncle Toad is replaced by **Mak Jenab**. She keeps the Nest Egg perk (start with $15 instead of $10): a thrifty stallholder who saves every sen. Her portrait is a human with a purple headscarf (tudung) and a teal baju, in the same pixel style. Her computer-shop lines are warm and thrifty. The other five characters and all perks are unchanged. The frog sprite is removed.

WHY HER

The owner left the choice of character and attributes open. The saver perk fits her, and keeping an already balanced perk avoids a rebalance.

NOT DONE OR NOT VERIFIED

Her portrait in the character pick and in a game on a phone; the full browser tests; the printed customer sheet and the character gallery still show the old six.

REP-062 · Jenab is a customer; character theming has no plan yet

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 22:54 (Asia/Kuala_Lumpur)
Type	Correction, plan implementation and a new proposal. **Game change**
Request	"sorry jenab is a customer" and "which plan will cover for all the character theming? design coherence"
Plans	PLAN-035 (approved; implemented; phone look and browser tests NOT RUN), PLAN-036 (proposed)

STATUS

The shopkeeper change was wrong. It is reverted: Uncle Toad is back as a shopkeeper, and DEC-036 and PLAN-034 are closed as superseded. Jenab is now a customer: **Auntie Jenab** replaces Chris the Clown and keeps his order, so demand does not change. Lint, types and 111 unit tests pass. Not pushed.

WHAT SHE IS

A lady customer with a purple headscarf (tudung) and a teal baju. Quirk: "Sniffs every mango twice, then asks for a discount anyway" (about what she does, as the cast rules require). Her order is mango-heavy, which fits. The character sheet (Characters_Gallery.html) is regenerated with her.

CHARACTER THEMING

No plan covered it. PLAN-027 and PLAN-033 handle palette, sizes and screens; the naming plan handles words. PLAN-036 is proposed to own portrait style, costume and theme rules, the diversity pass, and keeping the screen, cards and gallery the same.

NOT DONE OR NOT VERIFIED

Her card on a phone and on paper; browser tests.

REP-063 · Rara Jenab; cast coherence audit; PLAN-036 started

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 23:03 (Asia/Kuala_Lumpur)
Type	Plan implementation and design work. **Game change** (one customer's name and look)
Request	"change it to Rara Jenab, design it around the name"; the cast must be all human under one coherent theme; "approve the above and then work on plan 36"
Plans and decisions	PLAN-035 (approved; implemented), PLAN-036 (approved; in progress), DEC-038 (approved), DEC-039 and DEC-040 (proposed)

STATUS

Rara Jenab is built (lint, types and 111 unit tests pass; not pushed; phone look and browser tests NOT RUN). PLAN-036 is started as design work: the cast audit is done. No art or code changes for the cast happen until the owner decides the theme and the shopkeepers.

RARA JENAB

"Rara" gives the design: a cheerful "ra-ra" cheerleader (and in Javanese "rara" means a young lady). Purple headscarf, teal baju, gold-and-red pom-poms. Quirk: "Cheers every mango she buys." She keeps Chris's order, so demand is unchanged.

THE MIX-UP

Three kinds of cast sit under the one word "characters", drawn three ways: six animal shopkeepers with perks, four human start icons, and 39 customers (10 not human, 8 in costume, 21 everyday). See Cast_Coherence_Audit.html.

DECISIONS FOR THE OWNER

DEC-039 the cast theme (T1 townsfolk, recommended; T2 fancy dress); DEC-040 shopkeepers become humans (replaces the animal rivals rule); the portrait and naming rules in the creative bible (PLAN-036 T3); optional expression frames; the order of work (after PLAN-033, before PLAN-029).

ALREADY APPROVED, NOTHING NEW TO RECORD

PLAN-027 (built), PLAN-033 (restyle screens) and PLAN-031 (rename).

REP-064 · One human cast: T1 and human shopkeepers (first pass)

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 23:14 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (names, titles, quirks, portraits; no rules or perks)
Request	"t1 and proceed with dec040"
Plan	PLAN-036 (approved; in progress), DEC-039 and DEC-040 (approved)

STATUS

First pass built. Lint, types, 115 unit tests and the build pass. Browser tests, a phone look and a print did not run. Not pushed. The portraits are a first pass from the existing pixel parts; see Cast_After_T1.html.

WHAT CHANGED

- **Shopkeepers are human:** Sharp Rosa (Haggler), Big Benny (Big Basket), Charming Chen (Charmer), Thrifty Saleh (Nest Egg), Lucky Raj (Bargain Hunter), Kind Kiran (Local Favourite). Perks unchanged. New computer-shop lines without animal puns. The old names stay reserved so nobody can use them as a player name. The internal avatar keys are unchanged so saves and rooms keep working.
- **Customers follow T1:** the 10 non-humans and 8 costumes became everyday roles (Night Baker Robert, Diver Mei Ling, Magician Kumar, Paleontologist Hafiz, Florist Nurul, Pony Rider Sarah, Sleepy Zul, Engineer Vijay, Ranger Hui Min, Herbalist Mira, Rancher John, Scientist Aisyah, Courier Farid, Wedding Planner Siti, Sailor Jia Wen, Fencer Yuki, Delivery Rider Ravi, Chemist Daniel). Orders are unchanged, so demand is unchanged.
- **Creative bible:** a draft "Cast coherence" section (theme, naming, portrait rules tied to the style guide). DEC-018's animal rule is replaced.
- **Checks:** tests/cast.test.ts.

NOT DONE OR NOT VERIFIED

The owner's approval of the bible rules (PLAN-036 T3); portrait refinement (several faces with glasses read as sunglasses); the character pick and printed cards in the app and on paper; phone look; browser tests; the fox mascot on the home screen (PLAN-029); a check that every quirk says what someone does.

REP-065 · Cast review changes

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 23:24 (Asia/Kuala_Lumpur)
Type	Plan step. **Game change** (names, titles, quirks, portraits)
Request	"remove the manner nickname for shopkeepers, remove "cheerleader" from cheerleader Rara, replace the wedding planner, lumberjack, night baker, pony rider"
Plans	PLAN-036 (in progress), PLAN-035

STATUS

Done. Lint, types and 115 unit tests pass. Not pushed; phone look and browser tests NOT RUN.

WHAT CHANGED

- **Shopkeepers:** plain first names: Rosa, Benny, Chen, Saleh, Raj, Kiran. Perks unchanged. The rules, the daily-challenge line, the creative bible and the cast test follow.
- **Rara Jenab:** the title is just her name. The pom-poms and the quirk ("Cheers every mango she buys.") stay.
- **Four roles replaced** (names and orders kept, so demand is unchanged): Tailor Siti ("Measures every watermelon with her tape"), Fisherman Marcus ("Sells fish at dawn, buys apples by noon"), Librarian Robert ("Reads the label on every blueberry"), Photographer Sarah ("Photographs every mango before she buys it"). New portraits.
- **Side effect:** common first names like Raj and Chen are no longer reserved as player names, so real players can use them. The old animal-era names stay reserved.

STILL OPEN

The cast rules approval, the glasses that read as sunglasses, the phone look and the printed cards, browser tests.

REP-066 · Cast rules approved

Game	Fruit Market (rules version 5, unreleased; local branch workflow/plan-034, not pushed)
Report date	2026-10-28 23:26 (Asia/Kuala_Lumpur)
Type	Decision and a small art fix. **Game change** (one sprite part colour)
Request	"Approve rules"
Plan and decision	PLAN-036 (in progress), DEC-041 (approved)

STATUS

The Cast coherence rules in the creative bible are approved. The glasses on portraits now have a brown frame so they read as spectacles and not sunglasses (Rockstar Grace keeps real shades). Lint, types, unit tests and the build pass. Not pushed.

STILL OPEN IN PLAN-036

- T5: a check that every quirk says what someone does (today's check only blocks creature words and the shopkeeper name pattern).
- T6: the screen, the printed cards, the character pick and the gallery draw from the same data; only the sheet has been looked at.
- T7: phone look. T8: browser tests (CI).
- Optional happy and sad frames, not decided.

REP-067 · Pushed to the preview

Game	Fruit Market (rules version 5; preview updated after CI on main; live unchanged at 1.1.1)
Report date	2026-10-28 23:44 (Asia/Kuala_Lumpur)
Type	Delivery. **Game change** (preview)
Request	"Push to preview"

STATUS

CI on branch workflow/plan-034 passed in full, browser tests included (run 40716448881). The branch is merged to main by fast-forward; the preview updates after CI passes on main. The live game is not deployed: that needs you to run Deploy LIVE game and type PUBLISH.

WHAT IS ON THE PREVIEW

36 customers (DEC-032); blank name field that opens the keyboard (PLAN-022); one palette, no emoji, new size and emoji checks (PLAN-027); Rara Jenab (PLAN-035); an all-human cast, six human shopkeepers with plain first names and the approved cast rules (PLAN-036). Also the records for the art, naming and board decisions (text only).

NOT VERIFIED

Phone look of the new cast, the replaced marks and the name field keyboard (iPhone Safari may only open it from a tap); printed cards.

REP-068 · PLAN-023 built: online undo and lobby rename

Game	Fruit Market (rules version 5, message version 2; local branch workflow/plan-023, not pushed)
Report date	2026-10-29 00:18 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (online rooms; server authority)
Request	"ok" to Design 1 and to renaming in the lobby only
Plan	PLAN-023 (approved; implemented; browser tests and phones NOT RUN)

STATUS

Built. Lint, types, 122 unit tests (7 new) and the build pass. Not pushed, so the browser tests have not run.

HOW IT WORKS

- **Undo (Design 1).** The server saves the state a person's own buying turn began in (never sent to anyone). A new undo request is accepted only from the active player, in the buying phase, if something changed this turn, and on the current version. It puts the shelf, cash and market back, tells everyone (" took back their purchases"), and can be used again after buying again. A repeated request does nothing. It is cleared when a computer plays the seat, the turn ends or a new match starts. The button now shows online too, enabled once you have bought something.
- **Rename.** A rename request is accepted in the lobby only, with the same cleaning as joining; a refused or empty name keeps the old one. In the lobby your own name is an editable box.
- **Message version.** Phones and server now check a message version as well as the rules version, so an older build gets the update message instead of joining. The rules version did not change, so daily boards are not reset.

NOT DONE OR NOT VERIFIED

Browser tests (CI); two real phones in a room (how the market jumping back looks to the other player, the undo button, the name box); iPhone keyboard behaviour in the name box.

REP-069 · Approvals after the review list

Game	Fruit Market (no game change in this report)
Report date	2026-10-29 00:33 (Asia/Kuala_Lumpur)
Type	Decision. **No game change**
Request	"approve everything except plan 21, for the design related, require my approval on the design, else just run"
Decision	DEC-042 (approved)

WHAT WAS APPROVED

The order PLAN-031, PLAN-030, PLAN-033, PLAN-029; the lorry redrawn flat and smooth; no portrait frames; PLAN-032 and the PLAN-036 quirk check; and the rule that design work waits for the owner's approval of the design while everything else is just built.

NOT APPROVED

PLAN-021 (session length): the time target is still open. PLAN-003 (paid) and PLAN-018 (data counters) were not part of this approval.

WHAT RUNS NOW

The rename pass (words only), the PLAN-036 quirk check and the PLAN-032 drills. Design proposals for the board, the lorry, the restyle and the opening come to the owner as mock-ups first.

REP-070 · Rename pass, quirk check and free drills

Game	Fruit Market (rules version 5, message version 2; local branch workflow/plan-031, not pushed)
Report date	2026-10-29 00:42 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (words only)
Request	"approve everything except plan 21, for the design related, require my approval on the design, else just run"
Plans	PLAN-031 (approved; implemented), PLAN-036 T5, PLAN-032 (approved; in progress)

STATUS

Built. Lint, types, 126 unit tests (4 new) and the build pass. Not pushed, so the browser tests have not run. No rule or balance changed.

RENAME

The names in Naming_Audit are applied in the rules and the app: Wholesale Market, stall, shopkeeper, Tired, piece, Stocking, The rush, Start buying, Start game, Delivery card, Gossip, Tip Jar, perk, stall upgrade, Handcart, regular, special guest, the crier, Closing Day, Daily market, Basic game, Price drift and Fixed prices. A test blocks the retired names from coming back. **Good Riddance was not renamed (to Clearance)**: the owner asked why it is still in the game, and its future (replace with Scout, remove or keep) is a rule change that needs the owner's decision.

OTHER

PLAN-036 quirk check: added. PLAN-032: a logic-level restore test and a device checklist (docs/DEVICE-CHECKLIST.md). The real Durable Object export and restore, a forced restart on the preview server and the phone runs are NOT RUN.

NOT DONE OR NOT VERIFIED

Browser tests (CI); phone look of the renamed screens; a new-player check of the names; the board sign and printed cards (they take the names when built).

REP-071 · The table screen in look D

Game	Fruit Market (rules version 5, message version 2; local branch workflow/plan-030, not pushed)
Report date	2026-10-29 00:55 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (how the table screen looks)
Request	"okay approve this design"
Plan and decision	PLAN-030 (approved; in progress), DEC-043 (approved)

STATUS

The screen part of PLAN-030 is built. Lint, types, 126 unit tests and the build pass. Seen in a 390 px browser window (Table_Screen_Built_390px.png), not on a real phone. Not pushed, so the browser tests have not run.

WHAT CHANGED

The Wholesale Market is now the look-D board: a pastel lane per fruit, paper slots with a coin-shaped price and a freshness corner flag, a faint fruit in empty slots, and a gold outline on the slots the lorry delivers into. The lorry is a flat smooth drawing (an inline SVG) on a road strip under the queue; the old picture file is removed. Rival stalls and your stall are paper cards with striped awnings; the top bar, chips and the bar along the bottom are ink-outlined pills; the table sits on a cream mat. Trays and pop-ups keep their dark look until PLAN-033. Two scenery colours (street green, road) were added to the palette.

ALSO FIXED

The e2e, perf and smoke scripts clicked the old button labels from before the rename ("Open the shops", "Open the market"); they now click "Start game" and "Start buying". The tutorial text said "Tap Open the market"; it now says "Tap Start buying", and the retired-name test covers those labels.

NOT DONE OR NOT VERIFIED

The printed mat in look D and generating screen and mat from the same slot data (PLAN-030 T1); a real phone; browser tests (CI); the other screens (PLAN-033).

REP-072 · Pushed to the preview: undo, rename pass, new table screen

Game	Fruit Market (rules version 5, message version 2; preview updated after CI on main; live unchanged at 1.1.1)
Report date	2026-10-29 01:14 (Asia/Kuala_Lumpur)
Type	Delivery. **Game change** (preview)
Request	"Push to preview" (twice)

STATUS

PLAN-023 is on the preview (CI passed on the branch and on main). The rename pass, the quirk check, the free drills and the table screen in look D passed CI in full on branch workflow/plan-030 (run 40731340121) and are merged to main; the preview updates after CI passes on main. The live game is not deployed.

NOT VERIFIED

Phone looks (the new screen, the renamed text, the name box keyboard); undo on two real phones; the printed mat in look D; the other screens (PLAN-033).

REP-073 · PLAN-021 cancelled; continuing DEC-042

Game	Fruit Market (no game change in this report)
Report date	2026-10-29 01:21 (Asia/Kuala_Lumpur)
Type	Decision. **No game change**
Request	"cancel 21 and lets work on 42"

STATUS

PLAN-021 is cancelled (DEC-044). Work continues on the order in DEC-042: what is left is the printed mat and the shared slot data (PLAN-030), the restyle of the other screens (PLAN-033) and the animated opening (PLAN-029). The shared slot data is built; the three designs come to the owner as mock-ups first.

REP-074 · Shared slot data and two design proposals

Game	Fruit Market (rules version 5, message version 2; local branch workflow/plan-030b, not pushed)
Report date	2026-10-29 01:26 (Asia/Kuala_Lumpur)
Type	Plan step and design proposals. **Game change** (code only; no visible change)
Request	"cancel 21 and lets work on 42"
Plans	PLAN-030 (T1 PASS), PLAN-033 and PLAN-029 (designs to approve)

STATUS

PLAN-021 is cancelled (DEC-044). Shared slot data is built (trackSlots), with 4 new tests; lint, types, 130 unit tests and the build pass. Not pushed.

DESIGN PROPOSALS FOR THE OWNER

1. **Printed mat and tokens in look D** (Design_Proposal_Printed_Mat.html and .pdf): A3 landscape mat from the same slot data as the screen, two token sheets, a "what to test" page.
2. **Other screens in look D** (Design_Proposal_Screens.html): home, set-up, morning with a hint, results. The buying and pricing trays follow the same card and pill style.

STILL TO COME

The animated opening redrawn in the new look (PLAN-029). Nothing is built from the proposals until the owner approves them.

REP-075 · Restyle of pop-ups, trays and sheets; QA sweep

Game	Fruit Market (rules version 5; branch workflow/overnight-1)
Report date	2026-10-29 03:12 (Asia/Kuala_Lumpur)
Type	Plan step. **Game change** (looks only)
Request	DEC-045 and DEC-046
Plans	PLAN-033

STATUS

Built and green on branch CI. The pause pop-up, tips, coach notes, toast, buying and pricing trays, showdown, how-to dots and title, online error text and the page margin on wide screens are now in look D. Checked by screenshot at 360×640, 400×820 and 1280×800 (home, how-to, settings, trophies, online, quick match, set-up, morning, buying, pricing, showdown, evening, night, summary, results).

FOUND AND FIXED

- Pause pop-up was navy with an invisible title.
- Coach notes and toasts were cut off on the right (they used a transform that the pop animation overrode).
- Pricing and showdown trays were navy with pale text.
- Wide screens showed navy margins.

NOT DONE OR NOT VERIFIED

Phone look on a real phone; tablet width screenshots; the online lobby with a live server (the preview page shows only the offline message in a local file).

REP-076 · Balance study: Good Riddance and shopkeeper perks

Game	Fruit Market (rules version 5)
Report date	2026-10-29 03:12 (Asia/Kuala_Lumpur)
Type	Measurement only. **No game change**
Request	DEC-046 (balance simulations)

STATUS

600 computer games per row (normal level, 10 days, upgrades and events on). Nothing in the game was changed.

GOOD RIDDANCE

It is played rarely: 0.21 times a game with 2 players, 0.13 with 3, 0.10 with 4. A player who plays it wins less often than average (2 players 23% against 53%; 4 players 13% against 25%). That is mostly because the computer plays it when it is already stuck with old fruit, so it is a rescue card, not a winning move. Removing it, keeping one copy, or swapping it for Free Fruit or Fridge changes average cash by less than $1.5 and the winning margin by about 1 to 2 dollars: inside the noise. Seat order stays fair in all versions.

What this means: balance does not decide it. The question is feel: a card whose best use is dumping stock at $1 each. Scout (look at the front of one walk-in arriving today) cannot be simulated honestly because the computer does not use hidden information, so it needs a human playtest.

SHOPKEEPER PERKS (computer play, 2 to 4 stalls, win rate against a fair share)

Perk	2 stalls (fair 50%)	3 stalls (33%)	4 stalls (25%)
Big Basket	61%	50%	40%
Tip Jar	56%	40%	28%
Bargain Hunter	50%	33%	21%
Nest Egg	48%	28%	23%
Charmer	40%	25%	21%
Haggler	47%	24%	17%

Finding: Big Basket is clearly the strongest (40% at 4 stalls, against 25% fair) and Haggler the weakest (17%). Caution: the computer may use Haggler and Charmer less well than a person. This

is a rule change if acted on, so it is your decision: for example, change Big Basket to "buy 1 extra fruit on days 1 to 5", or make Haggler stronger.

OTHER CHECKS FROM THE EXISTING BALANCE RUN

- Seat order: 2 stalls 52/48, 3 stalls 31/33/36, 4 stalls 25/25/28/22. Fair enough.
- Skill: hard beats normal 78% of the time with 2 stalls; normal beats easy 79%.
- Dead days (nothing sold): 0.09 a game with 2 stalls, 0 with 4.
- Leader after day 5 wins about the fair share: comebacks are common.
- The 36-card customer deck: average demand per card matches the old 24 (4.4 pieces, $31.8).

NOT DONE OR NOT VERIFIED

Human play. Every number here is computer against computer.

REP-077 · Print kit, playtest pack and two opening designs

Game	Fruit Market (rules version 5; no game change)
Report date	2026-10-29 03:16 (Asia/Kuala_Lumpur)
Type	Deliverables outside the repo. **No game change**
Request	DEC-046
Plans	PLAN-030 (printed set), PLAN-029 (opening: designs only)

STATUS

Files are in the Drive folder trial_FruitMarket/Print_Kit (HTML and PDF of each) and trial_FruitMarket/Design_Proposal_Opening_Animation.html. Nothing is in the game repo.

PRINT KIT

Mat and tokens (A3 mat, 4 token sheets), 36 customer cards, 20 Delivery cards, 12 Favour cards, 13 Event cards, 6 Shopkeeper cards, and a playtest pack (rules page, three test sessions, day sheet, feedback questions). All numbers come from the same data as the app (slot costs, supply per stall count, favour deck, perks). Card backs are not made yet.

OPENING ANIMATION: YOUR CHOICE

1. **Market morning**: sun rises, sign drops, three stalls open, the lorry drives in and tosses fruit to the stalls, bell rings. Calm, like the home screen.
2. **The mat is laid**: the same board as the printed game is laid out, tokens thump on, the Delivery card flips. Ties the app to the table.

NOT DONE OR NOT VERIFIED

Nothing printed on paper yet; sizes were checked only in the PDF. No build of the opening until you choose.

REP-078 · Market opening built (option 1)

Game	Fruit Market (rules version 5; branch workflow/plan-029)
Report date	2026-10-29 07:59 (Asia/Kuala_Lumpur)
Type	Plan implementation. **Game change** (visual only)
Request	"1" (DEC-047)
Plans	PLAN-029

STATUS

Built. A new game now opens with the Market morning scene (about 8 seconds, tap to skip). It does not play on Continue or Rematch, and automated browser tests skip it. With reduced motion it shows a still frame for about a second. Screenshots at 2.5, 4.8 and 7.5 seconds match the approved design.

NOT DONE OR NOT VERIFIED

Browser tests (CI); how it feels on a real phone; whether 8 seconds is too long for repeat players (it could play only for the first few games).

REP-079 · Bulk session: QA sweep, accessibility, rulebook, concept tutorial

Game	Fruit Market (rules version 5; branch workflow/plan-037-rulebook-tutorial)
Report date	2026-10-29 (Asia/Kuala_Lumpur)
Type	Game change (readability, text) and tooling. **No rule or balance change**
Request	"yes do all 7 with preview pushes, together with updating the rulebook and including a tutorial explaining the concept of the game" (DEC-049)

STATUS

Done and on the preview once main CI passes. Local checks could not run (this computer was out of memory), so every check ran on GitHub.

SUMMARY

1. **Widths and layout.** A new sweep (scripts/qa-sweep.cjs) opened every screen at 7 sizes (small phone, phone, phone landscape, tablet both ways, laptop, desktop). No sideways overflow and no script errors anywhere. Screenshots looked right; on tablet and desktop the game is a centred column.
2. **Accessibility.** The sweep found 136 distinct text-contrast problems. Worst: the whole Trophies screen and the favour and upgrade chips in How to play were nearly invisible. Fixed. Now 3 left: "10 days" label, the SOLD stamp (fixed in the last commit) and a floating +$ number that is on screen for a moment. Also darker muted text, a slightly darker terracotta (so cream text on it passes), a darker teal on tips and the home buttons, and name labels on the set-up fields. Buttons that are 32 to 36 px tall (chips and market cells) were left as they are; they pass the 24 px minimum but not the 44 px ideal. Making them taller changes the board layout, which is your call.
3. **Online lobby.** Covered by the existing browser tests in CI (online match with a leaver, quick match, four players in one room, rejoin from a background tab). They passed. I did not add a new live-server test.
4. **Soak.** 1500 computer-only games over 2 to 4 players, 8, 10 and 14 days and nine option sets: none stuck, none crashed, no impossible numbers, same seed repeats the same game. Average length 10.6 days. Seat 2 wins most often (765 of 1500 against 399, 165, 171); that is the balance study's finding about computer levels, not a bug, and I changed nothing.
5. **Card backs.** Delivery, Favour and Event backs: Print_Kit/Card_Backs.pdf (draft design for your approval). Customer backs were already in the customer card PDF. The README is updated.
6. **Cleanup.** 19 unused style rules and one animation from the old look removed.
7. **Rulebook and tutorial.** RULES.md has a new "The big idea" section and a worked example (Hafiz buys 5 blueberries from three stalls), and two small typos fixed. How to play now

opens with "The big idea" and "One customer, start to finish". This is new public wording: please read it.

NOT DONE OR NOT VERIFIED

Real-phone check; taller tap targets (needs your call); the 1.2.0 release notes are in the CHANGELOG draft only, nothing is published.

NEXT ACTION

Owner: read the two new How to play slides and the "big idea" section; look at Card_Backs.pdf; check the preview on a phone.

REP-080 · Scout card (rules version 6) and release 1.2.0 prepared

Game	Fruit Market (rules version 6; branch workflow/plan-037-scout)
Report date	2026-10-29 (Asia/Kuala_Lumpur)
Type	**Game change** (one favour card) and release preparation
Request	Decision card answers "1c 2a 3a" (DEC-050, DEC-051)
Plans	PLAN-037

STATUS

Built; checks run on GitHub (local checks are unreliable on the owner's laptop). Release 1.2.0 is prepared as a candidate and is **not published**.

SUMMARY

- **Scout replaces Good Riddance.** On your buying turn, look at the front of one face-down walk-in who arrives today. Everyone sees you played it, not what you saw. Tap a face-down walk-in to choose (if only one is left it is chosen for you). Computer players do not play it, as with Gossip. The deck is still 12 cards, two of each.
- **Rules version 6.** Older phones cannot join newer online rooms, saved games from version 5 are not resumed, and daily scores start a fresh board.
- **Perks unchanged (2A).** No change to shopkeeper perks.
- **Release 1.2.0 prepared (3A).** Version 1.2.0 in the app, lockfile and Android; a release candidate entry with plain notes; the plans it contains are marked. Nothing is published.
- **New art:** a small magnifying-glass icon for Scout, drawn in the existing pixel style. Please say if you want it changed.

TO PUBLISH (owner)

1. Check the preview on a phone (the new look, the opening, the How to play slides, Scout in a solo game: you only get a favour card by serving a whole customer order alone).
2. GitHub, Actions, **Deploy LIVE game**, Run workflow, type `PUBLISH`.
3. Then ask Claude to mark 1.2.0 as published in the records.

NOT DONE OR NOT VERIFIED

Scout on a real phone; Scout in an online room (covered by the unit tests, not by a browser test); physical Scout card for the print kit (the favour card PDF still shows Good Riddance until regenerated).

REP-081 · New lorry and stalls in pixel art

Game	Fruit Market (rules version 6; branch workflow/plan-038-lorry-stalls)
Report date	2026-10-29 (Asia/Kuala_Lumpur)
Type	Game change (art only, no rules)
Request	"lets go for option tiles and sign for stall, lorry one use bunting" (DEC-052)
Plans	PLAN-038

STATUS

Built and checked locally (lint, types, 133 tests, build, screenshots). Browser tests run in CI.

SUMMARY

- **Why pixel art:** the owner pointed out the first mockups did not match the characters. The art guide was unclear about the lorry; it now says props are pixel art (DEC-052).
- **Lorry:** the Sunny Pickup with the wooden bunting roof, 54 × 31 art pixels shown at 3x (162 px wide, up from 150). Wheels turn and the flags flutter in two frames while it arrives and leaves.
- **Stalls:** the stall panels (rivals, yours, and a rival's opened stall) get a tiled roof band in the player's colour (so unlocked colours still work). The market opening shows the tiled stall with its sign and each shopkeeper's portrait.
- **Release:** this is part of the 1.2.0 candidate.

NOT DONE OR NOT VERIFIED

Real phone (size of the lorry in the street row: it is a little taller than before); tablet; the printed mat and card art still use the old look.

REP-082 · Full-screen option

Game	Fruit Market (rules version 6; branch workflow/fullscreen)
Report date	2026-10-29 (Asia/Kuala_Lumpur)
Type	Game change (interface only, no rules)
Request	"can we include an option to play the game full screen as well" (DEC-053)
Plans	PLAN-039

STATUS

Built. Lint, types and 134 tests pass locally, and a browser test entered and left full screen. CI runs the full browser suite.

SUMMARY

A **Full screen** button sits in Settings and in the pause menu. It uses the browser's Fullscreen API, so it shows on Android Chrome, desktop browsers and iPad. iPhone Safari has no full-screen mode for web pages, so there the button is hidden and Settings explains how to get full screen with Add to Home Screen. Browsers only allow full screen from a tap, so the game cannot start in full screen by itself. It is part of the 1.2.0 candidate.

NOT DONE OR NOT VERIFIED

Real Android phone and iPhone; the Android app (a separate build) is unchanged.

REP-083 · How to play rework and skippable tutorial

Game	Fruit Market (rules version 6; branch feat/how-to-play-rework)
Report date	2026-10-30 (Asia/Kuala_Lumpur)
Type	Game change (interface and wording only, no rules)
Request	"ok lets now work on the how to play/help section ... make the tutorial skippable please" (DEC-054)
Plans	PLAN-040

STATUS

Built. Lint, types and 135 unit tests pass locally. CI runs the browser suite. A copy of the cards was checked on a phone-sized screen.

SUMMARY

How to play is now seven cards in the order a game unfolds: **Welcome to Fruit Market** (what to expect, the $10 start, and the five stages of a game), **The seven phases of a day**, **Four fruits, four personalities**, **The rush: who gets the sale?**, **Evening and night**, **Market Days and Closing Day**, and **Tips from the market**. The first-game tips have a **Skip tutorial** button next to Got it. Settings can replay them.

Every tip was checked against the rules. One was wrong (it said later days get busier; customer numbers stay level) and was removed, and two more were removed at the owner's request.

NOT DONE OR NOT VERIFIED

The Skip tutorial button has not been tried in a real game, and the final seven cards have not been looked at on a real phone. The shopkeepers and upgrades are no longer listed in How to play (the upgrades appear on the Market Days card).

REP-084 · Tabletop review and fixes

Game	Fruit Market (rules version 6; branch feat/how-to-play-rework)
Report date	2026-10-30 (Asia/Kuala_Lumpur)
Type	Rules wording, card text, one small game fix, print kit
Request	"go thru the current game mechanics again ... ensure that everything can be done on a tabletop", then "Yes, fix all" (DEC-055)
Plans	PLAN-041

STATUS

Built. Lint, types and 135 unit tests pass locally. The new print pages were rendered and checked on screen, not printed on paper.

SUMMARY

The review compared RULES.md, the game engine, the printed kit and 600 computer-played games (2 to 4 stalls, 8, 10 and 14 days).

Found and fixed:

- **Guest orders were not printed.** The Hotel Chef, Influencer and Gym Club orders and top prices are now in RULES.md and on three new guest cards.
- **Bad Harvest said "scarce fruit"** but raises every fruit by $2. **Fridge said "all fruit"** but moves ripe fruit only. Both texts now match the game, in the app and on the printed cards. The Hotel Order card now lists the order.
- **Rules wording added:** how a tie on "fresher fruit" is decided (the younger freshest piece wins), that customers are drawn a day ahead, that the crate is a number and the pieces stay on the shelf, that the first-player token moves one seat each night, how decks run out and are reshuffled, rounding for Good Harvest, the Health Inspector cap, full favour hands, money, and which customers are generous in the Fixed prices option.
- **The favour deck could deal a third copy of a card** after it ran out, which a real deck cannot do. It ran out in about 1 game in 6. The game now shuffles back only the played cards. A new test fails on the old code and passes on the new.
- **Print kit:** a new Table_Extras pack (guest cards, stall sheet with crate numbers and price row, fold-up screen with reminders, and paper money for a $910 bank). Event cards and the playtest pack updated. Parts check: the most of one fruit ever in play was 25 against 30 tokens.

NOT DONE OR NOT VERIFIED

Nothing was printed or played on a table. The two ways of tracking age (sliding tokens on the printed mat, or day chips) are both described and both give the same ages; choosing one is left to

the first table test. The age slots on the stall sheet use the sliding method. Token counts come from computer players only.

REP-085 · Accounts, achievements and looks (release 1.3.0)

Game	Fruit Market (rules version 6, protocol 3; branch feat/accounts-looks)
Report date	2026-11-04 (Asia/Kuala_Lumpur)
Type	New feature release
Request	Optional accounts, more achievements, free stall colours, collectible looks, rivals see looks, wardrobe in rooms (DEC-056, DEC-057)
Plans	PLAN-042, PLAN-043

STATUS

Built. Lint, types and 171 unit tests pass, and CI run 41165517987 on the branch was green. The preview check on the staging worker is recorded below before the live deploy.

SUMMARY

- **Accounts:** an ID (3 to 20 letters, digits or underscore) and a password (8 to 72 characters). No email and no password recovery. Progress syncs and only grows. Passwords are stored as PBKDF2-SHA256 hashes, sessions as hashed tokens, and 5 wrong tries lock the ID for a growing time. Signing out clears the progress on that phone after syncing. Deleting the account needs the password.
- **Achievements:** 43, in ten themes. Every stall colour is free.
- **Looks:** 25 pieces unlocked by achievements. Stall roofs, trims and counter props show on the stall; hats, faces, neck pieces and badges are worn by the shopkeeper. They change no rule.
- **Online:** each player's look is sent to the room and shown on their seat, stall, showdown and results. The Wardrobe opens from the room lobby and the pause menu. Protocol 3: older apps cannot join new rooms.

KNOWN LIMITS

- The server checks that a look exists and fits its slot, not that the player earned it. A modified phone could wear a look it has not unlocked. It is cosmetic only.
- There is no way to recover a lost password.
- Owner still to do: update the Google Play data-safety form and add a privacy-policy link (accounts store an ID, a password hash and progress).

TESTS

- Lint, types, 171 unit tests, build: pass locally.
- Account, profile, achievement, look and room tests added (accounts.test.ts, account-client.test.ts, profile.test.ts, achievements.test.ts, looks.test.ts, room.test.ts).

- Not seen running before this report: the account endpoints on real Cloudflare and the online wardrobe in a room. See the preview result in the release record.

REP-086 · Full game audit and balance simulation

Game	Fruit Market 1.3.0 (rules version 6, protocol 3)
Report date	2026-11-05 (Asia/Kuala_Lumpur)
Type	Audit and recommendations. Nothing in the game's rules or balance was changed.
Request	"audit everything and suggest for improvements ... tabletop, UI and UX, game modes" and "a full simulation regarding balance ... suggest tweaks"
Status	Owner asked for the suggested changes (DEC-058, DEC-059). Built in 1.4.0 and 1.5.0: B1 to B4, B6, B7, U1 to U12, T1 to T4, T9, the rot stats, the print-kit items and a daily twist. Not built: B8 (no measurable gain), T3, T5, T8, and the Table Companion and other new modes.

1. The short version

1. **The core game is sound.** Seats are fair, the rules text matches the cards and numbers, and the basic game (no shopkeepers, favours or upgrades) is the tightest and fairest version.
2. **The extras are where the balance goes wrong.** Upgrades, shopkeeper perks and favour cards triple the number of lopsided games (11% in the basic game, 33% in the full game, 3 players).
3. **A rules mismatch:** the written rules, the set-up text and How to play all say "build one upgrade after each Market Day". The app lets a player build several in one evening. Skipping upgrades costs a player about $20 and nearly always loses (5.8% wins where 33% is fair).
4. **Two shopkeepers are off:** Big Basket (Benny) wins about 10 points above fair, Haggler (Rosa) about 9 points below. Extra starting cash does not help any perk, so cash perks do not work.
5. **Mango is the best fruit by 40%** (profit per fruit $5.30 against $3.7 for the others).
6. **Fruit almost never rots** when computers play (under 0.2 fruit per player per game). The headline tension, "sell it before it spoils", is not biting. Only human playtests can say whether this is true for people.
7. **Quick match is a 2-player game, the most lopsided size** (47% of games decided by $20 or more).
8. **The sales payout looked wrong on screen.** Totals jumped to their final value at the bell while customers were still being replayed. Fixed in a local branch (section 6).
9. **For a browser-first launch** the site has no install support (no manifest or service worker), no link-preview tags and no security headers. All three are small jobs.
10. **The printed kit cannot run the upgrade draft**: there are no upgrade cards, no day track and no queue strip, and the stall sheet cannot settle the tie-break (fruit sold).

2. How the simulation was run

- Every game is played start to finish by the computer players, using the same rules code as the app and server (all phases: buying, stocking, bell, rush, evening, night).
- **Baseline:** 3,000 games for each of 2, 3 and 4 players. Standard 10-day game, all players Hard, a different random shopkeeper for each stall (as at a table). Seats are fair, so any gap is the game, not the seat.

- **Other runs:** 500 to 1,500 games each. Game lengths, rule options, favour-card removal, upgrade policies, computer skill, and 11 changed-rule versions.
- **Reading the numbers:** win rates in 3-player games carry about +/-1.7 points of noise (500 games: +/-2.5). Treat gaps under 3 points as noise.
- **Limits:** these are computers, not people. They never play the Scout favour card, they buy only what they can sell, and their upgrade choices were tuned to the current app behaviour. Use the numbers to find structure, then confirm with human games.
- The test tooling was temporary and is not in the repo.

3. Baseline results (standard game)

	2 players	3 players	4 players
Win by seat	49.8 / 50.2	34.0 / 33.7 / 32.2	25.4 / 25.3 / 24.2 / 25.1
Median winning margin	$18	$13	$11
Margin as share of average final cash	12%	9%	7.6%
Middle half of margins	$9 to $32	$5 to $24	$5 to $21
Close games (margin $3 or less)	11%	17%	19%
Lopsided games (margin $20 or more)	47%	33%	28%
Average final cash	$150	$144	$144
Day-5 leader wins (fair: 50 / 33 / 25)	53.6%	33.7%	27.4%
Wins from last place at day 5	47%	34%	25%
Lead changes per game	3.4	4.8	5.6
Fruit sold per player	31.7	33.0	34.2
Demand served	76%	70%	69%
Fruit spoiled per player	0.14	0.08	0.05
Unsold ripe fruit at the end (jam maker pays $1)	3.2	2.1	1.6
Peak cash, one player / all together	$250 / $422	$258 / $639	$254 / $776

What this says:

- **Fair seats, no runaway leaders.** The leader on day 5 wins at chance in 3 and 4 players. The poorest-first buying order is a strong catch-up rule (see section 5, "Rotating order").
- **Smaller tables are swingier.** A 2-player day has one customer, so a few customers decide the game. Skill still shows (Hard beats Normal 79% in 2 players, 57% in 3 players against two Normals, and only 52% against three Easy in 4 players).
- **About 70% of demand is served in every setup.** Fruit supply, not cash, limits what stalls can do.
- **Paper money:** the printed bank (910)*coversthe*4 – *playerpeak*(776). It is tight if rich events stack up.

Shopkeepers (win rate, fair is 50 / 33.3 / 25)

Shopkeeper (perk)	2p	3p	4p	Gap to fair, 3p
Benny (Big Basket, +1 buy a day)	61.3	43.4	36.4	**+10.1**
Raj (Bargain Hunter)	52.1	36.2	27.7	+2.9
Kiran (Tip Jar)	49.4	34.9	24.3	+1.6
Saleh (Nest Egg, $15 start)	45.2	31.7	23.0	-1.6
Chen (Charmer, 3 favour cards)	48.5	29.9	22.6	-3.4
Rosa (Haggler, wins ties)	43.7	24.2	16.2	**-9.1**

The same order shows in all three table sizes, so it is the perks and not luck.

Fruit (profit per fruit bought, 3 players)

Blueberry $3.79, watermelon $3.73, apple $3.73, **mango $5.30**. Computers buy about 25 to 27 of each fruit per game, so everyone is fighting for mango. Mango customers pay up to $9 to $11 against a cost of $3 to $4.

Events (income per player on that day, against the average day, 3 players)

Festival +12.8, *HotelOrder* +7.2, Superfood +6.9, *MangoCraze* +5.4, Bad Harvest -0.3, *Heatwave* −1.4, Good Harvest -1.5, *PriceHike* −3.2, Minimum Price -3.3, *HealthInspector* −3.6, Blueberry Glut -3.8, *AppleBlight* −5.6, **Lorry Breakdown -$10.7**. Everyone feels an event, so events do not tilt a game. Without events total cash falls by 27%.

Customers (3 players)

- Richest per visit: Hotel Chef $46, Daniel $36, Gym Club $36, Rara Jenab $34, Ravi $34, John $33.
- Poorest per visit: Ben $19, Hafiz $20.5, Kumar $21, Aisyah $22, Hadi $22, Fatimah $22.
- Hardest to finish (share fully served): Ahmad 49%, Robert 51%, Wei Jie 57%, Yuki 57%, Hafiz 59%. All ask for 5 pieces.

Favour cards

- A player earns about 2 per game, and about 0.75 per player are never played.
- Win rate by cards earned: 0 to 1: 26%, 2 to 3: 35%, 4 to 5: 45% (3 players). Partly cause, partly good play.
- Removing all favour cards: lopsided games fall from 33% to 28% and average cash falls $2.4. No single card stands out.
- The computers never play Scout, so two of the 12 cards are dead in these numbers.

Game length and rule options (3 players unless marked, 500 to 600 games)

Option	Median margin	Lopsided	Close	Avg cash	Fruit spoiled
Standard (10 days)	$13	33.5%	16%	$146	0.08
8 days / 14 days	$13 / $15	31% / 38%	13% / 13%	$110 / $209	0.04 / 0.11

Option	Median margin	Lopsided	Close	Avg cash	Fruit spoiled
No events	$13	30%	16%	$107	0.03
No upgrades	$9	**16%**	21%	$146	**0.68**
No Market Days	$15	38%	16%	$121	0.13
Basic game (no perks, favours or upgrades)	**$7**	**11%**	**27%**	$141	0.69
Fixed prices	$14	36%	15%	$152	0.07
Price drift	$14	33%	14%	$160	0.08
No unripe fruit	$13	34%	15%	$149	0.08
2 players: standard / no upgrades / basic	$18 / $12 / $10	47% / 28% / 21%	12% / 19% / 18%	$152 / $153 / $148	0.18 / 1.17 / 1.23

Takeaways: upgrades add variance but no extra money on average (cash is the same with and without). The basic game is the fairest. Rot only appears once upgrades are gone.

Upgrades (3 players, one seat follows a fixed rule, 500 games)

Rule for that seat	Wins (fair 33.3)	Cash against the others
Computer's own choices	37.0%	+$1.5
Never builds any	**5.8%**	**-$20.2**
Builds everything it can	44.8%	+$6.2
Computer's choices, but never a Handcart	16.2%	-$12.6
... never Premium Display	27.6%	-$4.3
... never Cold Room	34.0%	+$0.4
... never Juice Bar	32.6%	-$0.7

- The Handcart is nearly mandatory. The Cold Room and Juice Bar change nothing, because computers almost never lose fruit to rot. Raising the Handcart price to $8 changed nothing (everyone can afford it by day 3).
- Players build 3.34 upgrades in 3 evenings. The rules say one per evening.

4. Rules and tabletop audit

Found now

1. **Upgrades per evening do not match the written rules.** RULES.md ("draft one stall upgrade"), the set-up screen ("build one") and How to play say one. The engine has no limit and the evening panel says "you may build upgrades". Fixing the app to one per evening is a one-line rule check, plus retuning the computer players. Measured effect: lopsided games 33% to 27%, never-build penalty -20*to*−10, fruit spoiled 0.08 to 0.33. Decision needed (section 8).
2. **The stall sheet has no "fruit sold" tally**, but the end-of-game tie-break is "whoever sold more fruit". Add a tally row.

3. **The printed kit is missing pieces the rules use:**
    - upgrade cards (4) and a draft procedure;
    - a day track with the Market Days and Closing Day marked;
    - the queue strip (dawn line-up) and the waiting pile area;
    - the fruit clock marker and day chips (the README says upgrade and Tip Jar tokens are "not printed yet");
    - a one-page crier card (the crier's top-minus-1 / top-minus-2 table) for the person who reads customers aloud.
4. **No solo or 2-player table rules** for a stall run by a card deck. The app has computer players; a table does not.
5. **No play-time estimate** for a table game. Time it at the first table test.
6. **Teaching ladder:** the simulation says the basic game is the fairest. Suggest teaching in four steps: basic, plus shopkeepers, plus favours, plus upgrades. The Playtest Pack could say so.
7. Stale code comment: `src/engine/game.ts` still says "rules version 4".

Checked and fine

Every count and number in RULES.md matches the app: 20 delivery, 36 customer, 13 event and 12 favour cards; start cash, shelf, buy limits, upgrade costs, jam price, Closing Day bonus, tip, generous customers list; Market Day days; tie-break order; walk-in counts.

5. Suggested balance changes (all need approval; effects are from the simulation)

#	Change	What the simulation showed	Notes
B1	**Upgrades: one per Market Day** (as written), and retune the computer to take the Handcart first	Lopsided 33% to 27%; median margin $13 to 11; *never* – *buildpenalty*−20 to -$10; rot 0.08 to 0.33. Average cash -5%	Matches every printed rule
B2	**Mango costs $1 more** (fresh $5, green $4)	Mango profit/fruit $5.30 to $4.57 (others $3.7); cash -4%; nothing else moves	$2 more would fully level it
B3	**Big Basket: the 5th piece costs $1 more**	Basket wins 43% to 39% (fair 33%); nothing else moves	Try $2 for fair
B4	**Rework Haggler, Nest Egg and Charmer so they give power, not cash**	Nest Egg at $10 start: 31.7% to 31.1% (no change). Haggler with $2 start: +2 points. Haggler's price counting $1 lower: no help	Cash is not scarce, so cash perks do nothing. Ideas, not yet simulated: Haggler "your first sale each day pays +$2"; Nest Egg "10-fruit shelf"
B5	**Keep the poorest-first buying order**	A rotating order doubles lopsided games (33% to 54%), halves close games (17% to 8%) and makes leaders win	The cost: an early lead means nothing (day-5 leader wins at chance)
B6	**2 players: add one walk-in**	Cash +45%; median margin unchanged in dollars, so 30% smaller as a share of cash	Does not cure lopsided games (47%)
B7	**Quick match and new players: offer the Basic**	Basic game: lopsided 11% (3p) / 21% (2p) against 33% / 47%	Easiest fairness win

#	Change	What the simulation showed	Notes
	game		
B8	**Cold Room and Juice Bar need a real job**	Neither changes win rate by more than +/-2 under any rule tested	Rot is rare, so insurance upgrades are worthless. Make them capacity or pricing upgrades
B9	Higher upgrade prices are not the answer	Prices x2.5: cash -16%, same upgrades built, Handcart still best	Do not use price as the lever

Not recommended: more starting cash, a bigger buy limit (3 to 5 changed nothing), and cheaper or dearer Handcart.

About rot: the rules say fruit goes bad, but computers lose under 0.2 fruit a game. Before changing any rule to make rot bite, record fruit spoiled and unsold per human game in the anonymous stats and look at a week of real play.

6. UI and UX audit

Checked on the live site at phone width (375 × 812) through home, set-up, morning, buying, stocking, pricing, the rush and the Wardrobe/Trophies/Settings screens.

Good

- Clear, warm art; the Showdown panels ("cheapest sells first") tell the story of each customer well.
- Market cells have full screen-reader labels ("Blueberry, fresh, $2, 6 left"); reduced motion is respected; undo purchases and Skip exist.
- The game screen has 327 page elements (it had 3,722 in October), so it is light.
- The page now has a "Skip tutorial" and a seven-card How to play.

To fix

#	Finding	Evidence	Suggested fix
U1	**Digits are misread in the pixel font at small sizes**	Rank "#2" looks like "#8"; the $2 blueberry price chip (9.9 px) looks like "$8"	Draw numbers under about 16 px in the rounded body font, and make price chips at least 12 px
U2	**Sales totals jumped to their final value at the bell**	Recorded cash: rivals' totals were final before the first customer; with the fix they climb customer by customer	**Done in branch `feat/payout-suspense` (local, not pushed)**; see below
U3	The rank chip has no spoken label and "#N" is unexplained	The accessibility tree shows no name for it	Label it "Rank 2 of 4"; show "2nd"
U4	Fast taps are dropped while buying	Three quick taps bought one blueberry	Queue taps, or shorten the lock and show a count on the cell
U5	Event preview wording	The morning card says "Tomorrow: Apple Blight. No apples arrive today"	Write event text that works for both days, or add a "tomorrow" version

#	Finding	Evidence	Suggested fix
U6	The set-up page is long and Start is at the bottom	Six shopkeepers, seats, length, four rules and three "Playtest options" in one scroll	Sticky Start button, three presets (Quick, Standard, With friends), and hide advanced options under "More rules"
U7	"Playtest options" is developer wording	Basic game, Fixed prices, Price drift	Rename "Beginner and expert rules"
U8	The "Psst..." hint on the pricing screen is partly under the Seal button	Screenshot	Add bottom padding equal to the button height
U9	Results have no cash-over-days chart and no share card (only the daily has one)	The data exists in the game history	Add a small chart and a share card
U10	The customer strip is clipped at the right edge with no hint to swipe	Screenshot	Add a fade or arrow
U11	No "change password" in Settings	AccountPanel has sign in, sign out, delete only	Add it, because there is no recovery
U12	Empty market slots draw ghost sprites (28 of them)	Screenshot	Show only slots that matter, or dim more

The payout change (U2)

- **What was wrong:** the game works out and pays every sale in one step at the bell (`resolve` in `src/engine/game.ts`). The screen drew that final cash straight away while the customers were replayed, so the numbers never built up.
- **What changed:** the cash chips and the rank badge are held back at their before-sales amount, and each customer's payment is added when its coins land, counting up a coin at a time. A Tip Jar tip lands at the end. Display only, no rule change.
- **Checked:** type check, lint, the whole test suite (174 tests, 3 new, including one that proves the held-back amounts equal what each stall had before the sales in 20 full games) and the build pass. In a real game the recorder showed the totals climbing in separate steps as each customer paid. Not yet tried online with a second player or on a phone.
- **Where:** commit `3394969` on branch `feat/payout-suspense` . Nothing is pushed.

7. Game modes, technology and operations

Modes today

Solo and pass-and-play (2 to 4 stalls), private online rooms, quick match (2 players), Daily market, Market League (6 stages, win to advance), Wardrobe and 43 achievements.

Mode ideas, best first

1. **Table Companion** (the app as the crier, bank and clock for a real table game). It uses the same rules code and fits the tabletop plan. It would also fix the table's biggest chore.

2. **League and Daily with objectives and stars** ("reach $120 with no upgrades", "sell 10 mangoes"), which adds goals without new rules.
3. **Daily with a twist of the day** (a rule option on rotation).
4. **Practice coach** (hints, "what would a good player do"), for new players.
5. **2 against 2** teams online; **async rooms** (a move a day); **replay links**.

Technology and operations

#	Finding	Fix
T1	No install support: no web manifest and no service worker	Add both (installable, full screen, works offline)
T2	No link-preview tags or page description	Add title, description and image tags so shared links look right
T3	Any unknown address (`/robots.txt` , `/sw.js` , `/manifest.webmanifest`) answers 200 with the game page	Add real files, and let unknown paths 404
T4	No security headers on the site	Add frame blocking, content-type and referrer headers, and a content security policy, through a `_headers` file
T5	Locking an account is per ID, so someone who knows an ID can lock it out for up to an hour	Lock by ID and address together, or add a delay for the same address
T6	Password hashing is 100,000 rounds, the Cloudflare maximum	Acceptable; no change possible
T7	Rate limit 12 account tries a minute per address	Fine; shared networks may hit it
T8	Dependencies: no known issues in shipped code; patch updates are available	Update in a quiet moment
T9	Tests and CI are strong (174 tests plus browser tests in CI)	Keep; add a test for the payout timing in the browser tests
T10	App code grew 40% since October (168 kB to 236 kB, 82 kB compressed)	Still small; watch it

8. Decisions needed from the owner

1. **Upgrades:** fix the app to one per Market Day (matches all printed text), or change the text to "as many as you can afford"? (B1)
2. **Which balance changes to try first:** B2 (mango), B3 (Basket) and B7 (Basic game for Quick match) are the safest.
3. **Shopkeeper rework (B4):** are you happy for me to propose new perks for Haggler, Nest Egg and Charmer?
4. **Rot:** add "fruit spoiled" and "unsold" to the anonymous stats so we can see how people really play?
5. **Table kit:** print upgrade cards, a day track, a queue strip and a crier card, and add a "fruit sold" row to the stall sheet?
6. **Push the payout change** (`feat/payout-suspense`) and the privacy-page change (`feat/privacy-page`), both still local.

7. **Public wording:** the About line in Settings still has your full name; keep or change?

9. Suggested order of work

1. Pixel-digit fix (U1), rank label (U3), tap queue (U4): small, no decisions.
2. Push the two finished branches and release 1.3.1 (privacy page and payout display).
3. Site basics (T1 to T4): install, link previews and headers.
4. Balance round 1 (B1, B2, B3, B7) after your approval, then re-run this simulation.
5. Table kit additions, then the first table test.
6. Table Companion mode.

REP-087 · Kickstarter art direction: audit, recommendation and pilot plan

Game	Fruit Market 1.7.0 (rules version 9). No game, app or print-kit change.
Report date	2026-11-07 (Asia/Kuala_Lumpur). Corrected the same morning after an internal review, before delivery.
Type	Audit and recommendation. No images were generated and no assets were built.
Request	"Audit the existing Fruit Market art direction and character requirements. Recommend the strongest visual identity for a Kickstarter-first tabletop release with a subsequent mobile adaptation. Propose the first two characters to develop … and present a concrete production plan. Do not generate or implement assets yet."
Status	Awaiting the owner's approval: DEC-063 (art direction) and PLAN-048 (first, unpaid part of the pilot). The proposed direction is in `docs/art/ART_DIRECTION.md` (v0.2).

1. The short version

1. **Recommendation: a controlled hybrid, "Look E".** The world, frames and graphics take the crate-label style of "The Great Fruit Exchange". The people take the personality of "Orchard & Oddballs", but stay ordinary townsfolk. Everything is drawn in one flat screen-print style in today's palette, plus a few additions for your approval.
2. **Three AI review passes agreed.** Each was written from one viewpoint (tabletop publisher, art-production lead, game identity) and scored the options out of 45: hybrid 35 to 36, Oddballs alone 25 to 27, Exchange alone 22 to 25. They are not outside experts. Treat them as a structured opinion. The scores per criterion are in appendix A.
3. **The pixel look you like stays on the phone's board.** In the rule "the board shows tokens; cards show portraits", illustrated portraits go on printed cards, the box, the store and a few card views in the app. **The trade-off:** during play the phone shows the same people as pixel tokens, not as the new pictures (question 2).
4. **Pilot characters: Benny (shopkeeper) and Sailor Jia Wen (customer).** They are the most different pair that is safe for hidden information.
5. **Biggest commercial risk: AI art on Kickstarter.** From search summaries, not yet checked on the source pages: Kickstarter allows AI art but requires a public disclosure section, and board-game backers react strongly. UK Games Expo bars products made "completely or in significant part" with AI and does not define "significant". AI drafts finished by a person may still be refused there.
6. **Biggest game risk: art giving away secrets.** A customer's freshness type, amounts and prices are fixed by who they are. A face, a back or an edge colour that varies by card would reveal them. ART_DIRECTION.md §9 sets nine hard rules, including a fixed fruit order on customer backs (today's backs already make 9 of 36 customers unique).
7. **Fix before anything is printed:** the Drive print kit is still rules version 8, and RULES.md disagrees with the game on the generous customers and the guest orders.
8. **Cost:** nothing is spent without a quote and your approval. Estimates: 45 card portraits by a human illustrator about USD 4,500 to 36,000, plus USD 1,000 to 6,000 for a box cover. The

recommended route (AI drafts plus a person's finishing) is about USD 2,250 to 6,750 for 45 portraits, plus the tool and the box. All of this is far above the RM500-a-month ceiling.

2. What was audited

- **Repo:** art code (`src/ui/pixel.tsx`, `propsArt.tsx`, `art.tsx`, `wearArt.ts`, `looks.ts`, `palette.ts`), the style guide, the creative bible, RULES.md, `src/engine/data.ts`, the online server's hidden-information code, tests and the print script.
- **Records:** every art, cast, look and print decision (DEC-014 to DEC-062) and reports REP-010 to REP-086.
- **Google Drive (read only):** the Print_Kit folder (51-page home-print PDF), Look_Comparison, Board_Look_Options, Cast_Coherence_Audit, and the Concept and Studio Lifecycle Companions.
- **Outside research:** Kickstarter's AI policy, backer and convention reactions, copyright of AI images (US, UK, EU, Malaysia), image-tool terms, factory print specifications, tool features and illustrator prices.
- **Method:** six parallel AI audits, then three AI review passes from different viewpoints, then a second set of three AI reviewers who tried to prove the drafts wrong. Their corrections are in this version. I re-checked the key repo claims against the code by hand.

3. What exists today

- **All art is drawn in code.** Fruit, icons, props and 50 people are text grids turned into small pictures. Portraits are 16 × 16 pixels from one shared face, told apart by hair, accessory, skin and shirt (DEC-041). There are no image files to replace.
- **Swapping art is clean.** Every screen asks for a portrait through one function (`Face` in `src/ui/art.tsx`), and names, quirks, perks and freshness types live separately in `data.ts`. A new portrait source would need no change to rules, saves or online play.
- **The 25 collectible looks** are 11 wearables drawn as pixel rows on the 16 × 16 face, plus 14 stall pieces. Wearables only work on the pixel face, which is why Look E keeps them there.
- **Print is thin and partly outside the repo.** Only the card-back script is in the repo (63 × 88 mm, no bleed, no crop marks, RGB). Customer, shopkeeper, mat and token pages were generated elsewhere. They are rules version 8 and lack the freshness type.
- **Printed portraits are small:** about 21 mm (customers) and 26 mm (shopkeepers), drawn from 16 × 16 pixels. They are the weakest asset for a polished Kickstarter box.
- **Your recorded taste:** you chose a smooth toy-box look (DEC-021, "go with plan b"), then moved back to pixel art twice (DEC-023, DEC-033 "Look C"), and praised the current look ("artstyle is awesome i love it", DEC-057). That is why Look E keeps pixel art on the board and shows every new design beside today's look before anything is built (DEC-042).
- **Precedent for AI art:** the lorry picture was made with Gemini, and you accepted that AI-made art may not be protected by copyright (history, PLAN-011). It was later replaced by pixel art for consistency (DEC-052).

4. The two directions, scored

- **The Great Fruit Exchange alone:** strong shelf appeal and a perfect match for lorries, events and the box. But many historic crate labels and fruit ads drew people as racist or "exotic"

caricatures. Copying real labels risks copyright and trademark claims: brands such as Sunkist and Chiquita appear to still use their marks, though no trademark search has been run. Detailed litho texture also blurs at phone sizes.
- **Orchard & Oddballs alone:** good personality and social energy. But "charming imperfection" is what AI tools keep least consistent. It slides toward exaggerated faces (against DEC-018 and DEC-041), and without a strong graphic frame it looks like generic mobile-game art.
- **Hybrid:** the Exchange supplies the graphics as code templates (no image generator), so every frame and edge is identical. Oddballs supplies the attitude, expressed only through pose, prop and expression. One flat style ties them together. Its one weakness is that each person has two renderings, and a shared character spec keeps them in step.

Totals out of 45 (the scores per criterion are in appendix A):

	Exchange alone	Oddballs alone	Hybrid
Publisher view	23	27	36
Production view	25	25	36
Identity view	22	27	35

5. The recommendation

Look E, as written in `docs/art/ART_DIRECTION.md`: principles, colour, type, proportions, line and texture, card layouts, print sizes, hidden-information rules, AI consistency rules and a banned list. It keeps the harbour world (DEC-017) and the townsfolk cast rules (DEC-039, DEC-040). Three changes need your approval:

- DEC-041's portrait rule becomes "one character spec, two renderings".
- Up to three skin tones and a few accent colours join the palette.
- One bold display face is added for printed titles.

ART_DIRECTION §12 lists every approved record it touches.

6. Pilot characters

Recommended: Benny (shopkeeper, Big Basket) + Sailor Jia Wen (customer, card 25).

	Benny	Sailor Jia Wen
Role	Shopkeeper; public perk; computer rival in the daily market	Everyday customer; wants 2 apples (top $6) and 2 mangoes (top $10)
Shape	Broad, round, bearded, steady	Slim, upright, quick
Colour	Warm brown shirt, skin tone c	Harbour navy, skin tone a
Head	Short dark hair and beard (a clean base for hats later)	Long black hair; the bible will propose a sailor's cap as her one accessory
Prop	The big basket (his perk is public, so showing it is fine)	A rolled "mango map" (her quirk), not coins, so no money cue

- **Why Benny:** a shopkeeper appears on the player card, box, Kickstarter page, app avatar and character pick. His broad build tests the bust template. His perk gives a natural prop, and his simple head makes the cleanest anchor for wearable looks later.
- **Why Jia Wen:** she carries the identity of "a harbour where the whole world comes to shop". Her back never singles her out. On today's backs (fruits in card order) it matches one Premium and one Bargain card. With the fixed order it would match two Premium, one Bargain and one other Everyday card. Her portrait tests the process: a prop chosen with no money cue, and the cue-balance check. She is Everyday and not generous in either list.
- **Contrast:** man and woman, broad and slim, warm and cool colours, bare head and cap, a shopkeeper's hero card and a small customer window.
- **What the pair does not cover:** guest cards, event-card people and the face-down walk-in figure. The walk-in figure is specified in the Character Bible.
- **Runners-up:** Rosa + Grandpa Tan (Rosa's bun and glasses are the hardest test for hats and the monocle). Raj + Grandpa Tan (the deepest skin tone tests printing and small sizes). Note: Tan's thrift quirk cannot be drawn literally until you decide its text, and his token has two accessories, so choosing him means settling both first.
- **Not a pilot yet:** Rara Jenab. She is your own approved design and a strong campaign face, but as a Premium card her back is unique today (section 11, item 4).

7. Production plan

Each stage ends with a stop for your approval. Nothing visual is built before you approve its design (DEC-042). PLAN-048 covers stages 1 to 3, which need no spending. Stages 4 and 5 become a second plan once dated quotes exist.

Stage	Your brief	What happens	Output	Gate
0 (now)	Phase 1	Audit and direction	This report, ART_DIRECTION.md v0.2	You approve Look E and PLAN-048, and answer three questions
1	Phase 2, in two steps	Character Bible: one line each for all 45 characters plus the walk-in figure (build, shape, hair, prop, colours, type-cue check, fixes needed); full entries and master prompts for the two pilots; style-lock sheet layout. The other full entries follow in waves (your rule: never the whole roster at once).	CHARACTER_BIBLE.md v0.1	You approve
2	Phase 5 (templates)	Card template design: a mock-up of one customer (densest card), one guest and one shopkeeper front, plus the customer back, beside today's card. Built only after you approve. Then rules-9 templates filled from `data.ts`, with bleed, crop marks and an empty portrait window, as home-print sheets and one-card-per-page PDFs.	Mock-up, then templates and print script	You approve the mock-up

Stage	Your brief	What happens	Output	Gate
3	Phase 3, steps 1 to 4	Pilot concepts: three directions per pilot with full prompts, scored on silhouette, personality, appeal and card readability, with a recommendation. The tool is chosen after a dated check of its terms.	Prompts, specs, scored concepts	You pick one per pilot
4	Phase 3, steps 5 to 8	Pilot masters: master reference, front, three-quarter and side views, six expression sheets (as drawing references), a person's finishing of the shipped pictures, script checks, manifest entries, and a mock-up of the matching pixel token	Masters on Drive, manifest	Quote approved first; you approve the masters
5	Phase 4	QA and style lock: eight scores per character, each at least 4/5; printed at 100% on the template; phone mock-up next to today's look	Style-lock sheet	You approve the style lock
6	Phases 2 to 4, repeated	Waves: five more shopkeepers (Rosa first), then customers in waves of 6 to 8, guests, the Health Inspector and the walk-in figure	Each wave	Each wave approved
7	Phase 6	Commercial readiness: box key art, factory quote and spec, printed proof, provenance and licence review, Kickstarter "Use of AI" section, font credits, name and logo trademark check	Launch pack	Separate approvals; spending needs a quote
8	Phase 5 (phone assets)	Phone adoption: card views show portraits; pixel tokens may be redrawn to match; tests; performance budget; old saves keep loading	App release	A separate plan

Images in this session: no image-generation tool is available here. For stage 3 I will prepare complete prompts and specifications for an outside tool. You (or I, in a session with a tool) make the images. I will not describe images I have not received.

QA scores (stage 5): identity consistency, distinctiveness, visual appeal, personality, tabletop readability, phone readability, style consistency and production readiness, each 1 to 5, all at least 4, scored by the reviewer and confirmed by you. Automatic flags: anatomy errors, clothing or face changes between views, an unclear silhouette at 44 px, details that only read when zoomed, and any text, symbol or type cue.

8. Asset structure (proposal; created at stage 4)

Where	What
`docs/art/`	ART_DIRECTION.md, CHARACTER_BIBLE.md, STYLE_LOCK.md
`art/index.json`	One short row per asset version: ID (the existing character key, never renamed), version, use, status, approval
`art/templates/`	Card frames and backs (SVG or HTML)
`scripts/print-cards.cjs`	Builds RGB print PDFs from templates, game data and portraits (the existing browser route). CMYK at factory hand-off, either by the factory

Where	What
	or with one added tool after a dependency decision.
Drive `Fruit Market/Art_Masters/<id>/v001/`	Accepted raw images, layered source, finished transparent master (2048 px or larger), turnarounds, expression sheet, and the full manifest row (tool, model version, prompt, references, terms checked, human changes). Never overwritten: a change makes v002.
Generated, never hand-made	Print PDFs (with bleed), phone images (WebP at 1x, 2x and 3x), Kickstarter images

Large masters and the full manifest live on Drive, so the repo, the Android build and the one-file web build stay small, and model names stay out of the repo. Game data, words, prices, symbols and frames stay separate from the pictures, so a rule change never means redrawing art.

9. AI, copyright and Kickstarter

All outside facts here come from web search summaries. Fetching the primary pages was blocked in this session, so each must be re-checked on its source page, with a date, before a decision depends on it.

- **Kickstarter** allows AI content but requires a public "Use of AI" section: which tools, which parts are AI, which are original, and whether the owners of the source works gave consent. Hiding AI use can get a project suspended.
- **Backers:** board-game buyers are among the most hostile audiences. Reported examples: review-bombing of an AI cover on BoardGameGeek (March 2026), UK Games Expo barring products made "completely or in significant part" with AI (from September 2026; "significant" undefined), BackerKit barring content made only by AI, and award and publisher bans. Observational studies of Kickstarter campaigns (all categories, 2022 to 2024) link AI disclosure to clearly lower funding. Reported effects range from about 40% less money to about half the odds of reaching the goal.
- **Copyright:** the US Copyright Office (January 2025) says prompts alone do not make you the author; human changes, and the selection and arrangement of material, can be protected. Separately, the courts confirmed that an AI cannot be an author (Thaler; the Supreme Court declined to hear it in March 2026). The UK may remove its computer-generated-works rule. Malaysia's Copyright Act 1987 has no AI provision and no court cases yet, and a consultation ran in 2026. Raw AI images may give no exclusive rights, so copycats could reuse them.
- **Tool terms:** most tools give you whatever rights exist in the output. Protection against infringement claims is offered mainly on business or API plans (for example Adobe Firefly for enterprise, OpenAI's business and API plans, Google Vertex), with conditions to check. Consumer plans (about USD 10 to 50 a month) do not include it. Midjourney offers none found, is being sued by major studios, and shows lower-tier images in a public gallery, so it is not recommended for shipped art.
- **Recommended policy:**
  - AI for ideas and drafts.
  - A person finishes every shipped character picture.
  - The box and key art come from a named, credited artist with no AI drafts. Without budget, use a type-and-frame box instead of an AI picture.
  - Frames and graphics are made as code templates.

- A manifest for every asset.
- A full, honest "Use of AI" section, which also mentions that an AI coding assistant wrote the template code and the existing pixel art.
- If showing at conventions such as UK Games Expo matters, the final character art needs to be made by a human artist, not only finished by one.

10. Costs (estimates only; no quotes yet)

Item	Estimate	Source
Image tool	about USD 10 to 50 a month (consumer plans), or about USD 0.03 to 0.24 per image through an API	Vendor pages, via search summaries
A person's finishing	about USD 50 to 150 per finished picture	An AI reviewer's estimate, not a quote
Recommended route, 45 card portraits	about USD 2,250 to 6,750, plus the tool and the box	45 × the row above
Full illustration per card	about USD 100 to 800	LaunchBoom, Stonemaier, Pine Island guides
45 card portraits by a human illustrator	about USD 4,500 to 36,000	45 × the row above; head-and-shoulders work usually sits in the lower half
Box cover	about USD 1,000 to 6,000	Same guides
Pilot (two shipped pictures)	about USD 110 to 350 (one month of a tool plus two finished pictures)	Arithmetic from the rows above; reference sheets are not repainted

The RM500-a-month ceiling (DEC-005) covers all games, so the pilot alone may use most or all of one month, depending on the exchange rate. A Kickstarter art budget would be a separate, one-off decision. The Studio Lifecycle Companion on Drive also recommends a separate launch budget. Nothing is spent until you approve a dated quote.

11. Problems found on the way (not art; for later)

1. **Generous customers disagree:** RULES.md lists 12 (including John, Grace and Lakshmi). The game computes 13 (including Daniel, Siti, Anna and Marcus). Checked by recomputing from `data.ts`.
2. **Guest orders disagree, and RULES.md disagrees with itself.** Its event table matches the game (Influencer 2 mangoes at $12, Gym Club 3 blueberries at $8, Hotel Chef 2/1/1, all at fixed prices). Its guest-card table says 3 at $11, 4 blueberries and 3/2/2. Its next line says the event's price change still applies, but the game treats guest prices as fixed.
3. **The Drive print kit is rules version 8:** no freshness type, the old "Not so" wording, old prices.
4. **Card backs list fruits in each card's own order,** in the printed kit, the app and the online server. That makes 9 cards unique from the back, including Premium Rara Jenab. A fixed fruit order would leave 4. (Separately, the fruit set alone already settles the type for 19 of 36 cards. That is how the rules work and is not changed here.)
5. **The online leak check** tests only the amounts and prices on face-down walk-ins. It does not test that the name, freshness type or generous flag are hidden. Only the server's allow-list protects them today, and Look E's app rule depends on the name staying hidden.

6. **The face-down "shopper" face** looks like Benny (same short hair and brown shirt).
7. **Likely bug (read in code, not reproduced):** online, any favour card a rival plays is announced as "Priority!" (`src/ui/game/animator.tsx` with the server's hidden-favour placeholder).
8. **Cast leftovers:** a fox sprite, 13 unused creature and costume parts, "animal traders" comments in `data.ts`, the creative bible's identity line still listing vampires and mermaids (DEC-039 removed creatures), and 11 sprites with two accessories against the one-accessory rule.
9. **Misleading quirks:** Bargain Wei Jie inspects fruit with a magnifier, Bargain Mei Ling splashes sellers of soggy fruit, Everyday Zul "wants fresh fruit".
10. **Store, launcher and splash art** use an older, off-palette style. No font licence or credit file exists.
11. **STATE.md was out of date** (it said 1.2.0 was live; the release records say 1.7.0 was published on 2026-11-07). Its top section is updated.

Items 1, 2, 4 and 5 need small plans of their own. Each changes text, app or test code, so none is inside PLAN-048.

12. Evidence quality and limits

- **Repo claims** were checked against the code. The generous list, guest orders, back uniqueness, walk-in redaction and the leak check were re-run or re-read by hand.
- **Outside figures** come from web search summaries (section 9). Prices and terms change often.
- **Records check:** `npm run records:check` PASS on 2026-11-07.
- **NOT RUN:** a physical print test; any image generation; a factory quote; a legal opinion on Malaysian copyright; a trademark search.
- **Not applicable:** lint, types, unit tests and build. No game code changed.

13. Questions for you

Approval asked: DEC-063 (Look E) and PLAN-048 (Character Bible, card-template design, pilot prompts: no spending).

1. **Final character art:** (A, recommended) AI drafts, with every shipped picture finished by a paid, credited artist; or (B) AI drafts finished only by you, with no artist. Both must be disclosed on Kickstarter. Answering A lets me collect dated quotes for a one-off art budget outside RM500 a month. Nothing is spent until you approve one.
2. **Phone game:** keep the pixel people on the board, home screen and wardrobe, with illustrated portraits only on cards, box, store and the character pick (recommended)? Or replace the pixel people everywhere? That means redrawing the 11 wearable looks for every shopkeeper and making each portrait readable at 18 to 34 px. The 14 stall looks are not affected.
3. **Pilots:** Benny + Sailor Jia Wen (recommended), Rosa + Grandpa Tan, or Raj + Grandpa Tan?

Assumptions unless you say otherwise:

- Kickstarter means a factory-made box (open since REP-038).
- The product name stays *Fruit Market*.

- The text mismatches in section 11 are fixed by a small separate plan. Until then, PLAN-048's test print is a marked draft using the game's data.
- Jia Wen is drawn as a young woman (her sprite has long hair).
- The six expressions in your brief are drawing references and marketing art, not app frames (DEC-042).

Appendix A · Scores per criterion

Each cell shows publisher / production / identity view, from 1 (poor) to 5 (strong).

Criterion	Exchange alone	Oddballs alone	Hybrid
Kickstarter appeal	4 / 4 / 4	3 / 3 / 4	5 / 5 / 5
Originality and IP safety	2 / 2 / 2	3 / 3 / 3	4 / 4 / 3
Fit with the world	3 / 3 / 3	3 / 3 / 3	4 / 4 / 4
Distinct across 45 characters	2 / 2 / 2	5 / 5 / 4	4 / 4 / 4
AI consistency	3 / 4 / 3	2 / 2 / 2	4 / 4 / 4
Print readability	4 / 4 / 3	3 / 3 / 3	4 / 4 / 4
Phone readability	2 / 2 / 2	3 / 3 / 3	4 / 4 / 4
Cost for a solo developer	2 / 3 / 2	2 / 1 / 2	3 / 3 / 3
Stereotype safety (5 = safest)	1 / 1 / 1	3 / 2 / 3	4 / 4 / 4
Total (of 45)	**23 / 25 / 22**	**27 / 25 / 27**	**36 / 36 / 35**

REP-088 · Character names: review and choices

Game	Fruit Market 1.7.0 (rules version 9). No game, app or print-kit change.
Report date	2026-11-07 (Asia/Kuala_Lumpur)
Type	Naming review, step 1 of PLAN-048 (design only)
Request	"before proceeding, i want to rework the character name and naming style, i dont like some names, then can proceed"
Status	Waiting for your choices (section 1). The Character Bible and the rest of the design package start after that.

1. How to answer

Reply with three things, for example: *"Style A. Change 6, 13, 20, S3. Also make your Fix changes. Name 13 'Ken'."*

1. **Which names you dislike:** customers by number (1 to 36), shopkeepers S1 to S6, guests G1 to G3. Add a new name if you have one in mind; otherwise I will suggest some.
2. **Which style:** A (recommended), B or C (section 3).
3. **My fixes:** should I also make the changes marked **Fix** in section 4, even where you didn't pick the name (recommended)? Or change only the ones you choose?

2. The short version

1. Most names are fine. 21 of the 45 need no change, and 6 more keep their name and only get a new one-line joke (quirk).
2. **Three pairs sound alike** when the crier reads names aloud: Ben and Benny, Wei Jie and Jia Wen (the same two sounds), and Hafiz and Hadi. One name in each pair should change. The pilots, Benny and Jia Wen, stay.
3. **Four titles hint at a hidden card type:** Captain (Premium), Rockstar (Bargain), Dr (generous) and Farid's "Or pay" joke (Bargain). Look E's rules ban that (ART_DIRECTION §9).
4. **Some jokes are about who someone is, not what they do:** Sleepy Zul (an echo of the "lazy" stereotype), a Japanese name given judo, and two Chinese grandfathers who both grumble about prices.
5. **The culture spread is lopsided:** none of the six Chinese-named customers is Premium, and four of them read as thrifty. A rename can fix this without moving any card's type.
6. **Changing a name never changes the game.** Each card keeps its type, wants and prices, because the old name stays as a hidden ID in the code.

3. Three naming styles

	A · Job + first name, tidied (recommended)	B · Name on top, job underneath	C · Crate-label sounds (light)
Rule	One everyday job a child can picture, then the first name. No describing words, no "the", no rank, fame or money words.	Big first name in the header, the job on a smaller second line	As A, but where a title must change anyway, pick a job with the same first sound
Examples	Sailor Jia Wen · Singer Grace · Stargazer Zul · Bellboy Theo	JIA WEN / Sailor · GRACE / Singer · ZUL / Stargazer	Sailor Jia Wen · Guitarist Grace · Hiker Hui Min · Trumpeter Theo
Good	Keeps your approved pattern and the pilot title. The job helps the crier tell names apart.	Quickest to read aloud; biggest letters on the card	Most memorable on a Kickstarter page; matches lorries like "Blueberry Bonanza"
Watch	A few titles are long (up to 18 letters) and need a fit test	Replaces your approved "Job Name" rule; bare names that sound alike stand out more	Lost in translation; easy to overdo; same-letter names cluster
Changes	13 titles (4 with new first names) and 6 quirks	Same names as A; every header changes layout	About 13 to 18 titles, the same 4 first names

4. Every customer

Type: **P** Premium, **E** Everyday, **B** Bargain; **g** marks a generous customer (Fixed prices option). The type never moves with a new name. Action: **Fix** (sounds alike, hints at a type, or a stereotype), **Tidy** (title pattern only), **Option** (your call), **Keep**.

#	Today	Type	What's wrong	Suggested (style A)	Action
1	Rancher John	E	Quirk: panning gold is a prospector's job, and "coins" is a money hint. "Rancher" and "Ranger" (#23) blur aloud.	Rancher John · "Lassoes a watermelon from across the street."	Fix (quirk)
2	Librarian Robert	E			Keep
3	Chemist Daniel	E g	One of five science jobs		Keep
4	Rara Jenab	P	Name-only title (your design, allowed)		Keep
5	Scientist Aisyah	E	The vaguest of the five science jobs	Any non-science job you like	Option
6	Delivery Rider Ravi	P g	Rhymes with Devi (also Premium) and opens like Raj. Same job as Courier Farid. 19 letters.	Lifeguard Nikhil · "Saves the day, one mango at a time."	Fix
7	Diver Mei Ling	B	Quirk punishes soggy fruit, but Bargain customers buy tired fruit	Diver Mei Ling · "Comes up from every dive hungry for blueberries."	Fix (quirk)

#	Today	Type	What's wrong	Suggested (style A)	Action
8	Courier Farid	B	"Or pay" is a joke about not paying, on a Bargain card	Courier Farid · "Rings once, drops a parcel, grabs four apples."	Fix (quirk)
9	Tailor Siti	E g			Keep
10	Magician Kumar	B g	"Three wishes" is a genie, and he has two orders	Magician Kumar · "Pulls a watermelon out of his top hat."	Fix (quirk)
11	Rockstar Grace	B	Fame word on a Bargain card; also a games and drinks brand	Singer Grace	Fix
12	Paleontologist Hafiz	B	Sounds like Hadi (#32). Longest title (20). "Hafiz" is also an honorific for someone who has memorised the Quran.	Geologist Irfan · "Fills his rock bag with five blueberries."	Fix
13	Detective Wei Jie	B	Same two sounds as pilot Jia Wen. A magnifying glass suggests a fussy buyer on a Bargain card.	Detective Wai Kit (the Cantonese form of the same name) · "Solves the case of the missing watermelon." Drawn with a notebook, not a magnifying glass.	Fix
14	Florist Nurul	E			Keep
15	Dr Priya	E g	"Dr" is a status title; an Indian doctor is a stock pairing; another science job	Teacher Priya · "Brings an apple for every pupil."	Fix
16	Captain Ahmad	P	Rank word on a Premium card; his navy look copies pilot Jia Wen	Tour Guide Ahmad · "Buying mangoes for the whole tour group."	Fix
17	Photographer Sarah	E g	18 letters, may shrink on the card; close to shopkeeper Saleh		Keep (fit test)
18	Grandpa Tan	E	Quirk about cheap prices is a thrift hint, and he's one of two Chinese elders grumbling about prices	Grandpa Tan · "Tells you about the old harbour, one apple at a time."	Fix (quirk)
19	Dancing Lina	P g	A describing word, not a job	Dancer Lina	Tidy
20	Sleepy Zul	E	"Sleepy" is who he is, not what he does; echoes the "lazy" stereotype	Stargazer Zul · "Names a star after every blueberry."	Fix
21	Anna the Artist	P g	The only "Name the Job" title	Painter Anna	Tidy
22	Engineer Vijay	E g	Mild: an Indian engineer is a stock pairing		Keep
23	Ranger Hui Min	E	"Ranger" and "Rancher" (#1) blur aloud; weak quirk	Hiker Hui Min · "Hikes down the hill hungry for watermelon."	Tidy
24	Firefighter Osman	E g	17 letters		Keep (fit test)
25	Sailor Jia Wen (pilot)	E	"Pays in coins" is a money hint	Sailor Jia Wen · "Reads the mango map twice."	Fix (quirk)

#	Today	Type	What's wrong	Suggested (style A)	Action
26	DJ Rizal	E			Keep
27	Gardener Devi	P	Rhymes with Ravi (solved if #6 changes)		Keep
28	Fisherman Marcus	E g	One of four sea jobs		Keep
29	Professor Fatimah	E	17 letters		Keep (fit test)
30	Bellboy Ben	E	Sounds like shopkeeper Benny (pilot)	Bellboy Theo	Fix
31	Fencer Yuki	E	"One clean stroke" reads like a samurai joke	Fencer Yuki · "Lunges for the last mango."	Option (quirk)
32	Builder Hadi	E	Sounds like Hafiz (solved if #12 changes)		Keep
33	Baker Lakshmi	E			Keep
34	Judoka Kenji	E g	A Japanese name with a Japanese sport ties a culture to a job	Juggler Kenji (quirk kept: "Flips a mango, catches it, pays for it.")	Fix
35	Herbalist Mira	E			Keep
36	Uncle Hock Seng	E g	The second Chinese elder with a price grumble	Barber Hock Seng · "Knows every customer's name, and their favourite apple."	Tidy

5. Shopkeepers and guests

#	Today	Perk	What's wrong	Suggested	Action
S1	Rosa	Haggler			Keep
S2	Benny (pilot)	Big Basket			Keep
S3	Chen	Charmer	Chen is usually a surname, and the cast rule asks for first names. You approved Chen (DEC-040).	Your call; for example Jun	Option
S4	Saleh	Nest Egg	The name is fine. His lines are the only Malay ones, all about saving, and "Every sen counts" uses a real currency.	Keep the name; rewrite his lines in the Character Bible	Fix (lines)
S5	Raj	Bargain Hunter	The name is fine. "Cheap and cheerful!" plus the bargain perk echoes a stingy-trader stereotype.	Keep the name; rewrite his lines in the Character Bible	Fix (lines)
S6	Kiran	Tip Jar			Keep
G1	The Influencer	guest			Keep
G2	The Gym Club	guest	A group, not a person (fine for a guest)		Keep

#	Today	Perk	What's wrong	Suggested	Action
G3	The Hotel Chef	guest			Keep

A wider world (optional): the cast is Malaysia's mix plus English names: among the 36 customers, 12 Malay, 8 English, 6 Chinese, 6 Indian, 2 Japanese and 2 that fit several cultures. For "a harbour where the whole world comes to shop", some new names could come from places not yet in the cast (Africa, Latin America, Korea, the Philippines, Europe). The suggestions above keep today's mix: each renamed character keeps a name from the same background.

6. Rules every new name must follow

- A job or hobby a child can picture and an artist can draw as one prop. The joke is what they do, never who they are (DEC-018, DEC-039, DEC-041).
- No rank, fame, money, thrift or fussy-taste words, so no name hints at Premium, Bargain or generous.
- Sounds clearly different from every other name when read aloud.
- About 16 letters or fewer, so the card header stays large.
- No culture tied to a job, and no famous people, brands or well-known fictional characters.
- The pilots keep their approved looks: Benny (broad, bearded, Big Basket) and Sailor Jia Wen (navy, sailor's cap, rolled mango map).

7. What happens after you choose

1. I record your choices as a decision with your exact words, and update the creative bible's cast list.
2. The design package continues with the final names: Character Bible v1.0 for the pilots, reference sheets and prompts, card mock-ups and mobile specifications (PLAN-048).
3. **Later, in a separate implementation plan:** the names change in the game, the printed kit and RULES.md. Old names stay as hidden IDs, so saved games, online rooms and every card's type, wants and prices stay the same. RULES.md's list of generous customers is rebuilt from the game at the same time (today it lists 12, and the game has 13; REP-087 §11).

8. Evidence and limits

- **Sources:** names, titles, quirks, types and generous customers were checked against `src/engine/data.ts` and the game's generous rule. Sprite details come from `src/ui/pixel.tsx`.
- **Method:** one AI audit of all 45 names, two AI style proposals (Kickstarter appeal; clarity read aloud), and one AI reviewer who tried to break the proposals. Their fixes are in this version.
- **NOT RUN:** a read-aloud test with a real crier, and a header-fit test on a printed card. Both are needed before names are final.